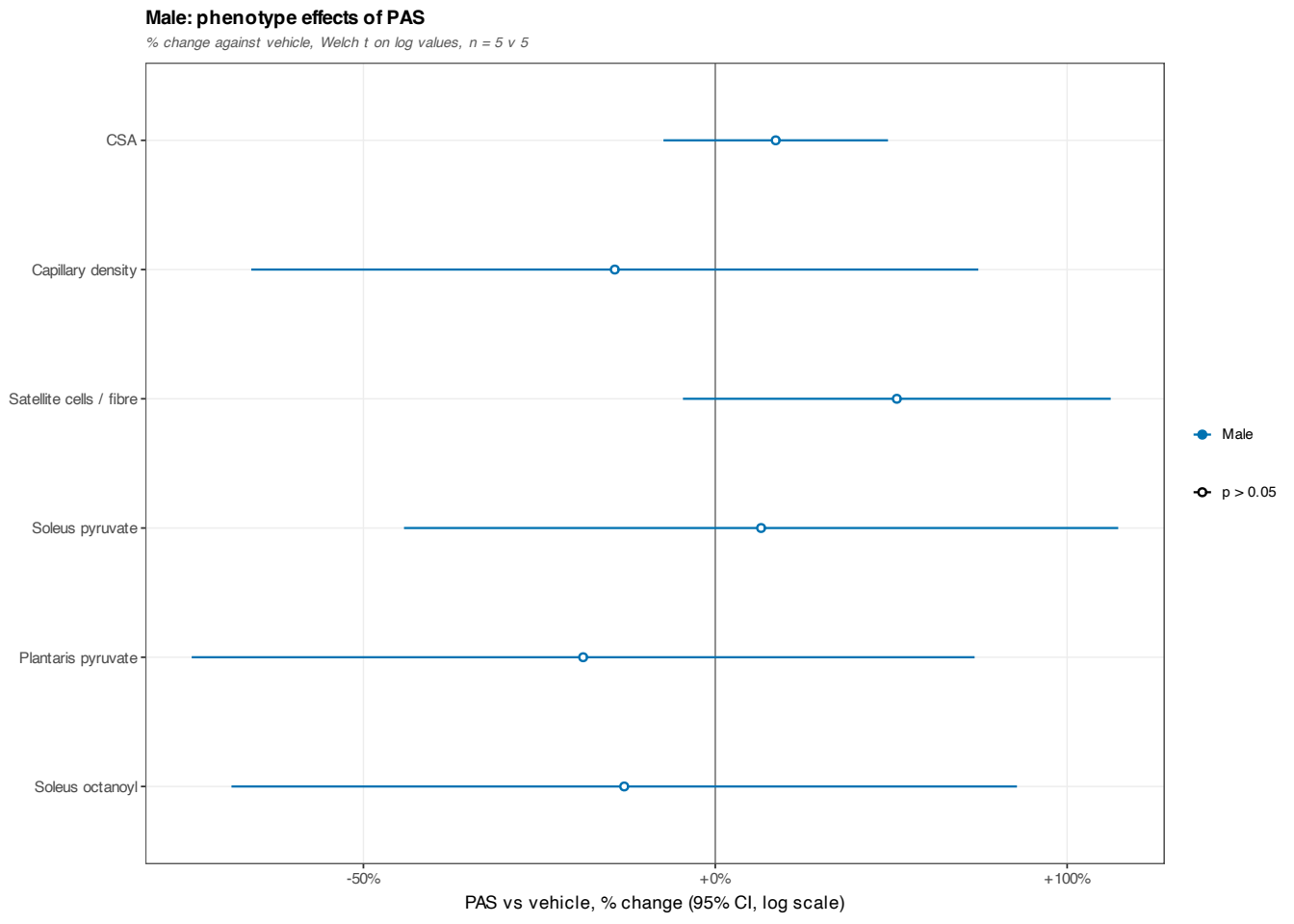
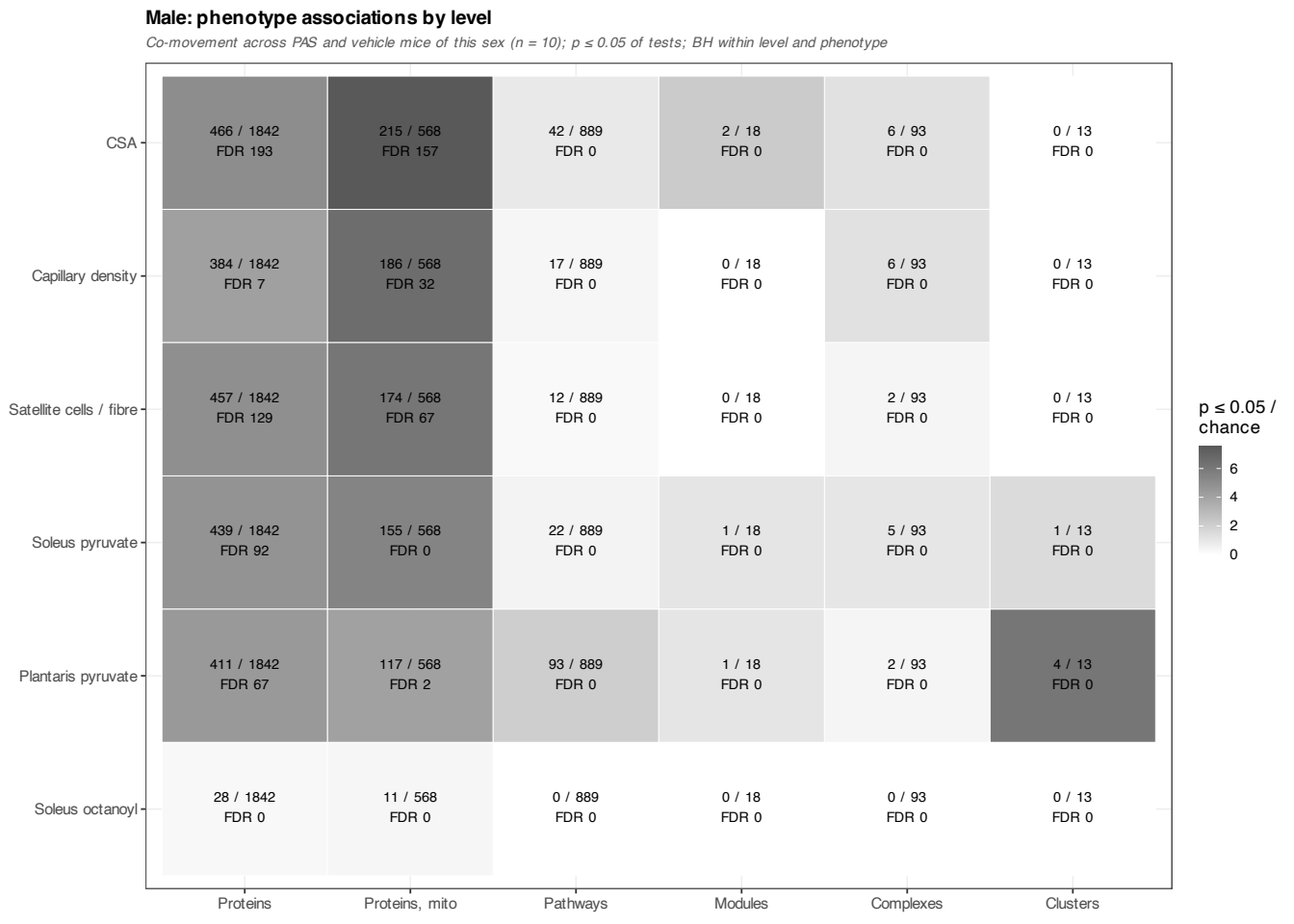
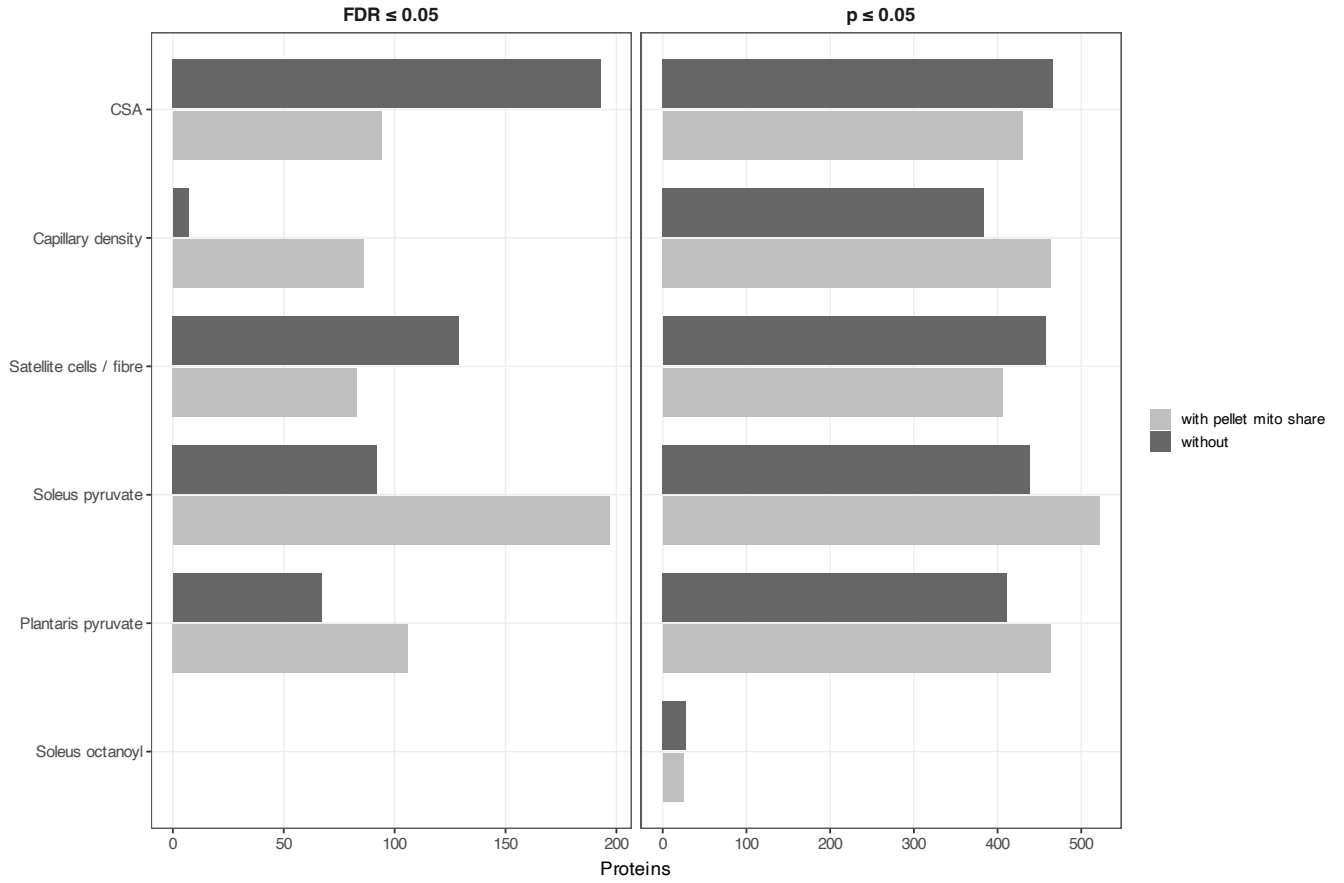

Male: phenotype effects of PAS
% change against vehicle, Welch t on log values, n = 5 v 5
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
-50%
+0%
+100%
PAS vs vehicle, % change (95% CI, log scale)
Male
p > 0.05


Male: phenotype associations by level
Co-movement across PAS and vehicle mice of this sex (n = 10); p ≤ 0.05 of tests; BH within level and phenotype
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
466 / 1842
FDR 193
215 / 568
FDR 157
42 / 889
FDR 0
2 / 18
FDR 0
6 / 93
FDR 0
0 / 13
FDR 0
384 / 1842
FDR 7
186 / 568
FDR 32
17 / 889
FDR 0
0 / 18
FDR 0
6 / 93
FDR 0
0 / 13
FDR 0
457 / 1842
FDR 129
174 / 568
FDR 67
12 / 889
FDR 0
0 / 18
FDR 0
2 / 93
FDR 0
0 / 13
FDR 0
439 / 1842
FDR 92
155 / 568
FDR 0
22 / 889
FDR 0
1 / 18
FDR 0
5 / 93
FDR 0
1 / 13
FDR 0
411 / 1842
FDR 67
117 / 568
FDR 2
93 / 889
FDR 0
1 / 18
FDR 0
2 / 93
FDR 0
4 / 13
FDR 0
28 / 1842
FDR 0
11 / 568
FDR 0
0 / 889
FDR 0
0 / 18
FDR 0
0 / 93
FDR 0
0 / 13
FDR 0
Proteins
Proteins, mito
Pathways
Modules
Complexes
Clusters
p ≤ 0.05 / chance
6
4
2
0

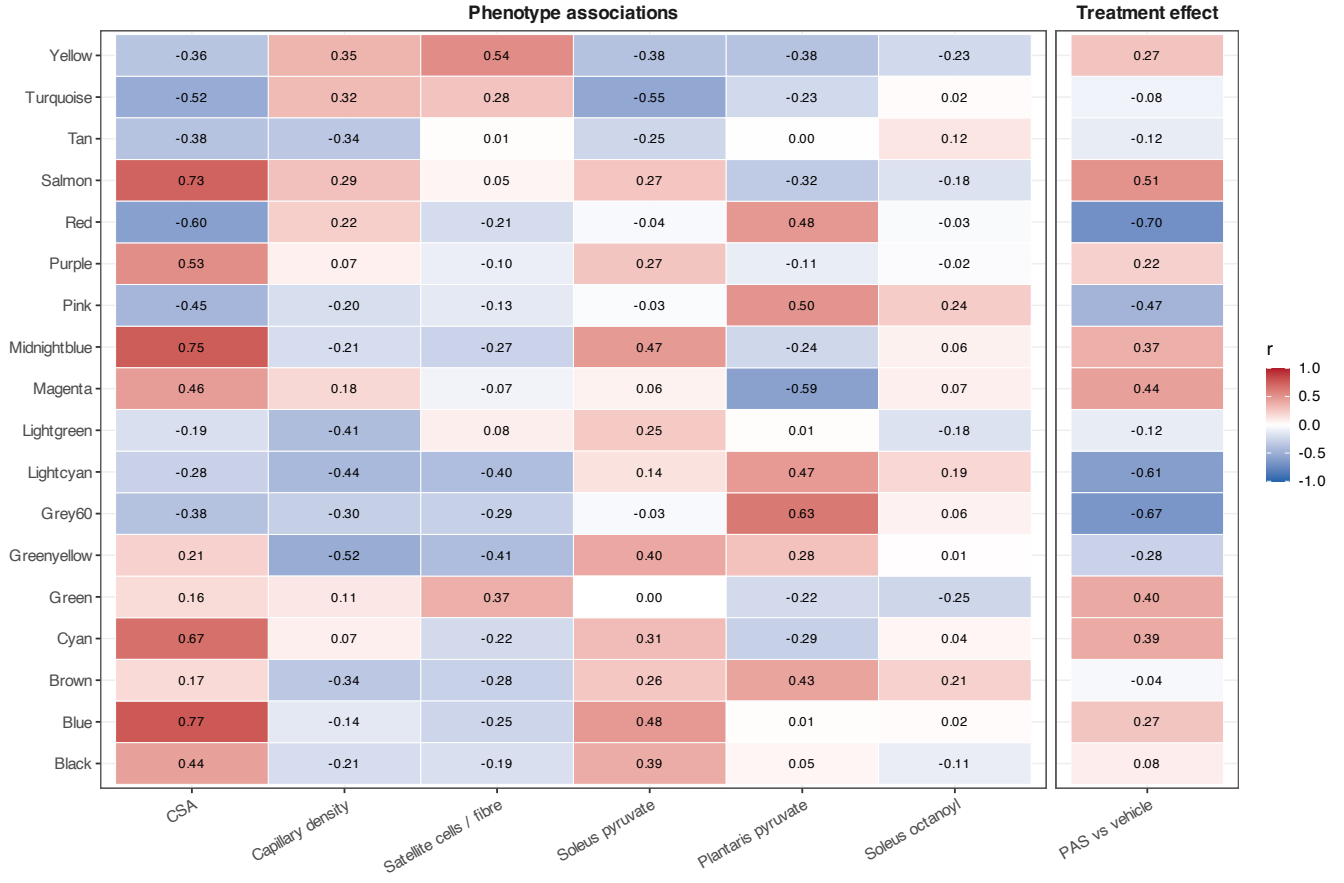
**Check: protein associations with and without pellet mitochondrial share**

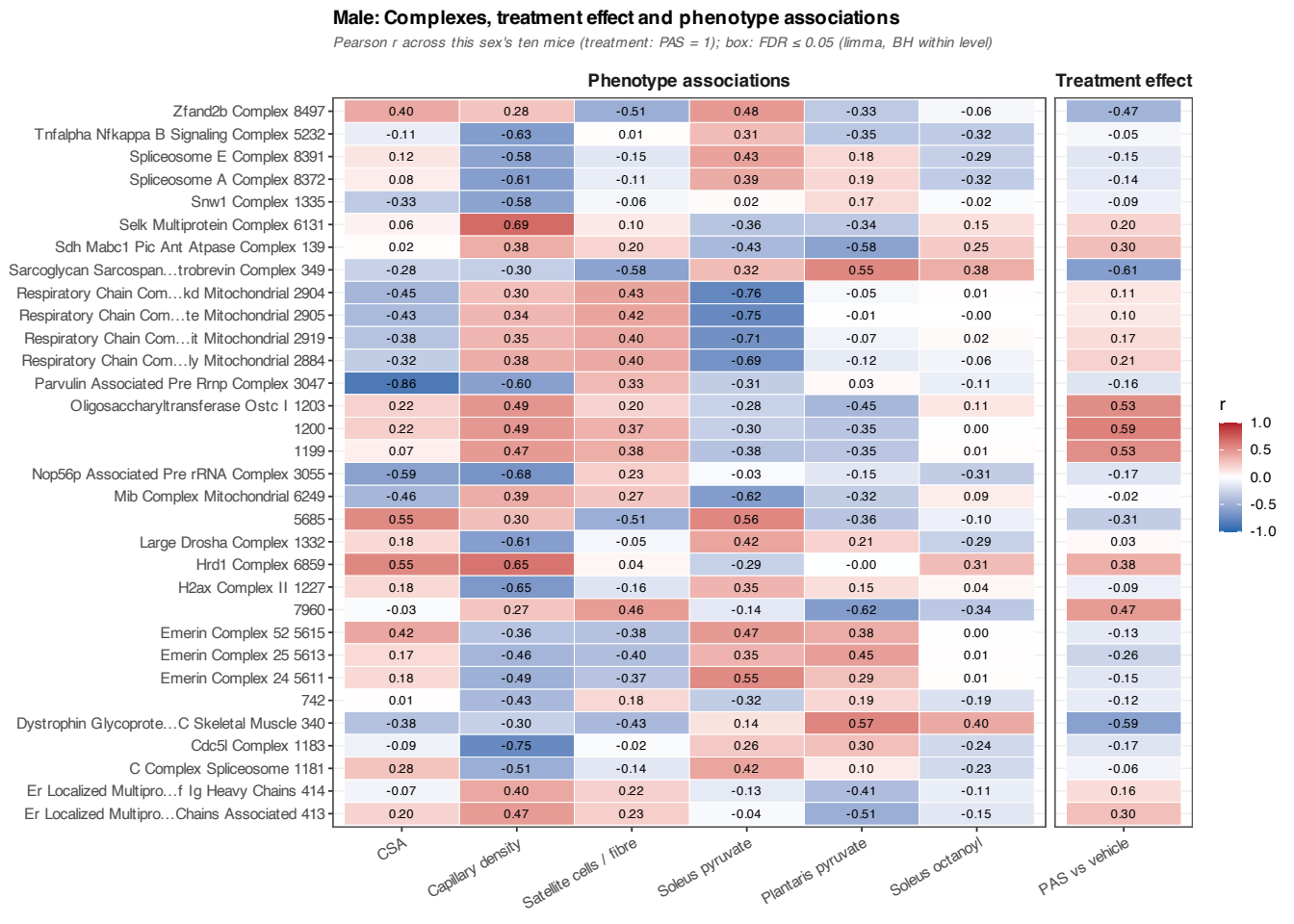
*Share centred within group, added to the association model*

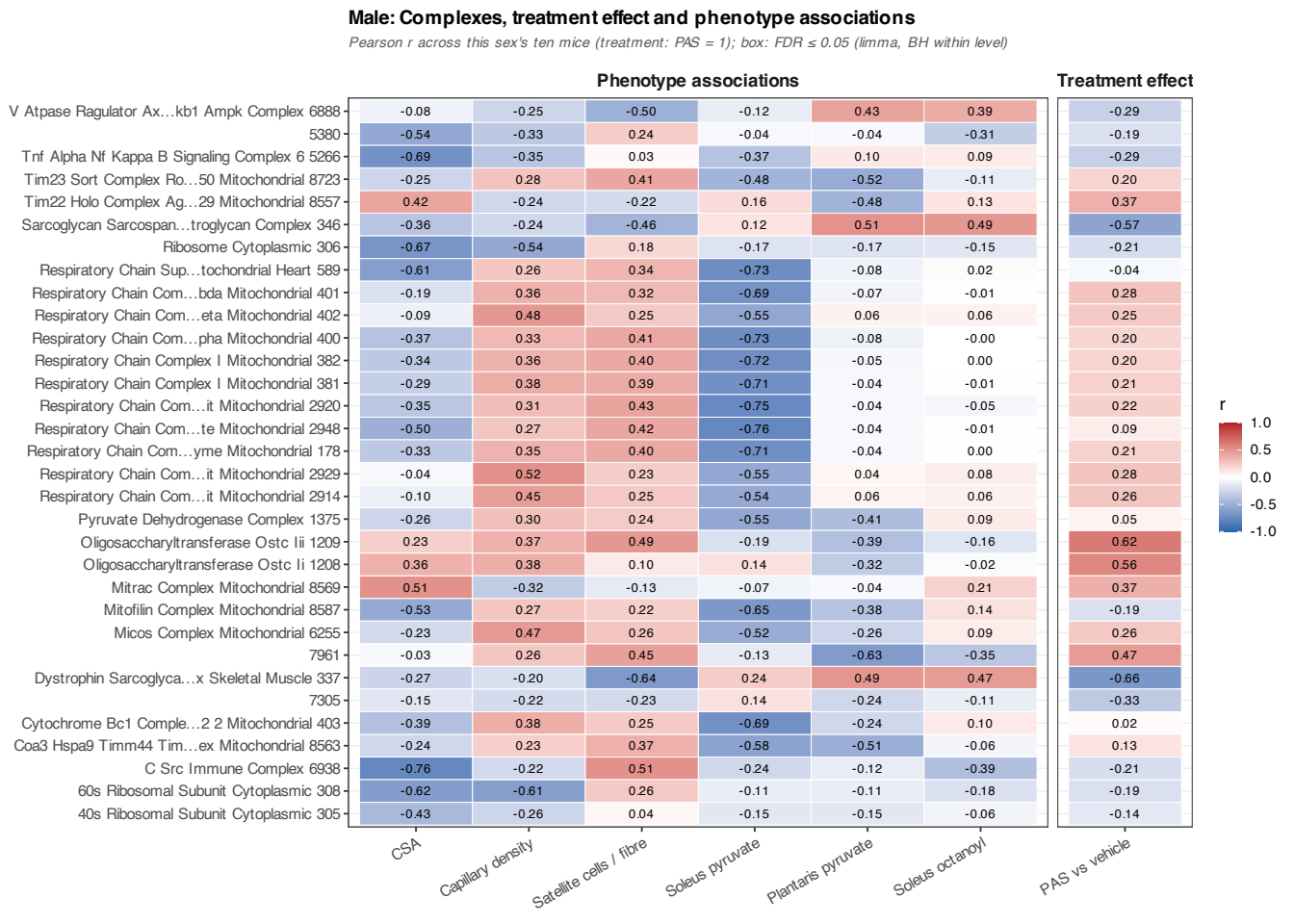


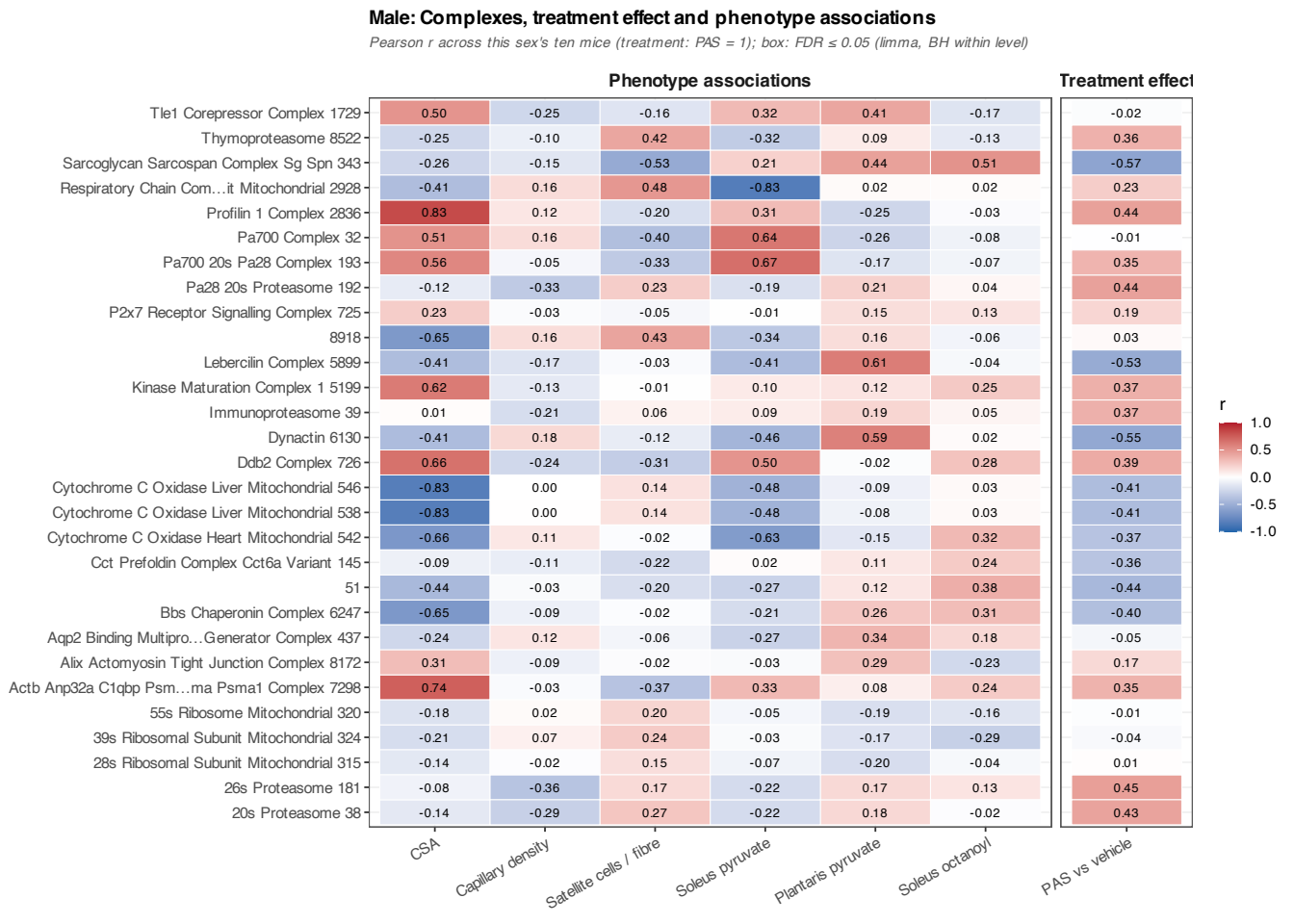
**Male: Modules, treatment effect and phenotype associations**

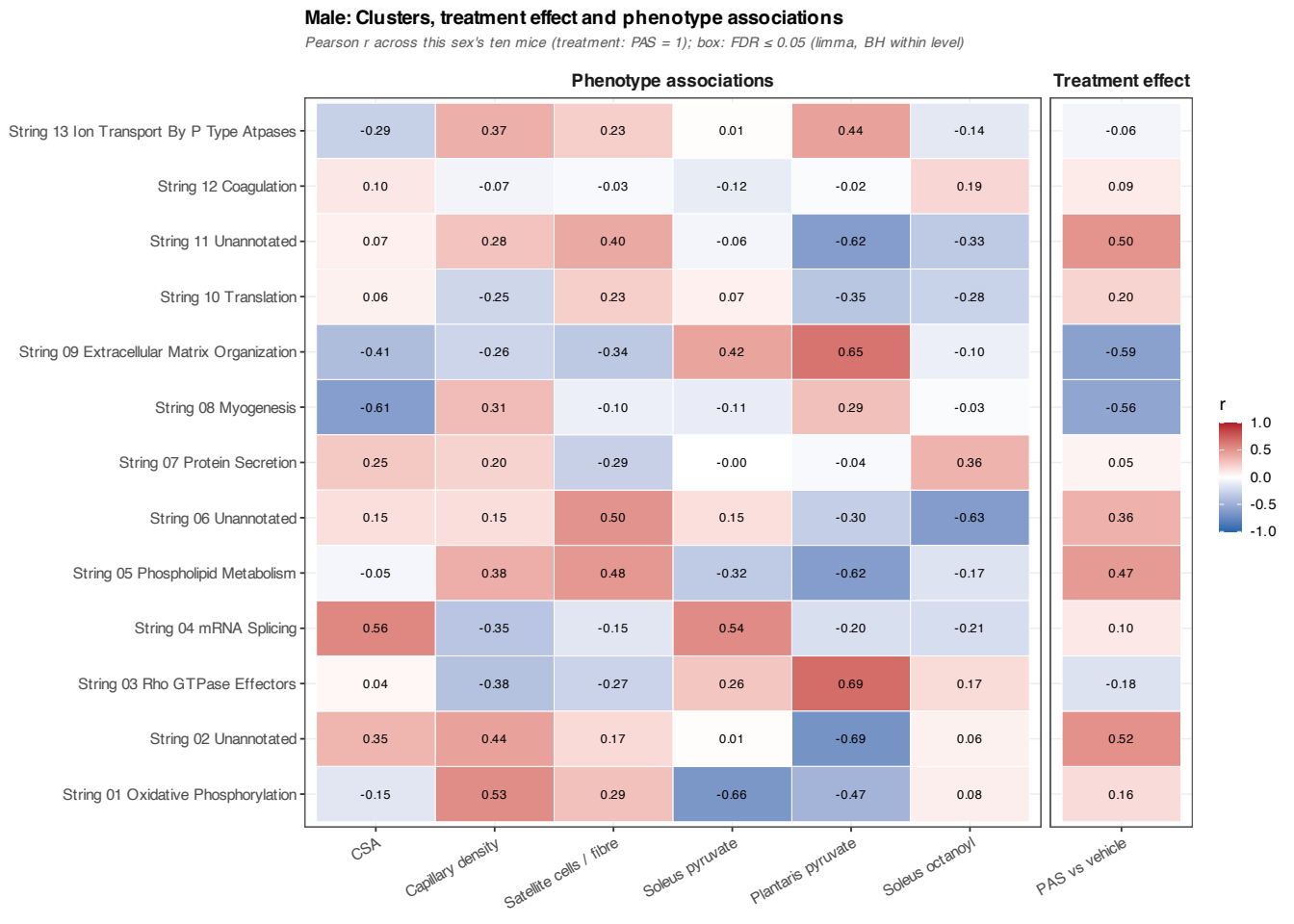
*Pearson r across this sex's ten mice (treatment: PAS = 1); box: FDR ≤ 0.05 (limma, BH within level)*




Male: Complexes, treatment effect and phenotype associations
Pearson r across this sex's ten mice (treatment: PAS = 1); box: FDR ≤ 0.05 (limma, BH within level)
Phenotype associations
Treatment effect
Zfand2b Complex 8497 0.40 0.28 -0.51 0.48 -0.33 -0.06 -0.47
Tnfalpha Nfkappa B Signaling Complex 5232 -0.11 -0.63 0.01 0.31 -0.35 -0.32 -0.05
Spliceosome E Complex 8391 0.12 -0.58 -0.15 0.43 0.18 -0.29 -0.15
Spliceosome A Complex 8372 0.08 -0.61 -0.11 0.39 0.19 -0.32 -0.14
Snw1 Complex 1335 -0.33 -0.58 -0.06 0.02 0.17 -0.02 -0.09
Selk Multiprotein Complex 6131 0.06 0.69 0.10 -0.36 -0.34 0.15 0.20
Sdh Mabc1 Pic Ant Atpase Complex 139 0.02 0.38 0.20 -0.43 -0.58 0.25 0.30
Sarcoglycan Sarcospan…trobrevin Complex 349 -0.28 -0.30 -0.58 0.32 0.55 0.38 -0.61
Respiratory Chain Com…kd Mitochondrial 2904 -0.45 0.30 0.43 -0.76 -0.05 0.01 0.11
Respiratory Chain Com…te Mitochondrial 2905 -0.43 0.34 0.42 -0.75 -0.01 -0.00 0.10
Respiratory Chain Com…it Mitochondrial 2919 -0.38 0.35 0.40 -0.71 -0.07 0.02 0.17
Respiratory Chain Com…ly Mitochondrial 2884 -0.32 0.38 0.40 -0.69 -0.12 -0.06 0.21
Parvulin Associated Pre Rrnp Complex 3047 -0.86 -0.60 0.33 -0.31 0.03 -0.11 -0.16
Oligosaccharyltransferase Ostc I 1203 0.22 0.49 0.20 -0.28 -0.45 0.11 0.53
1200 0.22 0.49 0.37 -0.30 -0.35 0.00 0.59
1199 0.07 0.47 0.38 -0.38 -0.35 0.01 0.53
Nop56p Associated Pre rRNA Complex 3055 -0.59 -0.68 0.23 -0.03 -0.15 -0.31 -0.17
Mib Complex Mitochondrial 6249 -0.46 0.39 0.27 -0.62 -0.32 0.09 -0.02
5685 0.55 0.30 -0.51 0.56 -0.36 -0.10 -0.31
Large Drosha Complex 1332 0.18 -0.61 -0.05 0.42 0.21 -0.29 0.03
Hrd1 Complex 6859 0.55 0.65 0.04 -0.29 -0.00 0.31 0.38
H2ax Complex II 1227 0.18 -0.65 -0.16 0.35 0.15 0.04 -0.09
7960 -0.03 0.27 0.46 -0.14 -0.62 -0.34 0.47
Emerin Complex 52 5615 0.42 -0.36 -0.38 0.47 0.38 0.00 -0.13
Emerin Complex 25 5613 0.17 -0.46 -0.40 0.35 0.45 0.01 -0.26
Emerin Complex 24 5611 0.18 -0.49 -0.37 0.55 0.29 0.01 -0.15
742 0.01 -0.43 0.18 -0.32 0.19 -0.19 -0.12
Dystrophin Glycoprote…C Skeletal Muscle 340 -0.38 -0.30 -0.43 0.14 0.57 0.40 -0.59
Cdc5l Complex 1183 -0.09 -0.75 -0.02 0.26 0.30 -0.24 -0.17
C Complex Spliceosome 1181 0.28 -0.51 -0.14 0.42 0.10 -0.23 -0.06
Er Localized Multipro…f Ig Heavy Chains 414 -0.07 0.40 0.22 -0.13 -0.41 -0.11 0.16
Er Localized Multipro…Chains Associated 413 0.20 0.47 0.23 -0.04 -0.51 -0.15 0.30
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
PAS vs vehicle
r
1.0
0.5
0.0
-0.5
-1.0

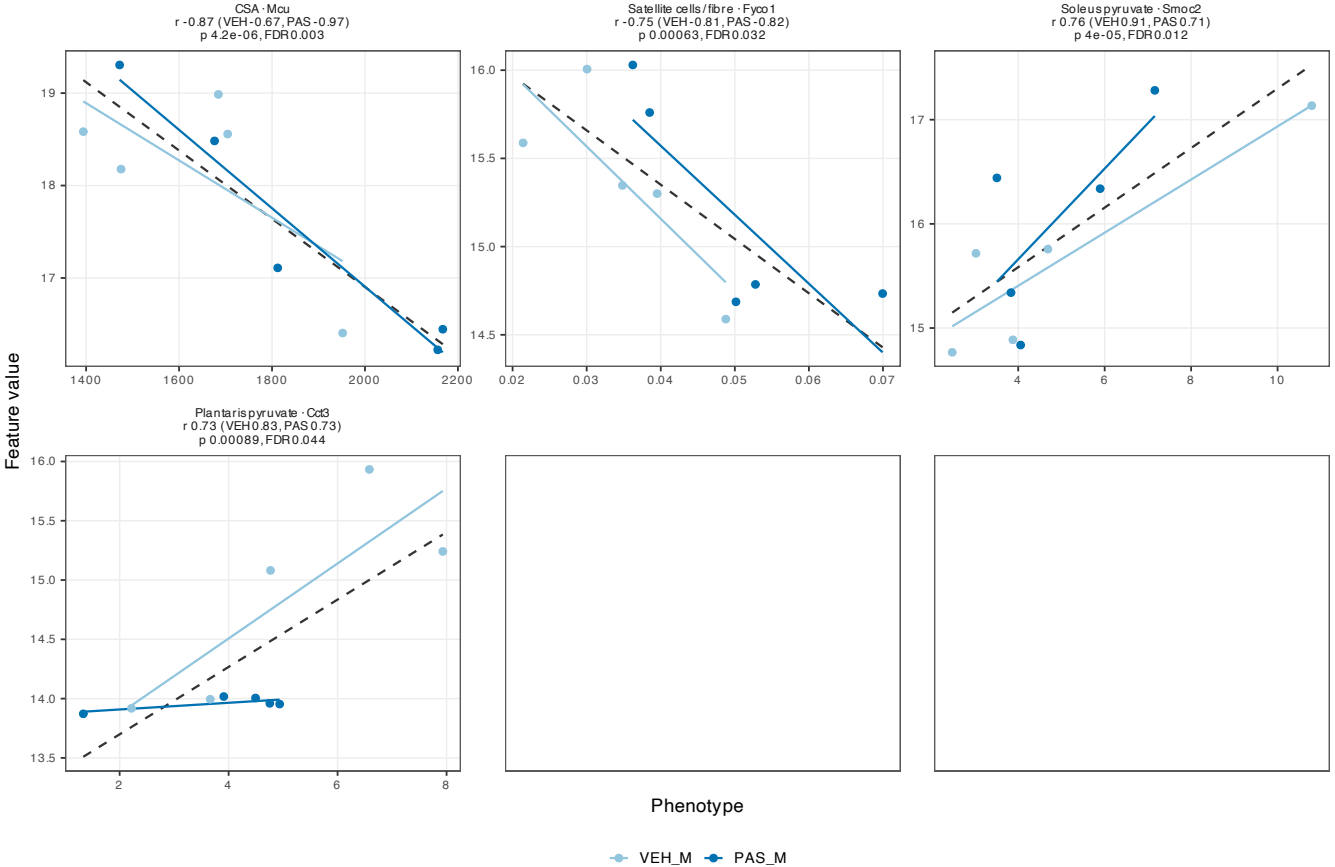

Male: Complexes, treatment effect and phenotype associations
Pearson r across this sex's ten mice (treatment: PAS = 1); box: FDR ≤ 0.05 (limma, BH within level)
Phenotype associations
Treatment effect
V Atpase Ragulator Ax…kb1 Ampk Complex 6888 -0.08 -0.25 -0.50 -0.12 0.43 0.39 -0.29
5380 -0.54 -0.33 0.24 -0.04 -0.04 -0.31 -0.19
Tnf Alpha Nf Kappa B Signaling Complex 6 5266 -0.69 -0.35 0.03 -0.37 0.10 0.09 -0.29
Tim23 Sort Complex Ro…50 Mitochondrial 8723 -0.25 0.28 0.41 -0.48 -0.52 -0.11 0.20
Tim22 Holo Complex Ag…29 Mitochondrial 8557 0.42 -0.24 -0.22 0.16 -0.48 0.13 0.37
Sarcoglycan Sarcospan…troglycan Complex 346 -0.36 -0.24 -0.46 0.12 0.51 0.49 -0.57
Ribosome Cytoplasmic 306 -0.67 -0.54 0.18 -0.17 -0.17 -0.15 -0.21
Respiratory Chain Sup…tochondrial Heart 589 -0.61 0.26 0.34 -0.73 -0.08 0.02 -0.04
Respiratory Chain Com…bda Mitochondrial 401 -0.19 0.36 0.32 -0.69 -0.07 -0.01 0.28
Respiratory Chain Com…eta Mitochondrial 402 -0.09 0.48 0.25 -0.55 0.06 0.06 0.25
Respiratory Chain Com…pha Mitochondrial 400 -0.37 0.33 0.41 -0.73 -0.08 -0.00 0.20
Respiratory Chain Complex I Mitochondrial 382 -0.34 0.36 0.40 -0.72 -0.05 0.00 0.20
Respiratory Chain Complex I Mitochondrial 381 -0.29 0.38 0.39 -0.71 -0.04 -0.01 0.21
Respiratory Chain Com…it Mitochondrial 2920 -0.35 0.31 0.43 -0.75 -0.04 -0.05 0.22
Respiratory Chain Com…te Mitochondrial 2948 -0.50 0.27 0.42 -0.76 -0.04 -0.01 0.09
Respiratory Chain Com…yme Mitochondrial 178 -0.33 0.35 0.40 -0.71 -0.04 0.00 0.21
Respiratory Chain Com…it Mitochondrial 2929 -0.04 0.52 0.23 -0.55 0.04 0.08 0.28
Respiratory Chain Com…it Mitochondrial 2914 -0.10 0.45 0.25 -0.54 0.06 0.06 0.26
Pyruvate Dehydrogenase Complex 1375 -0.26 0.30 0.24 -0.55 -0.41 0.09 0.05
Oligosaccharyltransferase Ostc Iii 1209 0.23 0.37 0.49 -0.19 -0.39 -0.16 0.62
Oligosaccharyltransferase Ostc Ii 1208 0.36 0.38 0.10 0.14 -0.32 -0.02 0.56
Mitrac Complex Mitochondrial 8569 0.51 -0.32 -0.13 -0.07 -0.04 0.21 0.37
Mitofilin Complex Mitochondrial 8587 -0.53 0.27 0.22 -0.65 -0.38 0.14 -0.19
Micos Complex Mitochondrial 6255 -0.23 0.47 0.26 -0.52 -0.26 0.09 0.26
7961 -0.03 0.26 0.45 -0.13 -0.63 -0.35 0.47
Dystrophin Sarcoglyca…x Skeletal Muscle 337 -0.27 -0.20 -0.64 0.24 0.49 0.47 -0.66
7305 -0.15 -0.22 -0.23 0.14 -0.24 -0.11 -0.33
Cytochrome Bc1 Comple…2 2 Mitochondrial 403 -0.39 0.38 0.25 -0.69 -0.24 0.10 0.02
Coa3 Hspa9 Timm44 Tim…ex Mitochondrial 8563 -0.24 0.23 0.37 -0.58 -0.51 -0.06 0.13
C Src Immune Complex 6938 -0.76 -0.22 0.51 -0.24 -0.12 -0.39 -0.21
60s Ribosomal Subunit Cytoplasmic 308 -0.62 -0.61 0.26 -0.11 -0.11 -0.18 -0.19
40s Ribosomal Subunit Cytoplasmic 305 -0.43 -0.26 0.04 -0.15 -0.15 -0.06 -0.14
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
PAS vs vehicle
r
1.0
0.5
0.0
-0.5
-1.0


Male: Complexes, treatment effect and phenotype associations
Pearson r across this sex's ten mice (treatment: PAS = 1); box: FDR ≤ 0.05 (limma, BH within level)
Phenotype associations
Treatment effect
Tle1 Corepressor Complex 1729 0.50 -0.25 -0.16 0.32 0.41 -0.17 -0.02
Thymoproteasome 8522 -0.25 -0.10 0.42 -0.32 0.09 -0.13 0.36
Sarcoglycan Sarcospan Complex Sg Spn 343 -0.26 -0.15 -0.53 0.21 0.44 0.51 -0.57
Respiratory Chain Com…it Mitochondrial 2928 -0.41 0.16 0.48 -0.83 0.02 0.02 0.23
Profilin 1 Complex 2836 0.83 0.12 -0.20 0.31 -0.25 -0.03 0.44
Pa700 Complex 32 0.51 0.16 -0.40 0.64 -0.26 -0.08 -0.01
Pa700 20s Pa28 Complex 193 0.56 -0.05 -0.33 0.67 -0.17 -0.07 0.35
Pa28 20s Proteasome 192 -0.12 -0.33 0.23 -0.19 0.21 0.04 0.44
P2x7 Receptor Signalling Complex 725 0.23 -0.03 -0.05 -0.01 0.15 0.13 0.19
8918 -0.65 0.16 0.43 -0.34 0.16 -0.06 0.03
Lebercilin Complex 5899 -0.41 -0.17 -0.03 -0.41 0.61 -0.04 -0.53
Kinase Maturation Complex 1 5199 0.62 -0.13 -0.01 0.10 0.12 0.25 0.37
Immunoproteasome 39 0.01 -0.21 0.06 0.09 0.19 0.05 0.37
Dynactin 6130 -0.41 0.18 -0.12 -0.46 0.59 0.02 -0.55
Ddb2 Complex 726 0.66 -0.24 -0.31 0.50 -0.02 0.28 0.39
Cytochrome C Oxidase Liver Mitochondrial 546 -0.83 0.00 0.14 -0.48 -0.09 0.03 -0.41
Cytochrome C Oxidase Liver Mitochondrial 538 -0.83 0.00 0.14 -0.48 -0.08 0.03 -0.41
Cytochrome C Oxidase Heart Mitochondrial 542 -0.66 0.11 -0.02 -0.63 -0.15 0.32 -0.37
Cct Prefoldin Complex Cct6a Variant 145 -0.09 -0.11 -0.22 0.02 0.11 0.24 -0.36
51 -0.44 -0.03 -0.20 -0.27 0.12 0.38 -0.44
Bbs Chaperonin Complex 6247 -0.65 -0.09 -0.02 -0.21 0.26 0.31 -0.40
Aqp2 Binding Multipro…Generator Complex 437 -0.24 0.12 -0.06 -0.27 0.34 0.18 -0.05
Alix Actomyosin Tight Junction Complex 8172 0.31 -0.09 -0.02 -0.03 0.29 -0.23 0.17
Actb Anp32a C1qbp Psm…ma Psma1 Complex 7298 0.74 -0.03 -0.37 0.33 0.08 0.24 0.35
55s Ribosome Mitochondrial 320 -0.18 0.02 0.20 -0.05 -0.19 -0.16 -0.01
39s Ribosomal Subunit Mitochondrial 324 -0.21 0.07 0.24 -0.03 -0.17 -0.29 -0.04
28s Ribosomal Subunit Mitochondrial 315 -0.14 -0.02 0.15 -0.07 -0.20 -0.04 0.01
26s Proteasome 181 -0.08 -0.36 0.17 -0.22 0.17 0.13 0.45
20s Proteasome 38 -0.14 -0.29 0.27 -0.22 0.18 -0.02 0.43
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
PAS vs vehicle
r
1.0
0.5
0.0
-0.5
-1.0


Male: Clusters, treatment effect and phenotype associations
Pearson r across this sex's ten mice (treatment: PAS = 1); box: FDR ≤ 0.05 (limma, BH within level)
Phenotype associations
Treatment effect
String 13 Ion Transport By P Type Atpases
String 12 Coagulation
String 11 Unannotated
String 10 Translation
String 09 Extracellular Matrix Organization
String 08 Myogenesis
String 07 Protein Secretion
String 06 Unannotated
String 05 Phospholipid Metabolism
String 04 mRNA Splicing
String 03 Rho GTPase Effectors
String 02 Unannotated
String 01 Oxidative Phosphorylation
-0.29 0.37 0.23 0.01 0.44 -0.14 -0.06
0.10 -0.07 -0.03 -0.12 -0.02 0.19 0.09
0.07 0.28 0.40 -0.06 -0.62 -0.33 0.50
0.06 -0.25 0.23 0.07 -0.35 -0.28 0.20
-0.41 -0.26 -0.34 0.42 0.65 -0.10 -0.59
-0.61 0.31 -0.10 -0.11 0.29 -0.03 -0.56
0.25 0.20 -0.29 -0.00 -0.04 0.36 0.05
0.15 0.15 0.50 0.15 -0.30 -0.63 0.36
-0.05 0.38 0.48 -0.32 -0.62 -0.17 0.47
0.56 -0.35 -0.15 0.54 -0.20 -0.21 0.10
0.04 -0.38 -0.27 0.26 0.69 0.17 -0.18
0.35 0.44 0.17 0.01 -0.69 0.06 0.52
-0.15 0.53 0.29 -0.66 -0.47 0.08 0.16
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
PAS vs vehicle
r
1.0
0.5
0.0
-0.5
-1.0

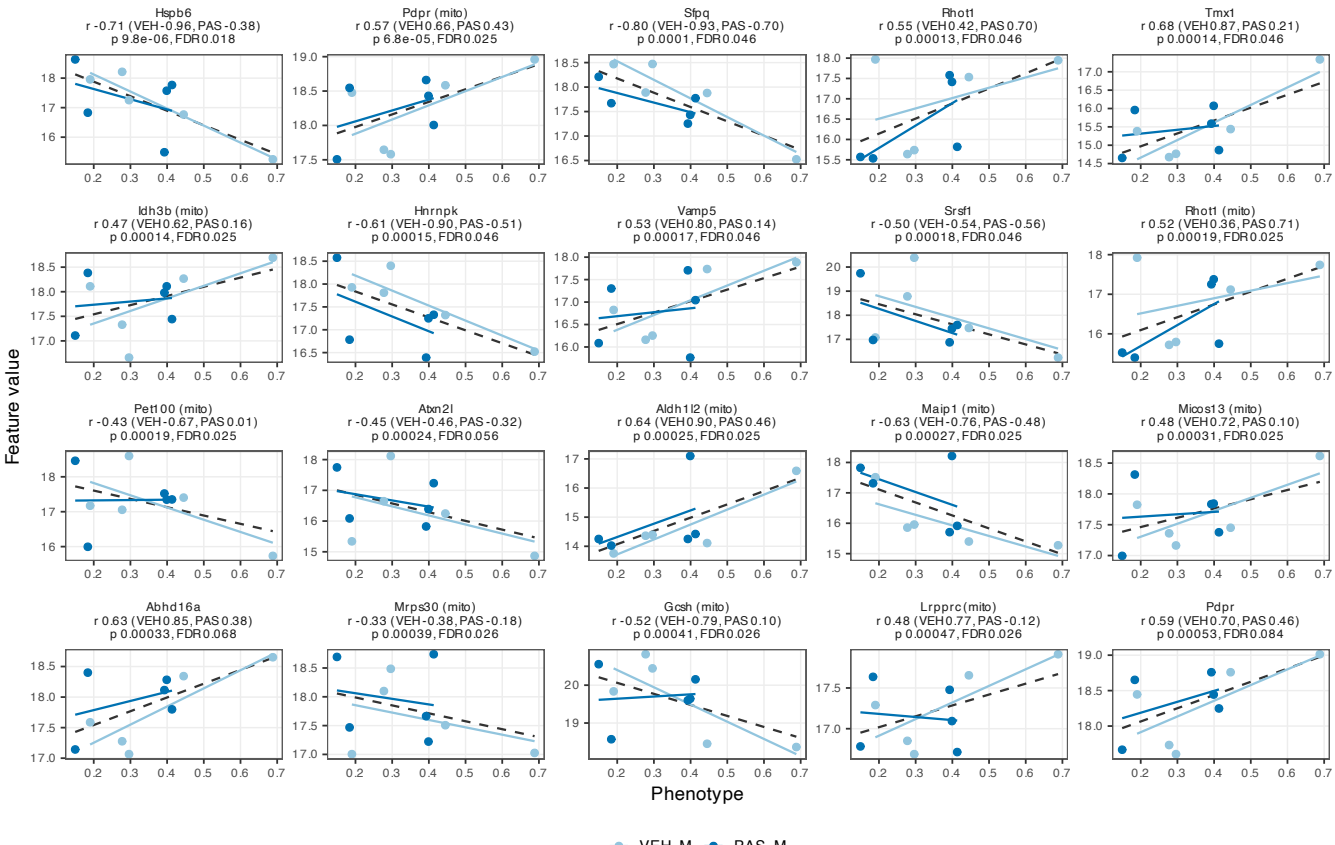
**Male: strongest robust protein per phenotype**

*Smallest association p that keeps its sign and p ≤ 0.05 with any one mouse left out*



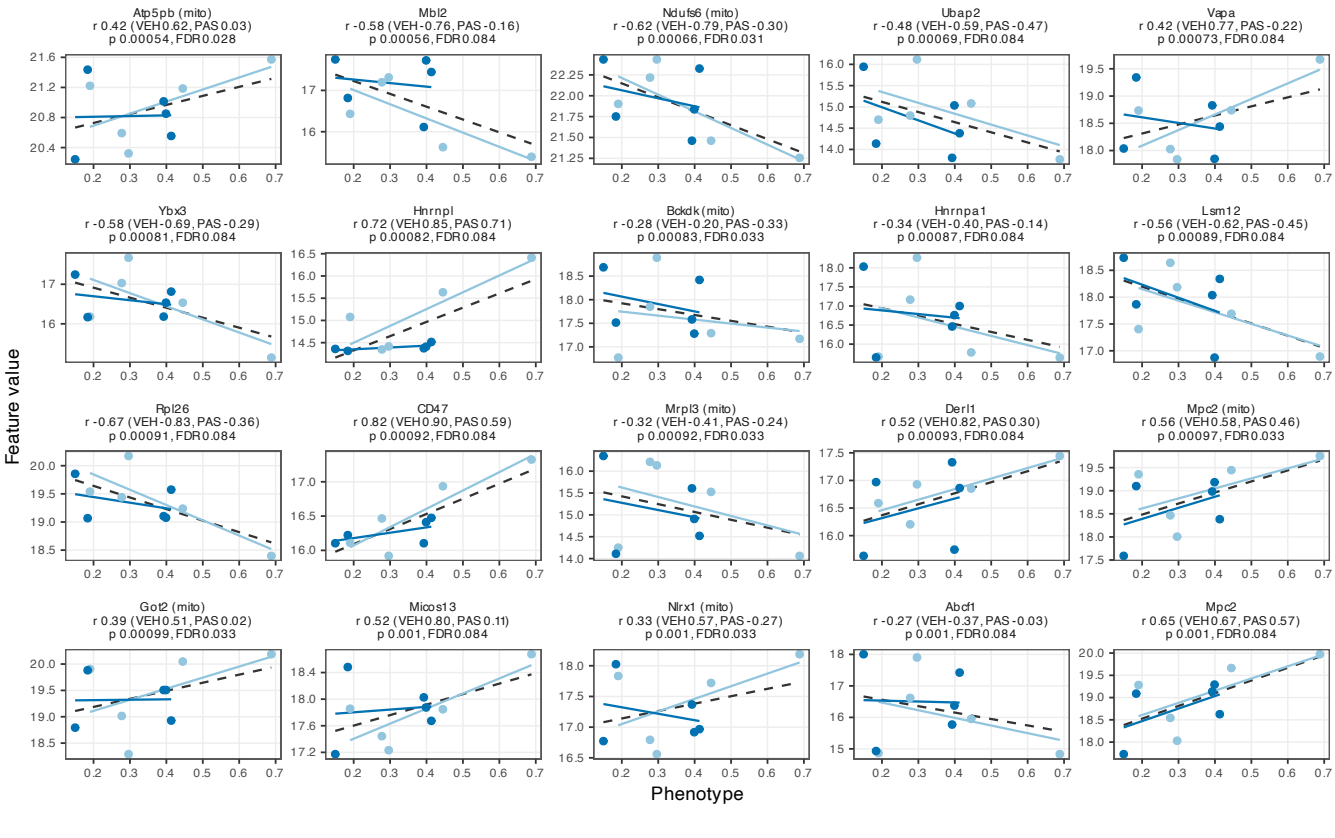
**Male: Proteins associated with Capillary density, every association at p ≤ 0.05 (n = 570)**

*Phenotype effect of PAS in this sex: -18% (p 0.54), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



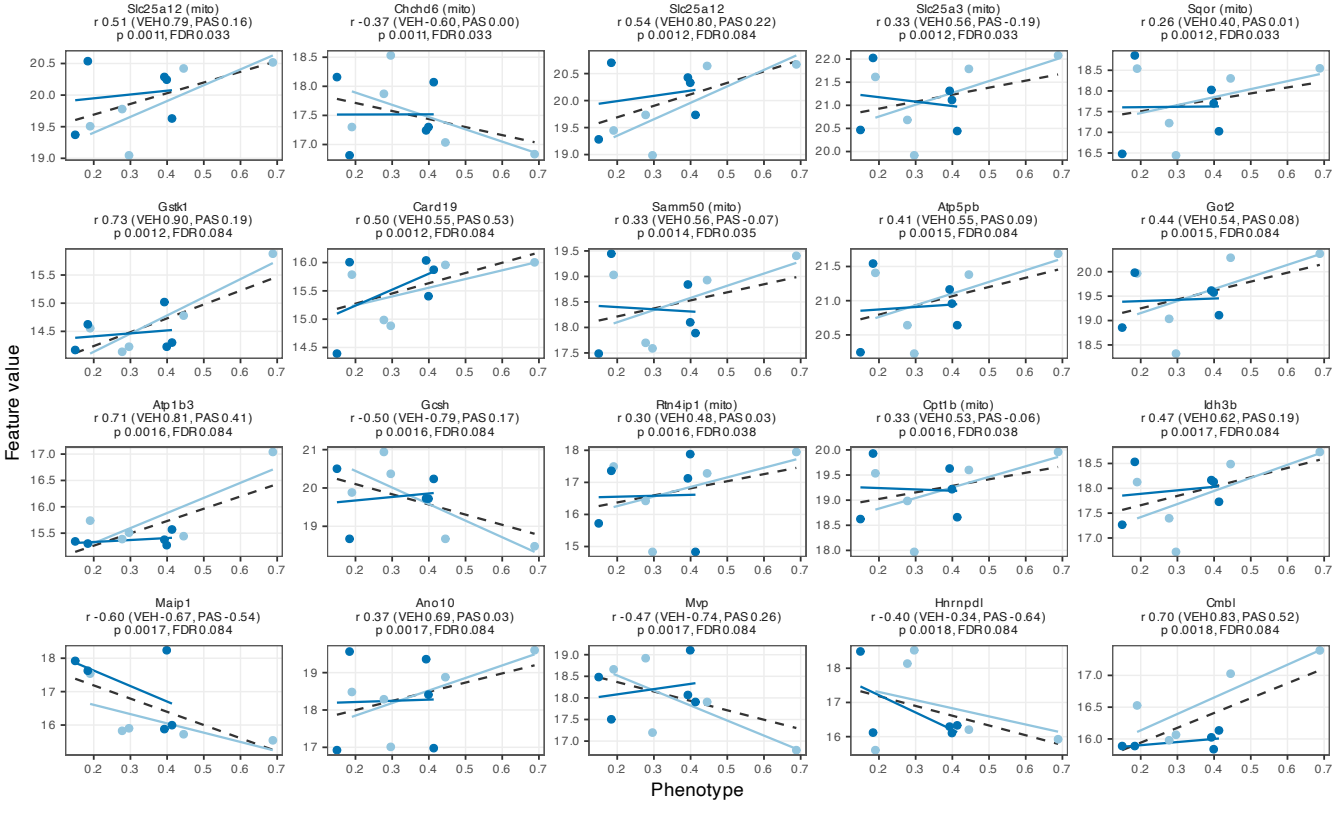
**Male: Proteins associated with Capillary density, every association at p ≤ 0.05 (n = 570)**

*Phenotype effect of PAS in this sex: -18% (p 0.54), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



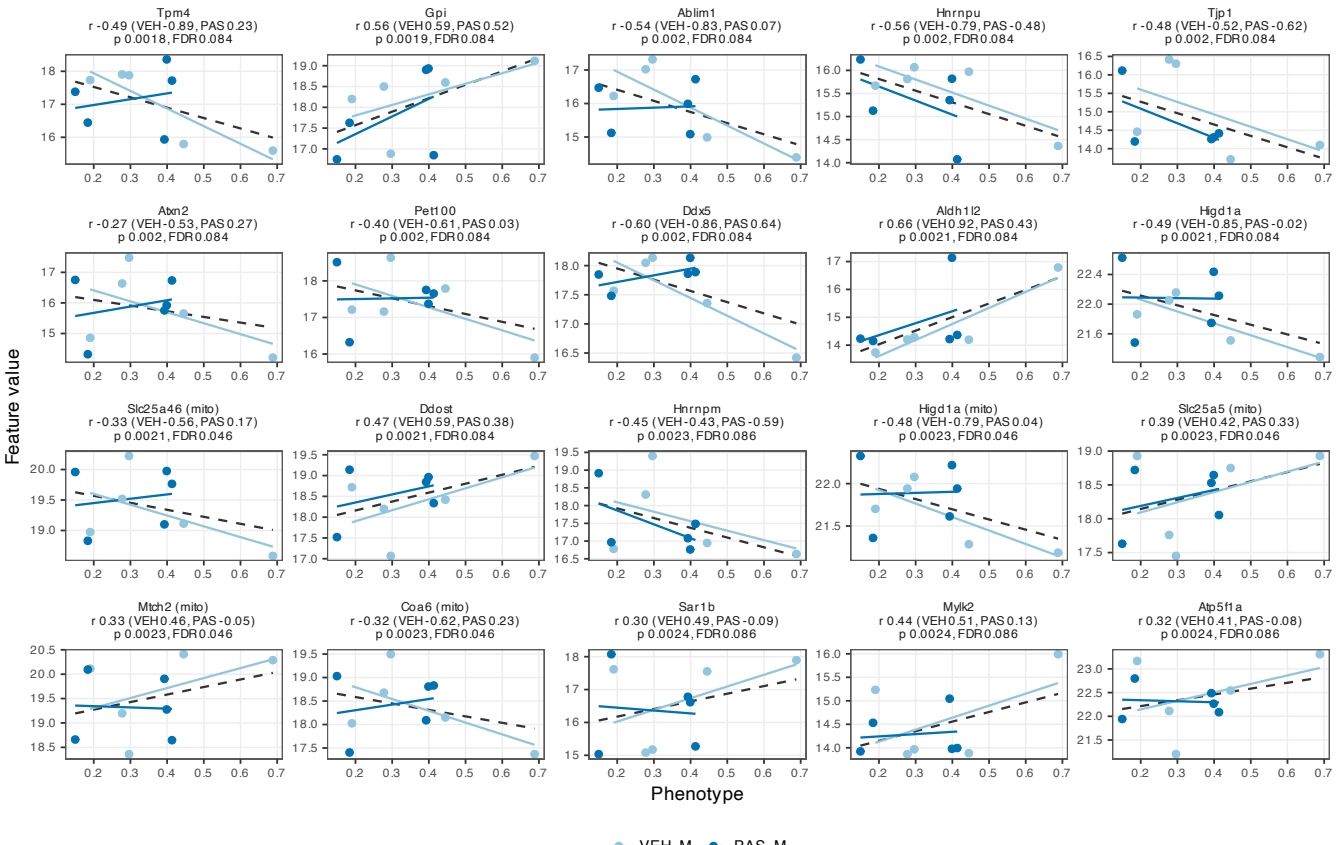
**Male: Proteins associated with Capillary density, every association at p ≤ 0.05 (n = 570)**

*Phenotype effect of PAS in this sex: -18% (p 0.54), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



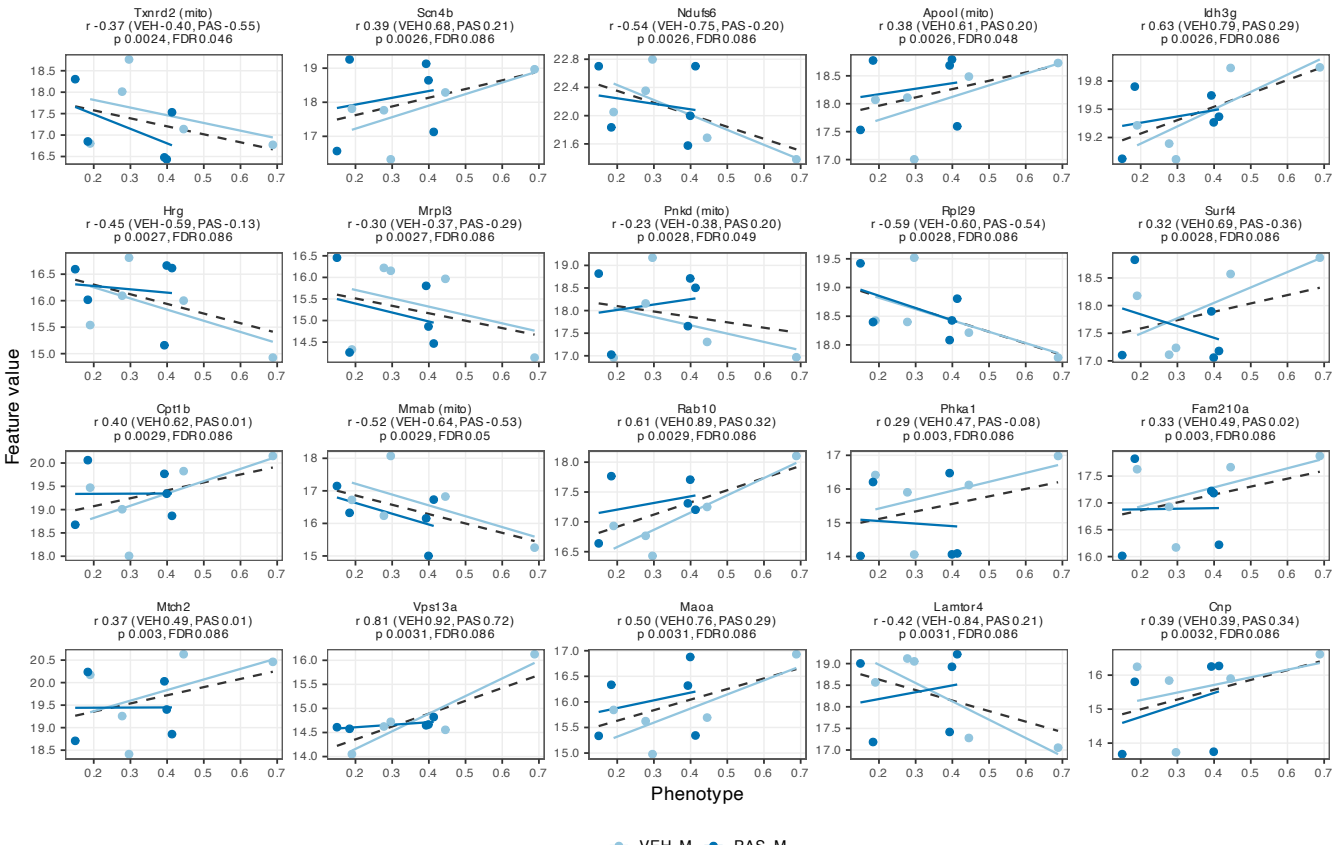
**Male: Proteins associated with Capillary density, every association at p ≤ 0.05 (n = 570)**

*Phenotype effect of PAS in this sex: -18% (p 0.54), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



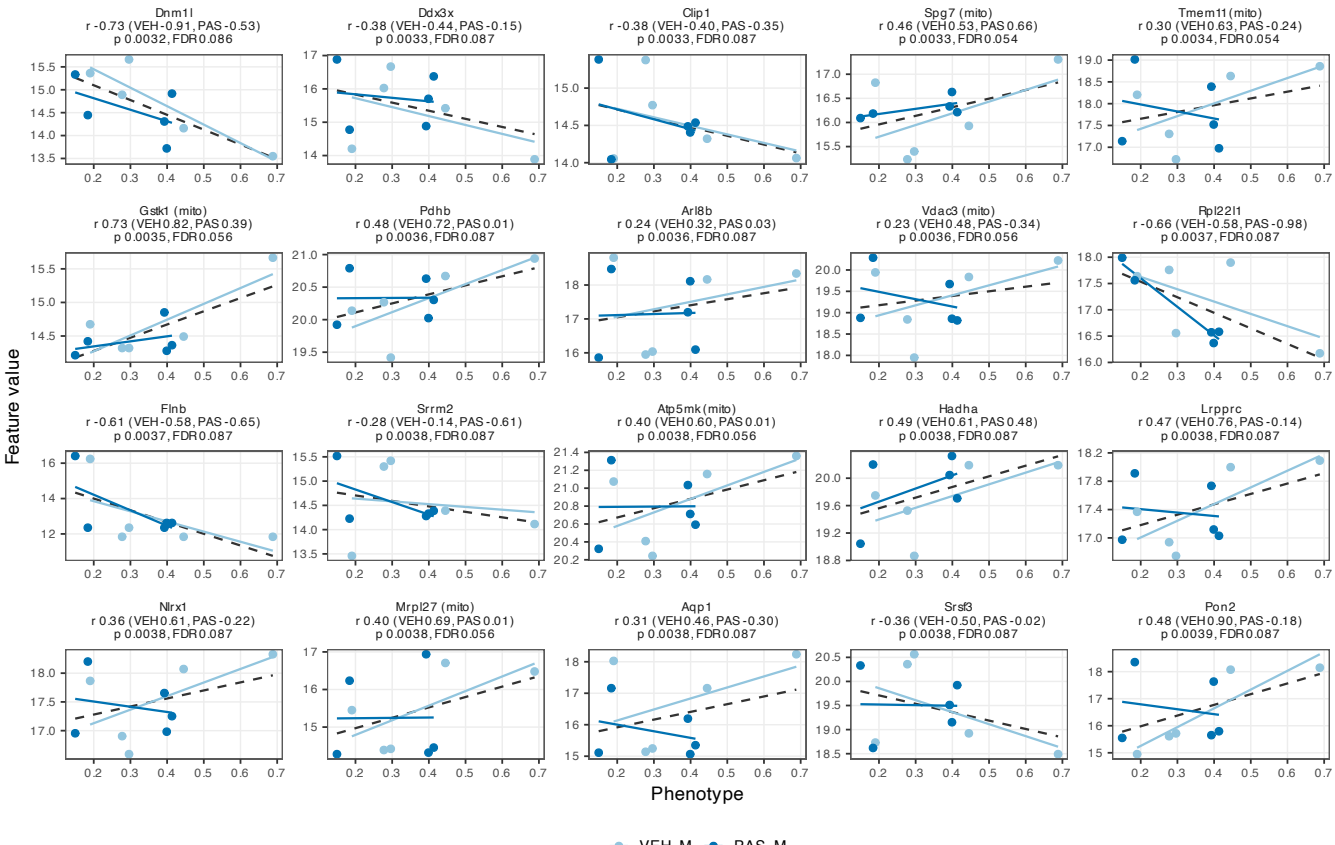
**Male: Proteins associated with Capillary density, every association at p ≤ 0.05 (n = 570)**

*Phenotype effect of PAS in this sex: -18% (p 0.54), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



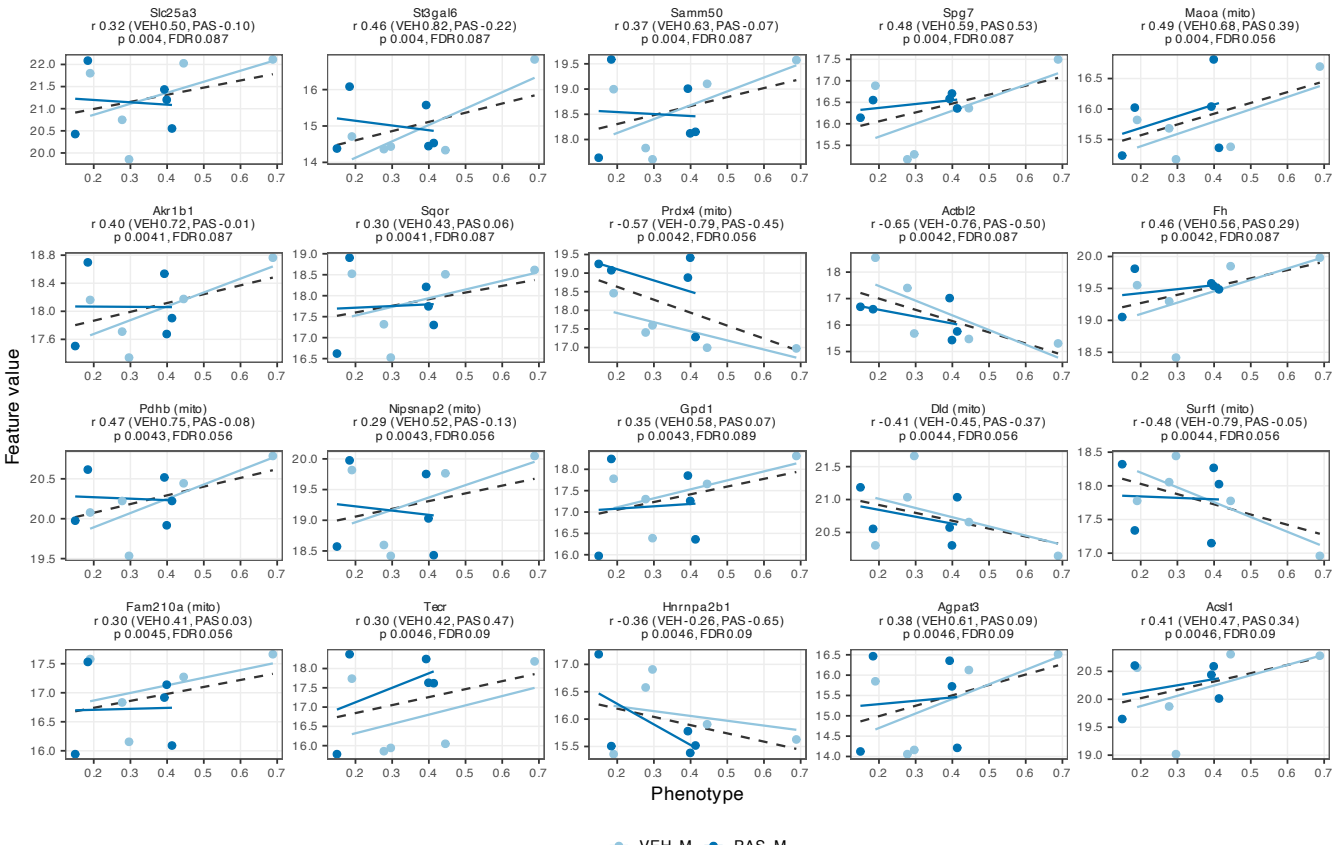
**Male: Proteins associated with Capillary density, every association at p ≤ 0.05 (n = 570)**

*Phenotype effect of PAS in this sex: -18% (p 0.54), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



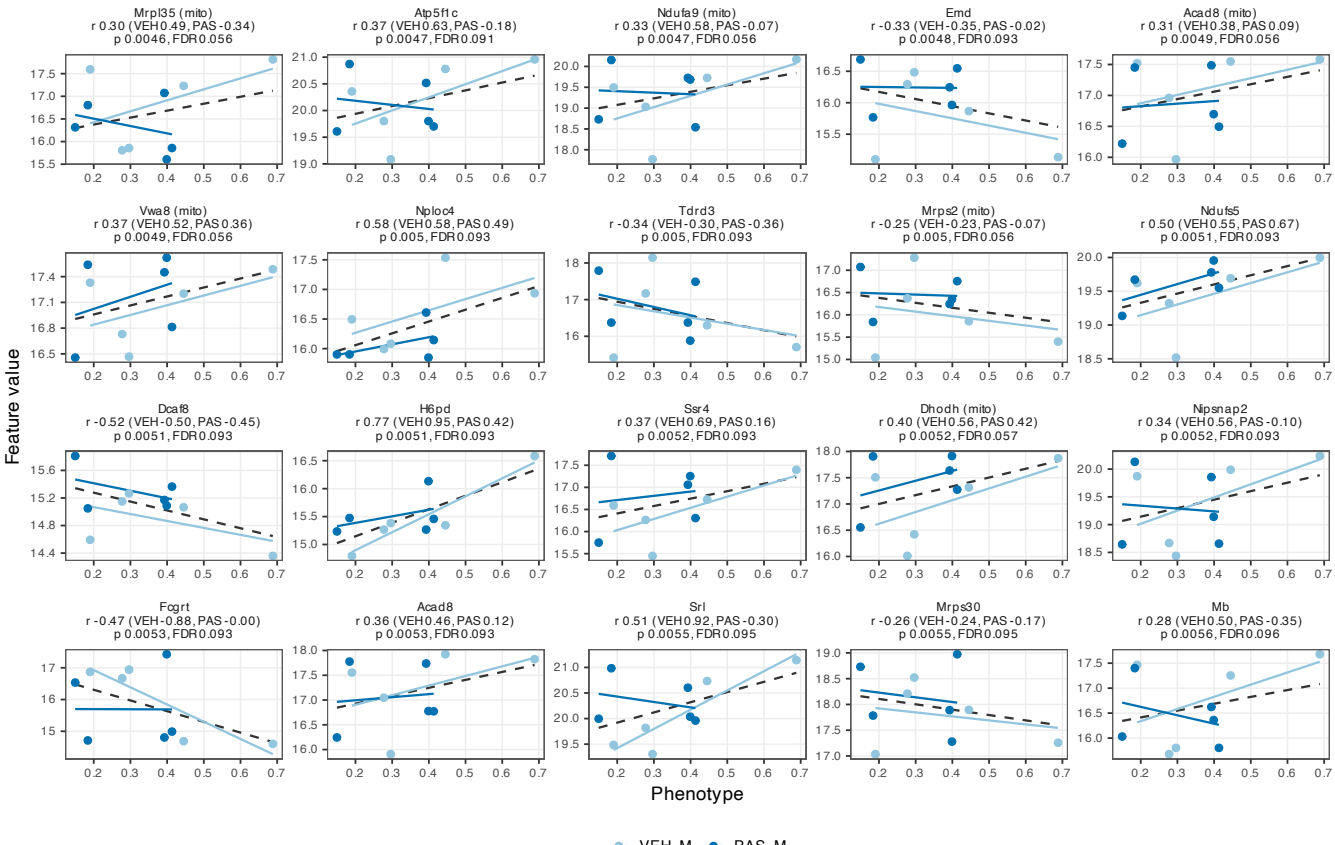
**Male: Proteins associated with Capillary density, every association at p ≤ 0.05 (n = 570)**

*Phenotype effect of PAS in this sex: -18% (p 0.54), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



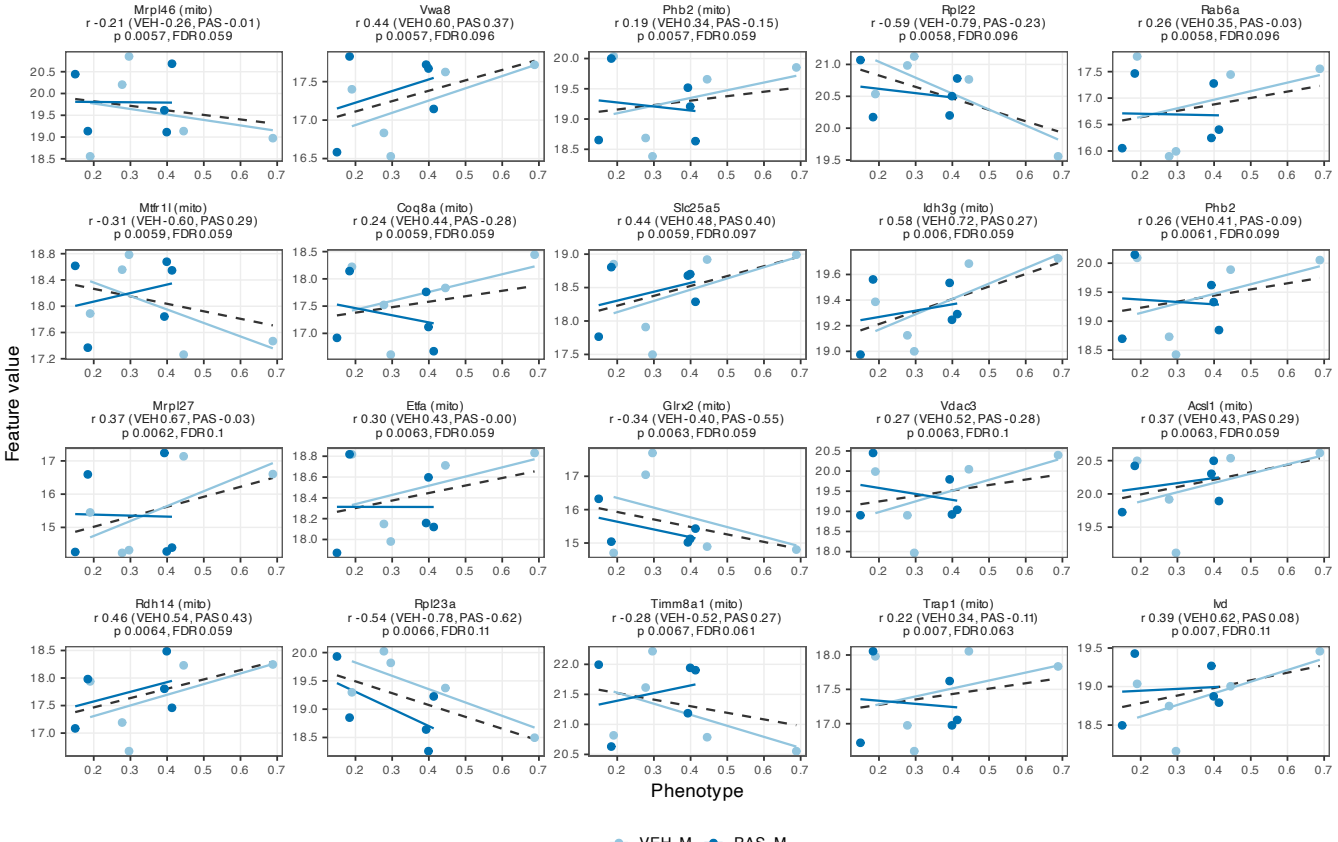
**Male: Proteins associated with Capillary density, every association at p ≤ 0.05 (n = 570)**

*Phenotype effect of PAS in this sex: -18% (p 0.54), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



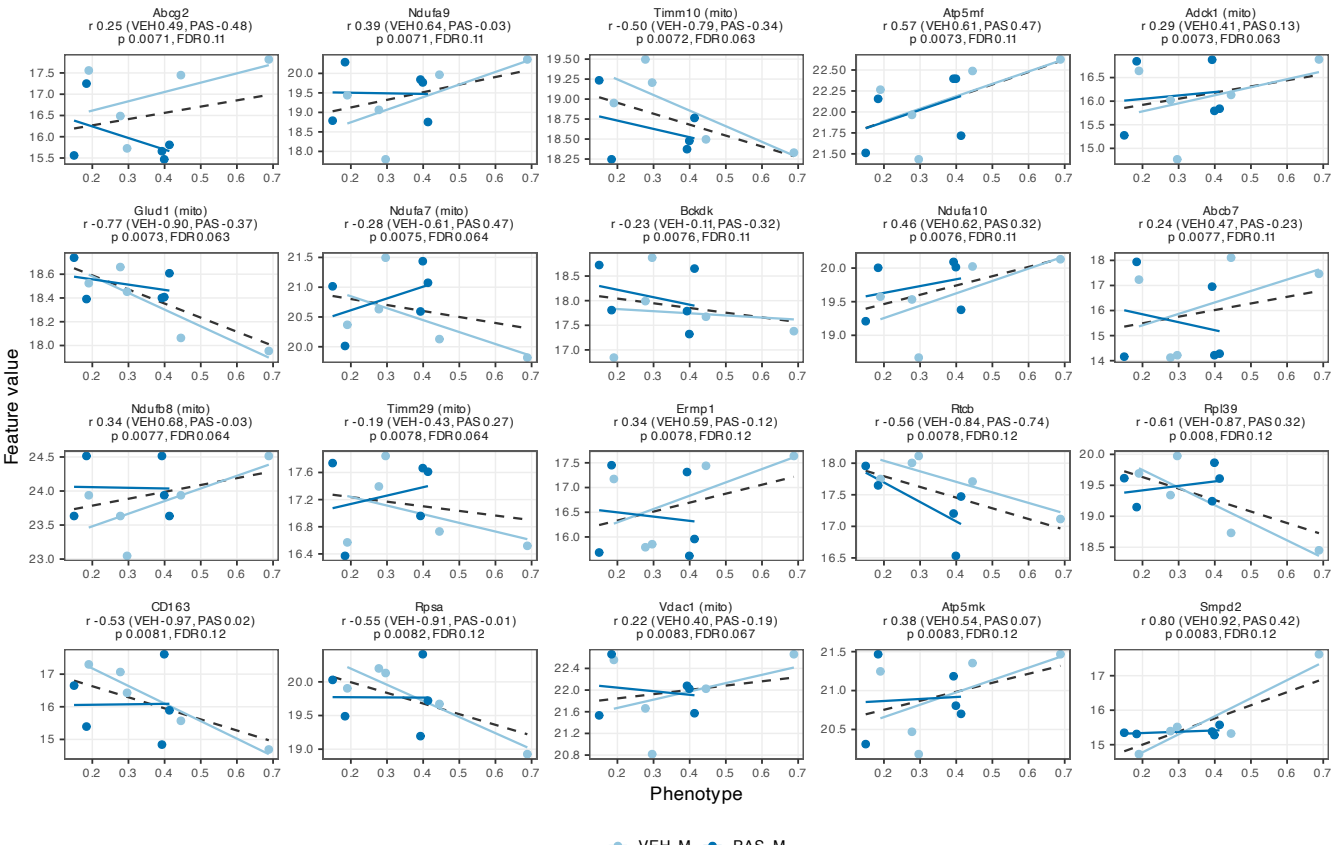
**Male: Proteins associated with Capillary density, every association at p ≤ 0.05 (n = 570)**

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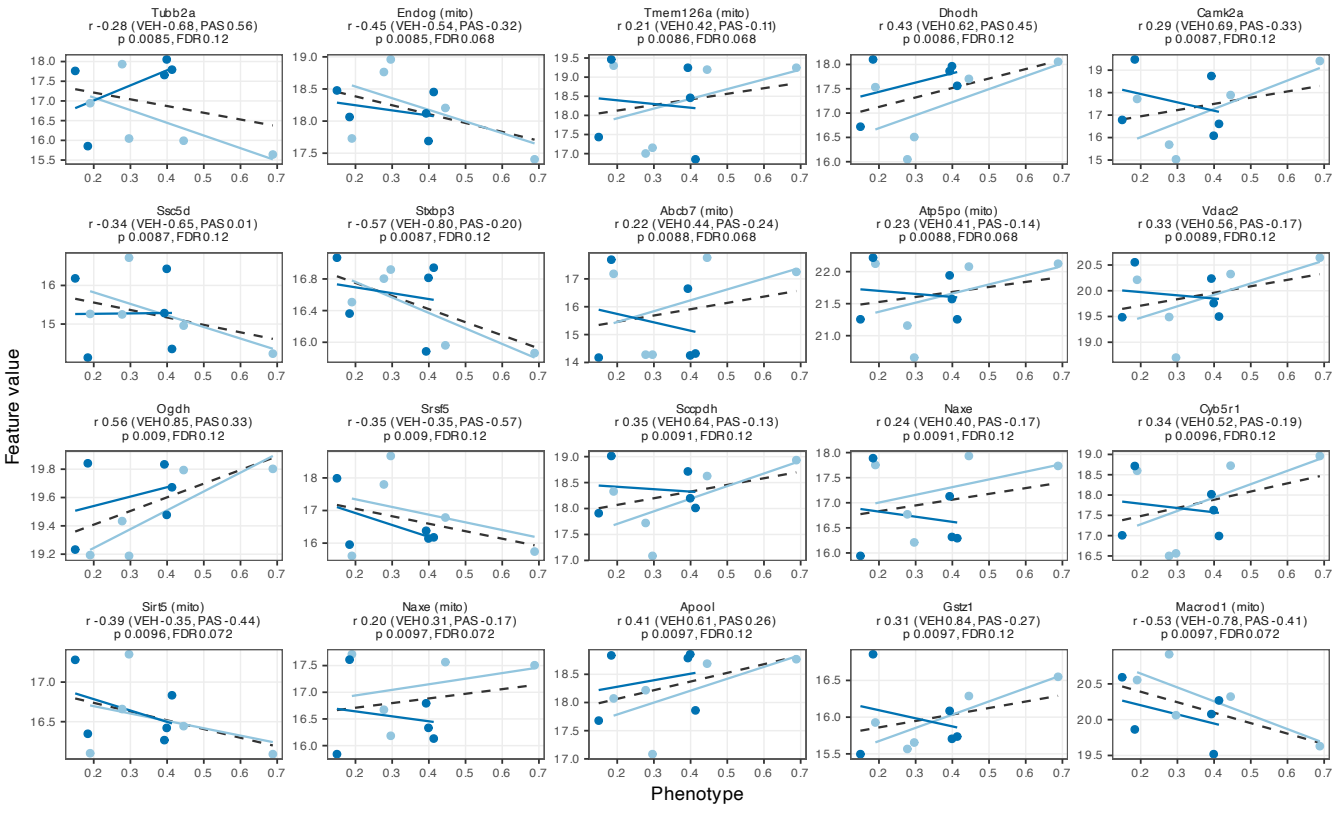
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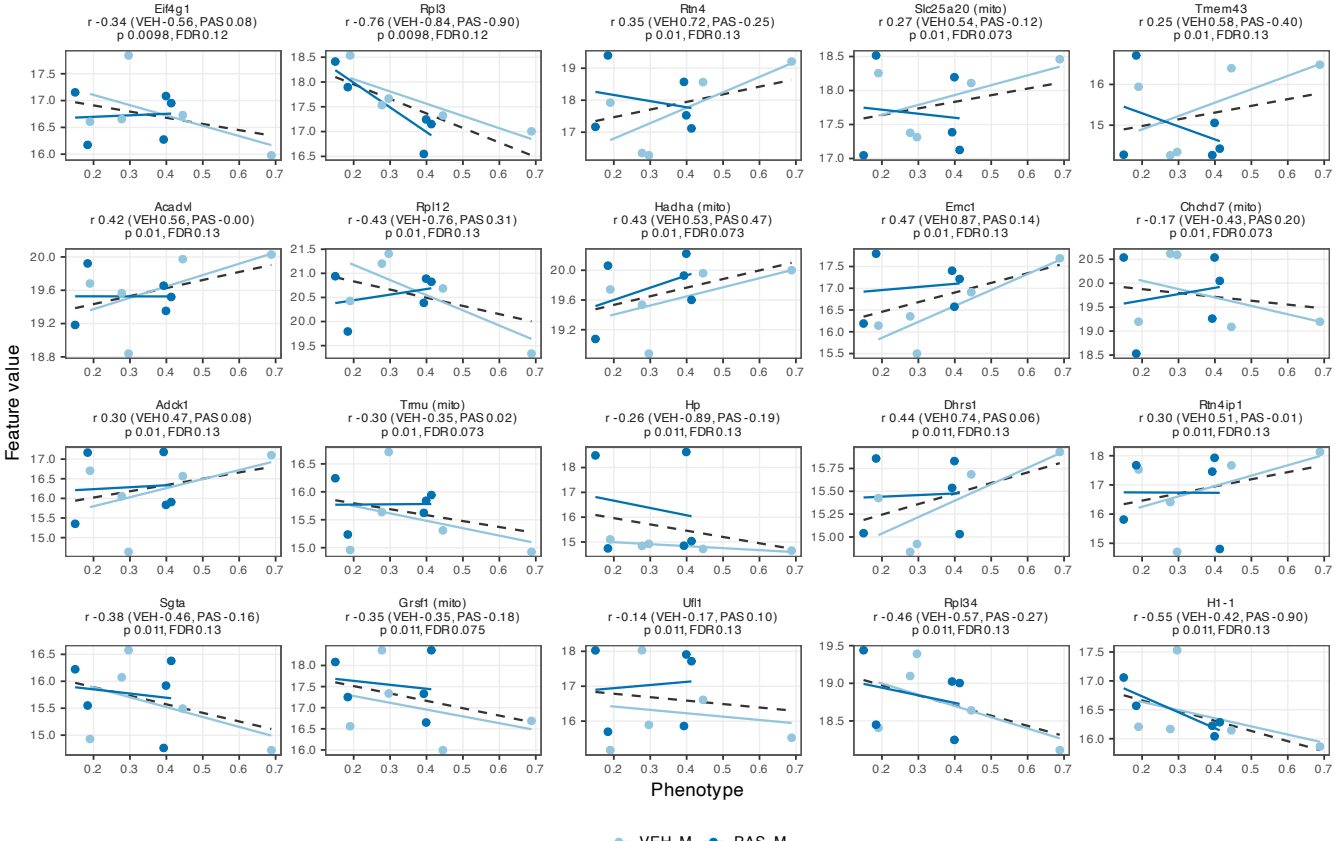
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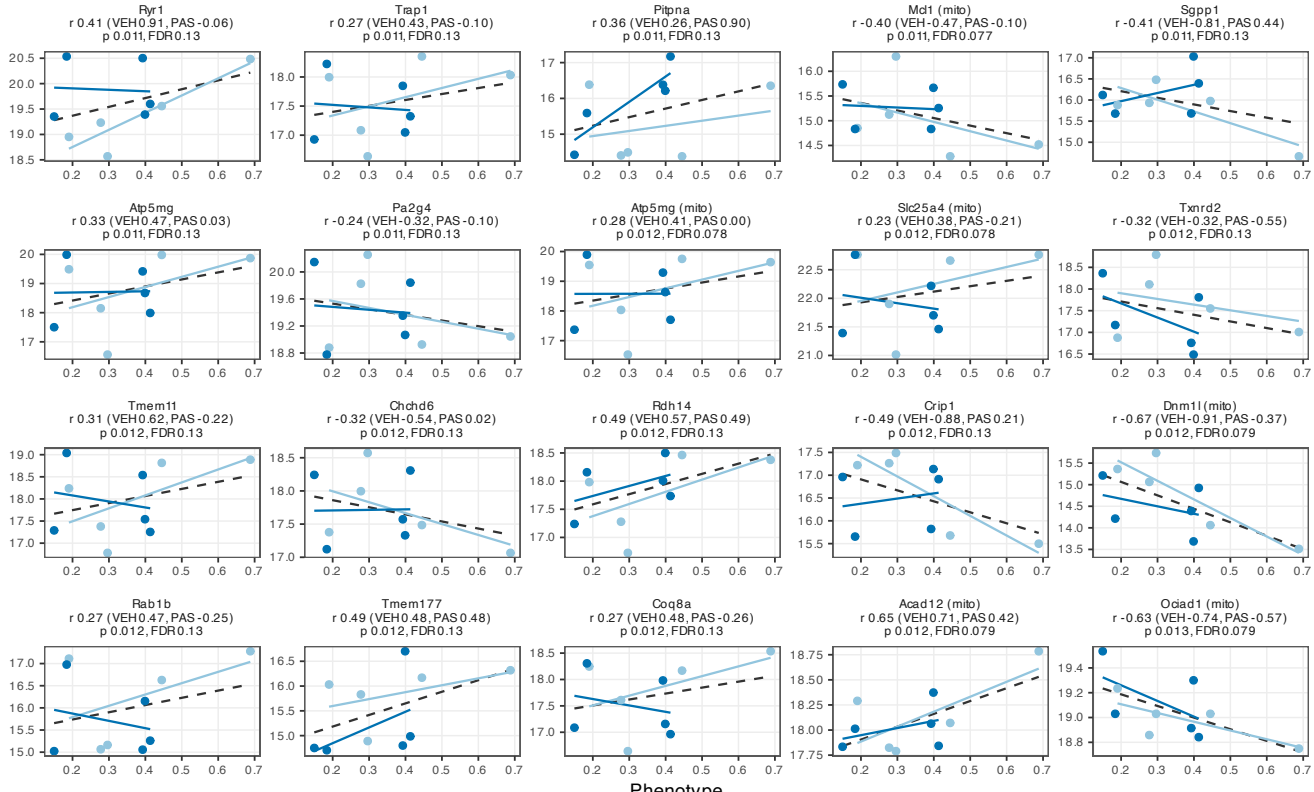
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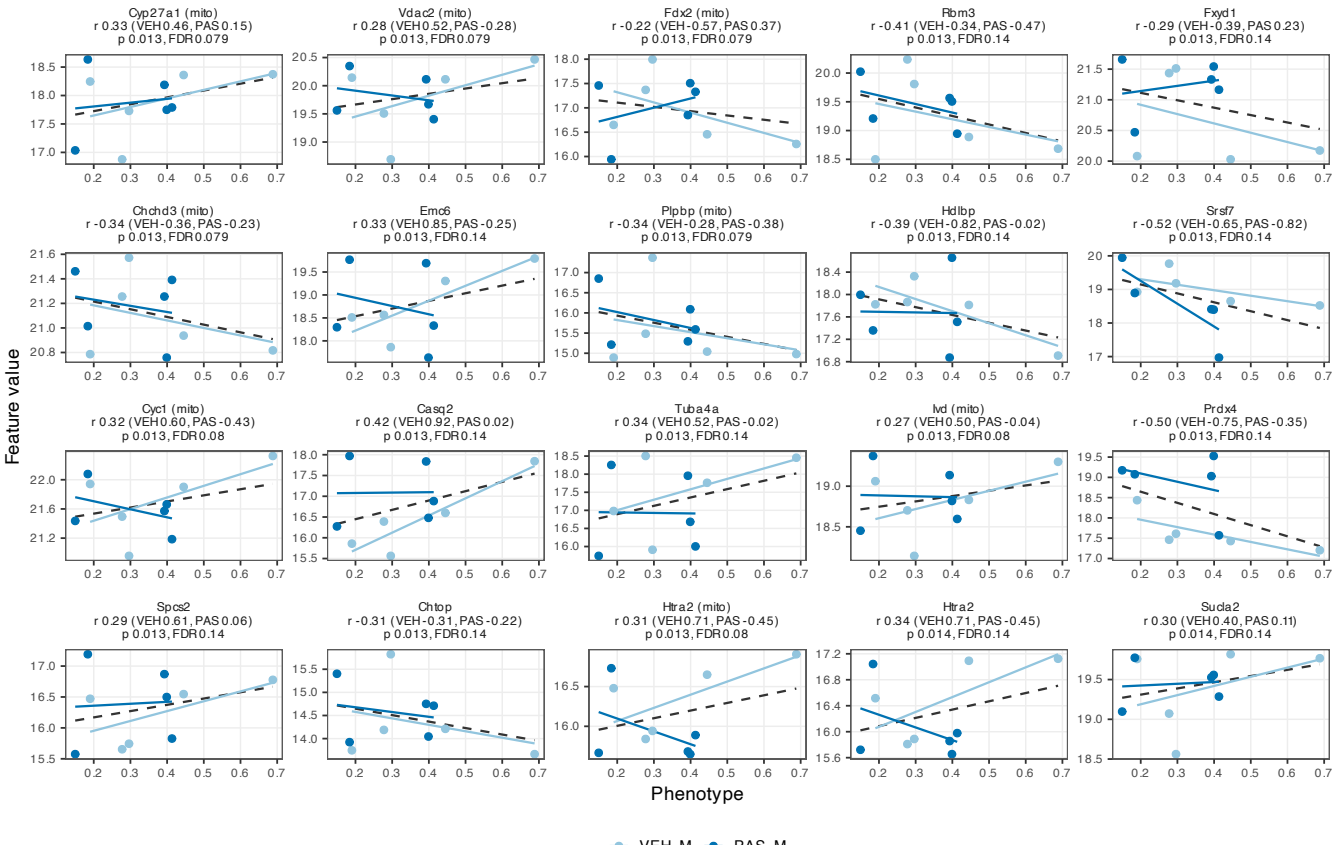
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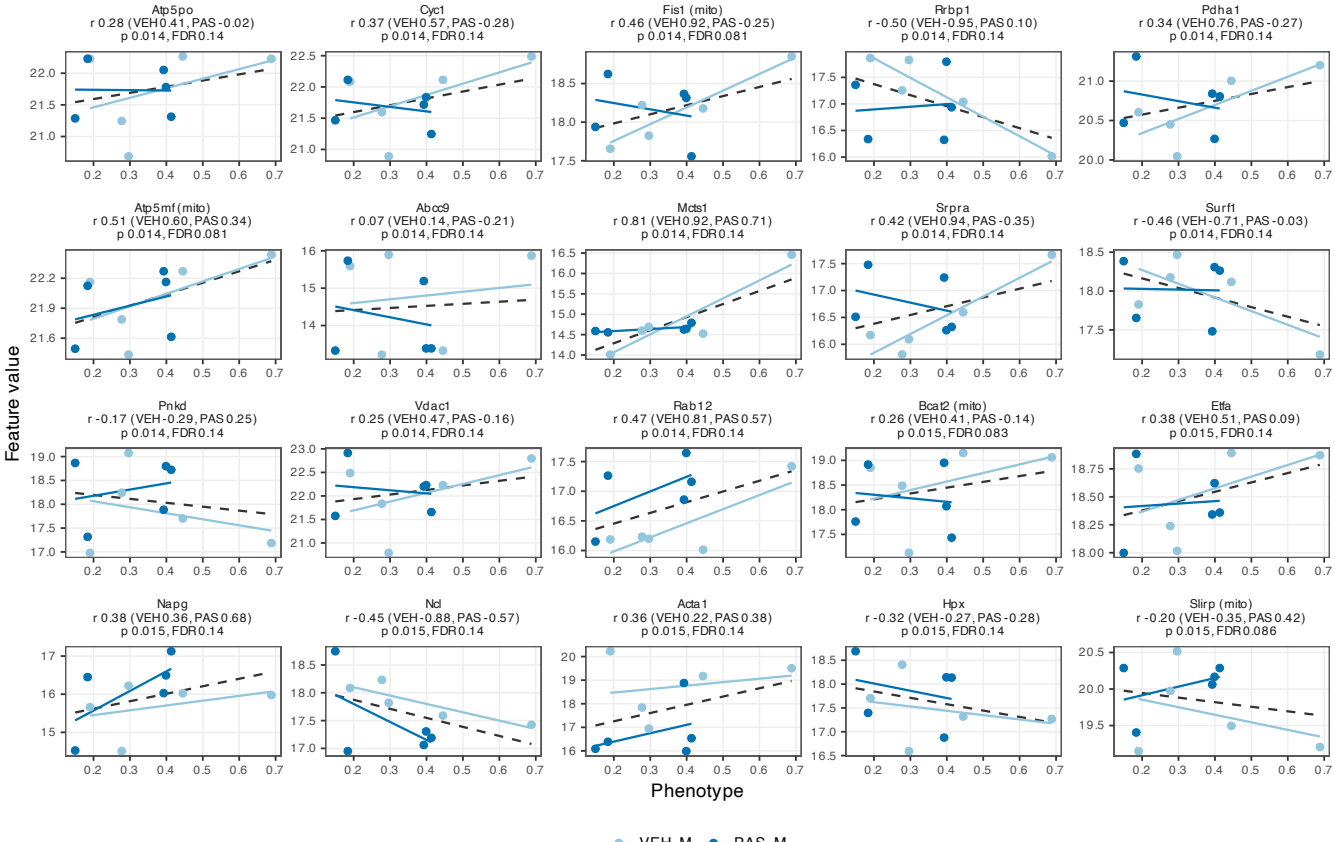
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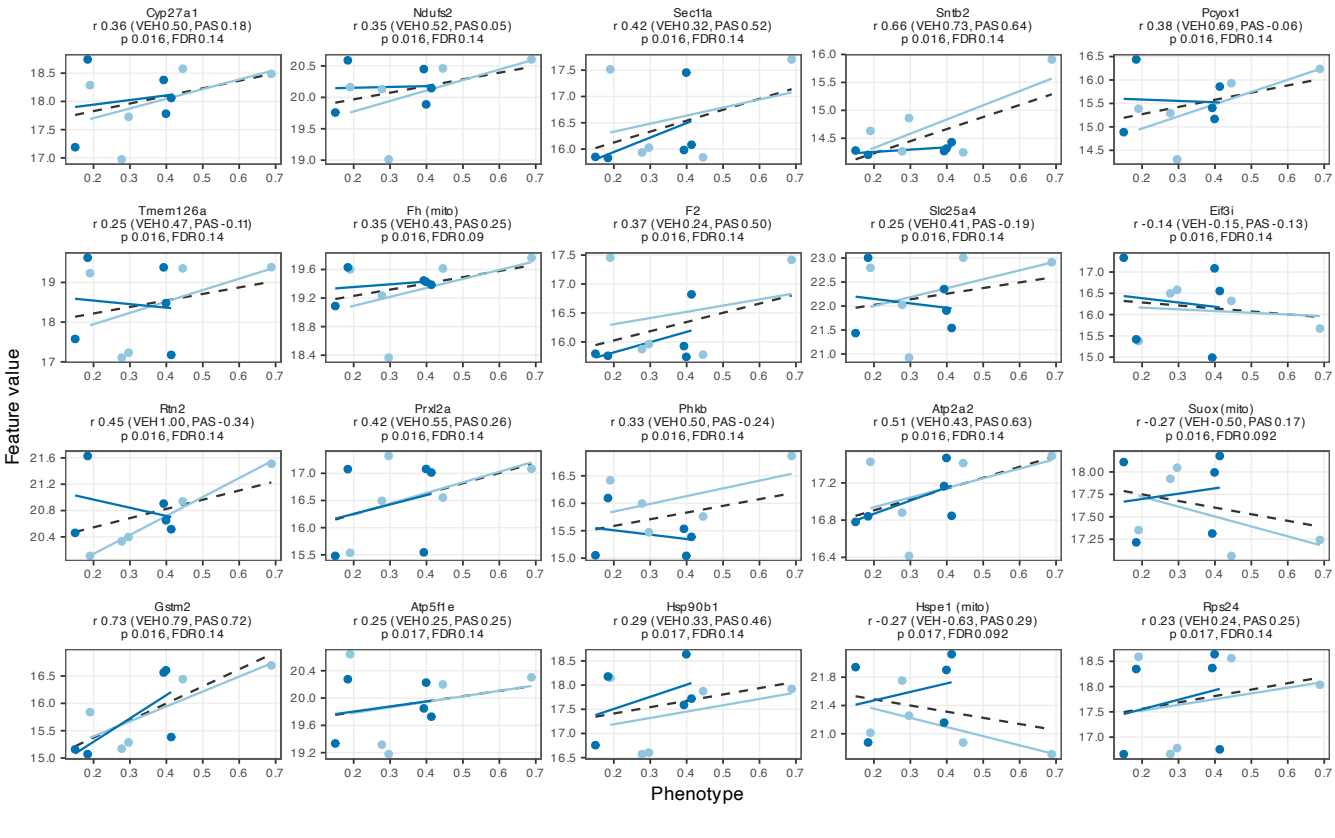
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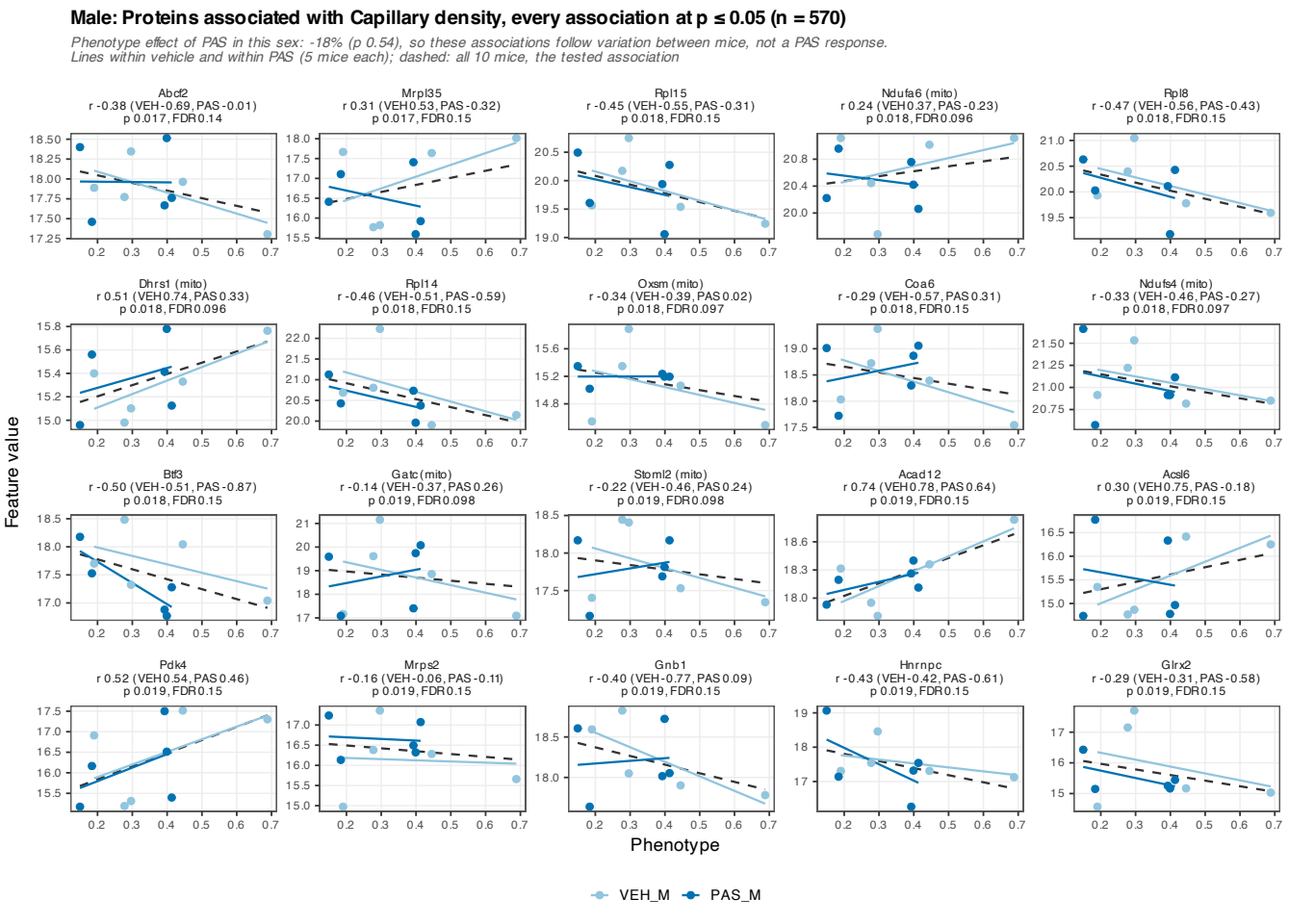
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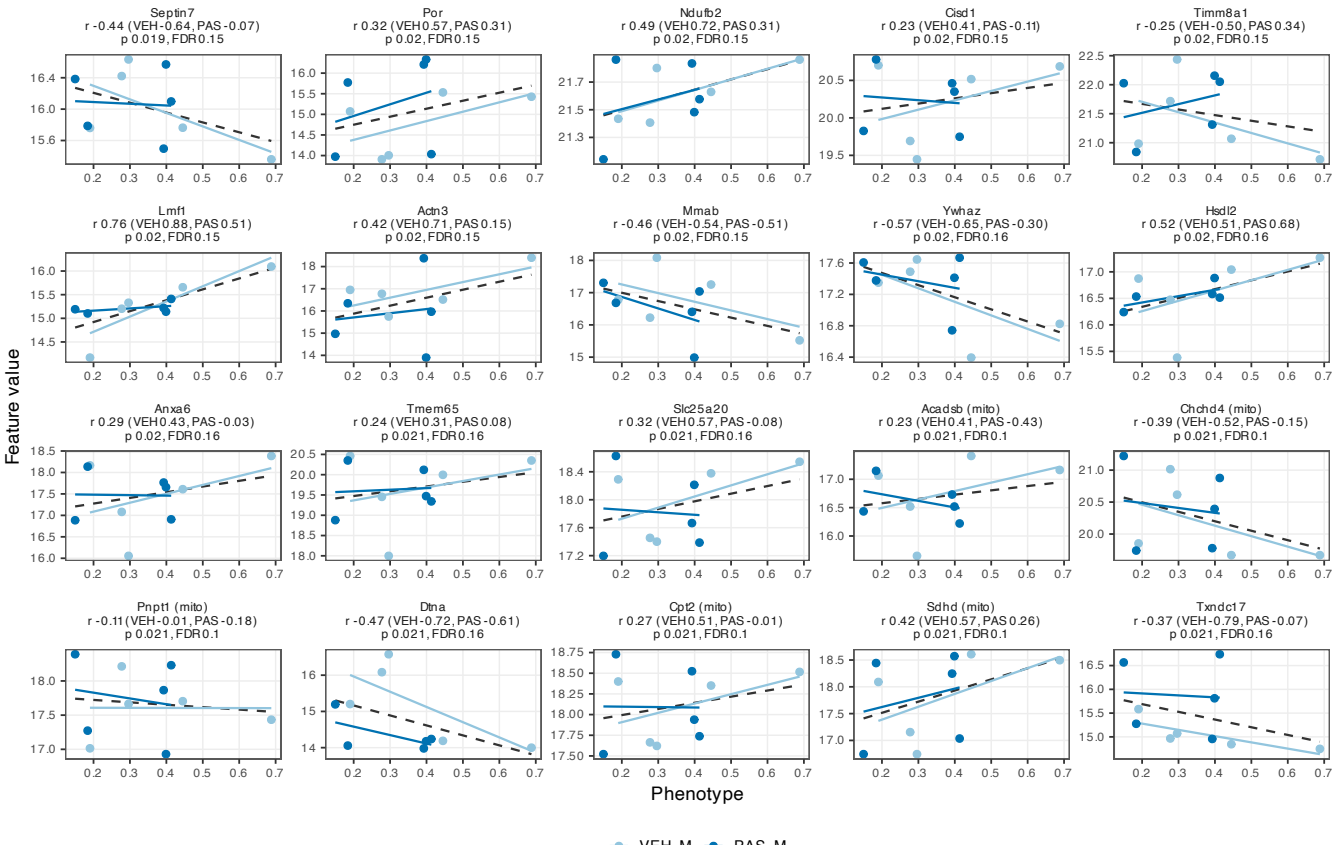
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Phenotype effect of PAS in this sex: -18% (p 0.54), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association
Abcf2
r -0.38 (VEH -0.69, PAS -0.01)
p 0.017, FDR 0.14
Mrpl35
r 0.31 (VEH 0.53, PAS -0.32)
p 0.017, FDR 0.15
Rpl15
r -0.45 (VEH -0.55, PAS -0.31)
p 0.018, FDR 0.15
Ndufa6 (mito)
r 0.24 (VEH 0.37, PAS -0.23)
p 0.018, FDR 0.096
Rpl8
r -0.47 (VEH -0.56, PAS -0.43)
p 0.018, FDR 0.15
Dhrs1 (mito)
r 0.51 (VEH 0.74, PAS 0.33)
p 0.018, FDR 0.096
Rpl14
r -0.46 (VEH -0.51, PAS -0.59)
p 0.018, FDR 0.15
Oxsm (mito)
r -0.34 (VEH -0.39, PAS 0.02)
p 0.018, FDR 0.097
Coa6
r -0.29 (VEH -0.57, PAS 0.31)
p 0.018, FDR 0.15
Ndufs4 (mito)
r -0.33 (VEH -0.46, PAS -0.27)
p 0.018, FDR 0.097
Btf3
r -0.50 (VEH -0.51, PAS -0.87)
p 0.018, FDR 0.15
Gatc (mito)
r -0.14 (VEH -0.37, PAS 0.26)
p 0.019, FDR 0.098
Stoml2 (mito)
r -0.22 (VEH -0.46, PAS 0.24)
p 0.019, FDR 0.098
Acad12
r 0.74 (VEH 0.78, PAS 0.64)
p 0.019, FDR 0.15
Acsl6
r 0.30 (VEH 0.75, PAS -0.18)
p 0.019, FDR 0.15
Pdk4
r 0.52 (VEH 0.54, PAS 0.46)
p 0.019, FDR 0.15
Mrps2
r -0.16 (VEH -0.06, PAS -0.11)
p 0.019, FDR 0.15
Gnb1
r -0.40 (VEH -0.77, PAS 0.09)
p 0.019, FDR 0.15
Hnrnpc
r -0.43 (VEH -0.42, PAS -0.61)
p 0.019, FDR 0.15
Glrx2
r -0.29 (VEH -0.31, PAS -0.58)
p 0.019, FDR 0.15
Feature value
Phenotype
VEH_M
PAS_M

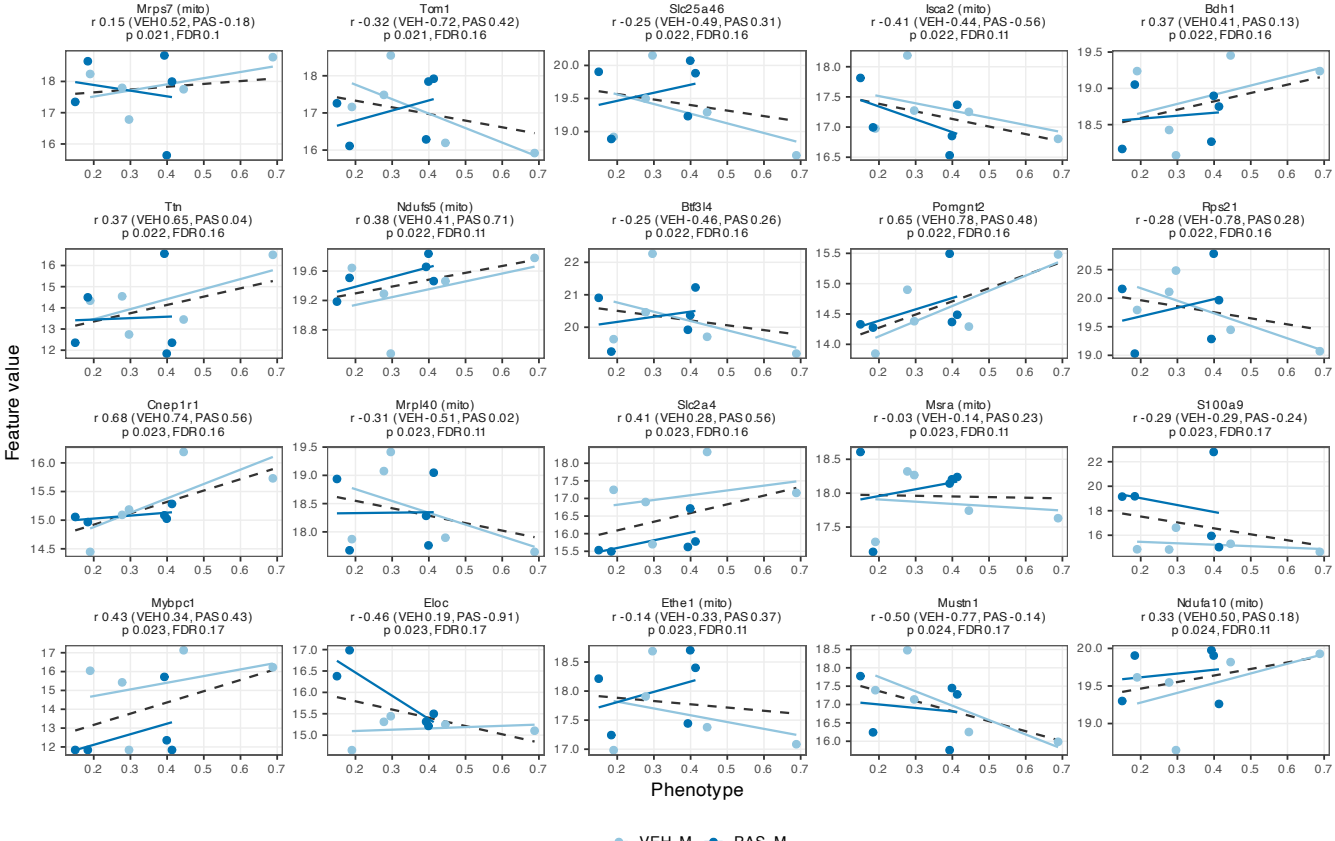
**Male: Proteins associated with Capillary density, every association at p ≤ 0.05 (n = 570)**

*Phenotype effect of PAS in this sex: -18% (p 0.54), so these associations follow variation between mice, not a PAS response.*
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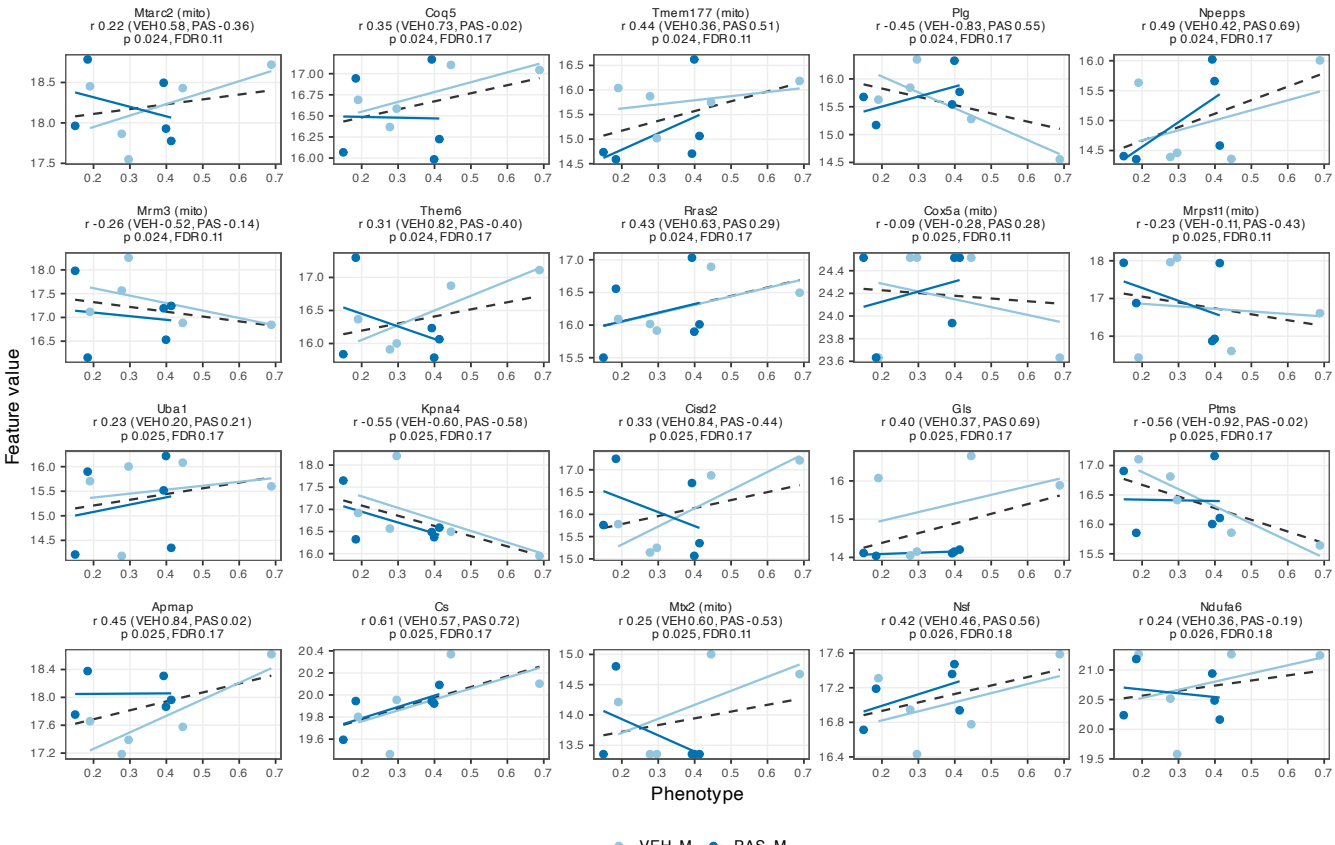
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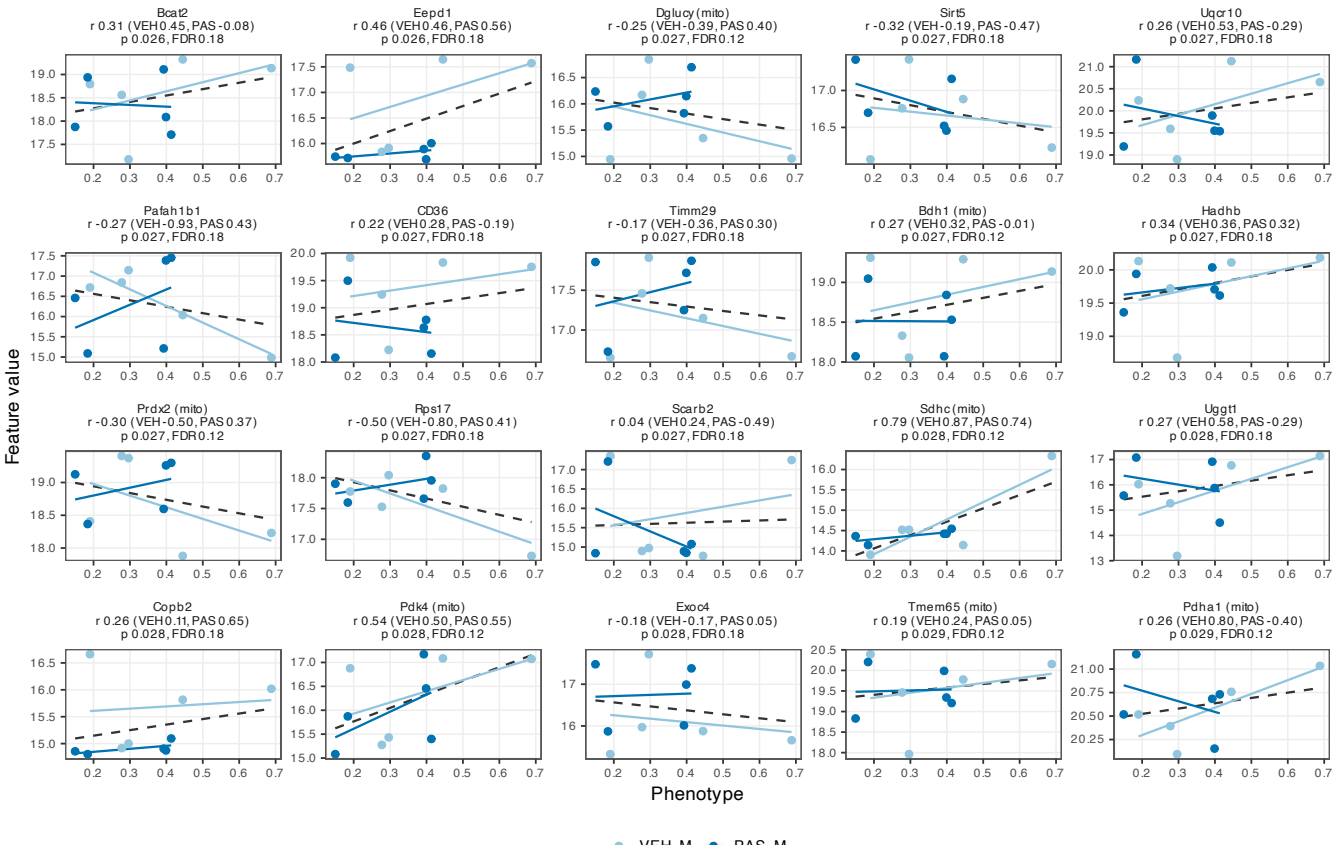
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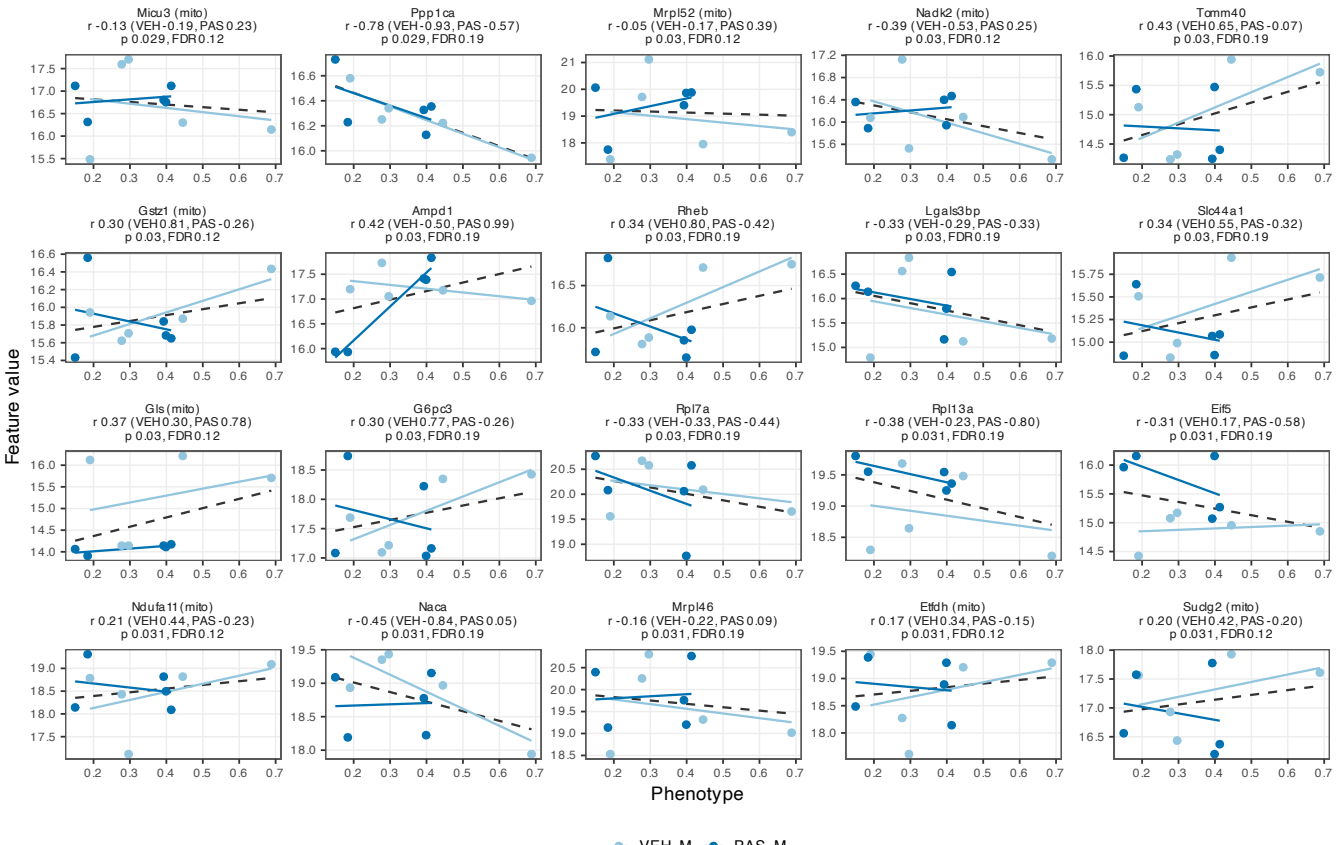
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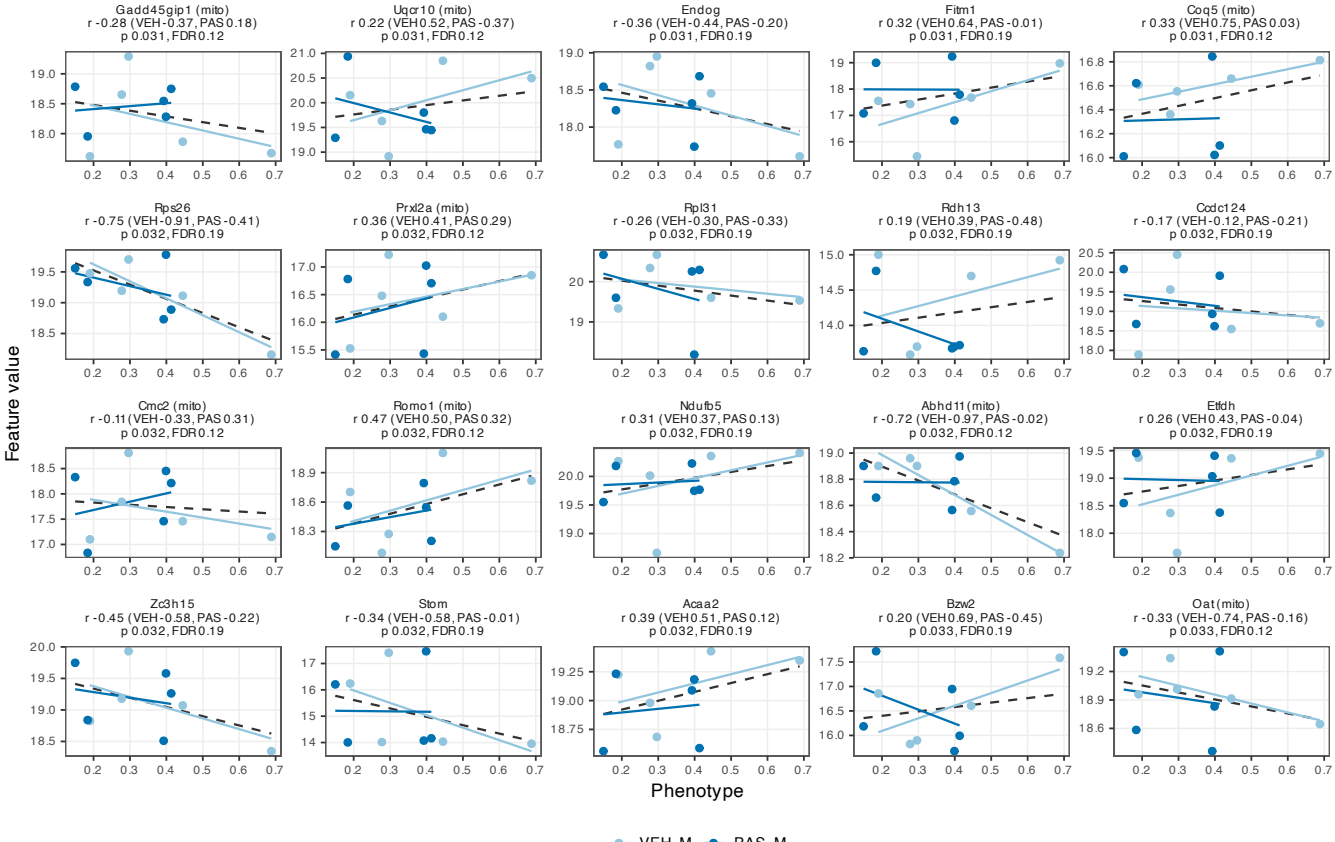
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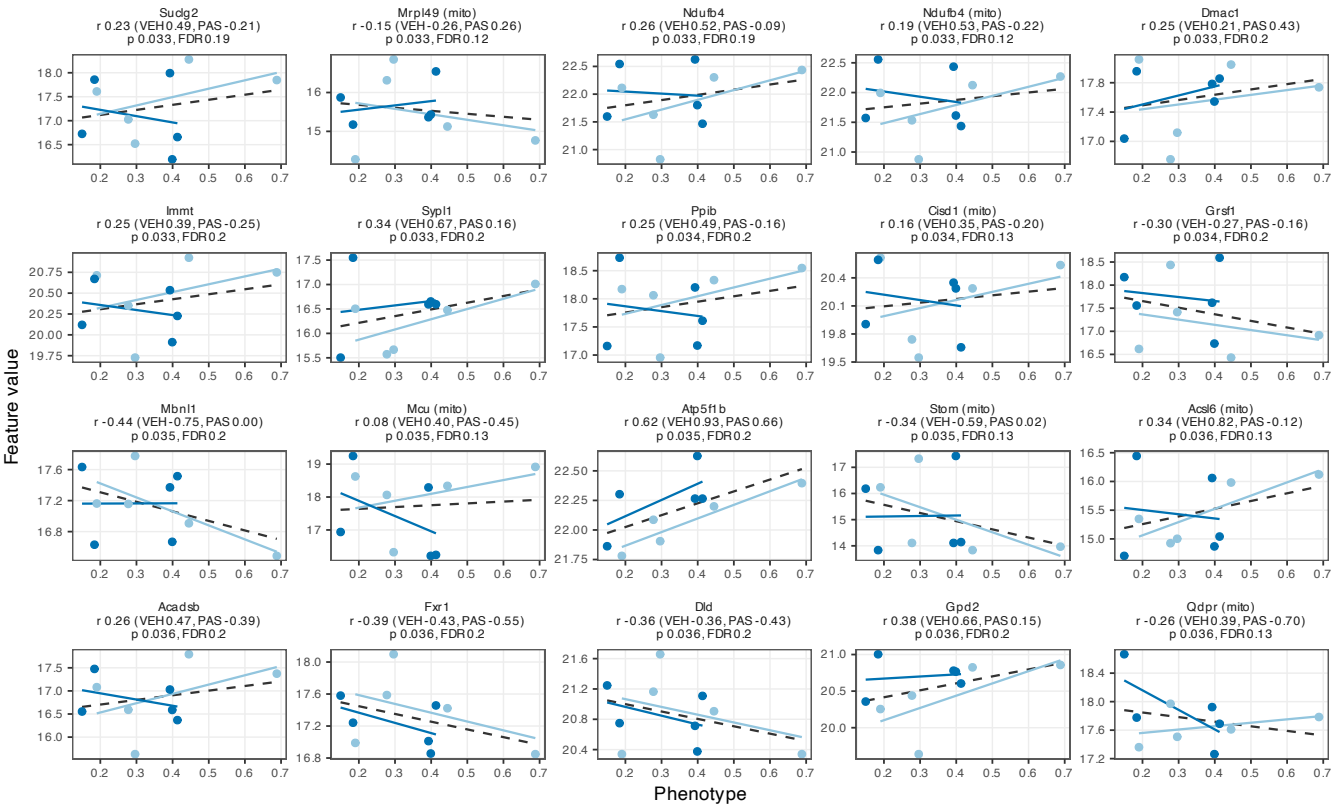
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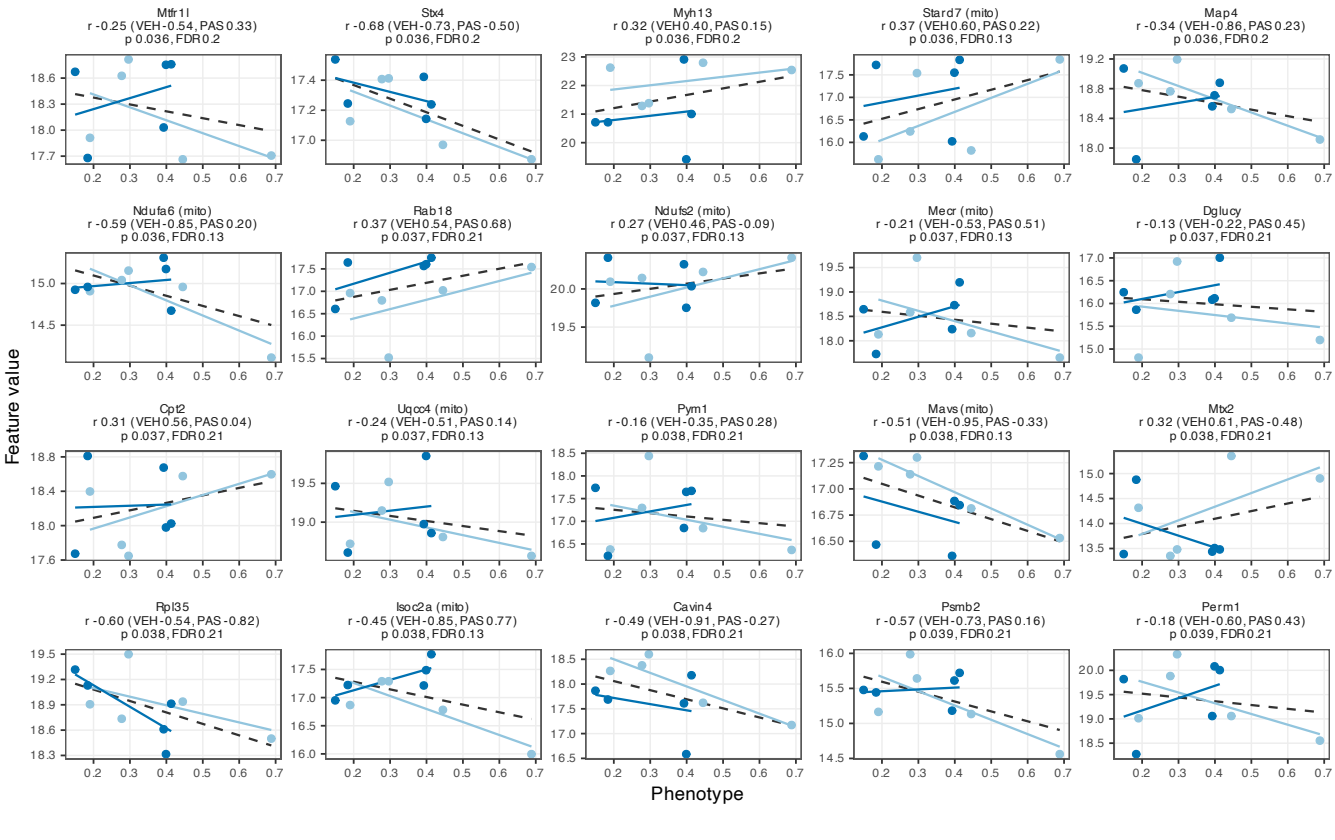
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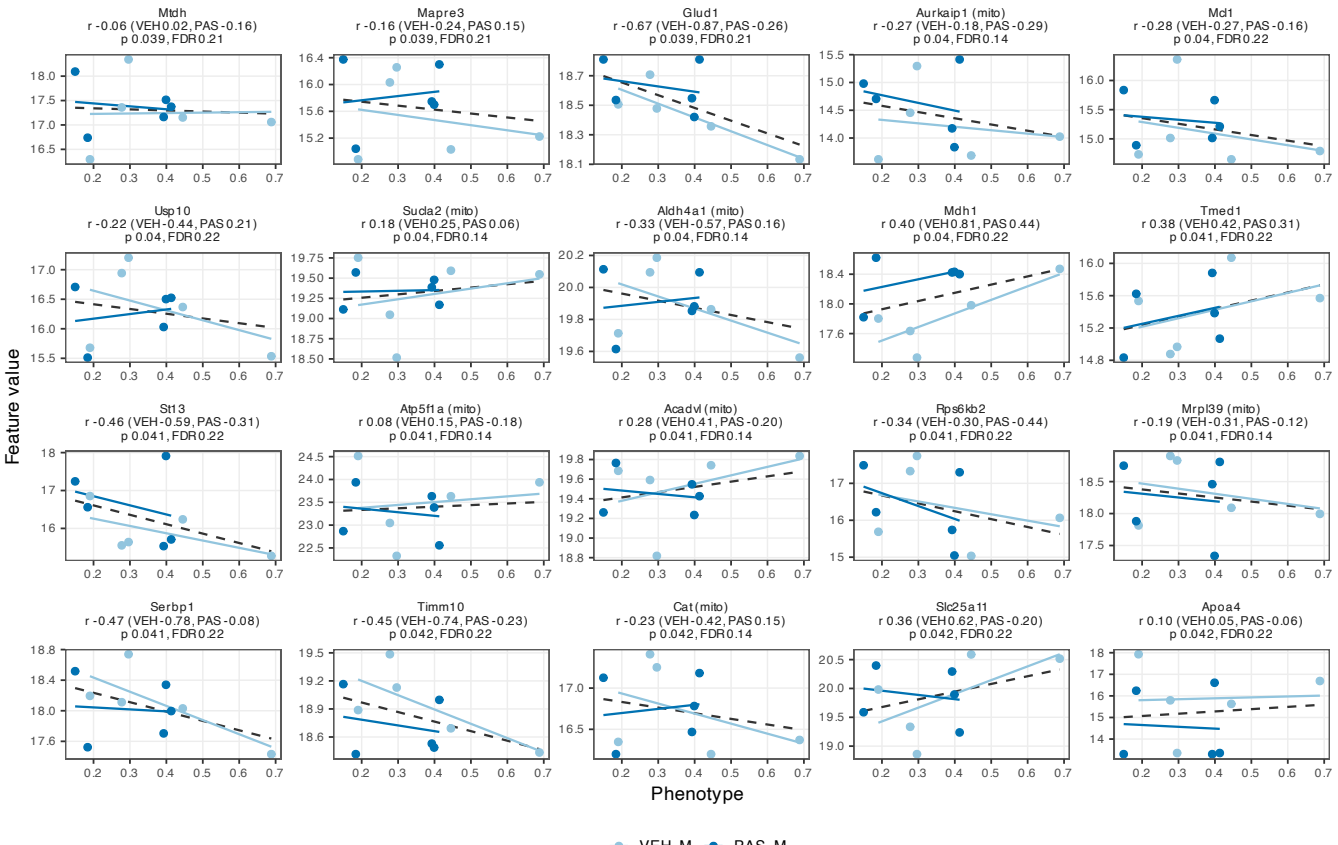
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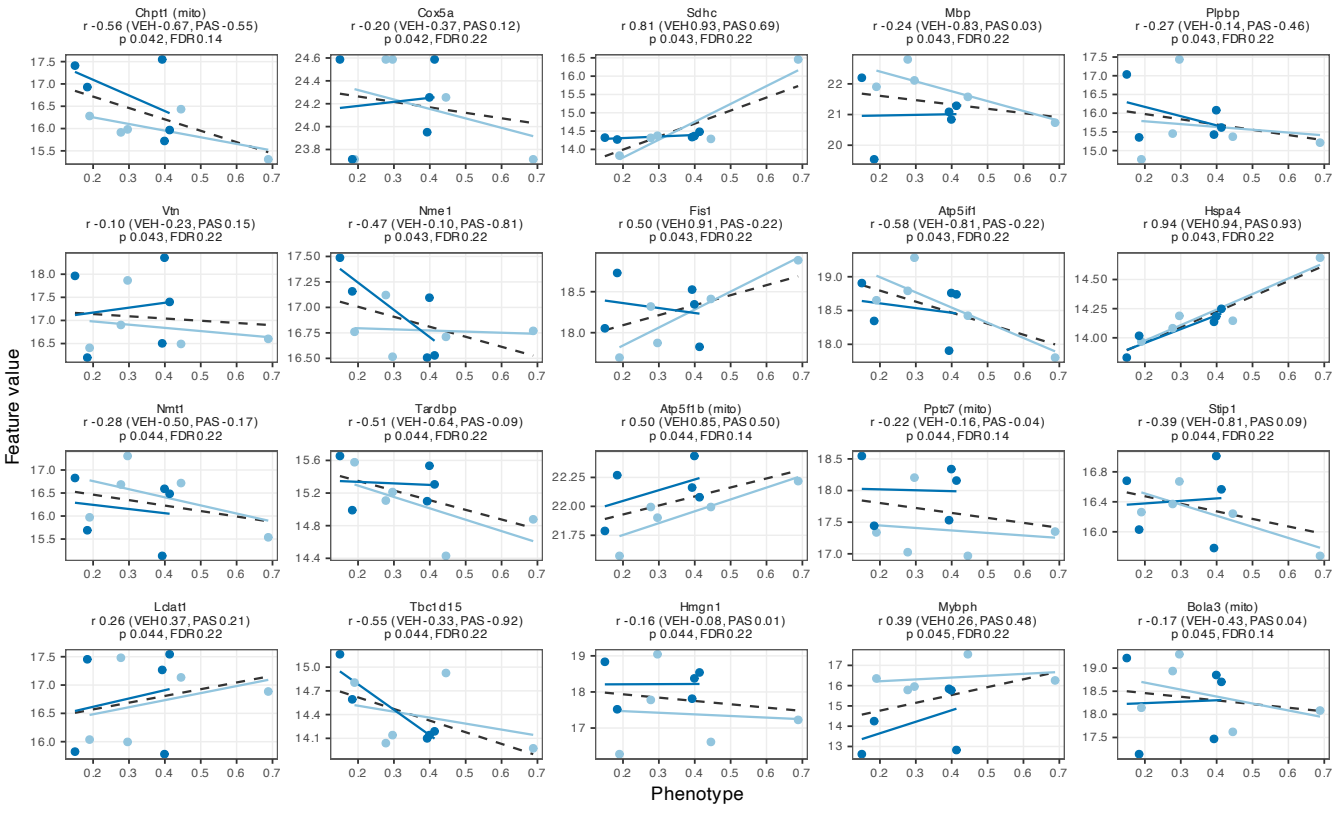
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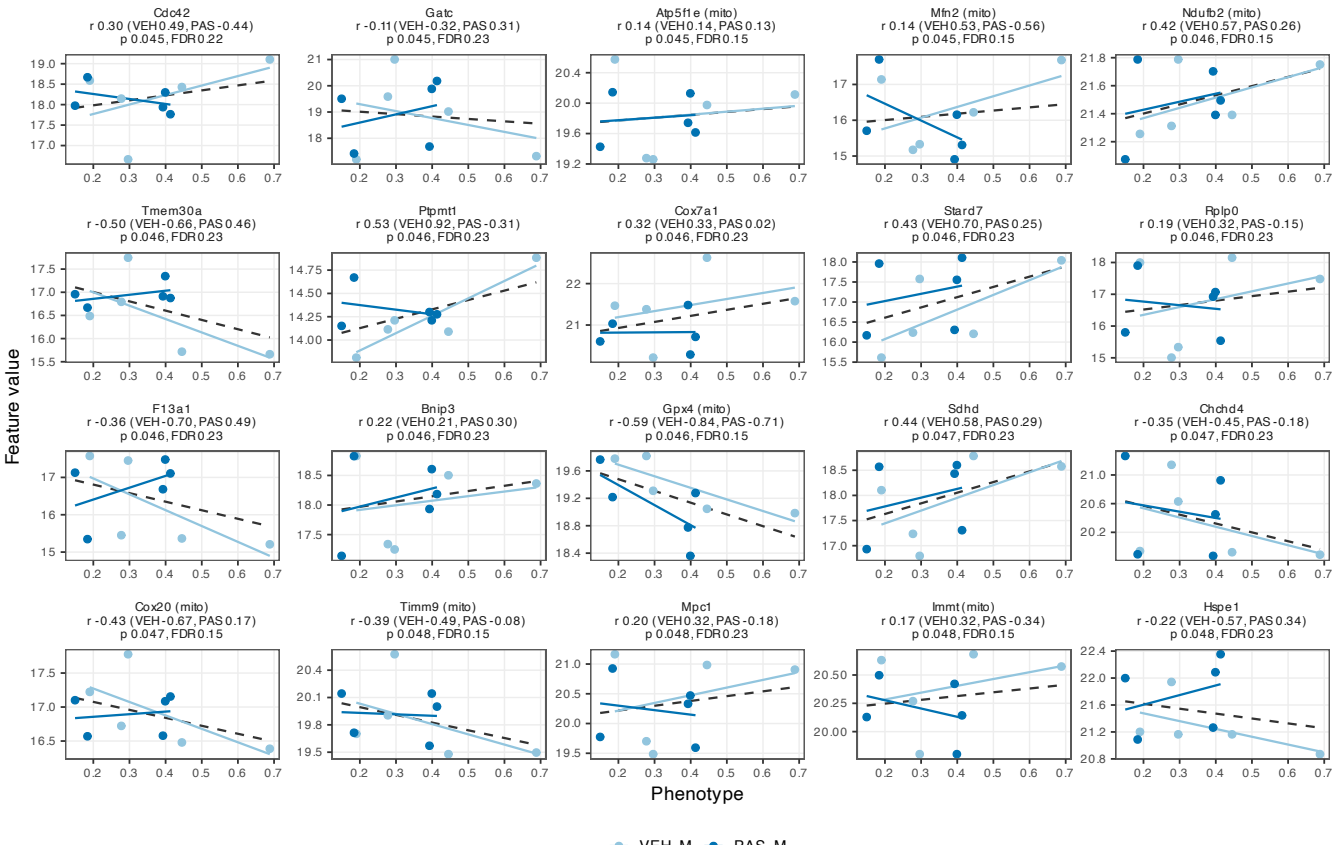
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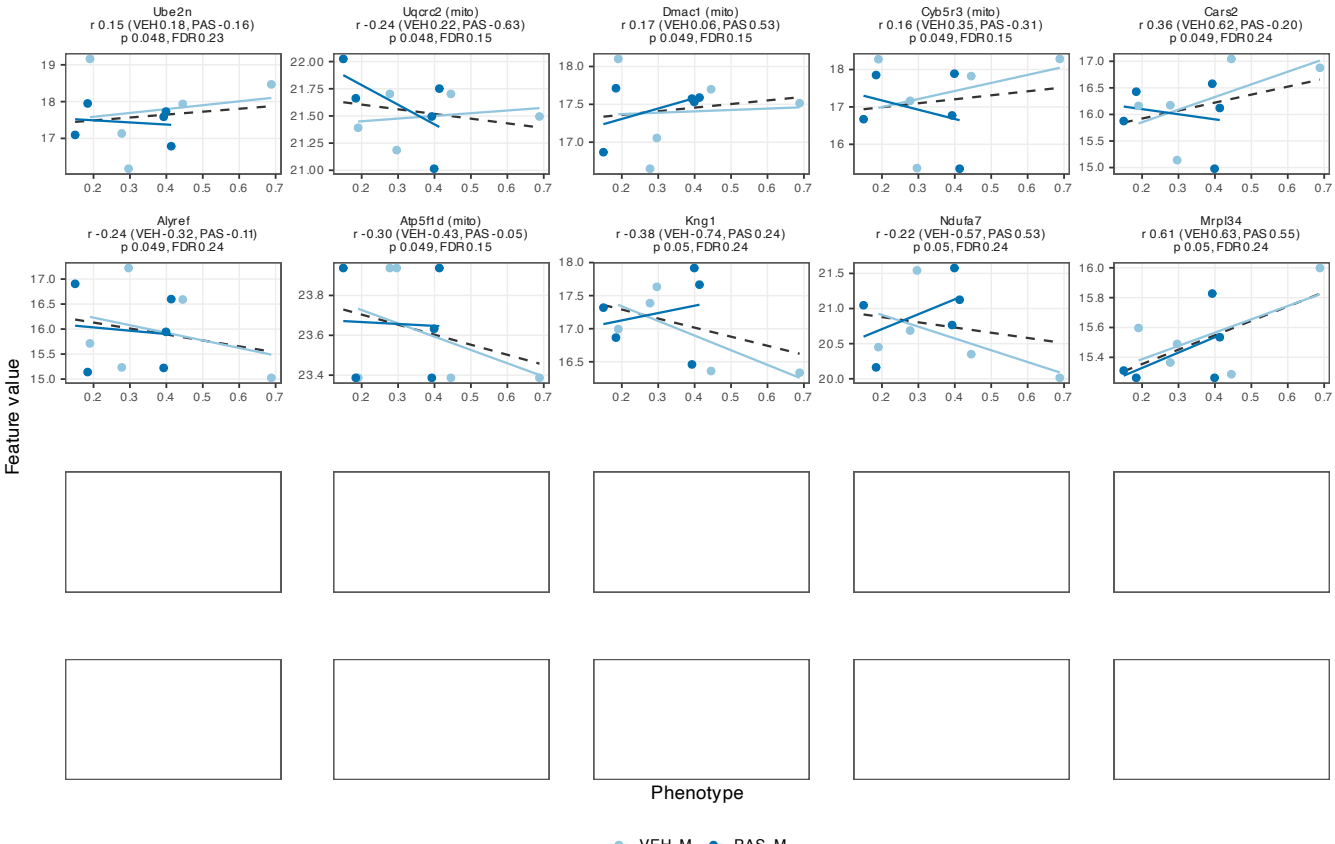
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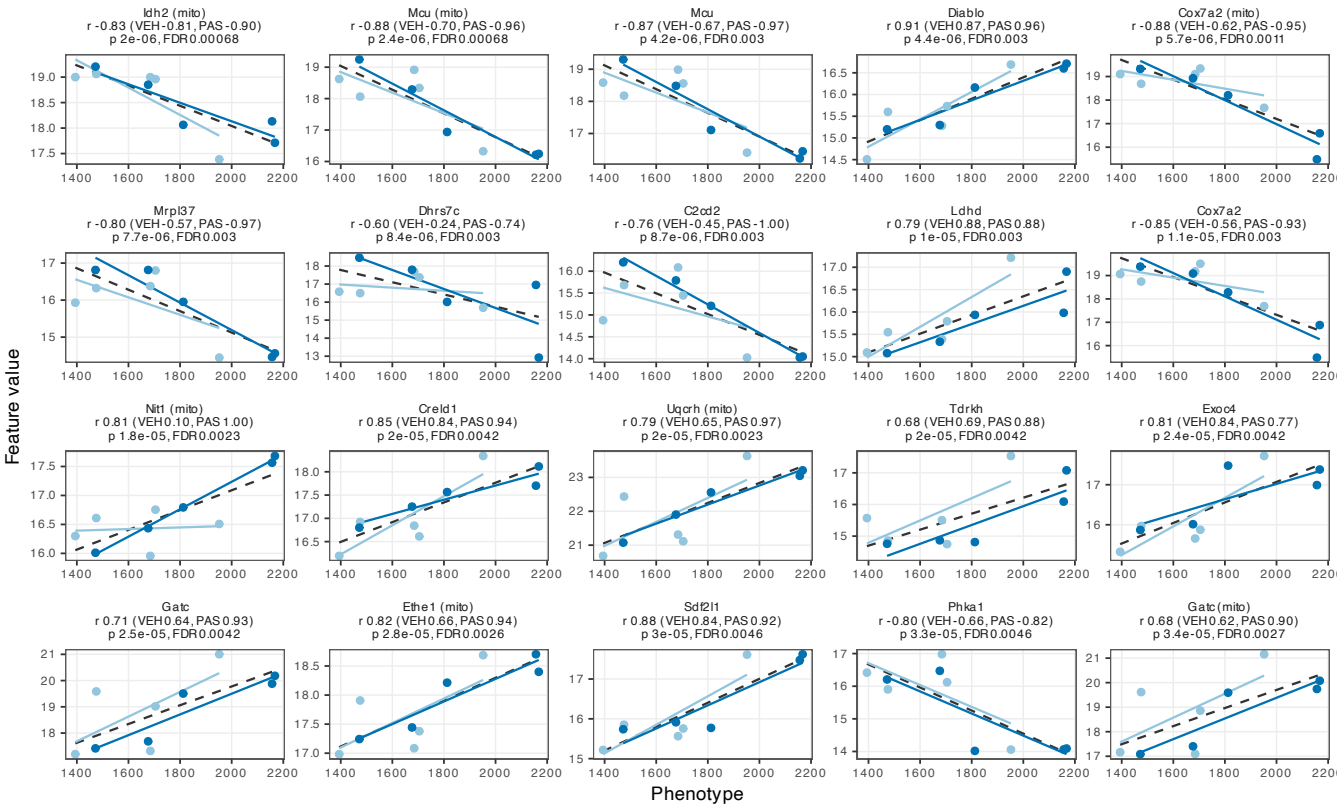
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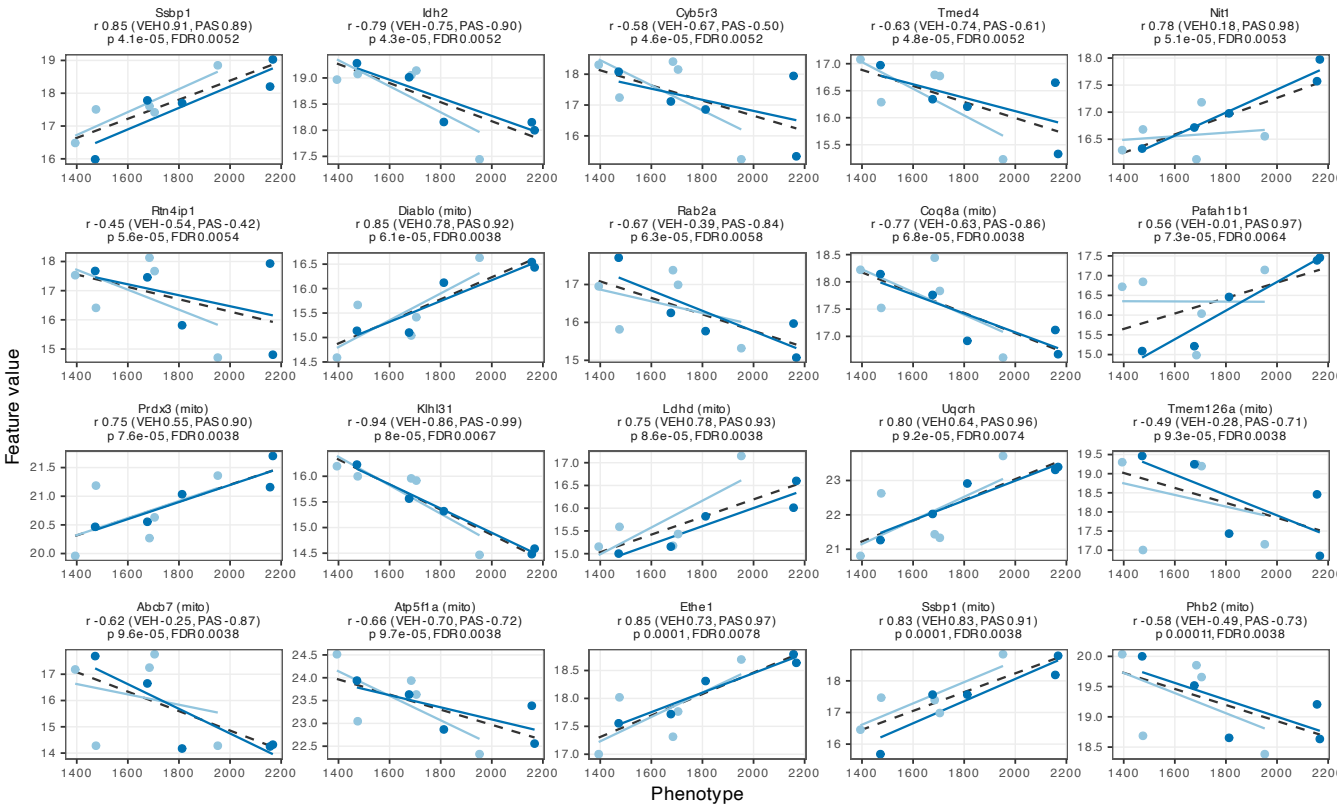
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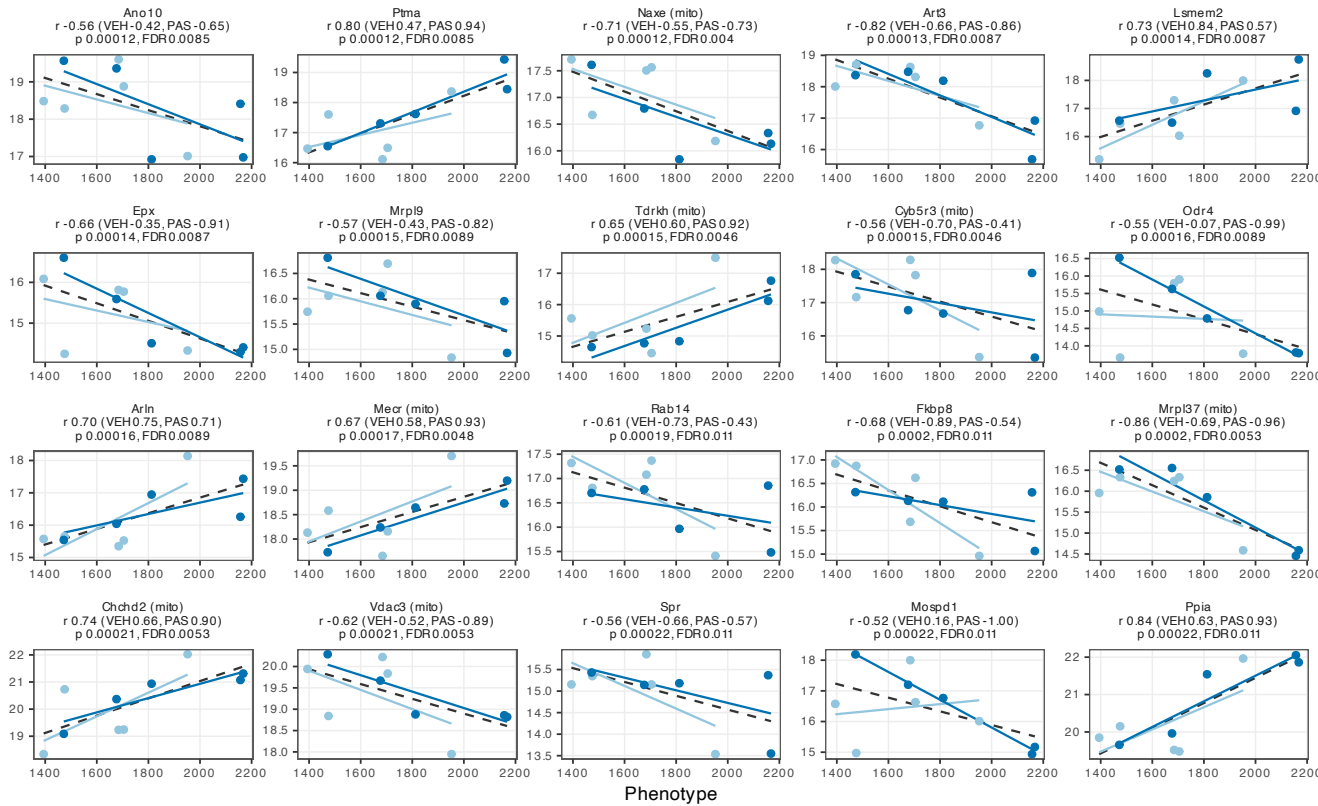
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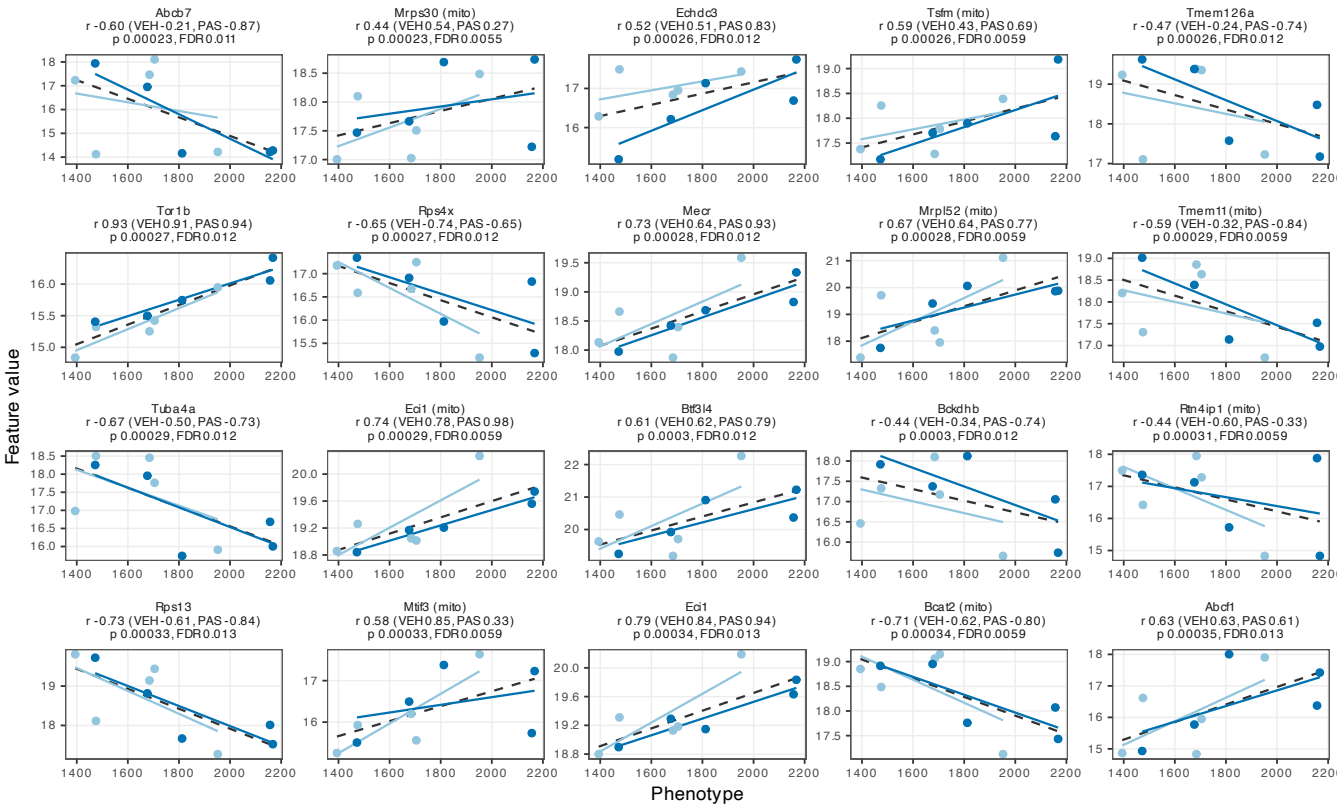
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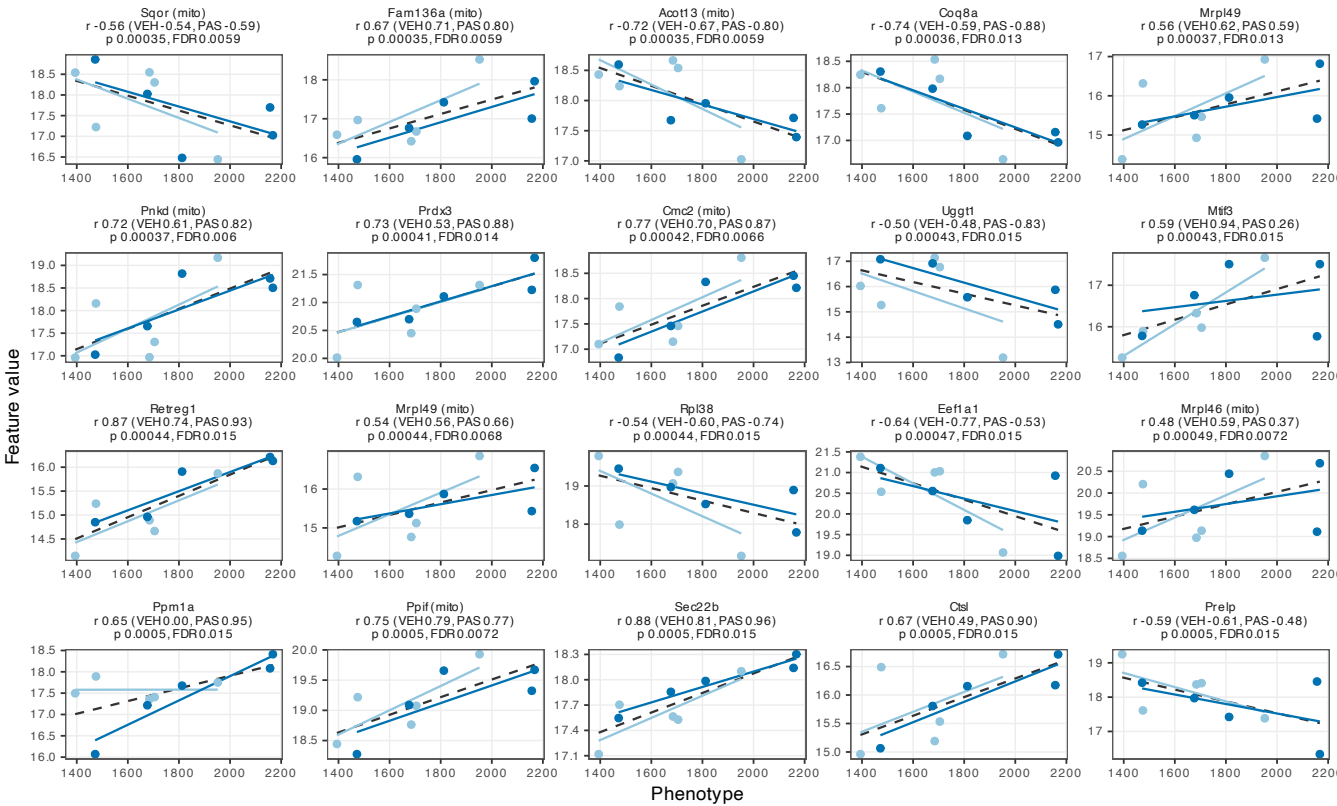
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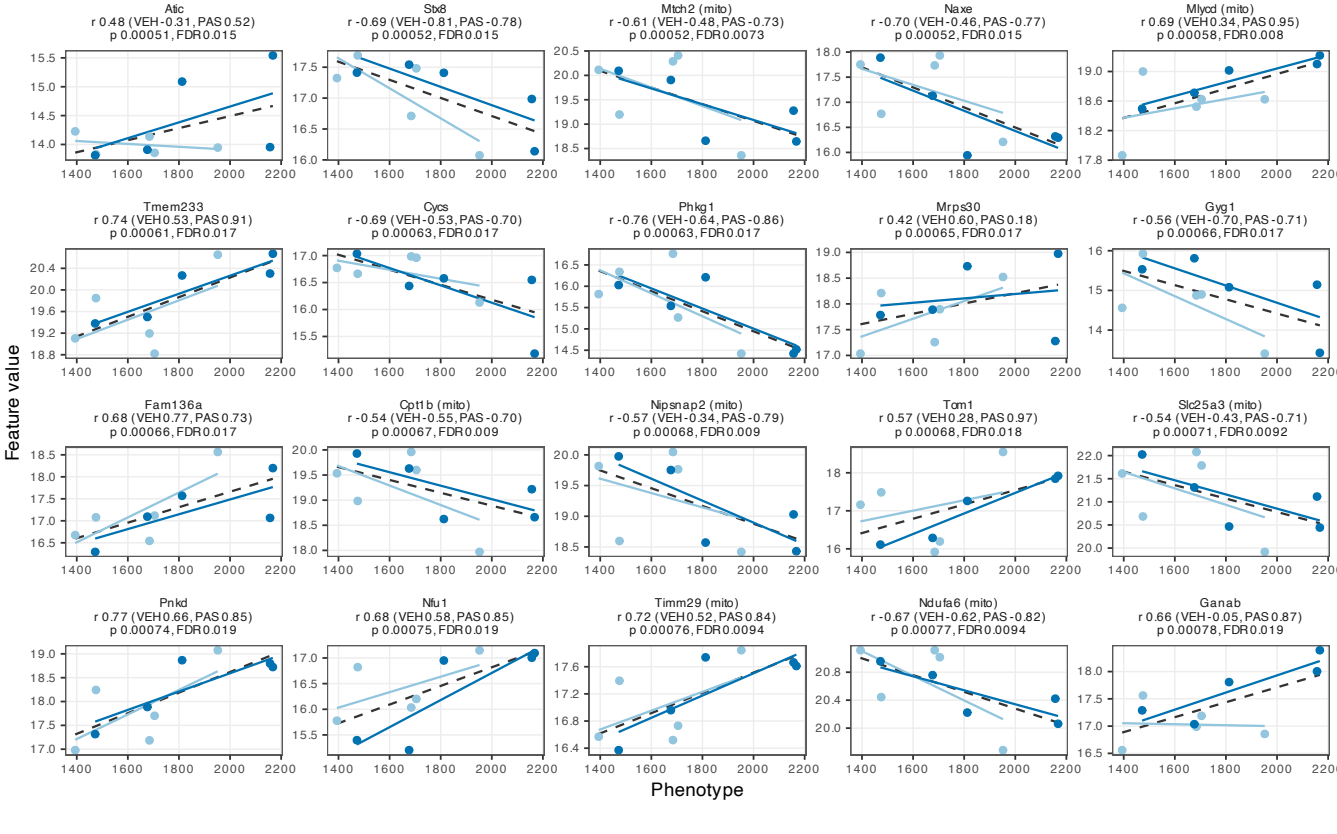
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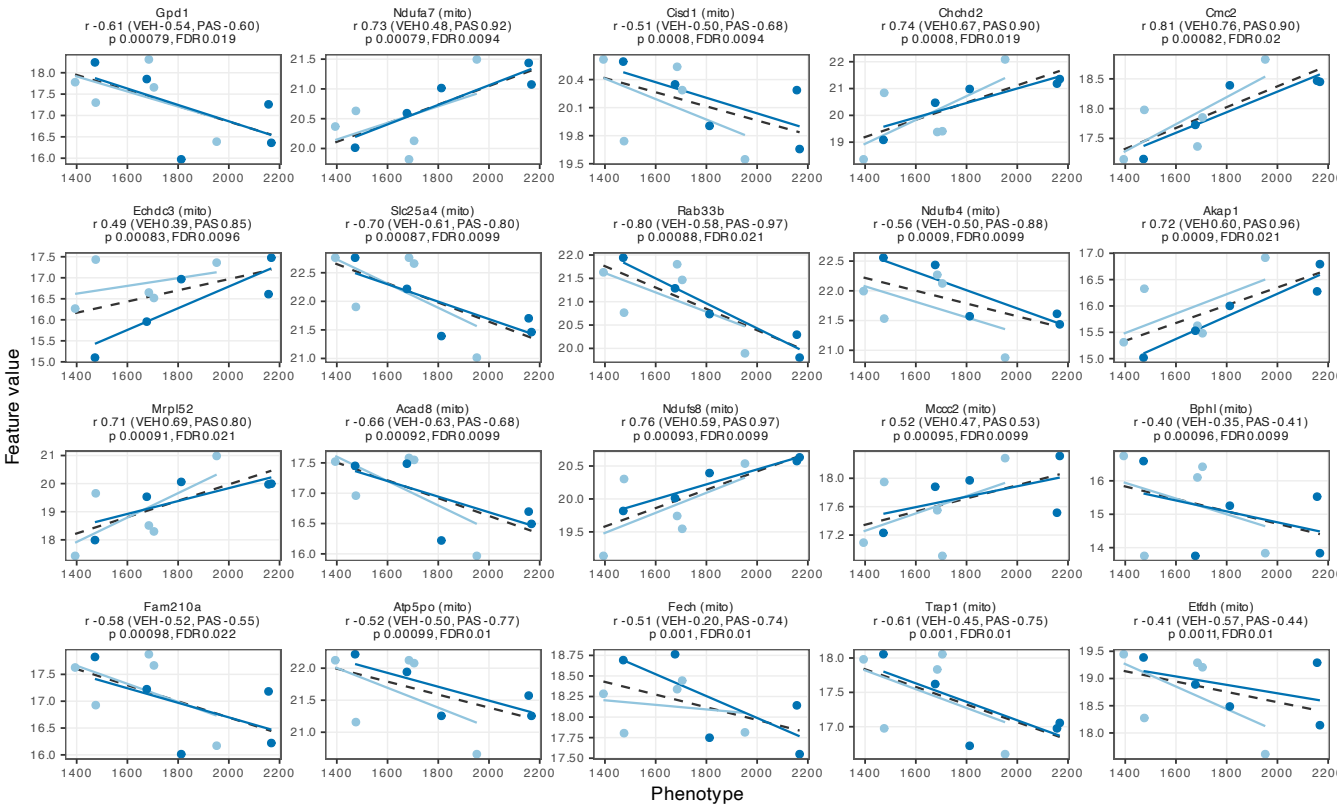
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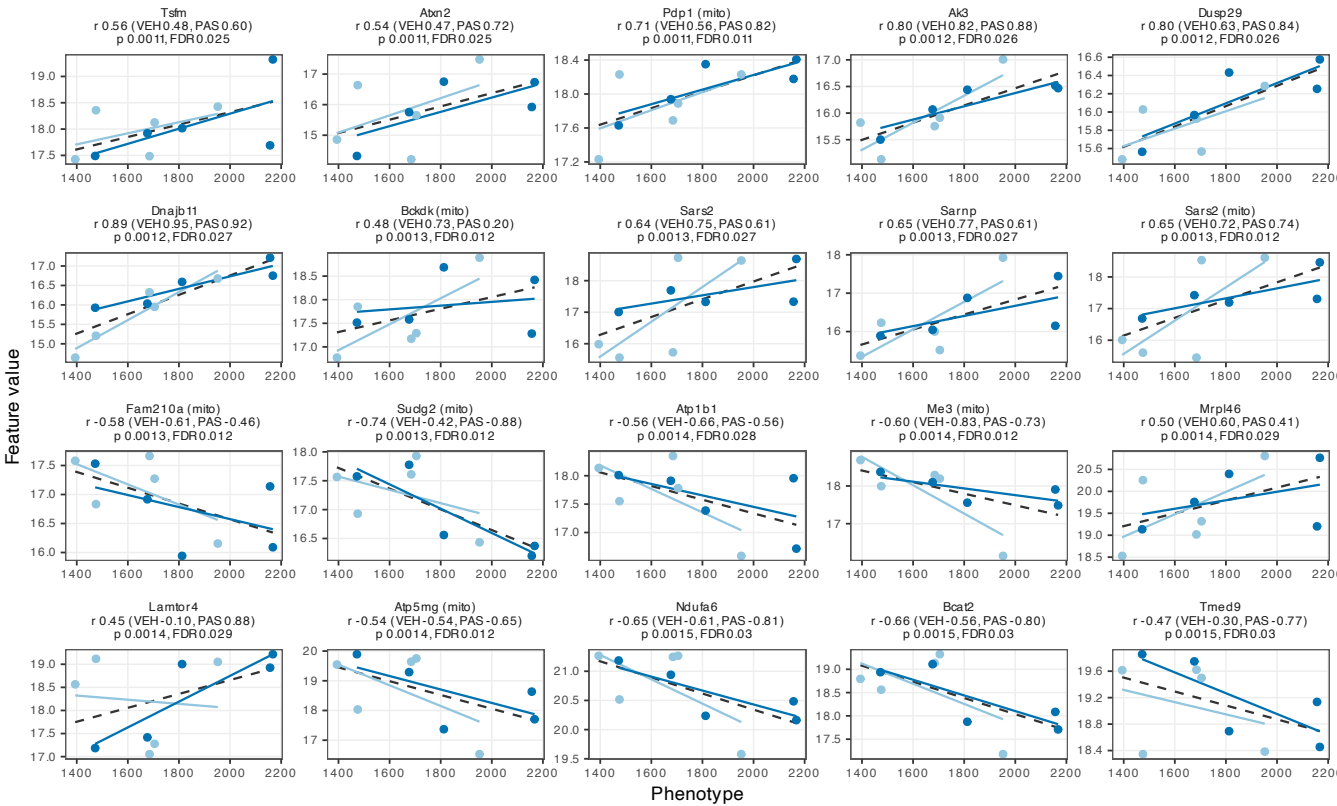
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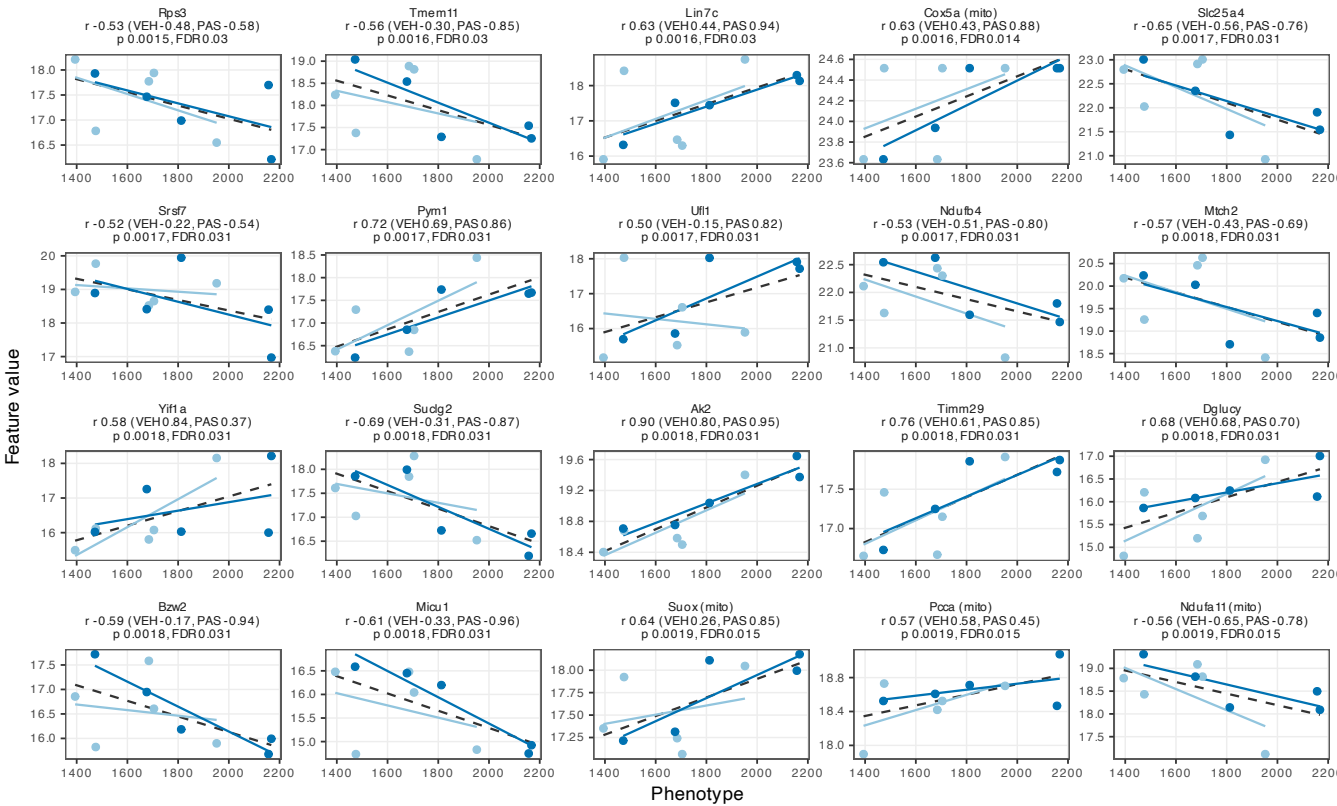
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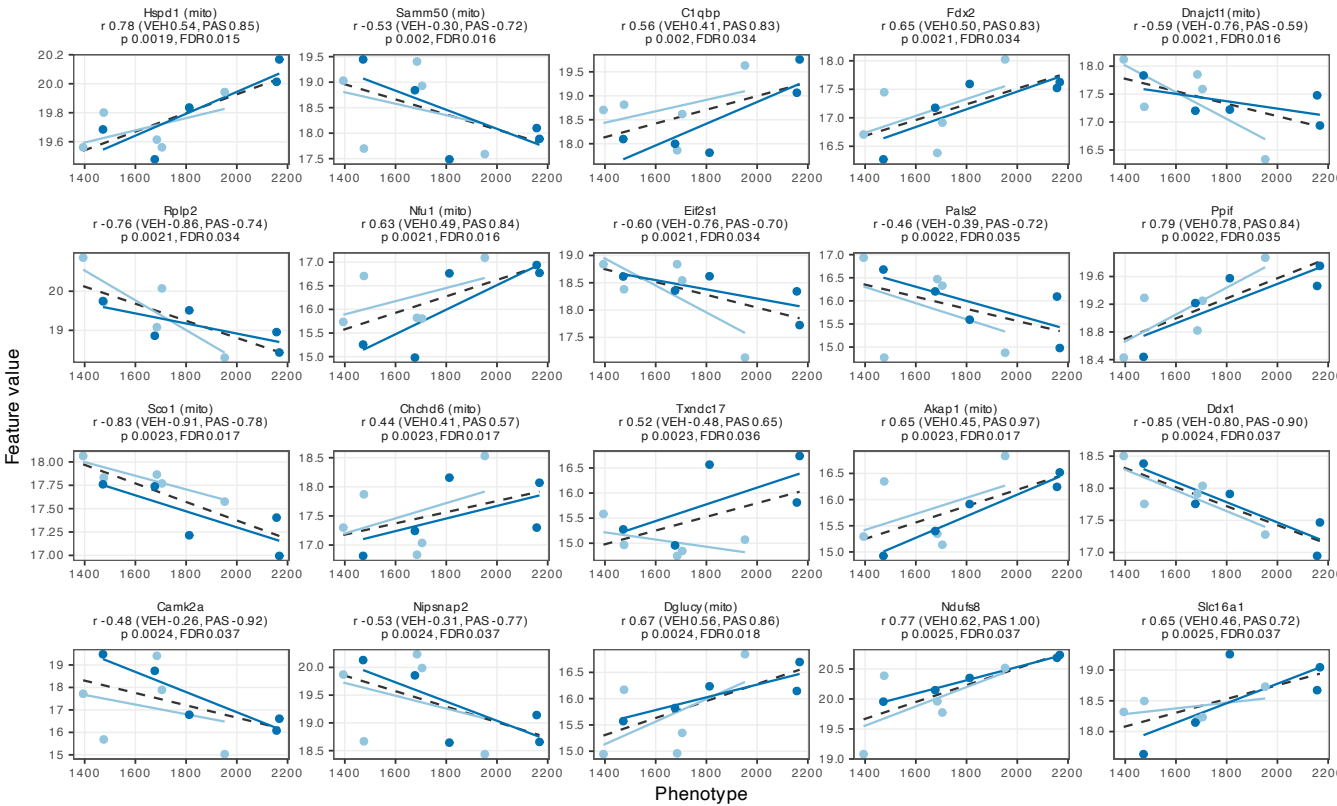
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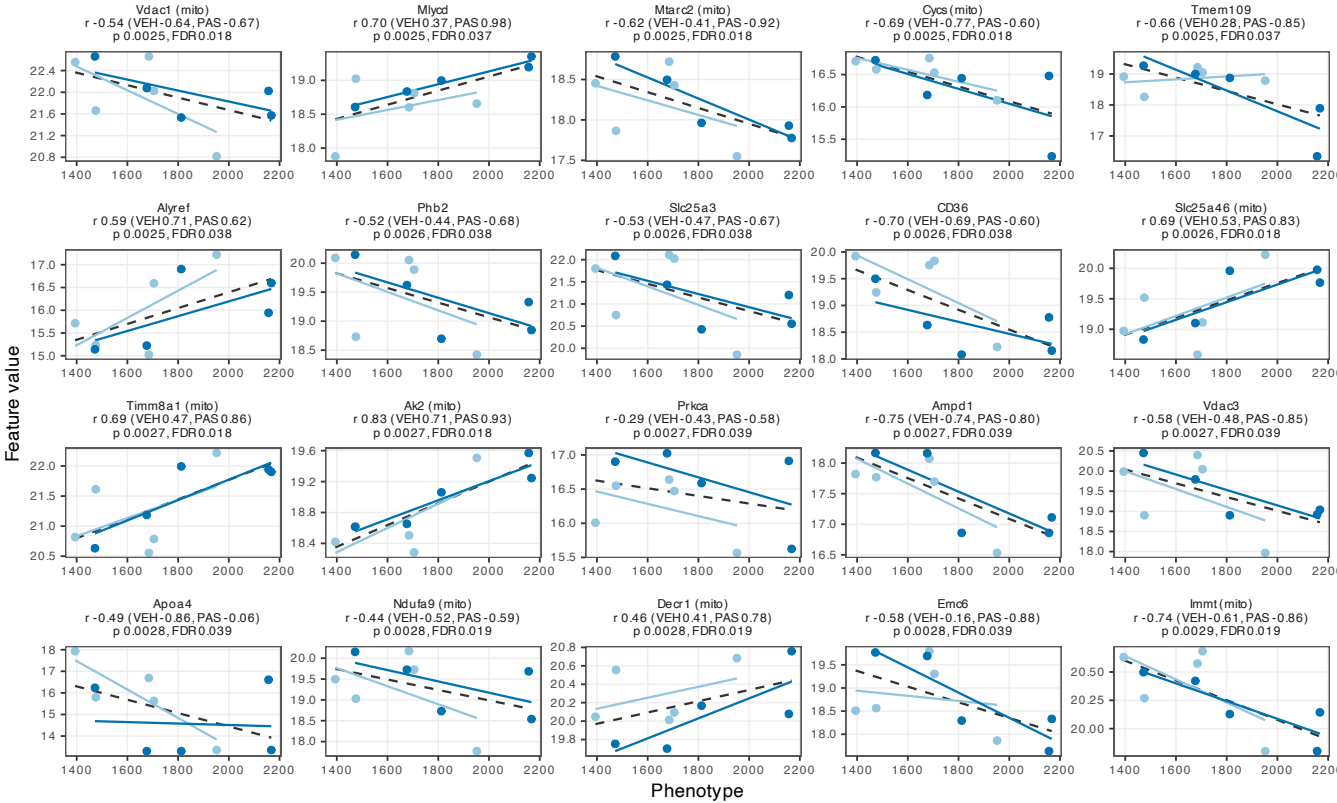
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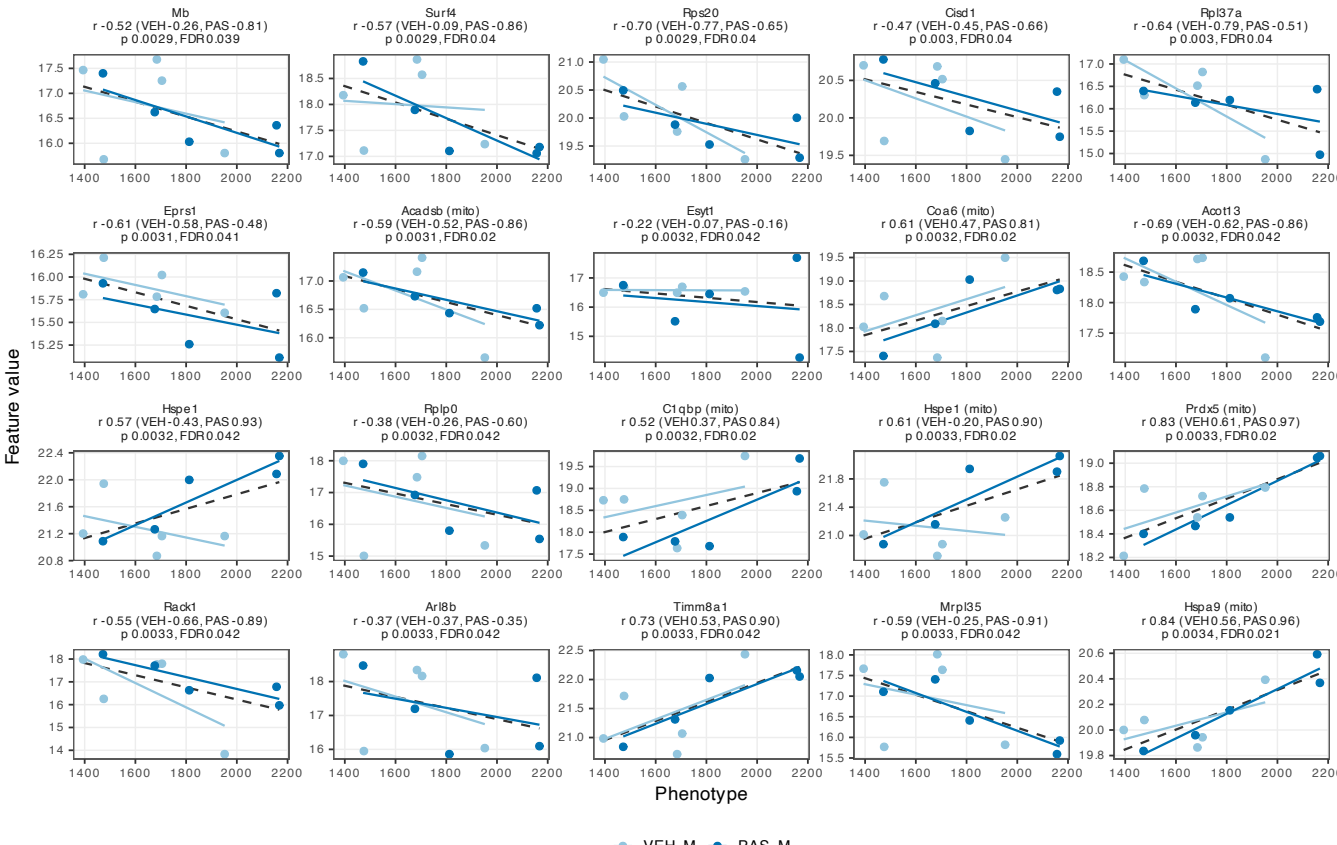
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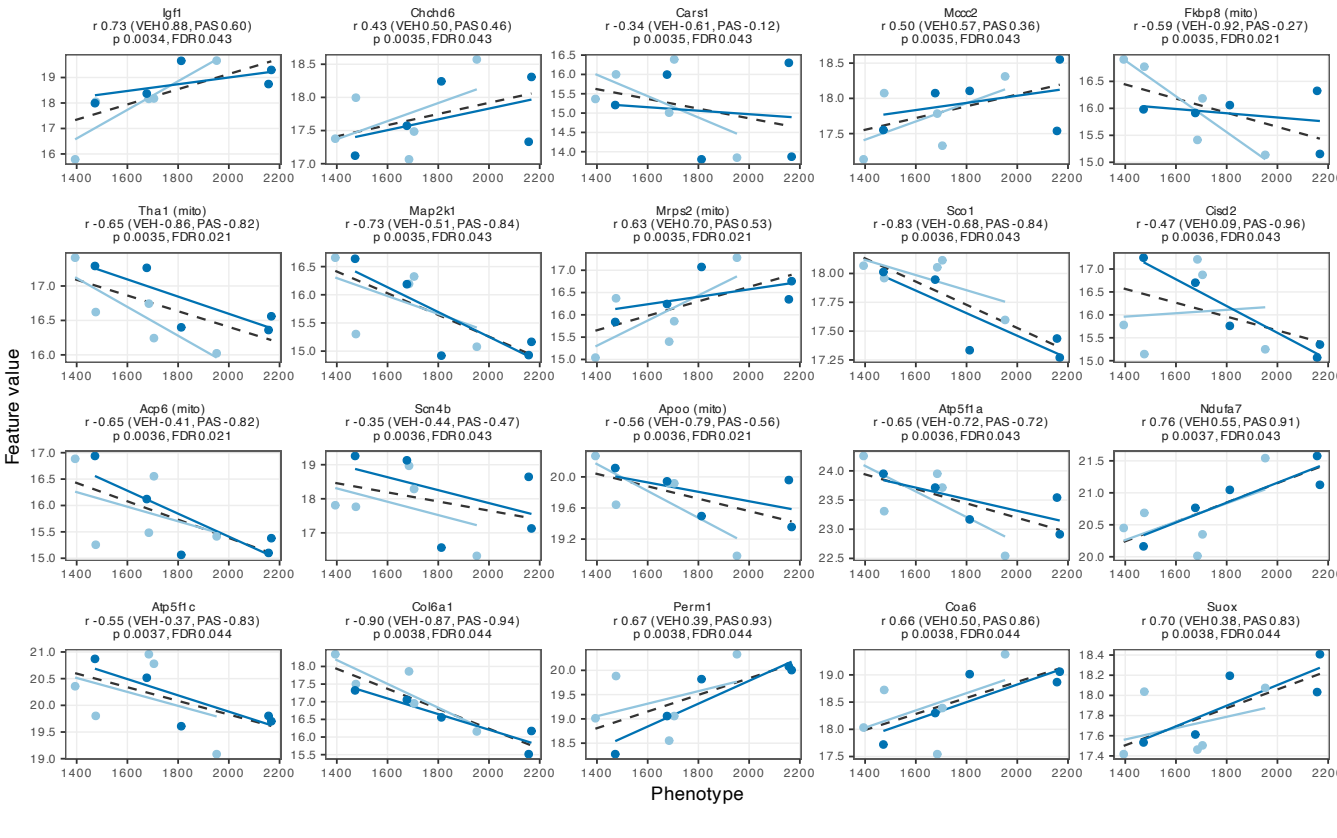
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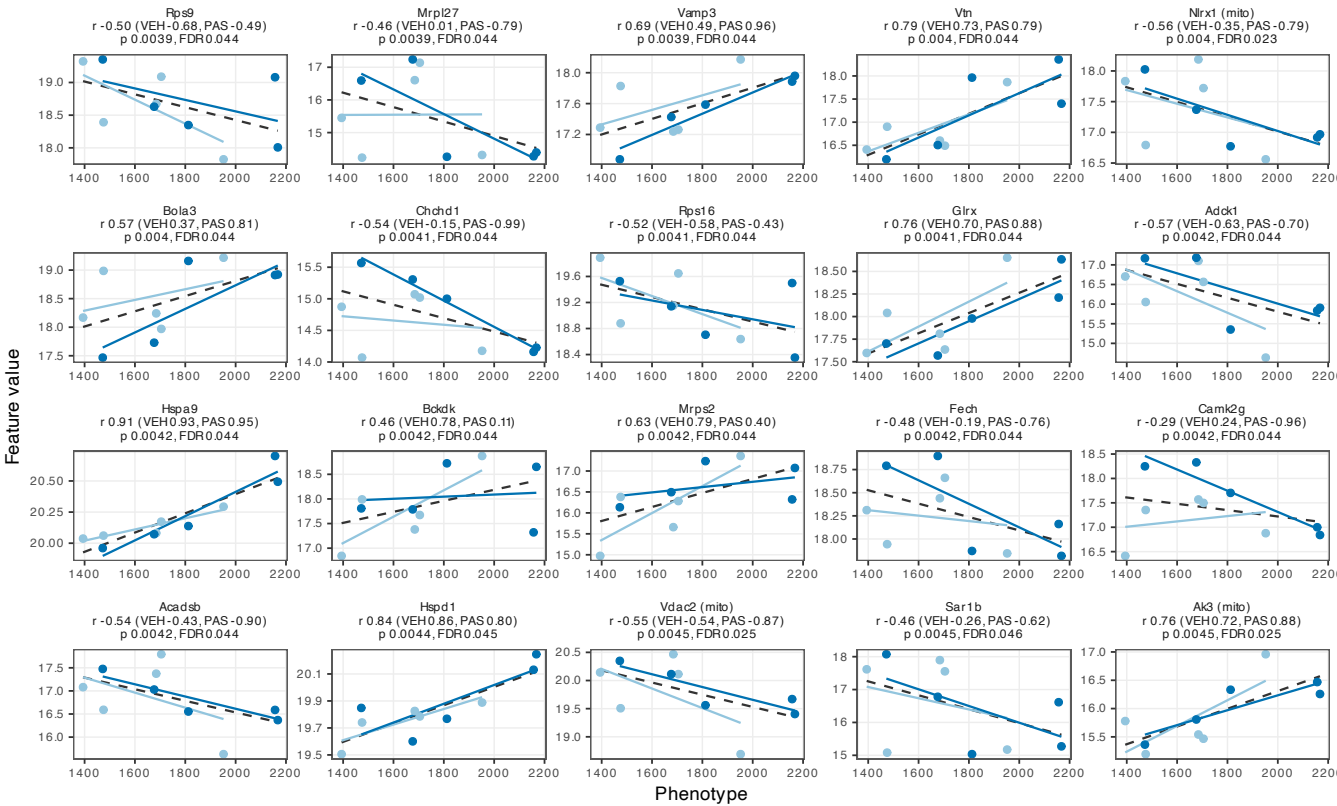
**Male: Proteins associated with CSA, every association at p ≤ 0.05 (n = 681)**

*Phenotype effect of PAS in this sex: +13% (p 0.25), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



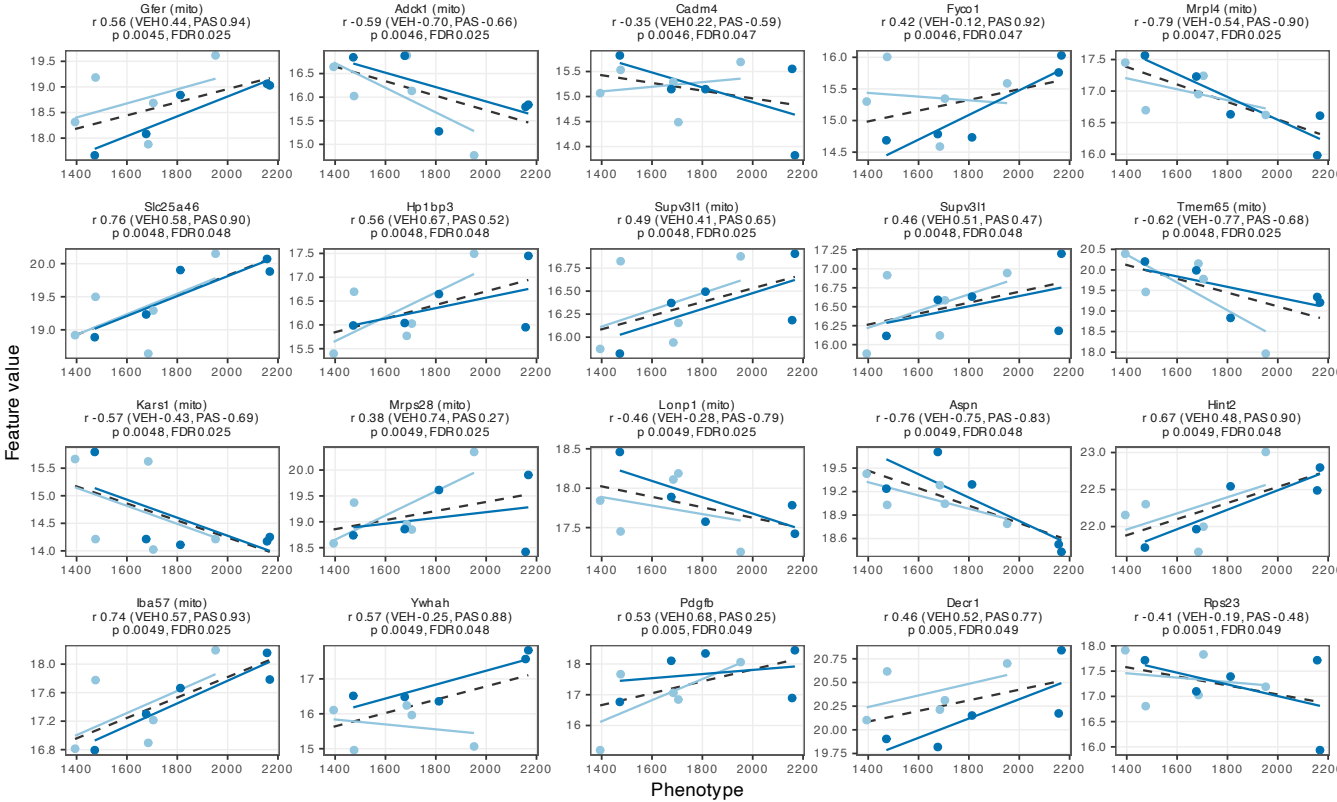
**Male: Proteins associated with CSA, every association at p ≤ 0.05 (n = 681)**

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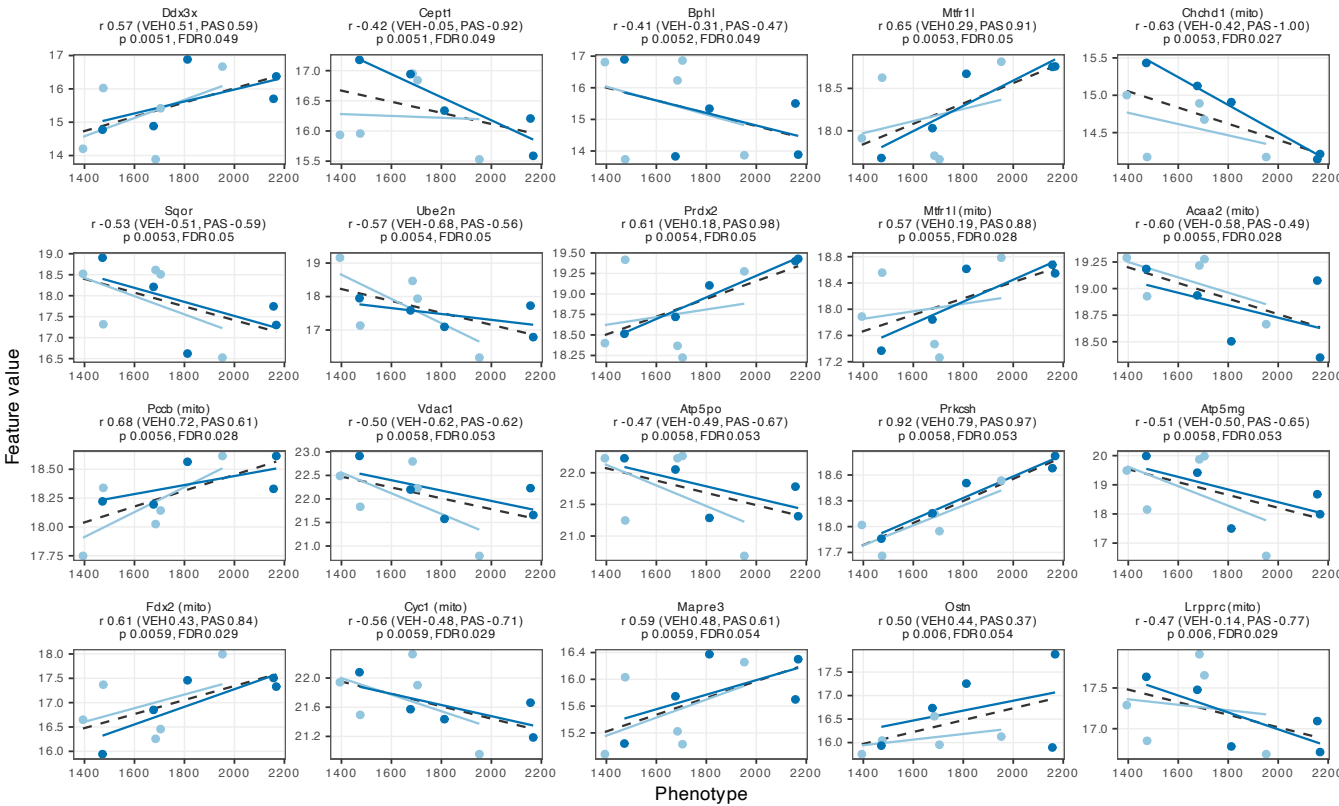
**Male: Proteins associated with CSA, every association at p ≤ 0.05 (n = 681)**

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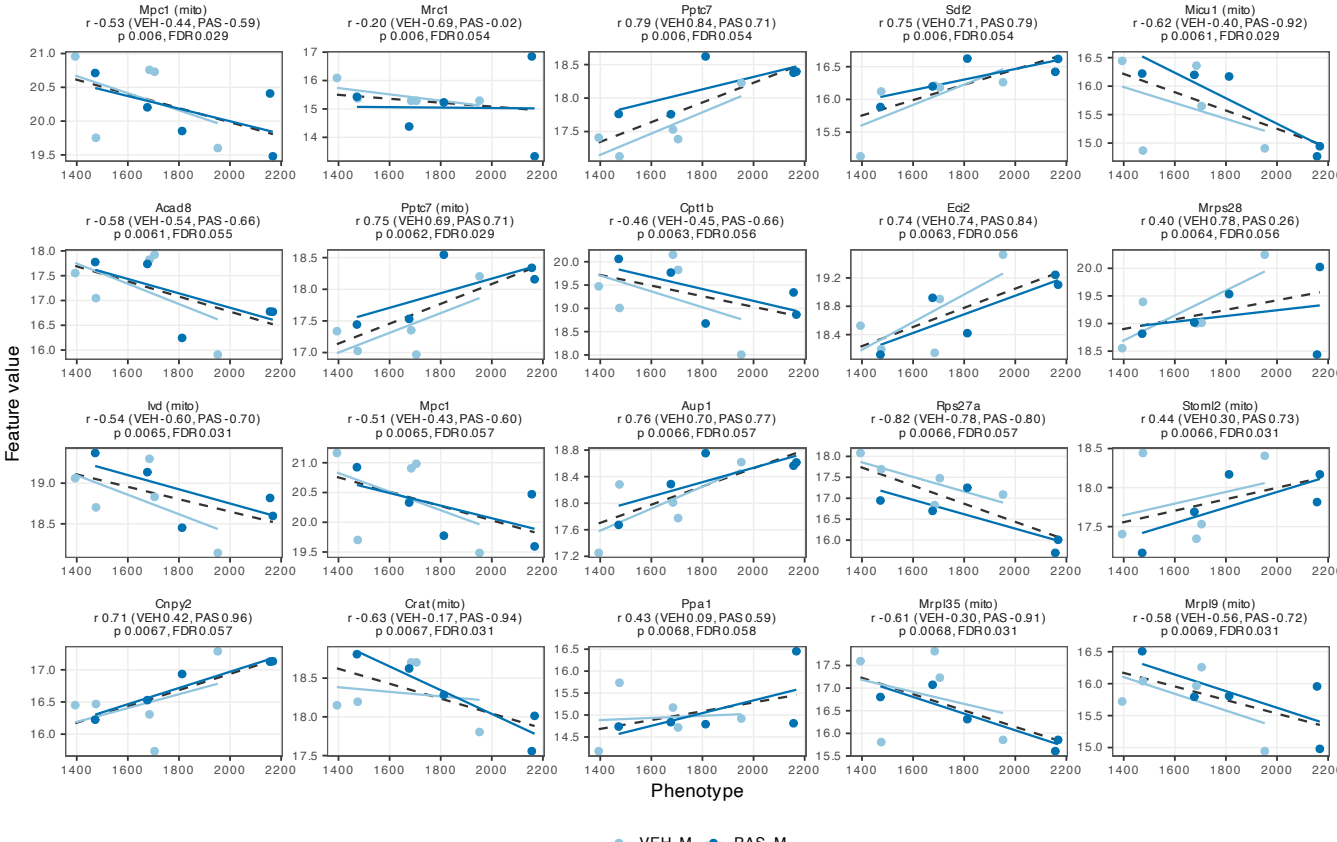
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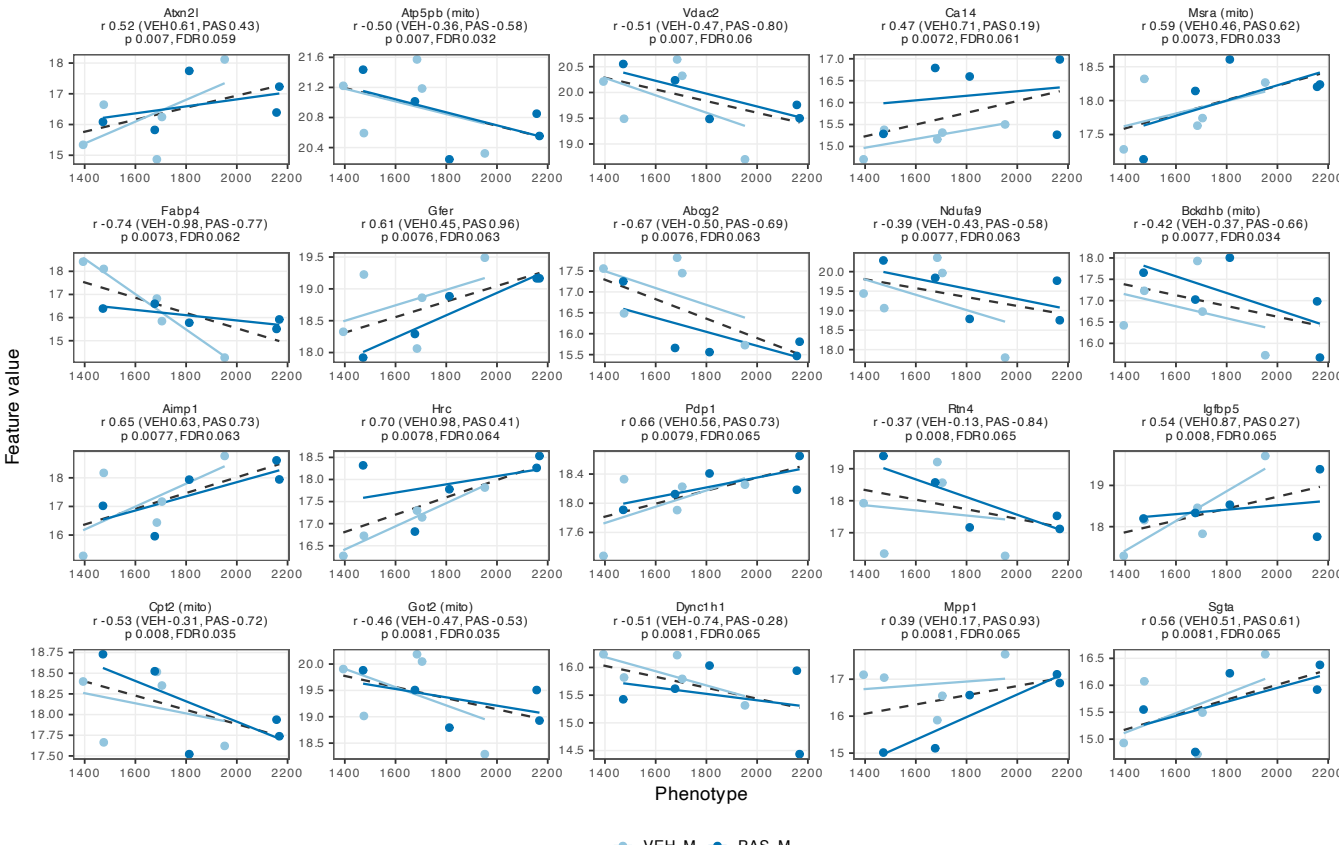
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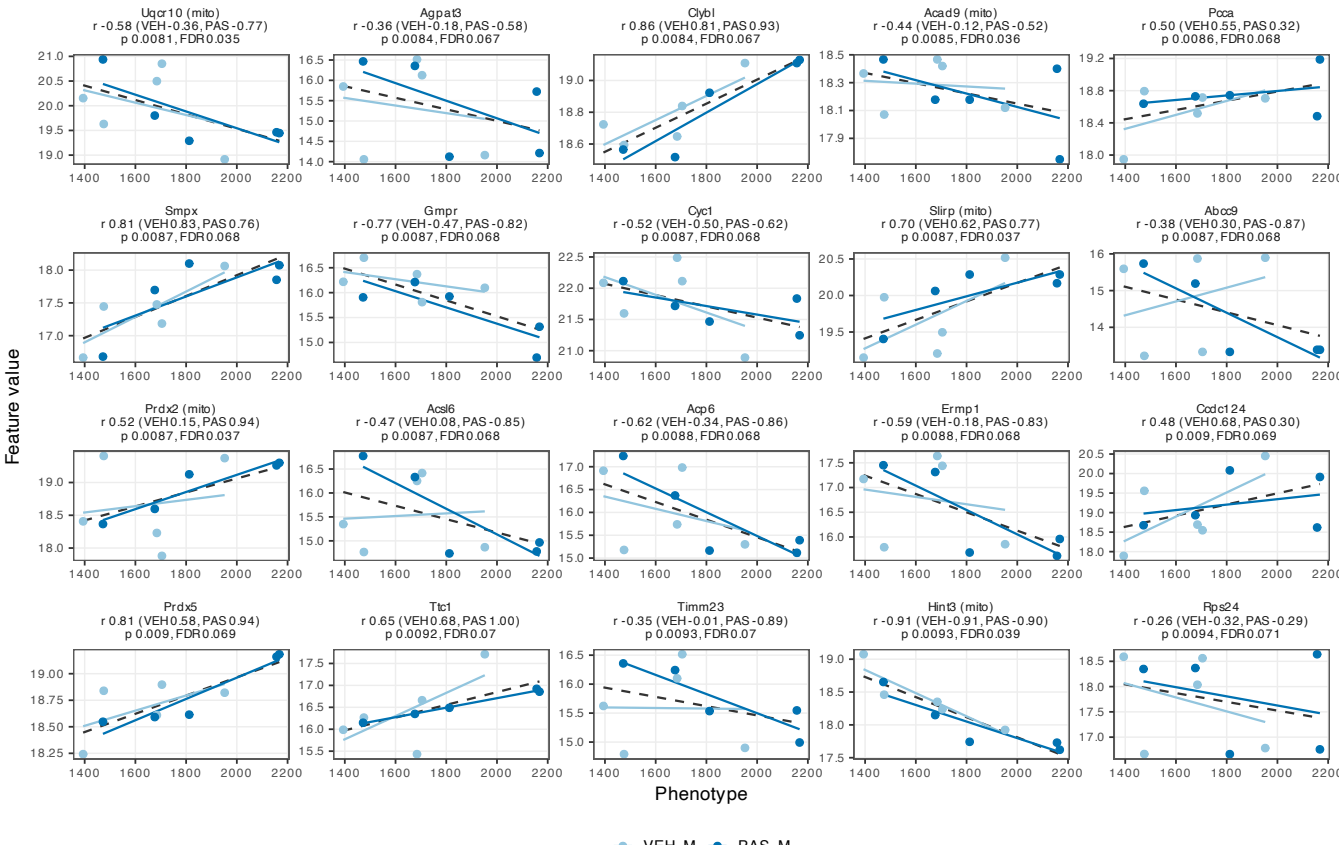
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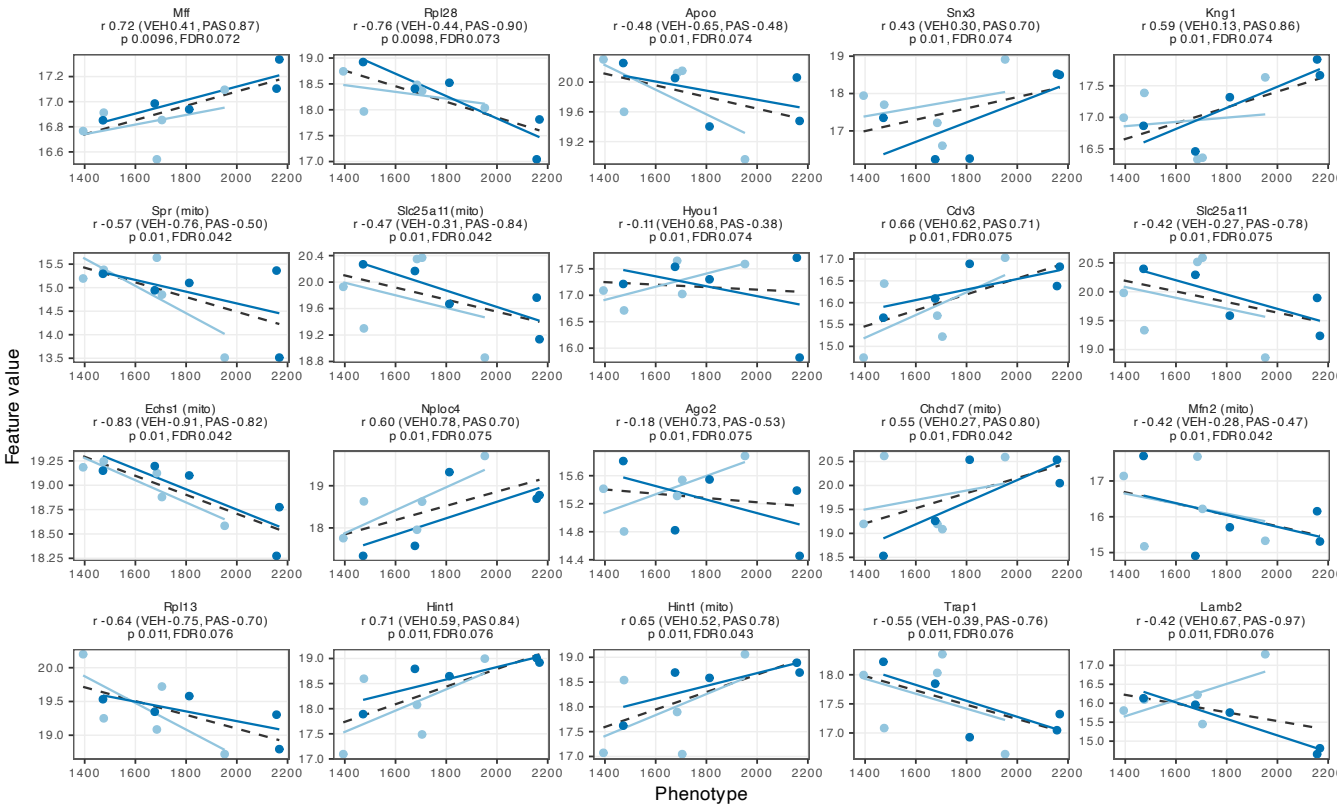
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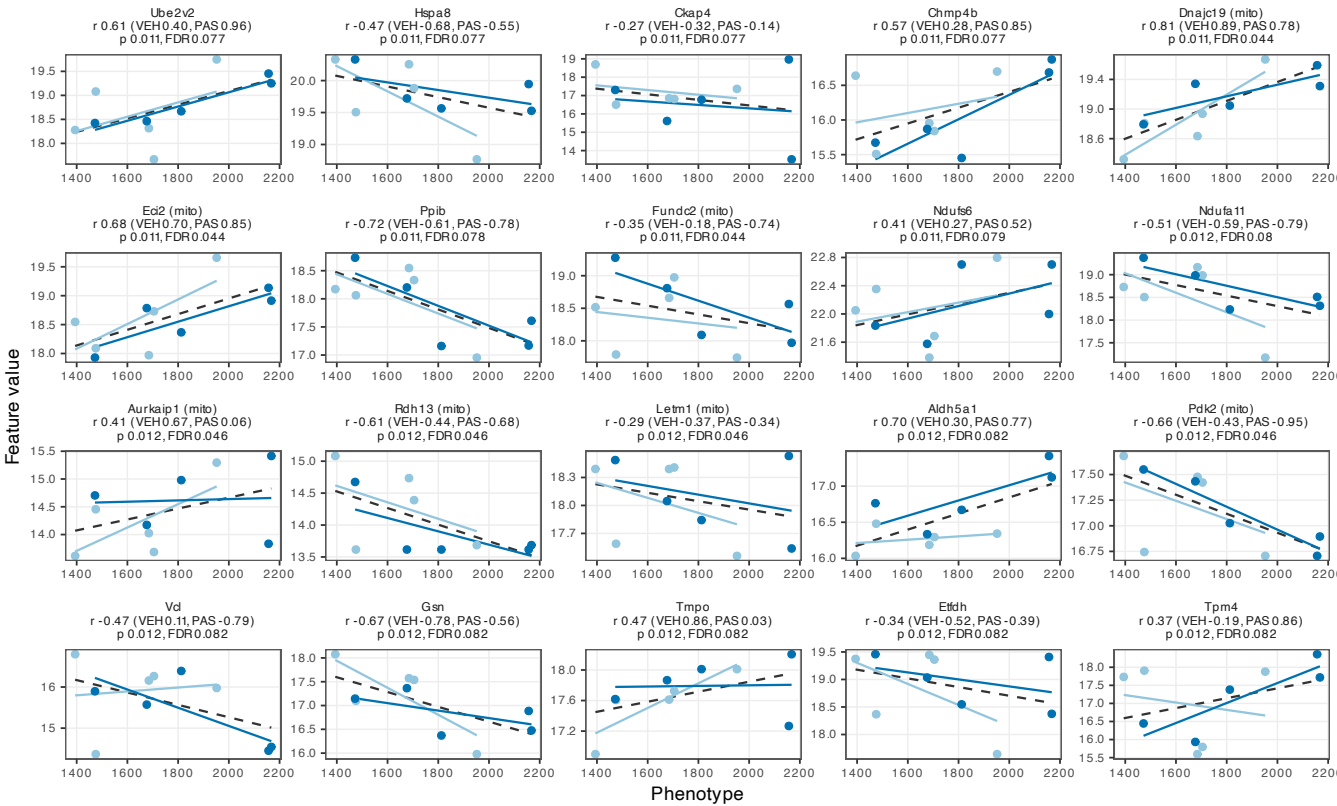
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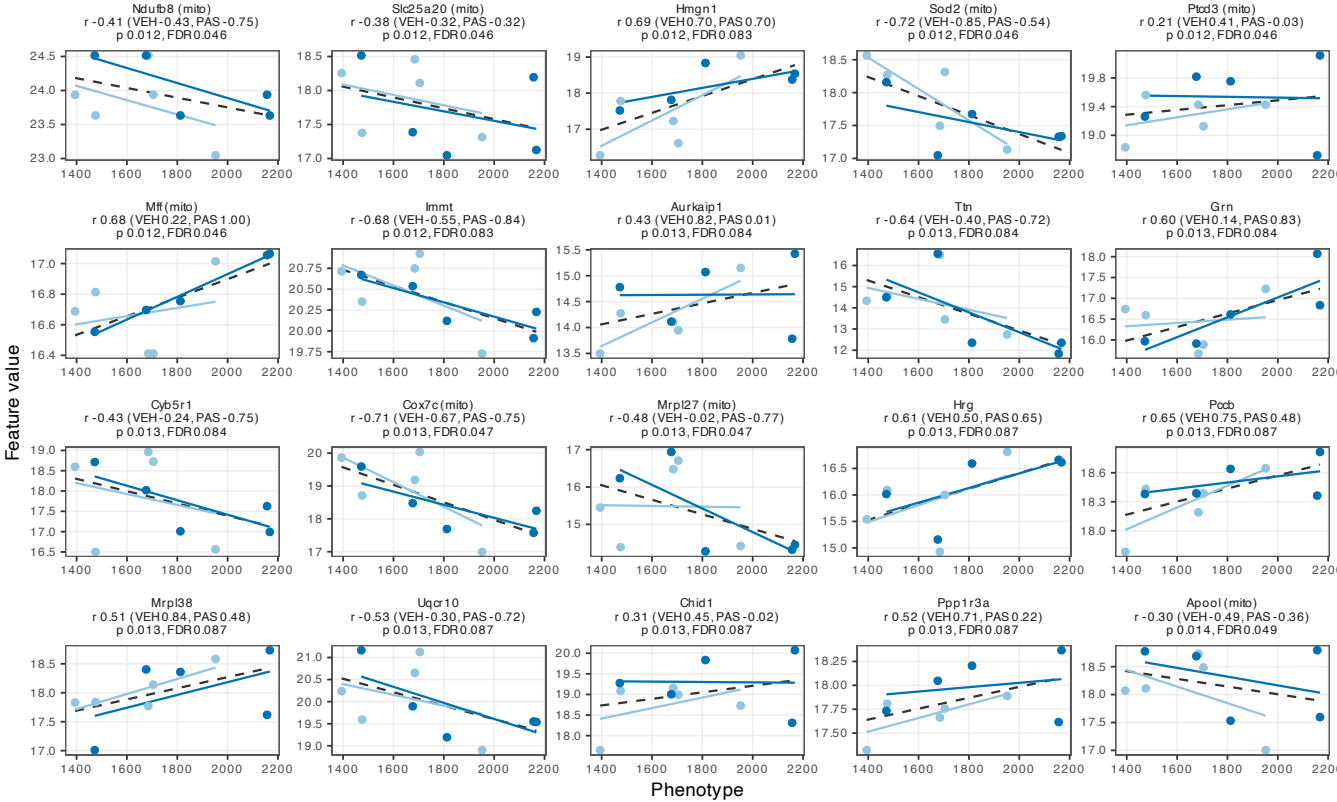
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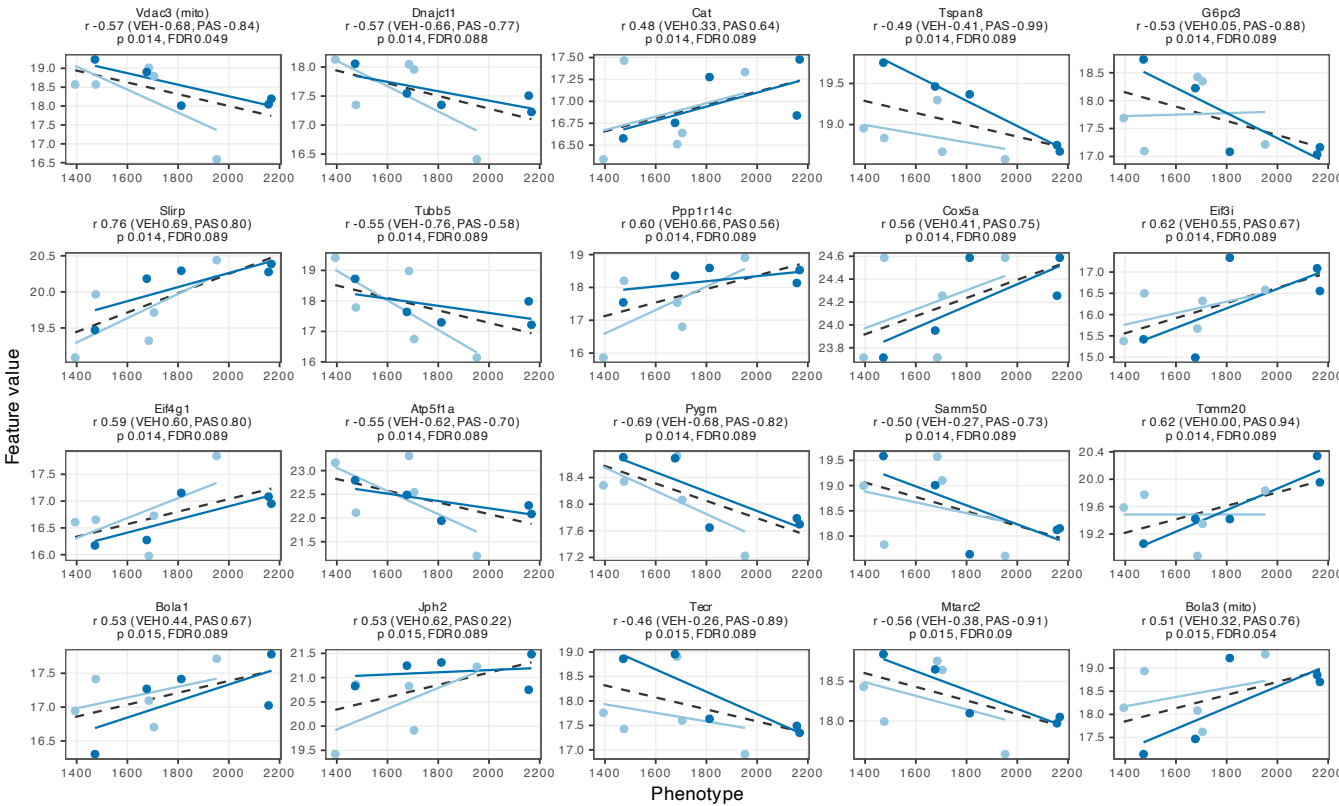
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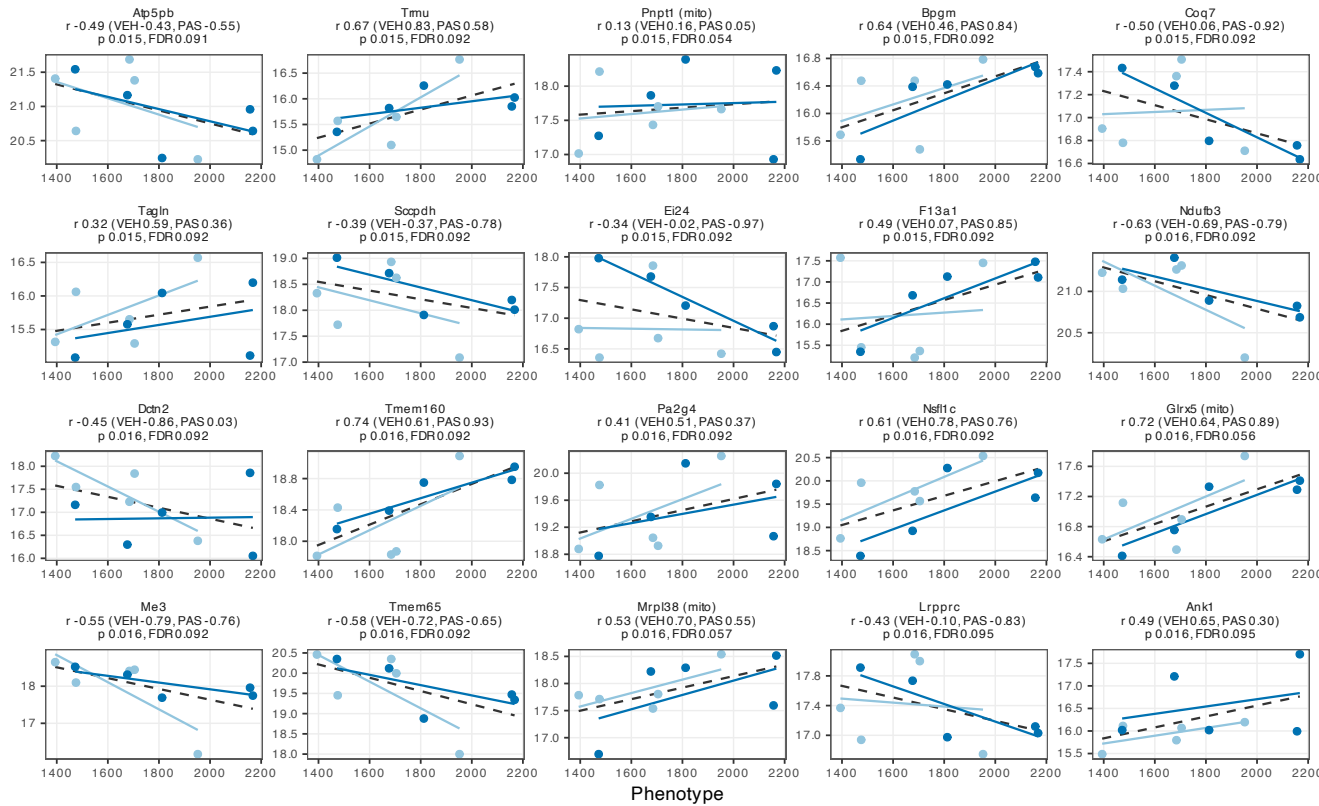
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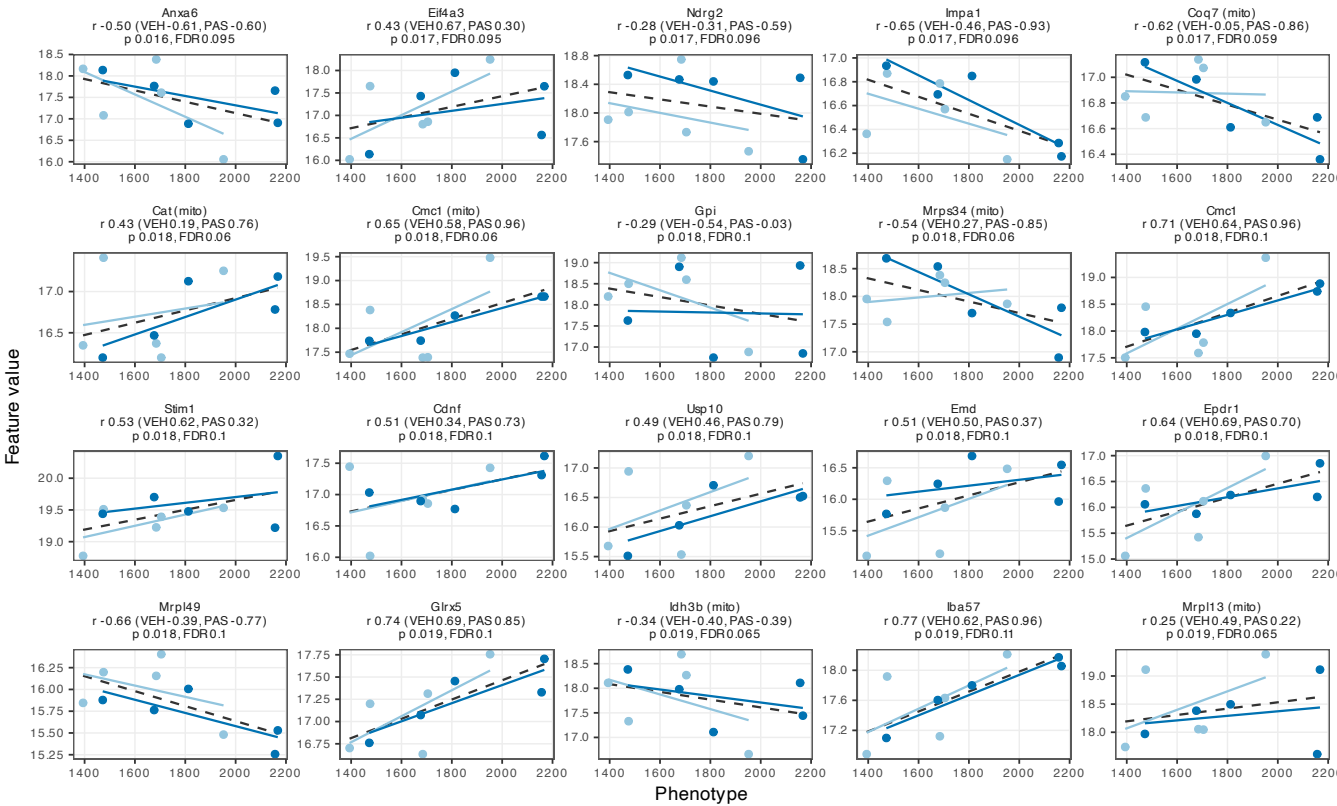
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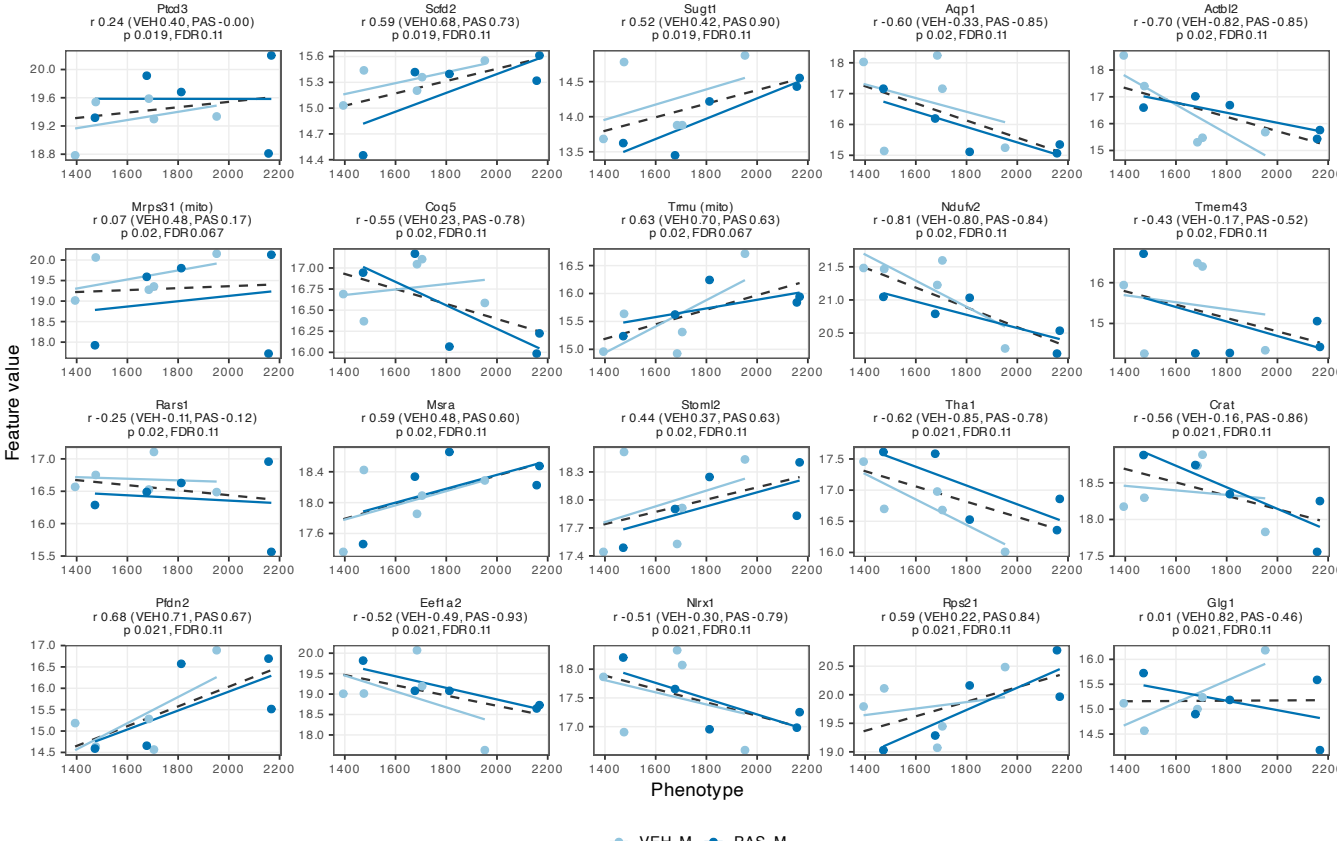
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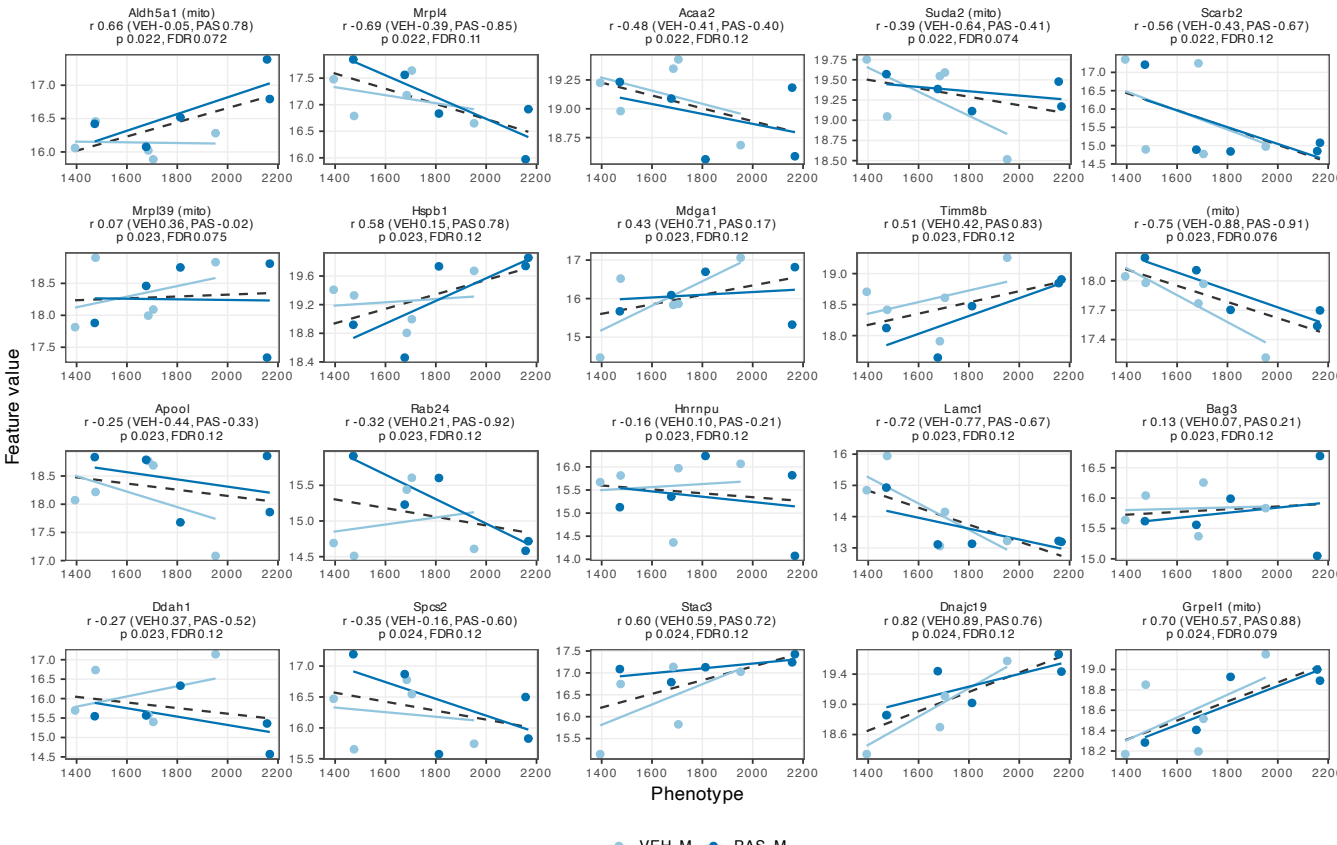
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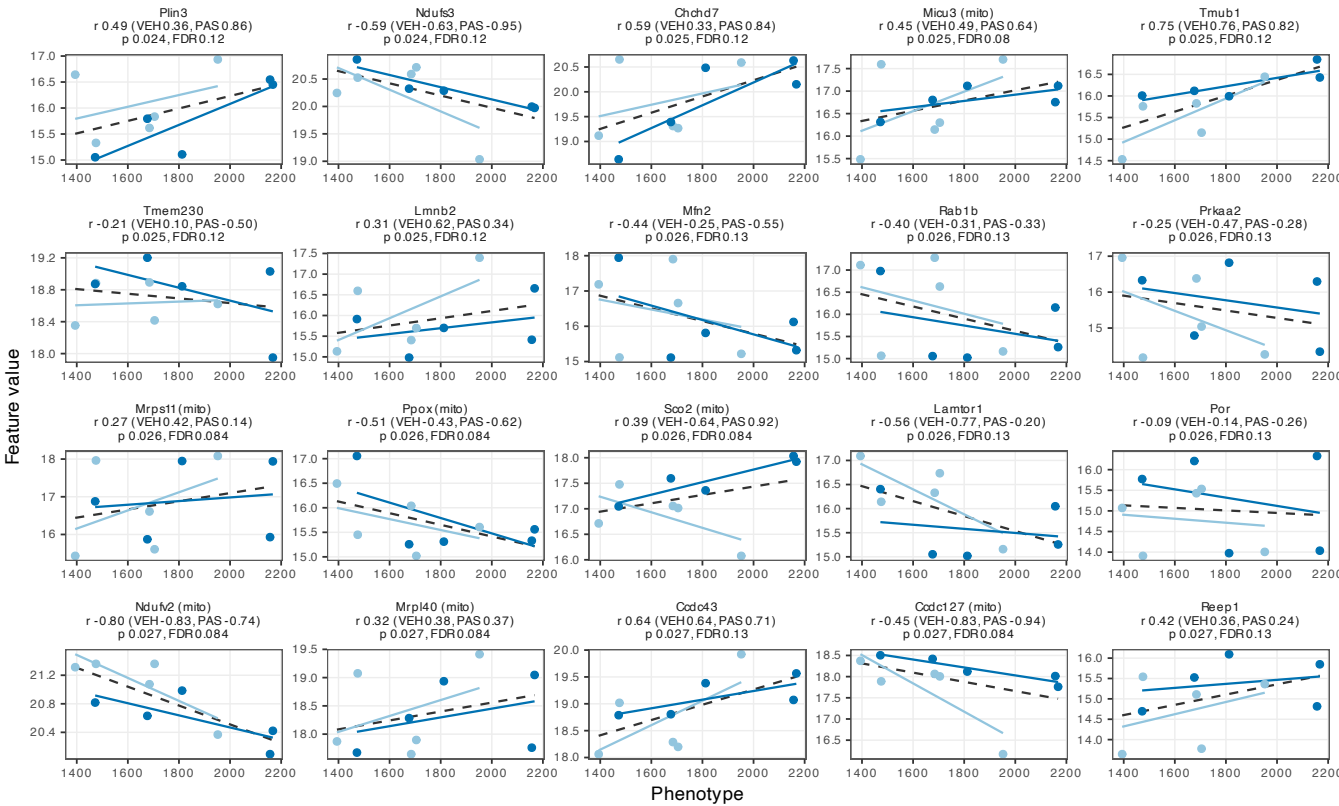
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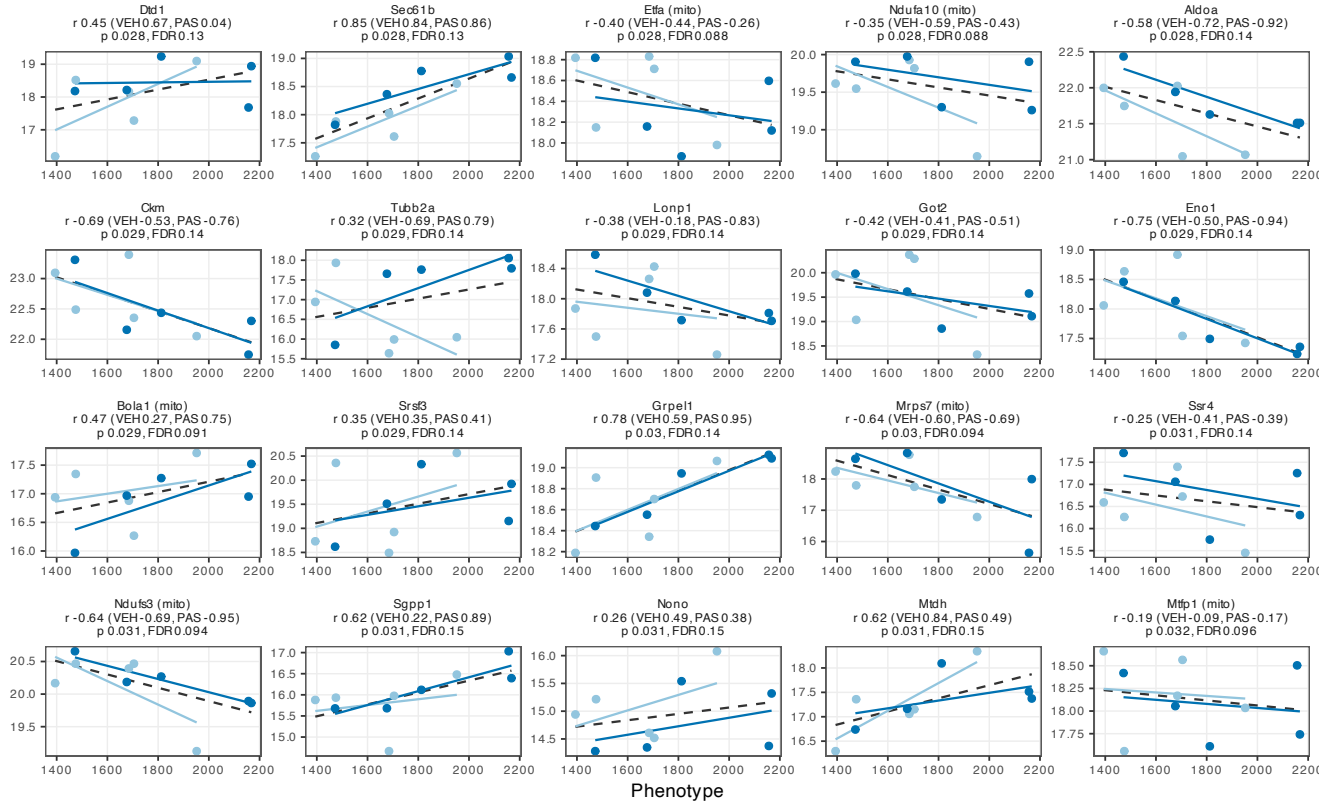
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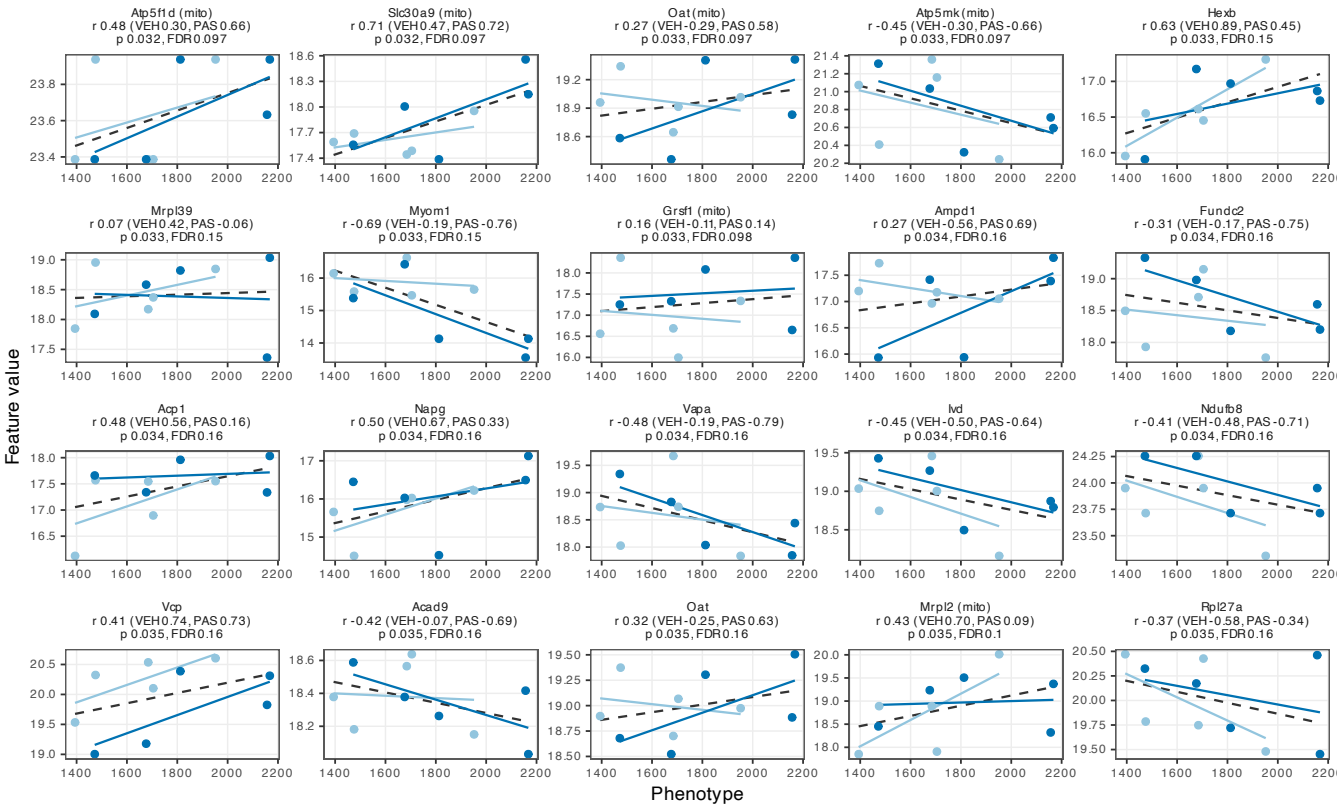
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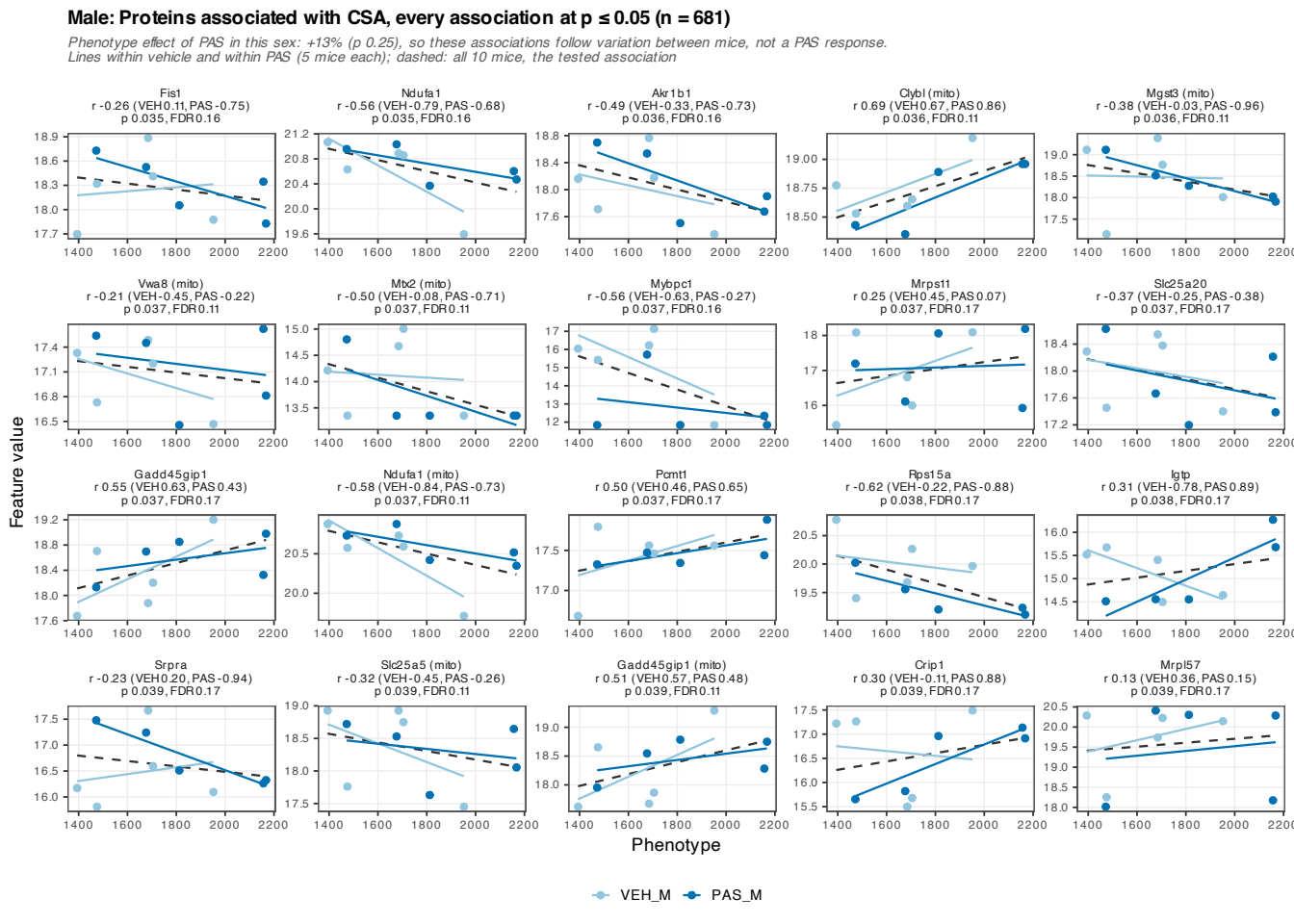
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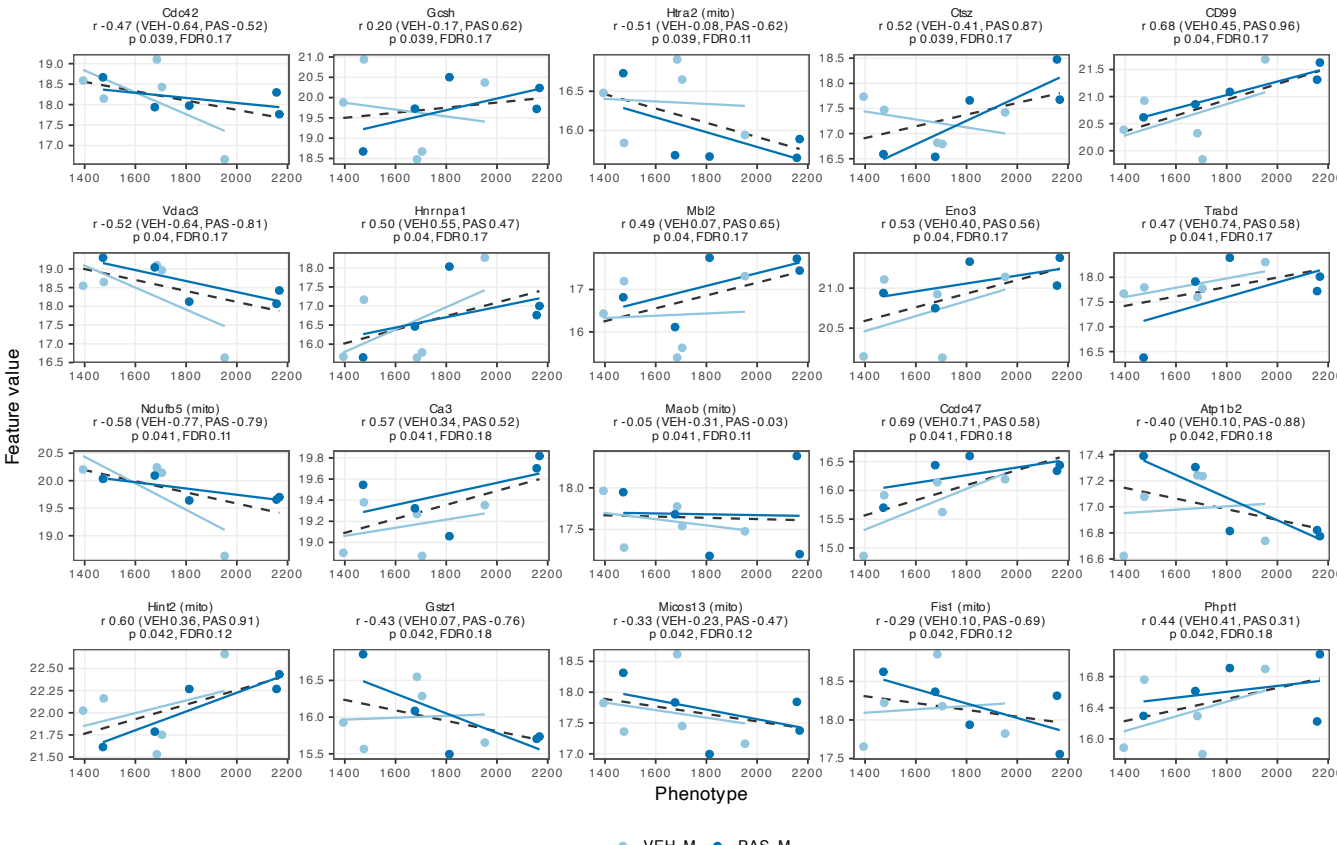
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Male: Proteins associated with CSA, every association at p ≤ 0.05 (n = 681)
Phenotype effect of PAS in this sex: +13% (p 0.25), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association
Fis1
r -0.26 (VEH 0.11, PAS -0.75)
p 0.035, FDR 0.16
Ndufa1
r -0.56 (VEH -0.79, PAS -0.68)
p 0.035, FDR 0.16
Akr1b1
r -0.49 (VEH -0.33, PAS -0.73)
p 0.036, FDR 0.16
Clybl (mito)
r 0.69 (VEH 0.67, PAS 0.86)
p 0.036, FDR 0.11
Mgst3 (mito)
r -0.38 (VEH -0.03, PAS -0.96)
p 0.036, FDR 0.11
Vwa8 (mito)
r -0.21 (VEH -0.45, PAS -0.22)
p 0.037, FDR 0.11
Mtx2 (mito)
r -0.50 (VEH -0.08, PAS -0.71)
p 0.037, FDR 0.11
Mybpc1
r -0.56 (VEH -0.63, PAS -0.27)
p 0.037, FDR 0.16
Mrps11
r 0.25 (VEH 0.45, PAS 0.07)
p 0.037, FDR 0.17
Slc25a20
r -0.37 (VEH -0.25, PAS -0.38)
p 0.037, FDR 0.17
Gadd45gip1
r 0.55 (VEH 0.63, PAS 0.43)
p 0.037, FDR 0.17
Ndufa1 (mito)
r -0.58 (VEH -0.84, PAS -0.73)
p 0.037, FDR 0.11
Pcmt1
r 0.50 (VEH 0.46, PAS 0.65)
p 0.037, FDR 0.17
Rps15a
r -0.62 (VEH -0.22, PAS -0.88)
p 0.038, FDR 0.17
Igtp
r 0.31 (VEH -0.78, PAS 0.89)
p 0.038, FDR 0.17
Srpra
r -0.23 (VEH 0.20, PAS -0.94)
p 0.039, FDR 0.17
Slc25a5 (mito)
r -0.32 (VEH -0.45, PAS -0.26)
p 0.039, FDR 0.11
Gadd45gip1 (mito)
r 0.51 (VEH 0.57, PAS 0.48)
p 0.039, FDR 0.11
Crip1
r 0.30 (VEH -0.11, PAS 0.88)
p 0.039, FDR 0.17
Mrpl57
r 0.13 (VEH 0.36, PAS 0.15)
p 0.039, FDR 0.17
Feature value
Phenotype
VEH_M
PAS_M

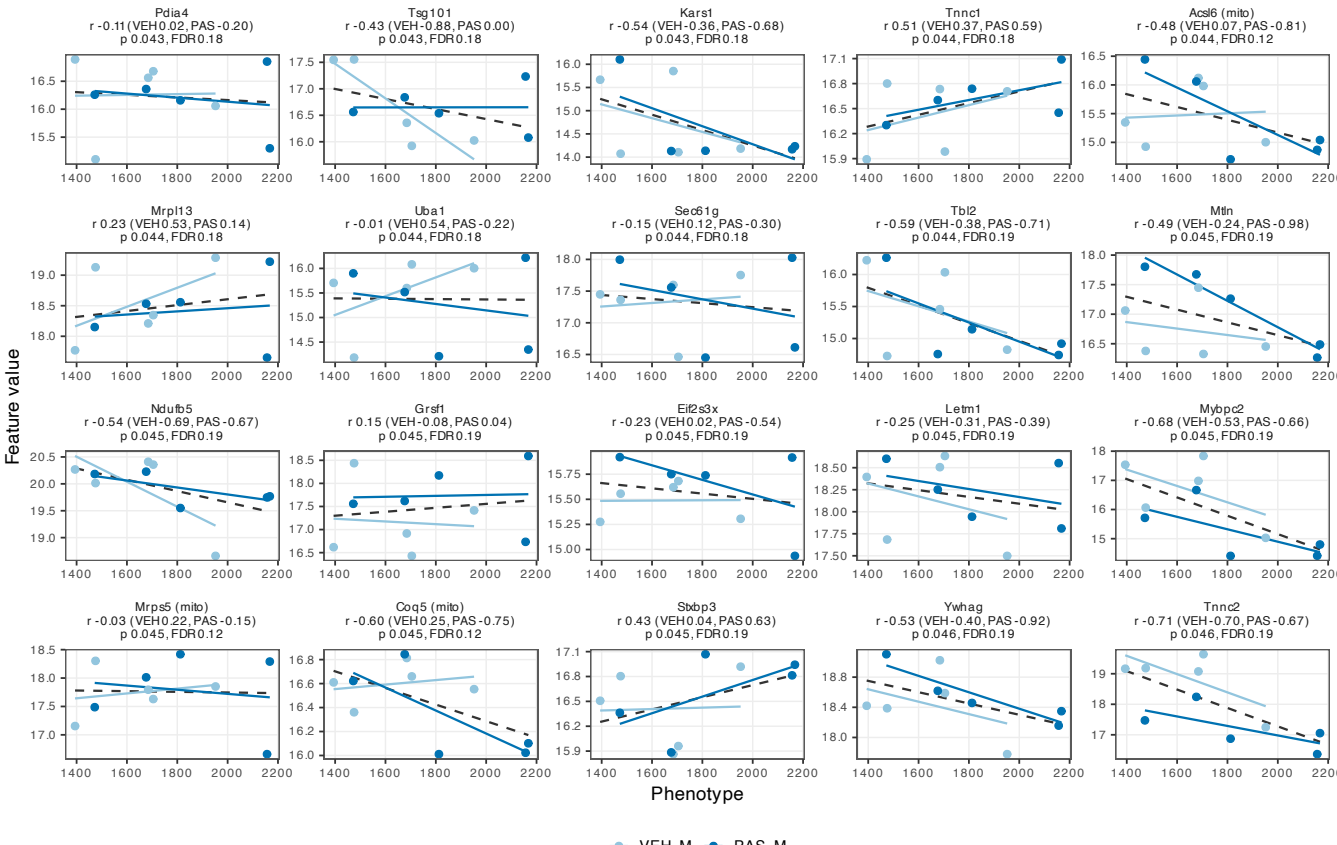
**Male: Proteins associated with CSA, every association at p ≤ 0.05 (n = 681)**

*Phenotype effect of PAS in this sex: +13% (p 0.25), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



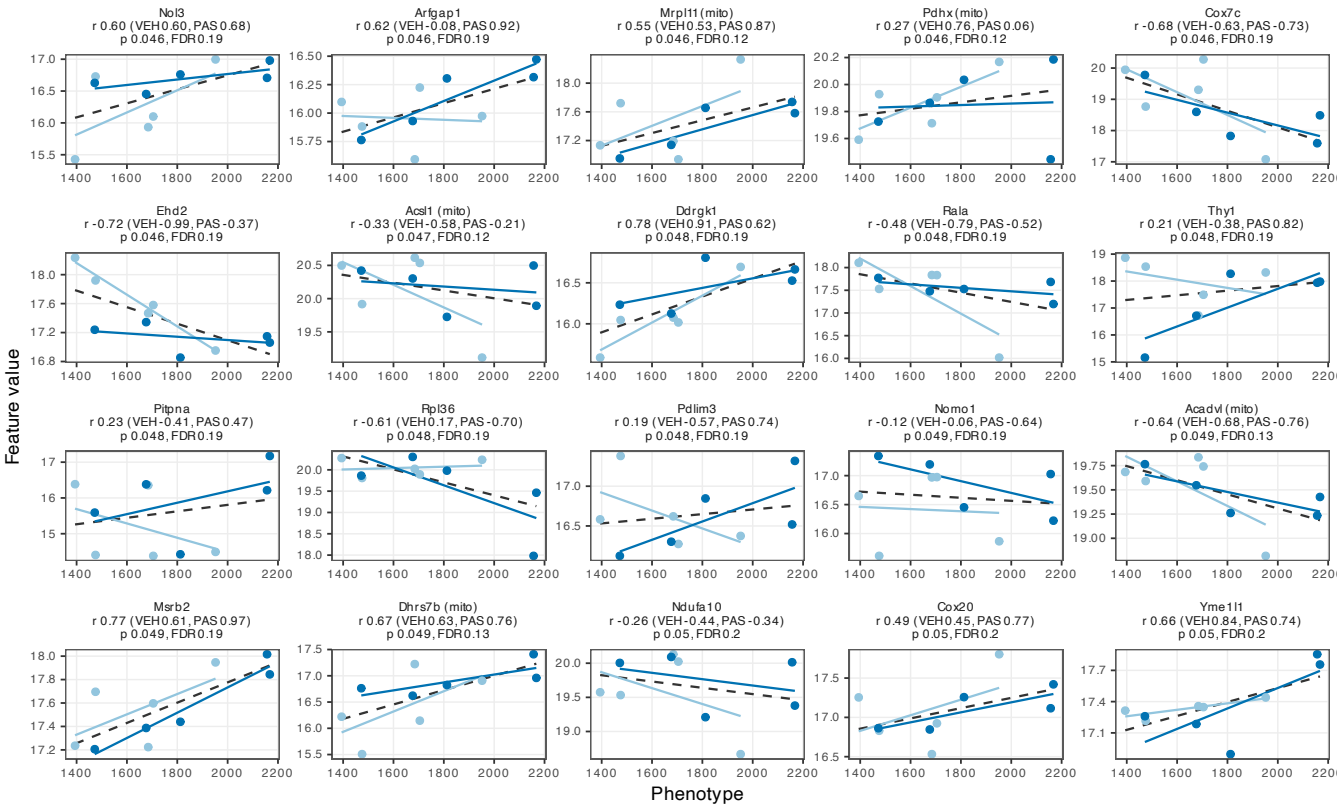
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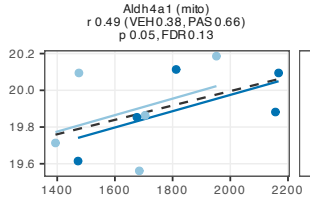
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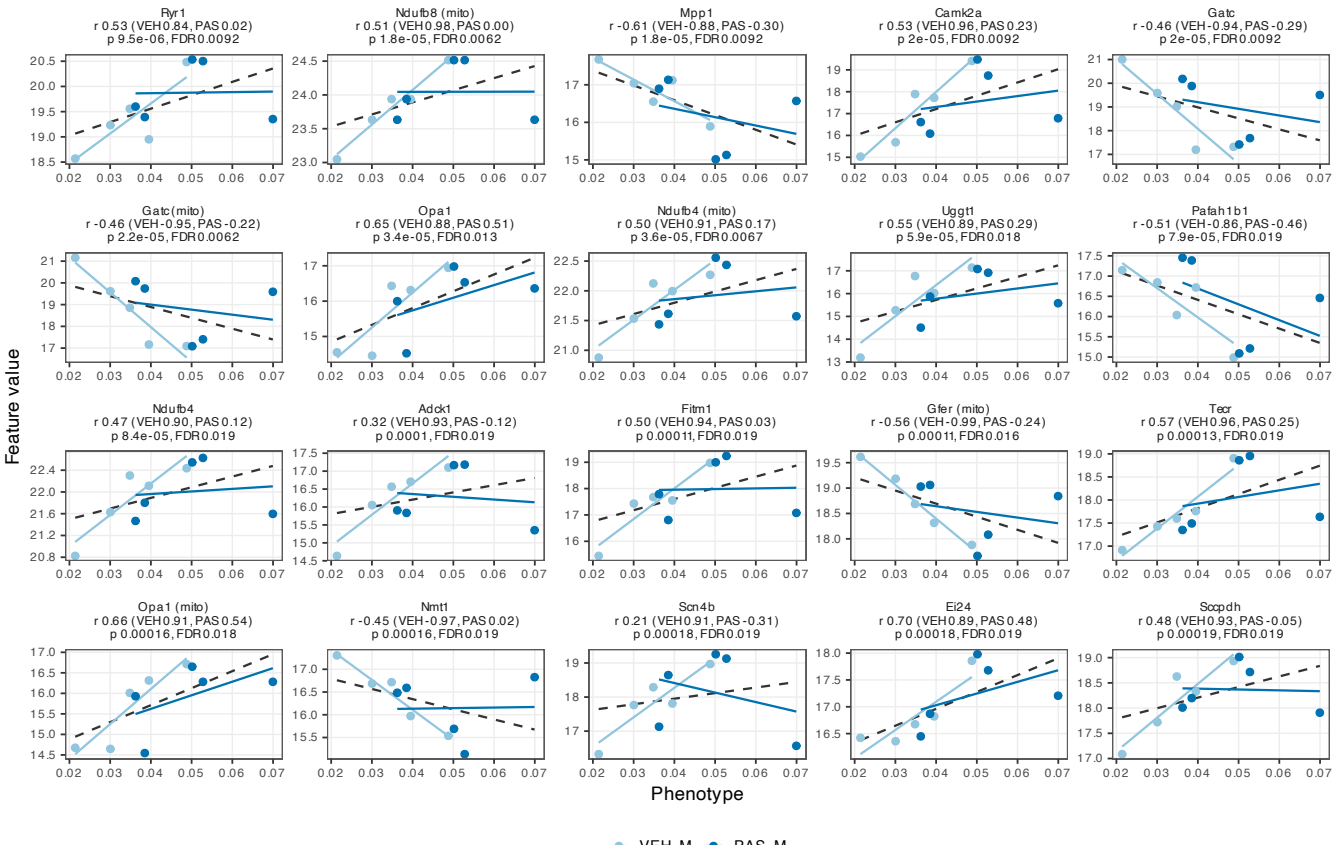

Feature value

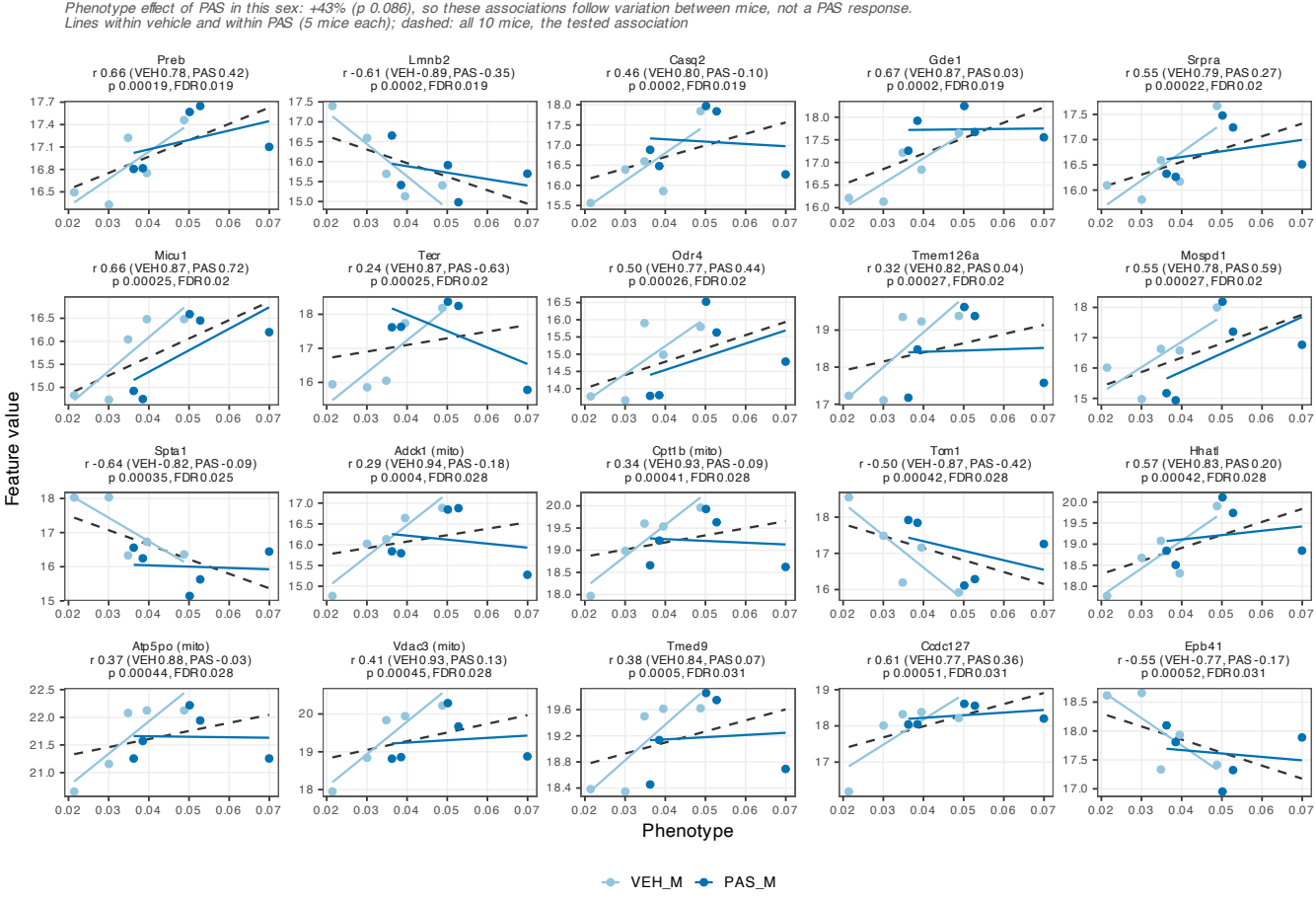
Phenotype

VEH_M PAS_M

**Male: Proteins associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 631)**

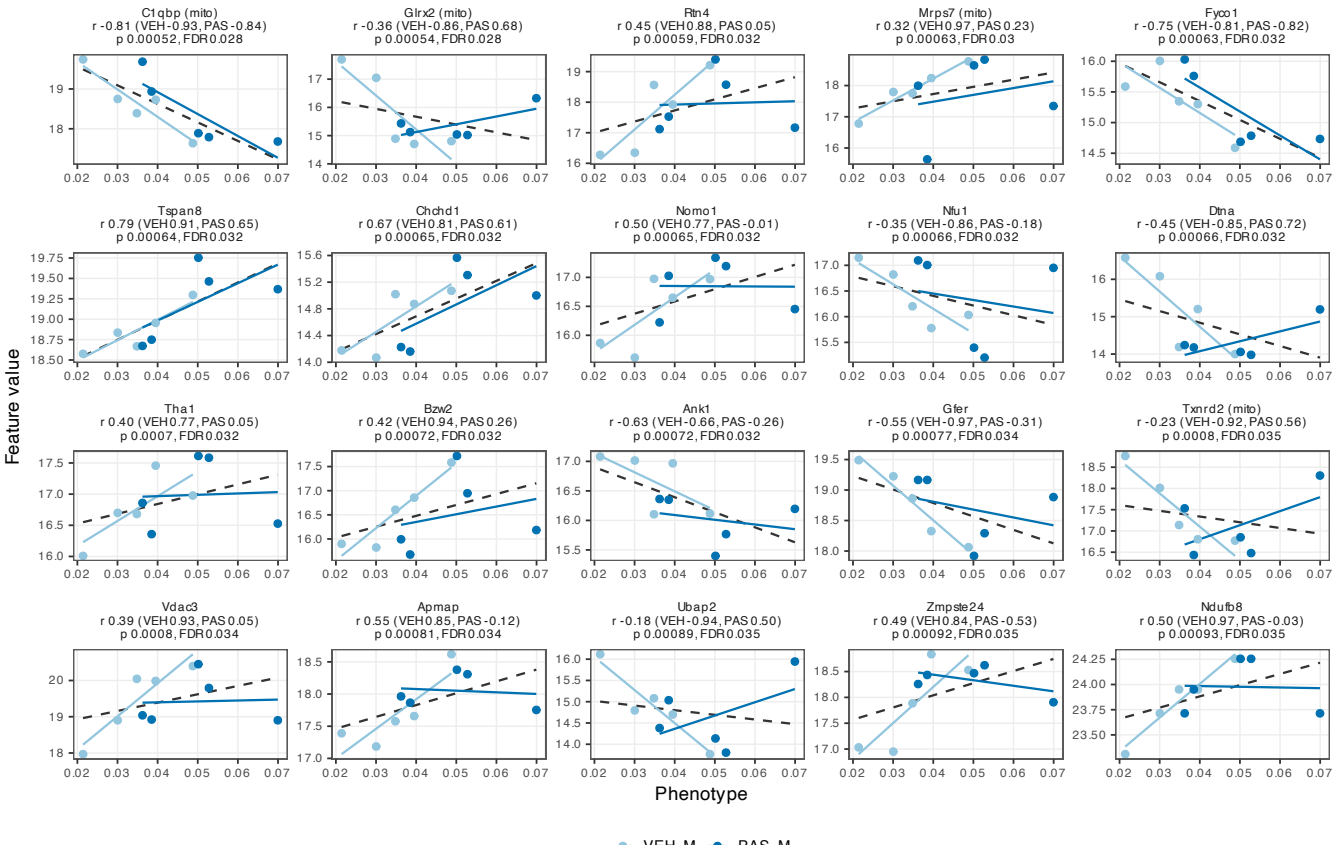
*Phenotype effect of PAS in this sex: +43% (p 0.086), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*




Male: Proteins associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 631)
Phenotype effect of PAS in this sex: +43% (p 0.086), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association
Preb
r 0.66 (VEH 0.78, PAS 0.42)
p 0.00019, FDR 0.019
Lmnb2
r -0.61 (VEH -0.89, PAS -0.35)
p 0.0002, FDR 0.019
Casq2
r 0.46 (VEH 0.80, PAS -0.10)
p 0.0002, FDR 0.019
Gde1
r 0.67 (VEH 0.87, PAS 0.03)
p 0.0002, FDR 0.019
Srpra
r 0.55 (VEH 0.79, PAS 0.27)
p 0.00022, FDR 0.02
Micu1
r 0.66 (VEH 0.87, PAS 0.72)
p 0.00025, FDR 0.02
Tecr
r 0.24 (VEH 0.87, PAS -0.63)
p 0.00025, FDR 0.02
Odr4
r 0.50 (VEH 0.77, PAS 0.44)
p 0.00026, FDR 0.02
Tmem126a
r 0.32 (VEH 0.82, PAS 0.04)
p 0.00027, FDR 0.02
Mospd1
r 0.55 (VEH 0.78, PAS 0.59)
p 0.00027, FDR 0.02
Spta1
r -0.64 (VEH -0.82, PAS -0.09)
p 0.00035, FDR 0.025
Adck1 (mito)
r 0.29 (VEH 0.94, PAS -0.18)
p 0.0004, FDR 0.028
Cpt1b (mito)
r 0.34 (VEH 0.93, PAS -0.09)
p 0.00041, FDR 0.028
Tom1
r -0.50 (VEH -0.87, PAS -0.42)
p 0.00042, FDR 0.028
Hhatl
r 0.57 (VEH 0.83, PAS 0.20)
p 0.00042, FDR 0.028
Atp5po (mito)
r 0.37 (VEH 0.88, PAS -0.03)
p 0.00044, FDR 0.028
Vdac3 (mito)
r 0.41 (VEH 0.93, PAS 0.13)
p 0.00045, FDR 0.028
Tmed9
r 0.38 (VEH 0.84, PAS 0.07)
p 0.0005, FDR 0.031
Ccdc127
r 0.61 (VEH 0.77, PAS 0.36)
p 0.00051, FDR 0.031
Epb41
r -0.55 (VEH -0.77, PAS -0.17)
p 0.00052, FDR 0.031
Feature value
Phenotype
VEH_M
PAS_M

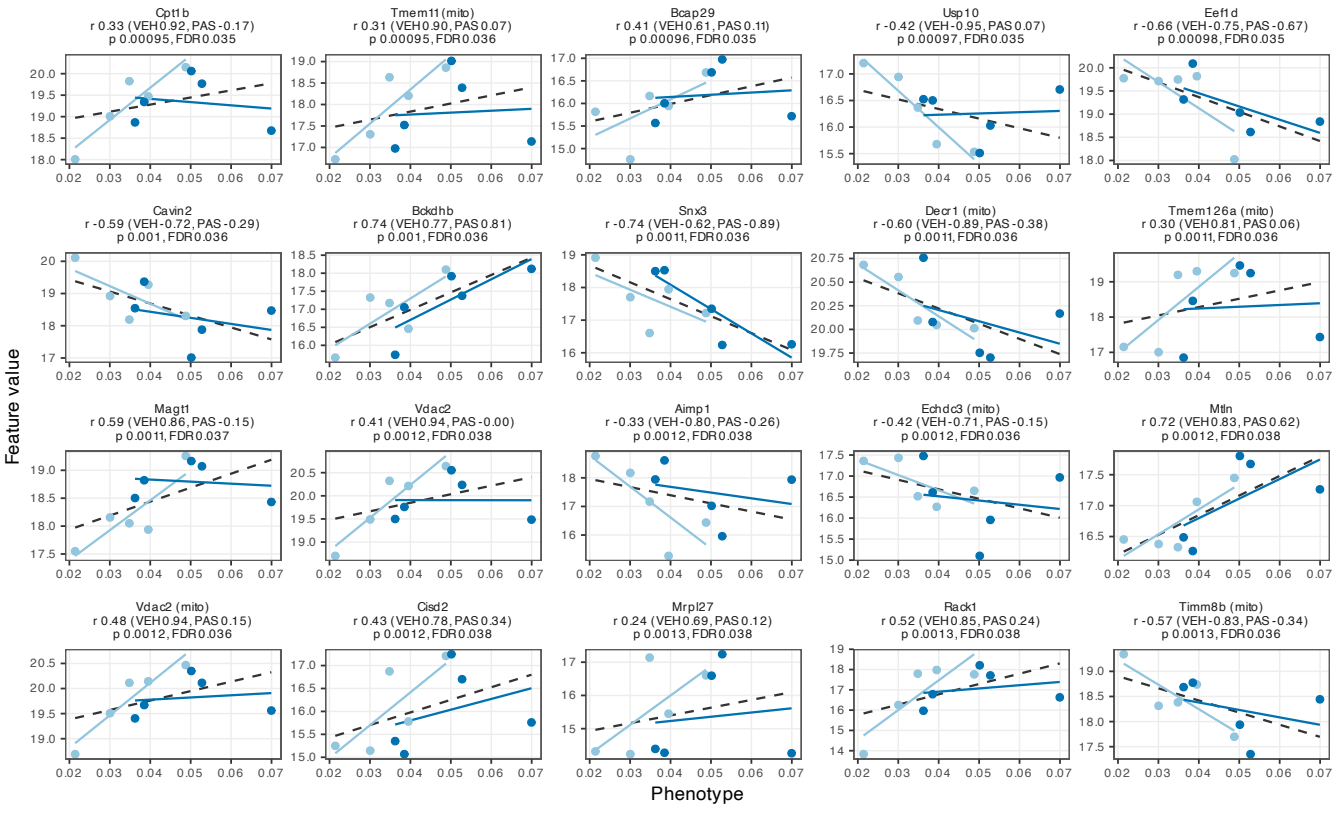
**Male: Proteins associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 631)**

*Phenotype effect of PAS in this sex: +43% (p 0.086), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



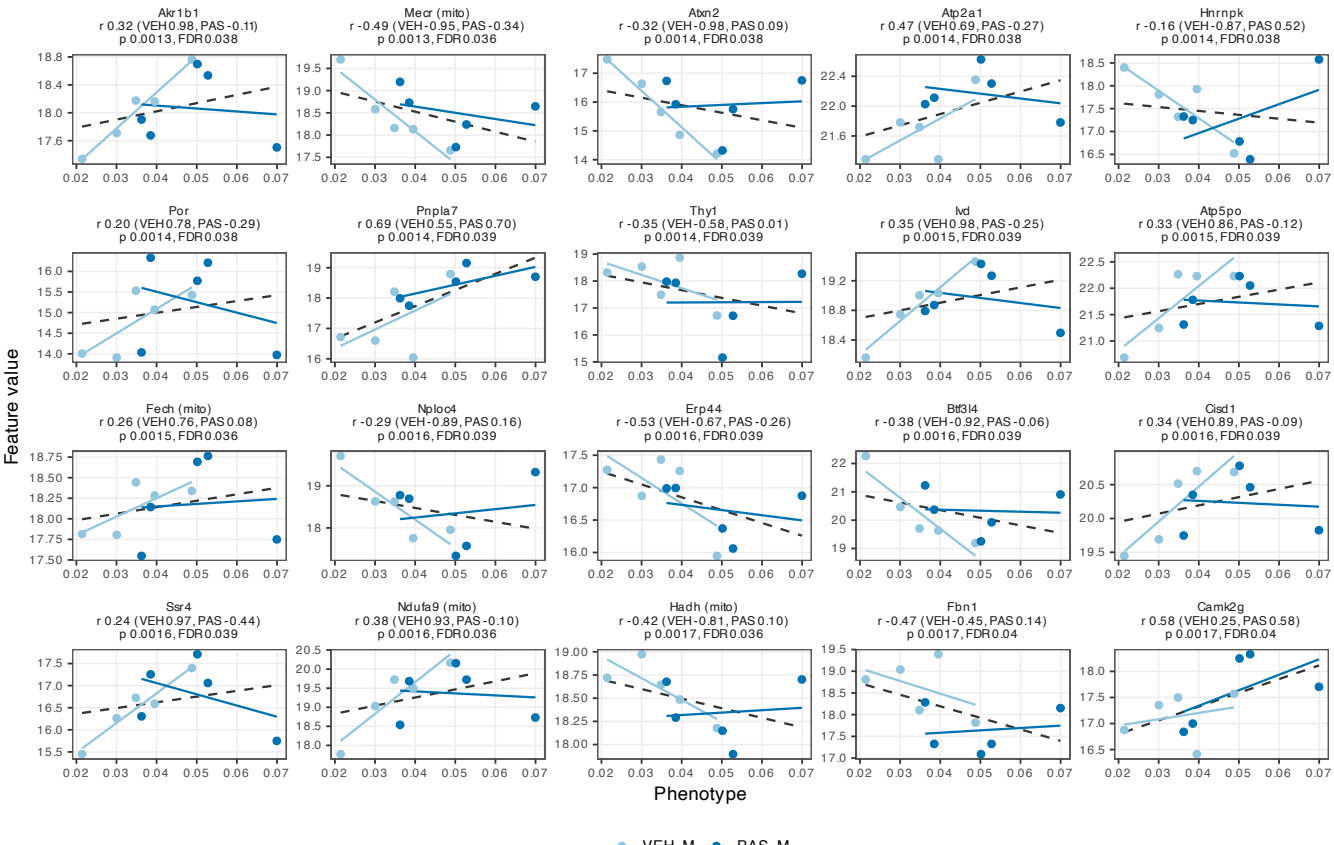
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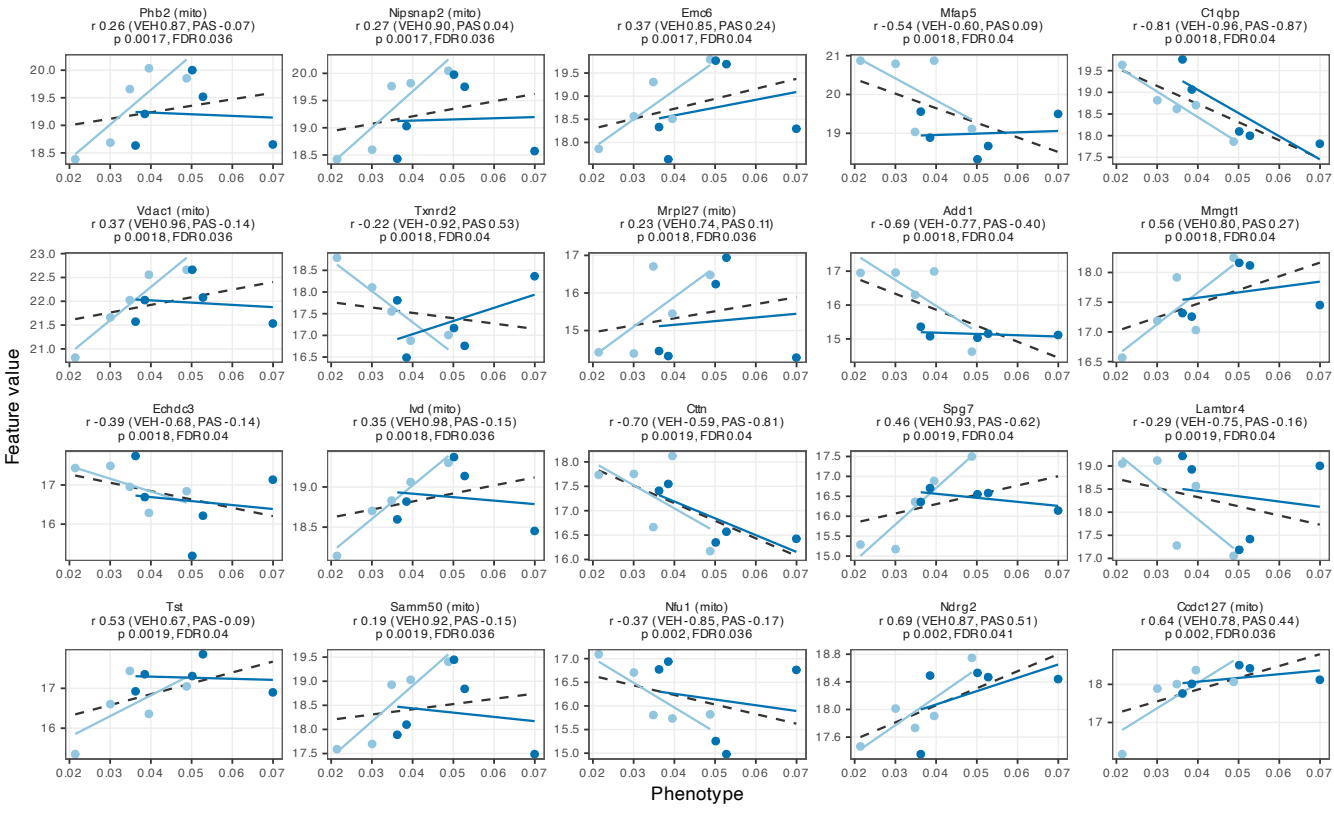
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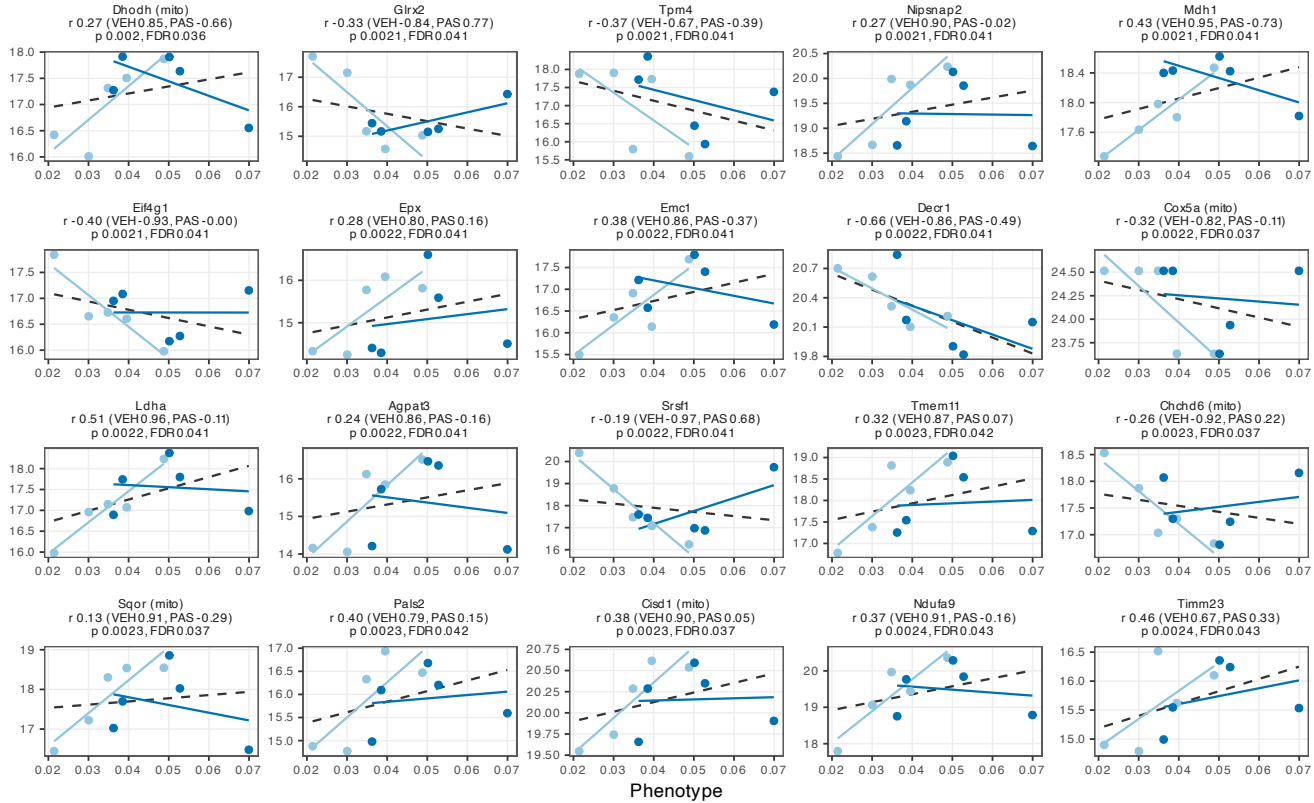
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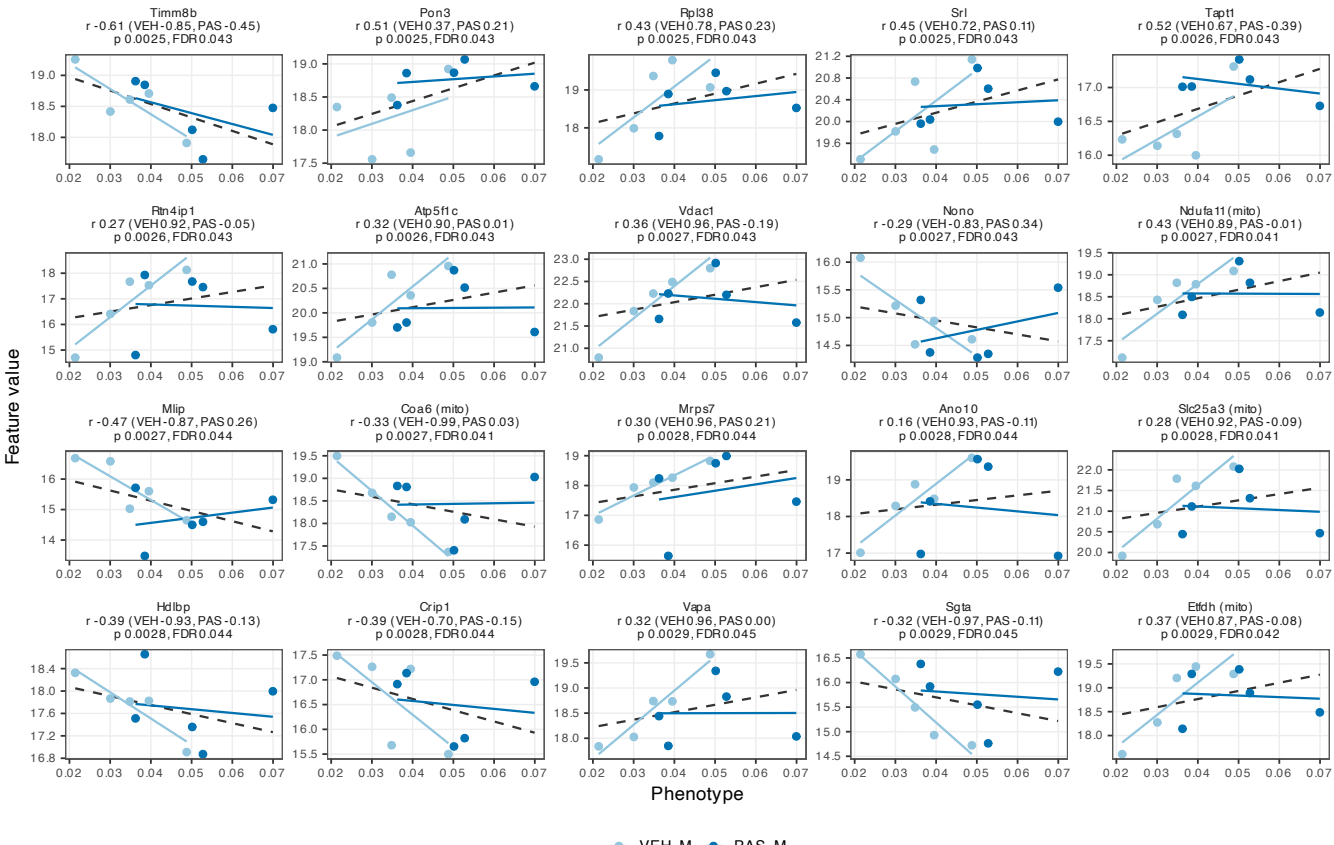
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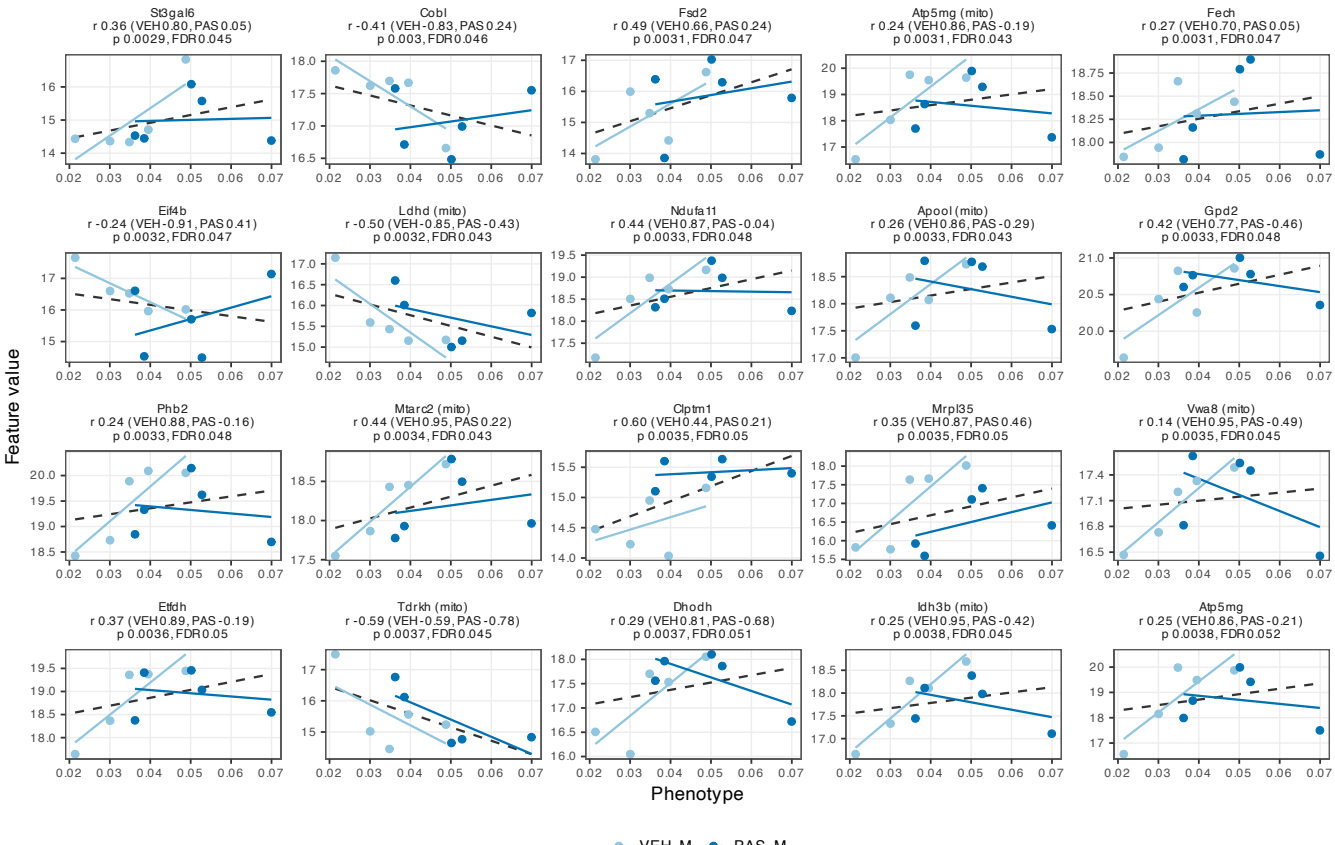
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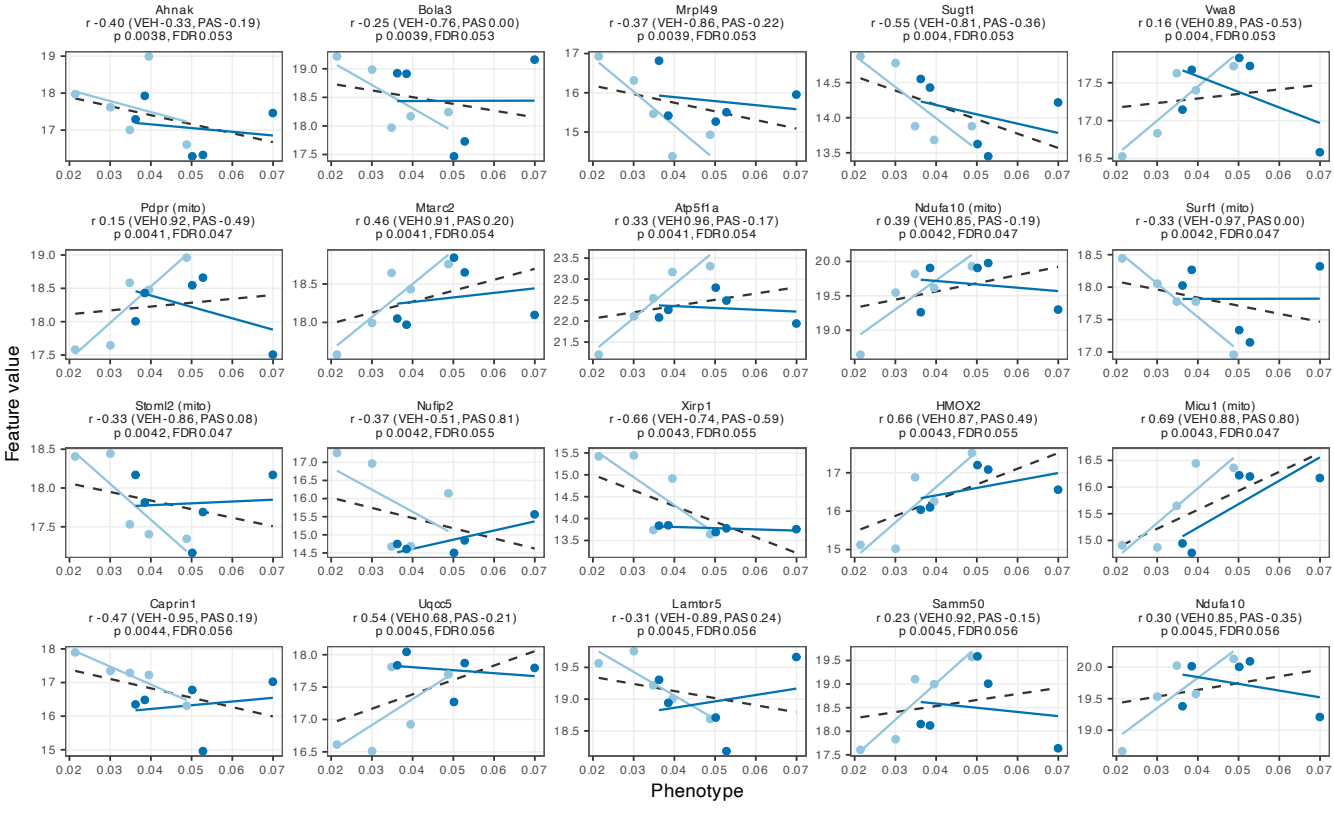
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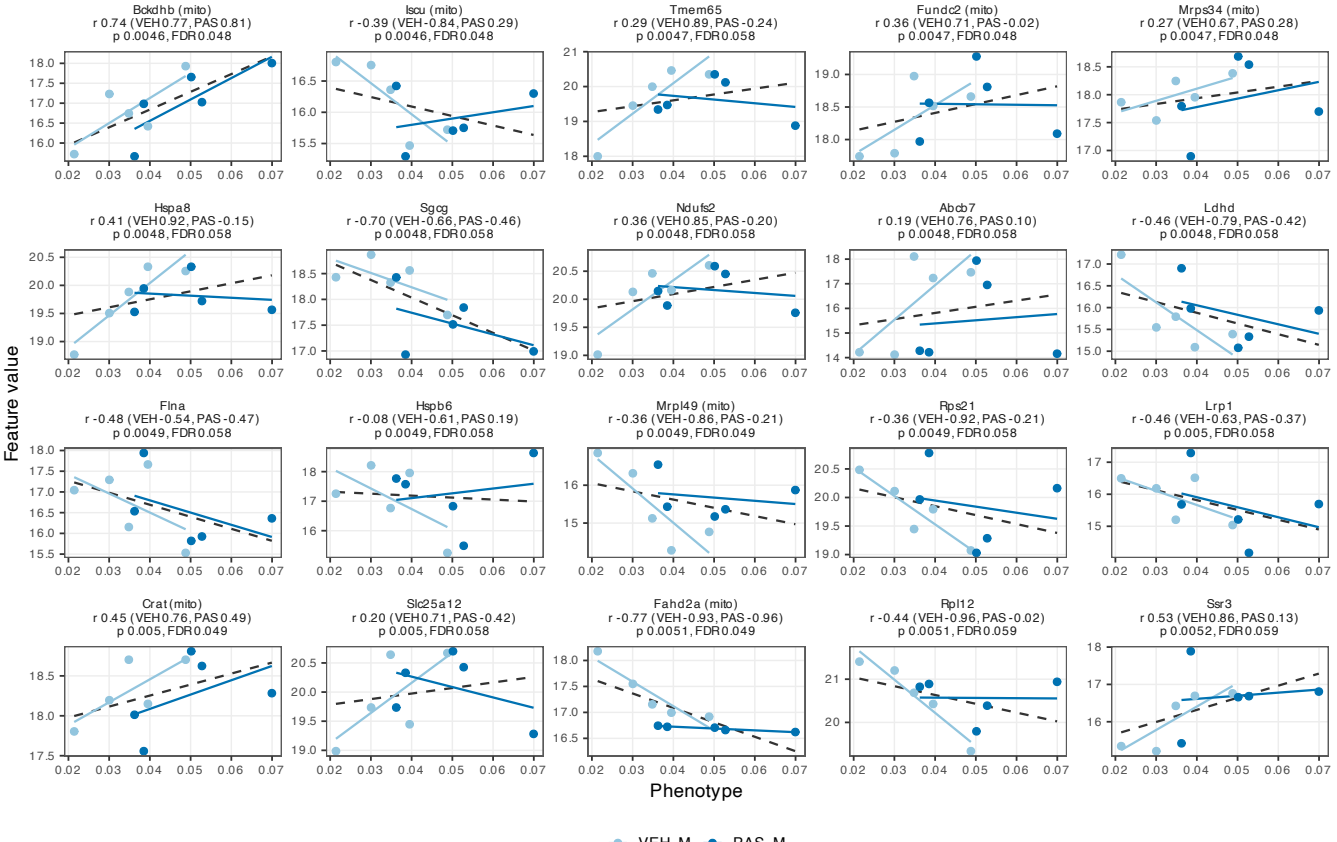
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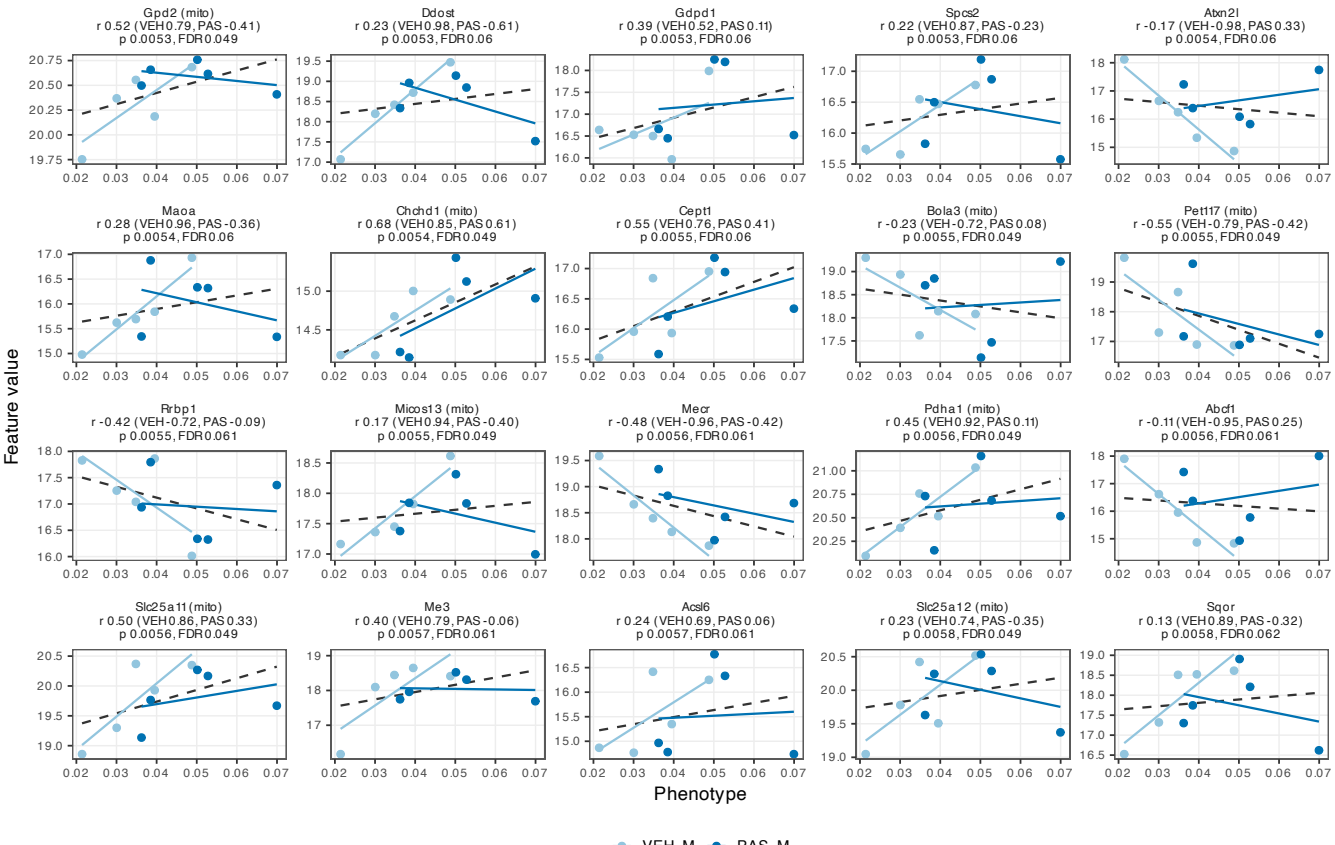
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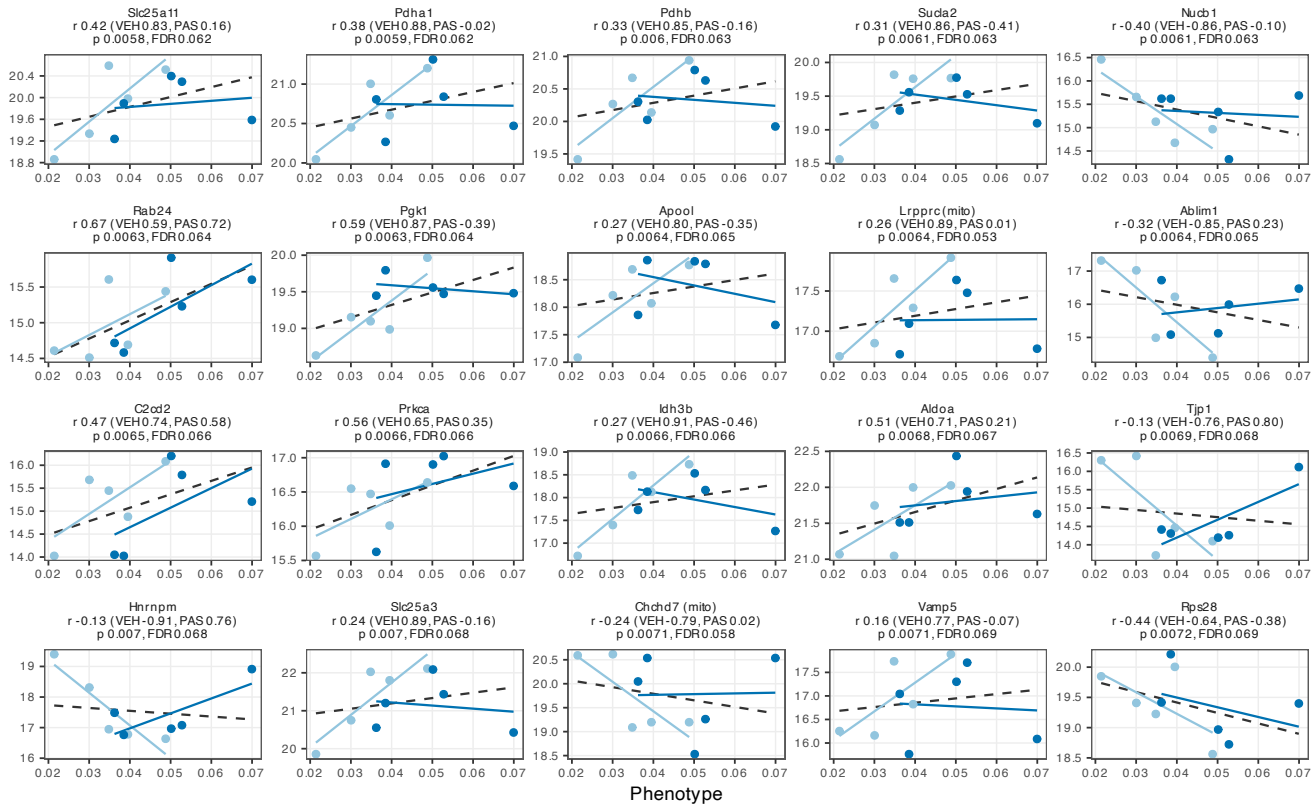
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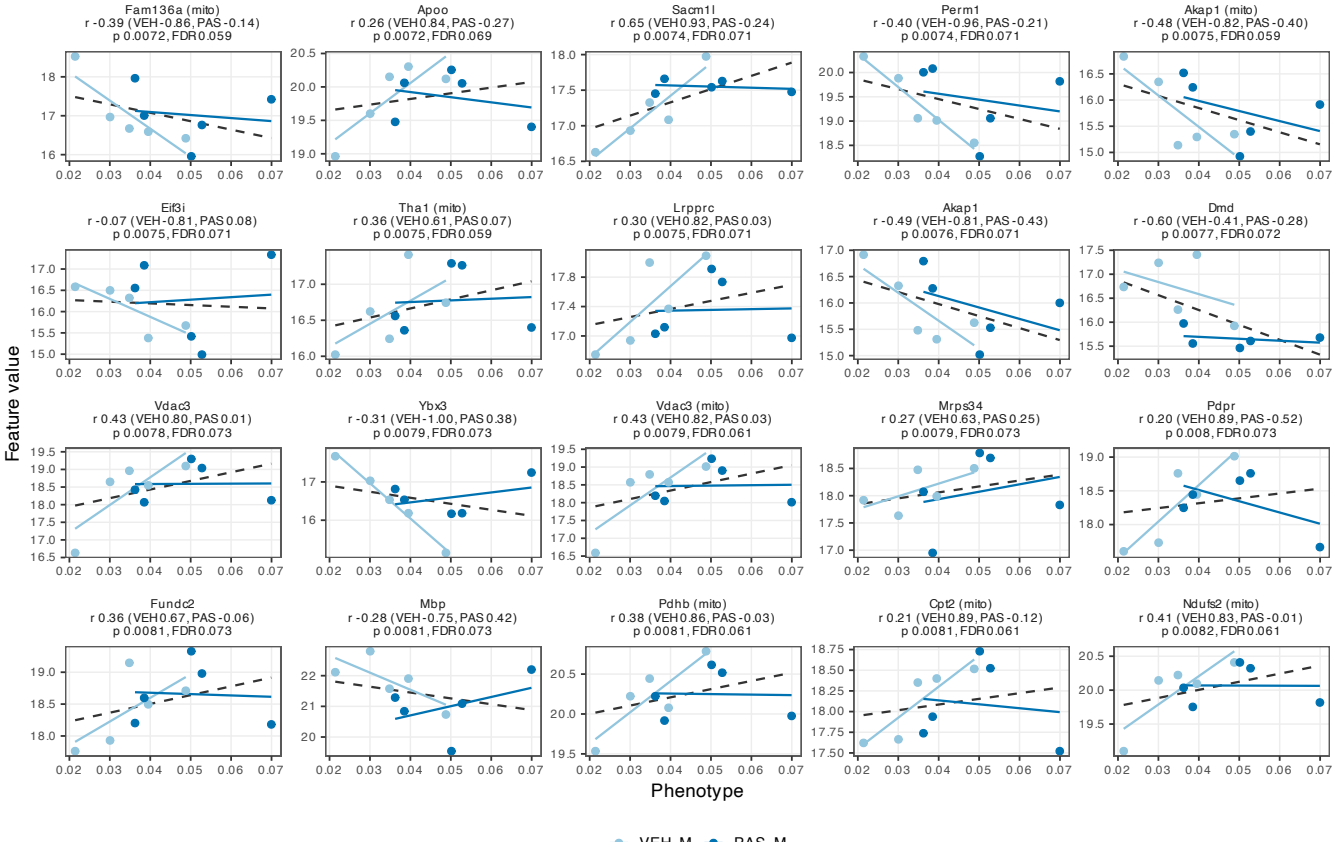
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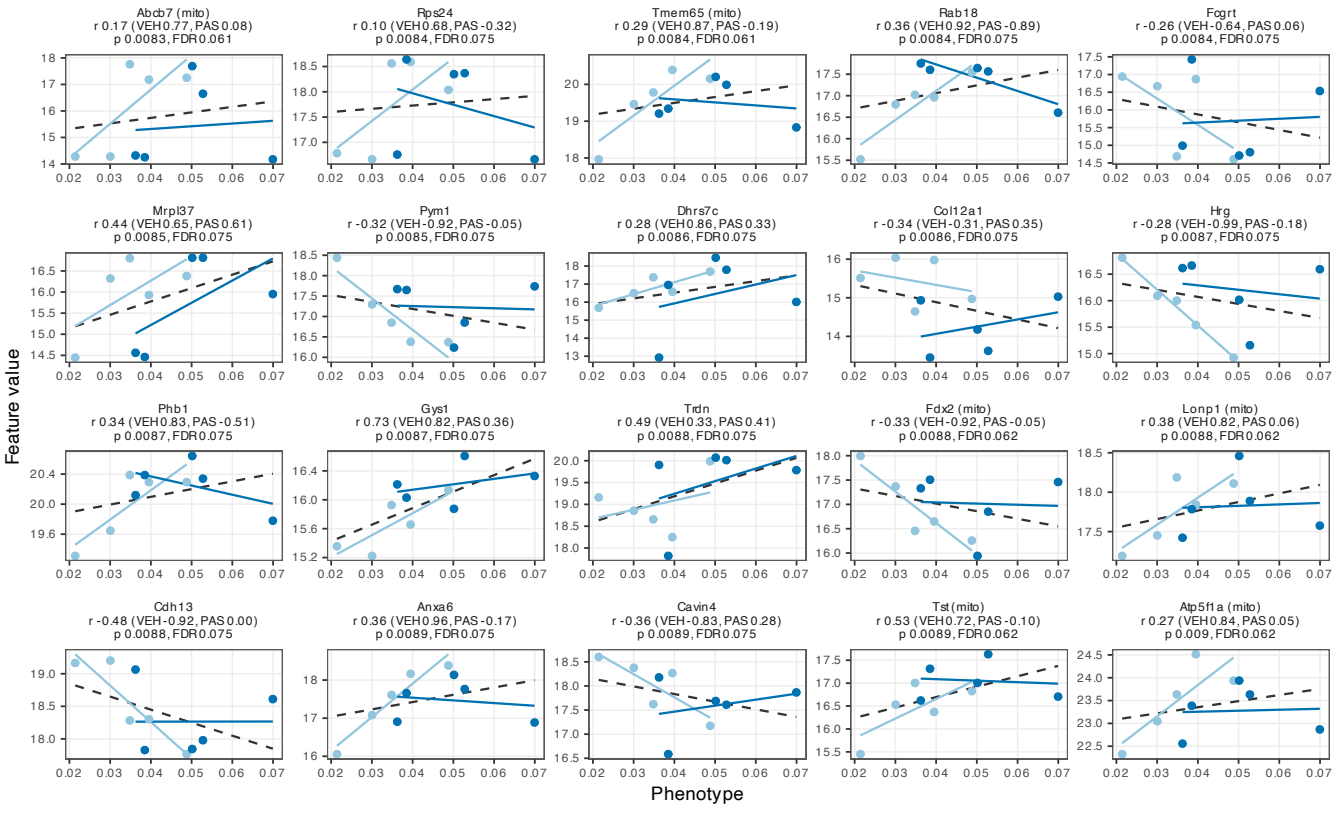
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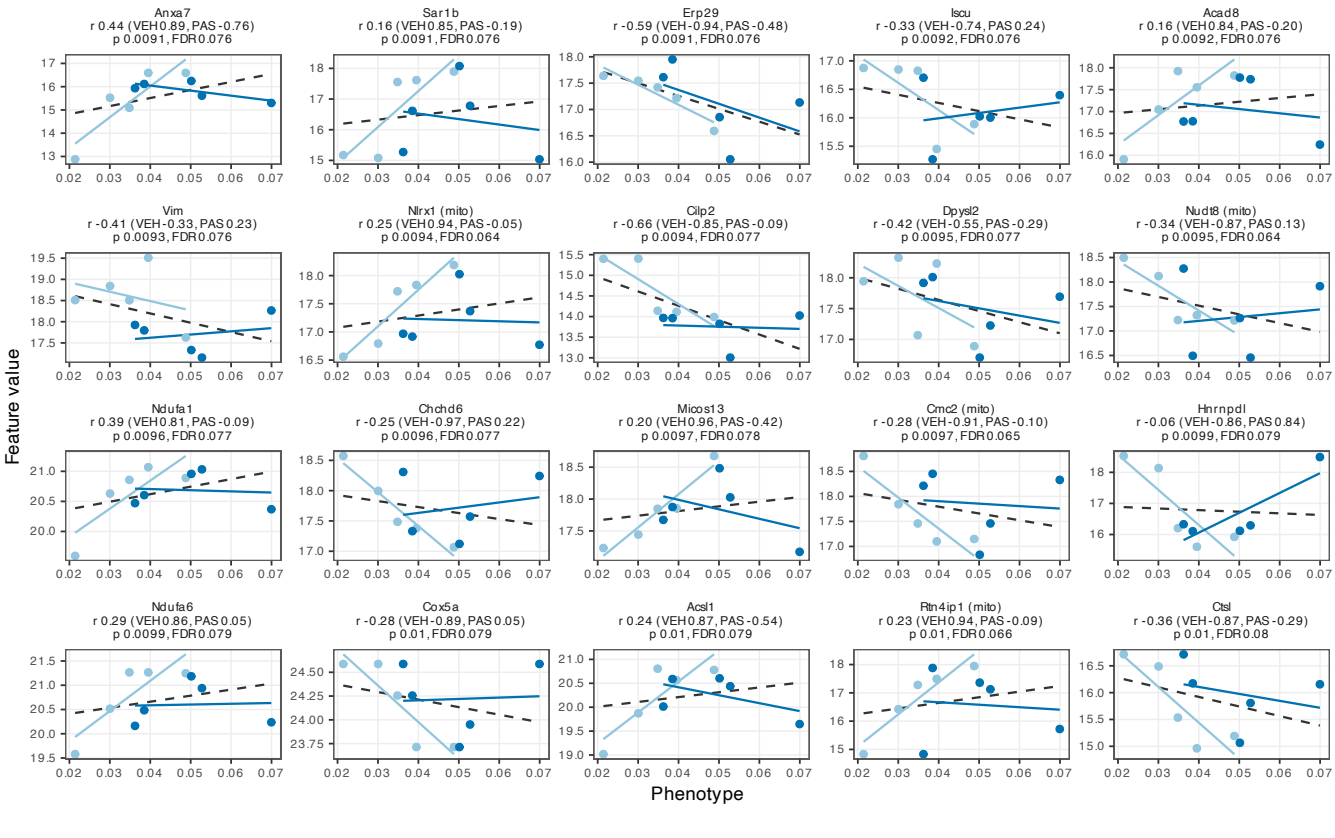
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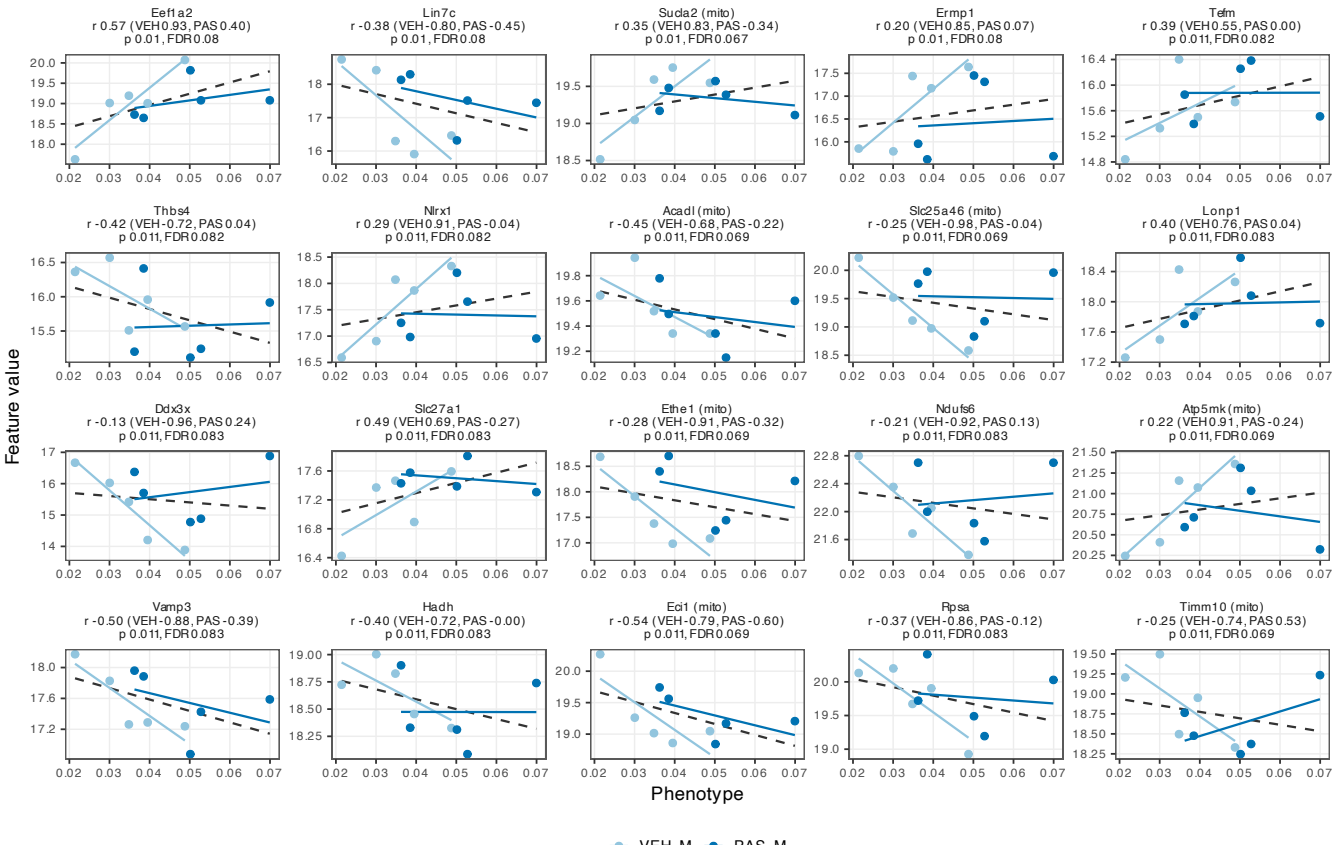
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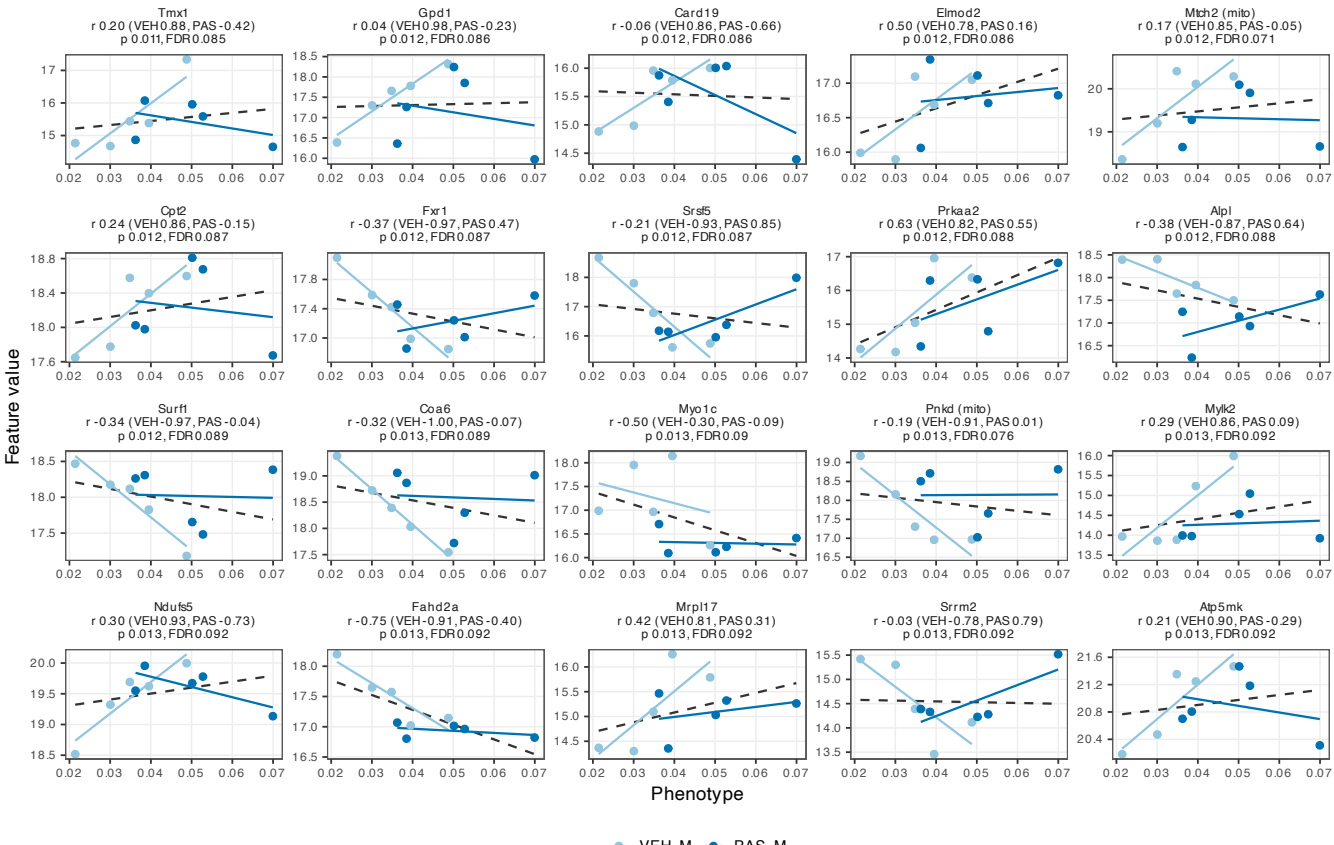
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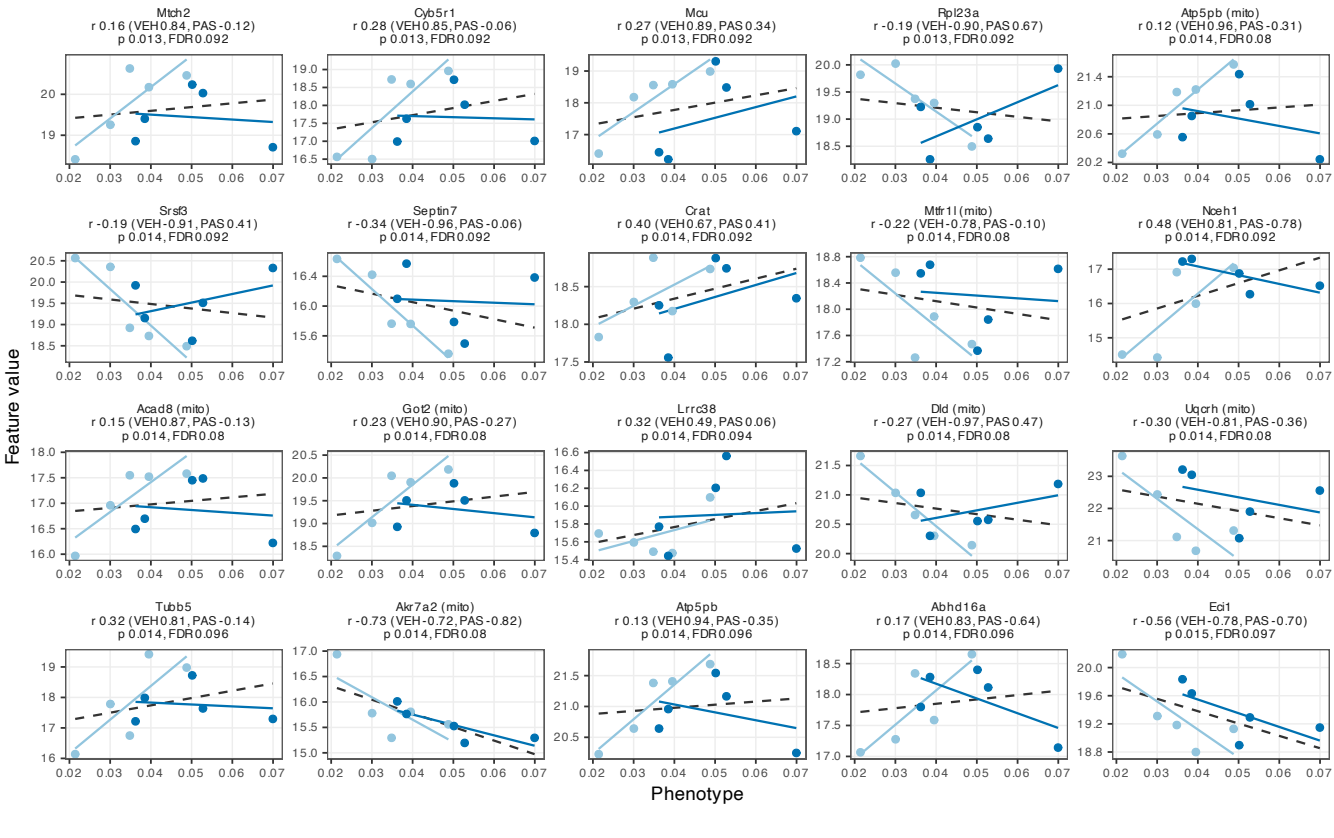
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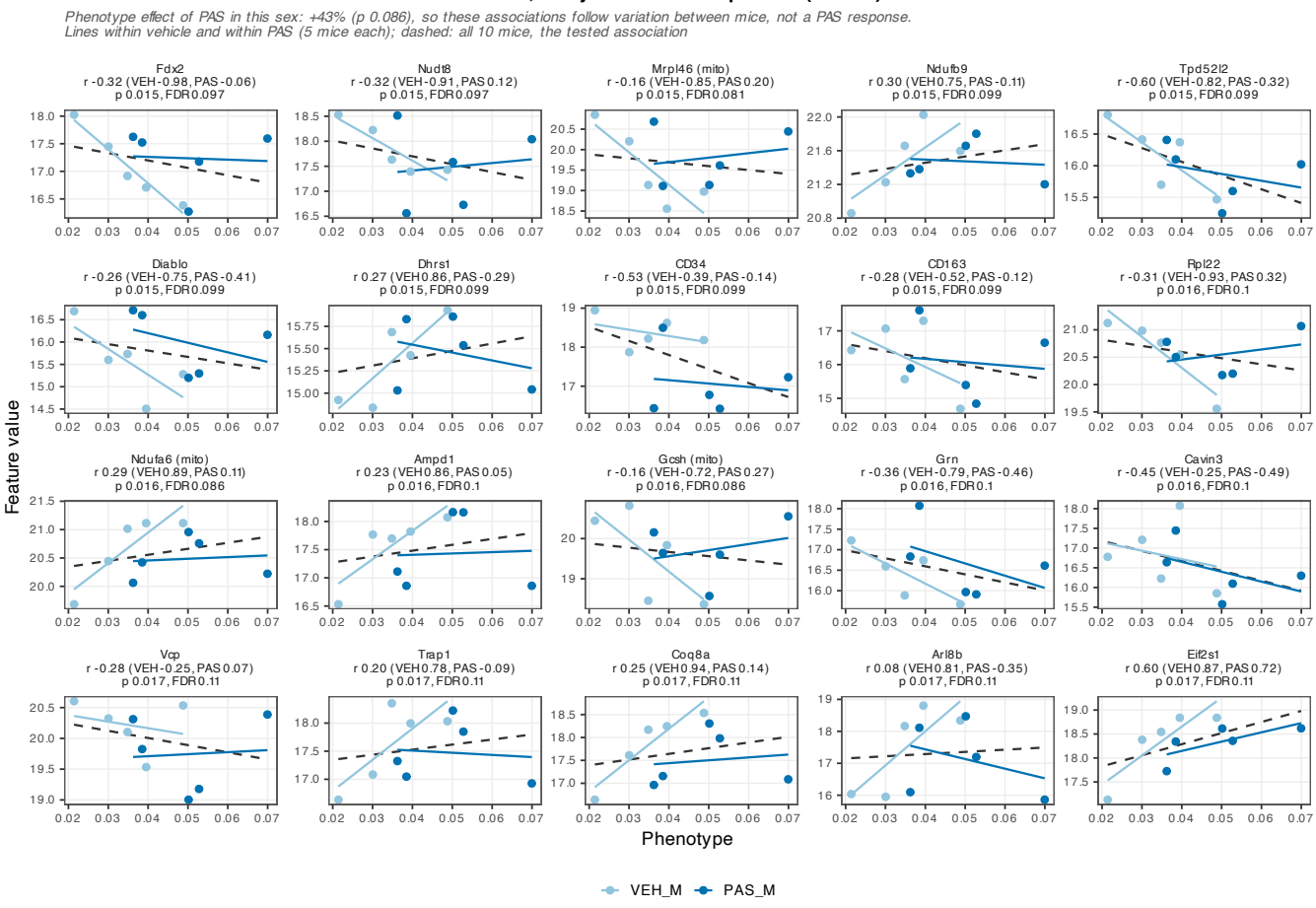
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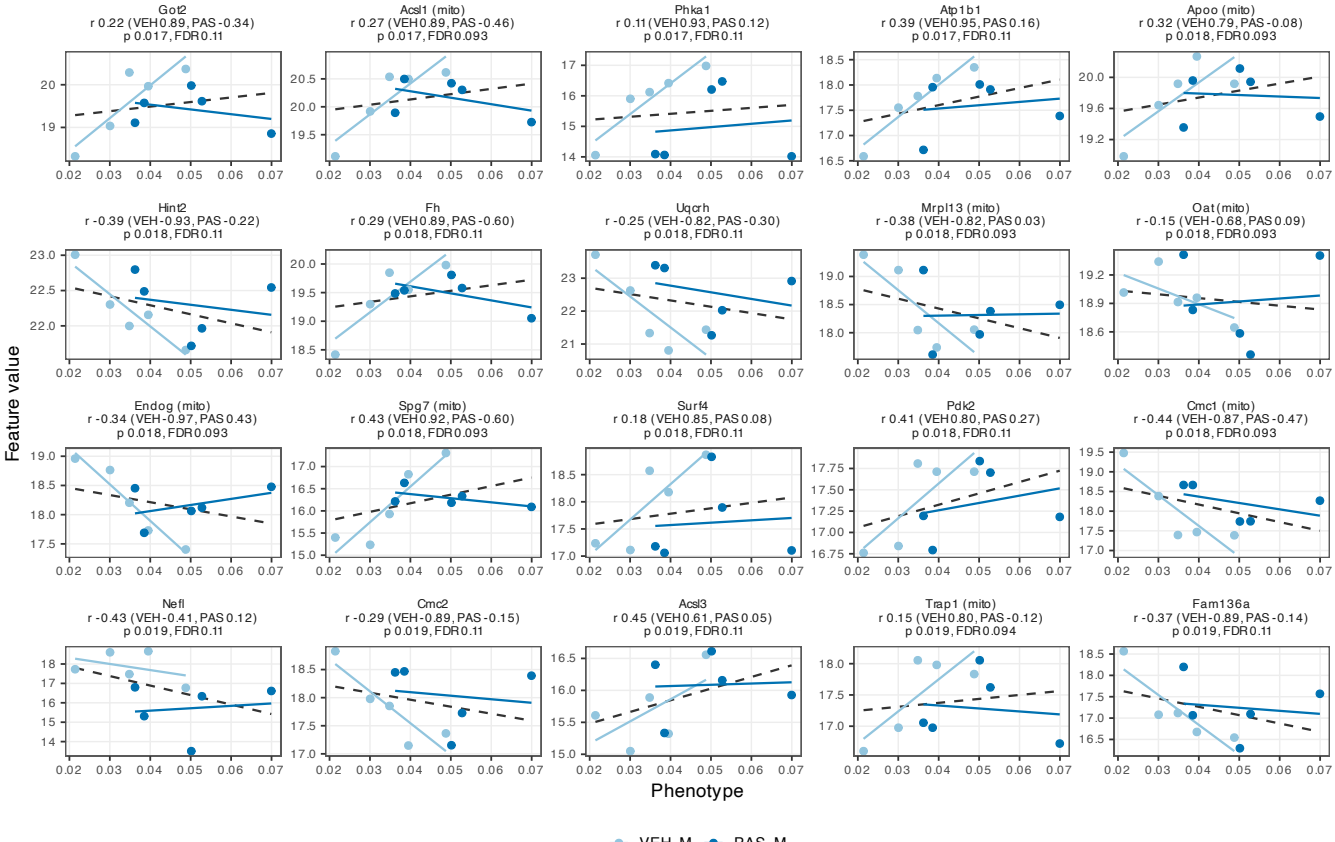
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Phenotype effect of PAS in this sex: +43% (p 0.086), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association
Fdx2
r -0.32 (VEH -0.98, PAS -0.06)
p 0.015, FDR 0.097
Nudt8
r -0.32 (VEH -0.91, PAS 0.12)
p 0.015, FDR 0.097
Mrpl46 (mito)
r -0.16 (VEH -0.85, PAS 0.20)
p 0.015, FDR 0.081
Ndufb9
r 0.30 (VEH 0.75, PAS -0.11)
p 0.015, FDR 0.099
Tpd52l2
r -0.60 (VEH -0.82, PAS -0.32)
p 0.015, FDR 0.099
Diablo
r -0.26 (VEH -0.75, PAS -0.41)
p 0.015, FDR 0.099
Dhrs1
r 0.27 (VEH 0.86, PAS -0.29)
p 0.015, FDR 0.099
CD34
r -0.53 (VEH -0.39, PAS -0.14)
p 0.015, FDR 0.099
CD163
r -0.28 (VEH -0.52, PAS -0.12)
p 0.015, FDR 0.099
Rpl22
r -0.31 (VEH -0.93, PAS 0.32)
p 0.016, FDR 0.1
Ndufa6 (mito)
r 0.29 (VEH 0.89, PAS 0.11)
p 0.016, FDR 0.086
Ampd1
r 0.23 (VEH 0.86, PAS 0.05)
p 0.016, FDR 0.1
Gcsh (mito)
r -0.16 (VEH -0.72, PAS 0.27)
p 0.016, FDR 0.086
Grn
r -0.36 (VEH -0.79, PAS -0.46)
p 0.016, FDR 0.1
Cavin3
r -0.45 (VEH -0.25, PAS -0.49)
p 0.016, FDR 0.1
Vcp
r -0.28 (VEH -0.25, PAS 0.07)
p 0.017, FDR 0.11
Trap1
r 0.20 (VEH 0.78, PAS -0.09)
p 0.017, FDR 0.11
Coq8a
r 0.25 (VEH 0.94, PAS 0.14)
p 0.017, FDR 0.11
Arl8b
r 0.08 (VEH 0.81, PAS -0.35)
p 0.017, FDR 0.11
Eif2s1
r 0.60 (VEH 0.87, PAS 0.72)
p 0.017, FDR 0.11
Feature value
Phenotype
VEH_M
PAS_M

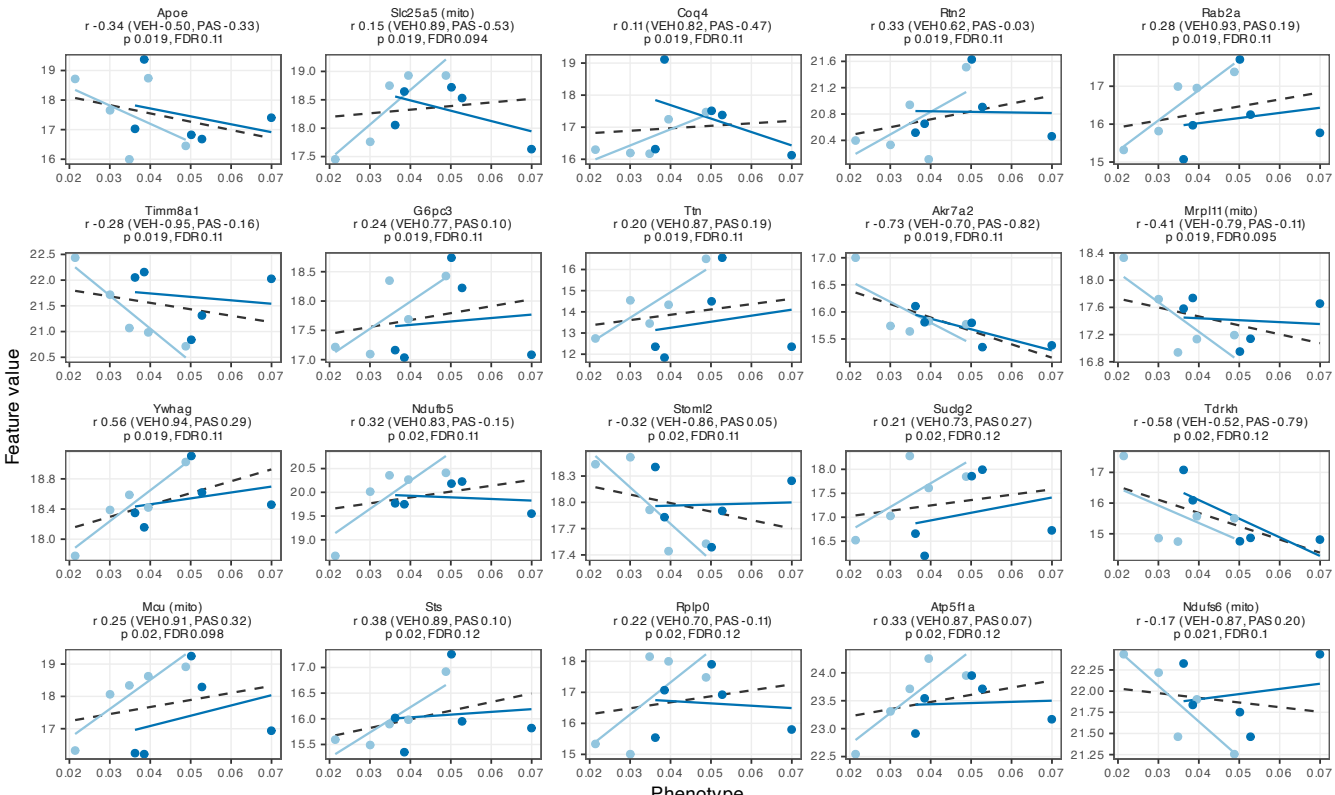
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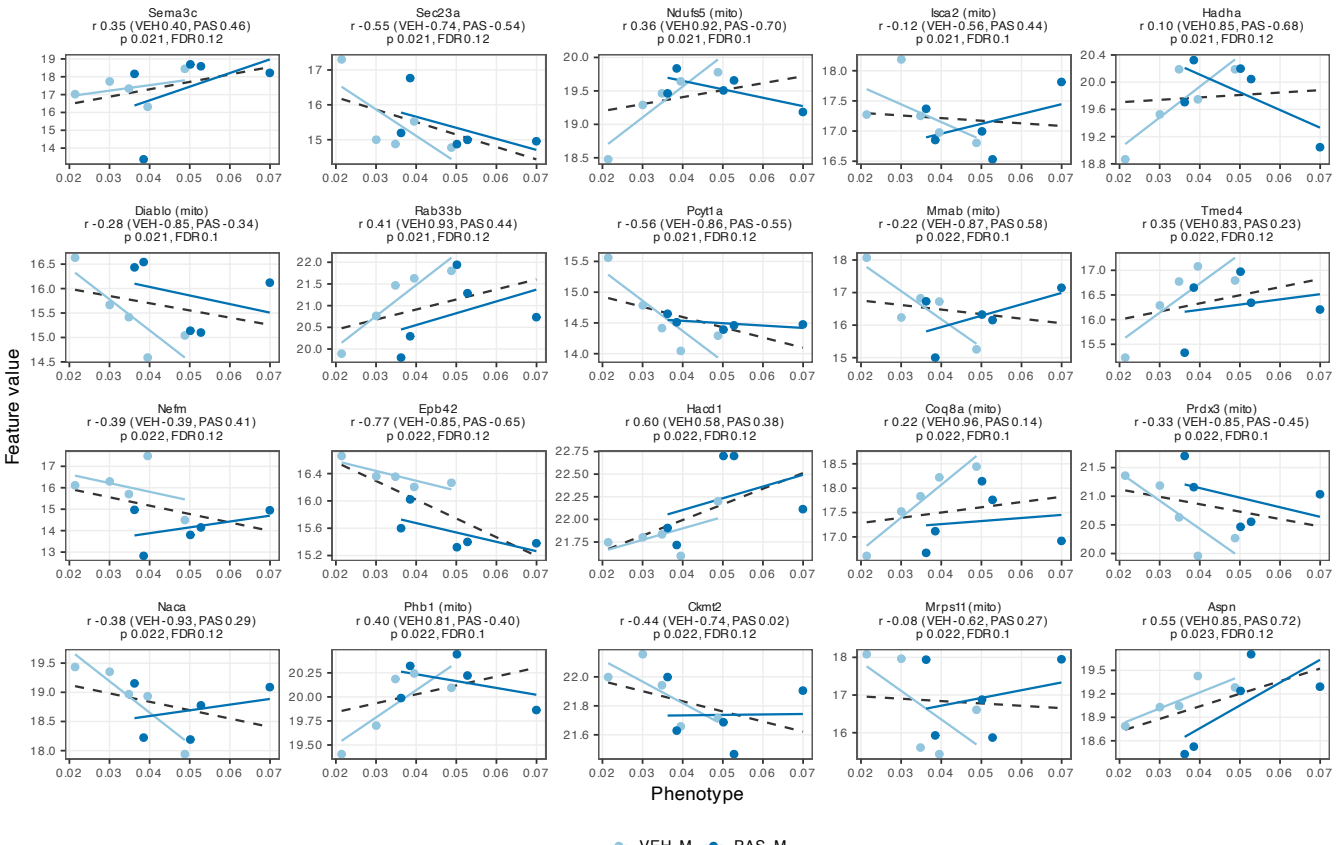
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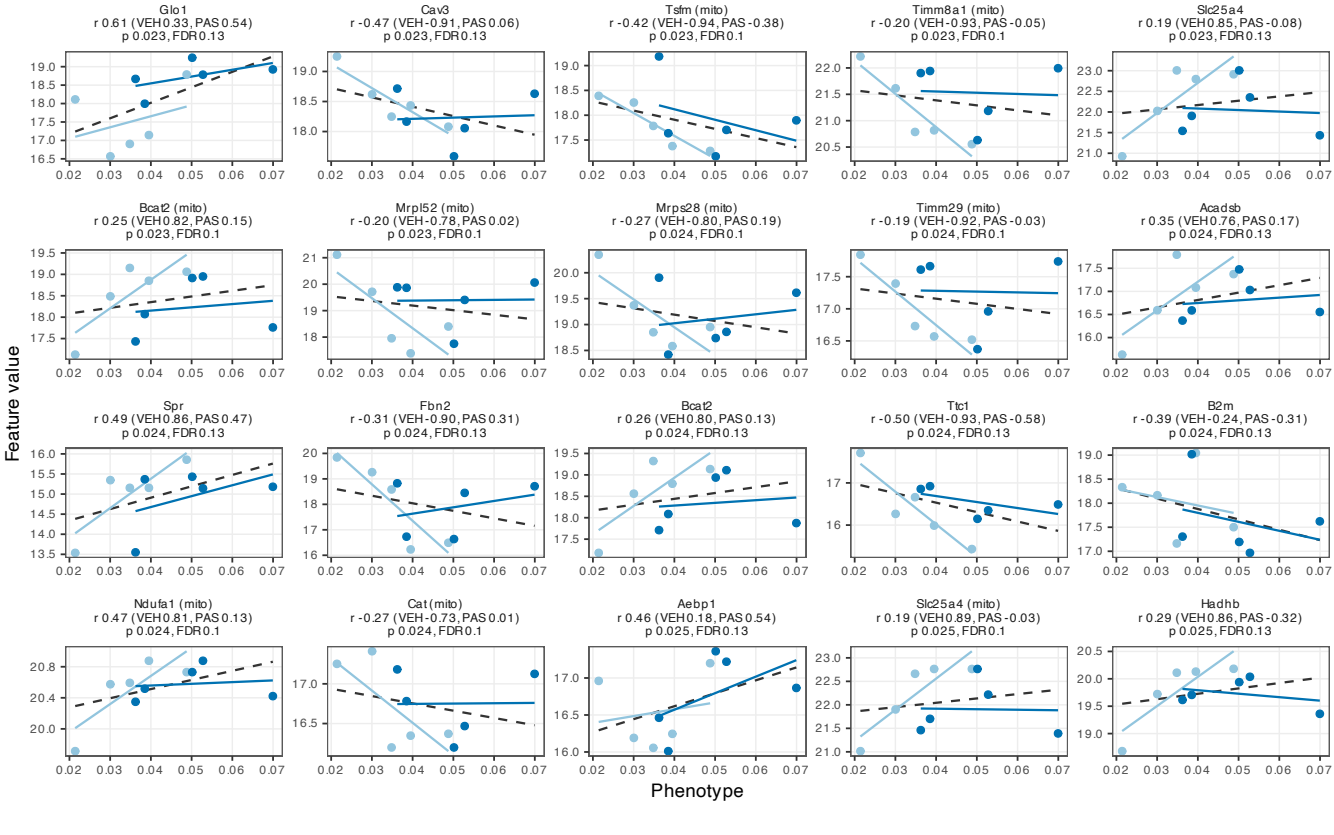
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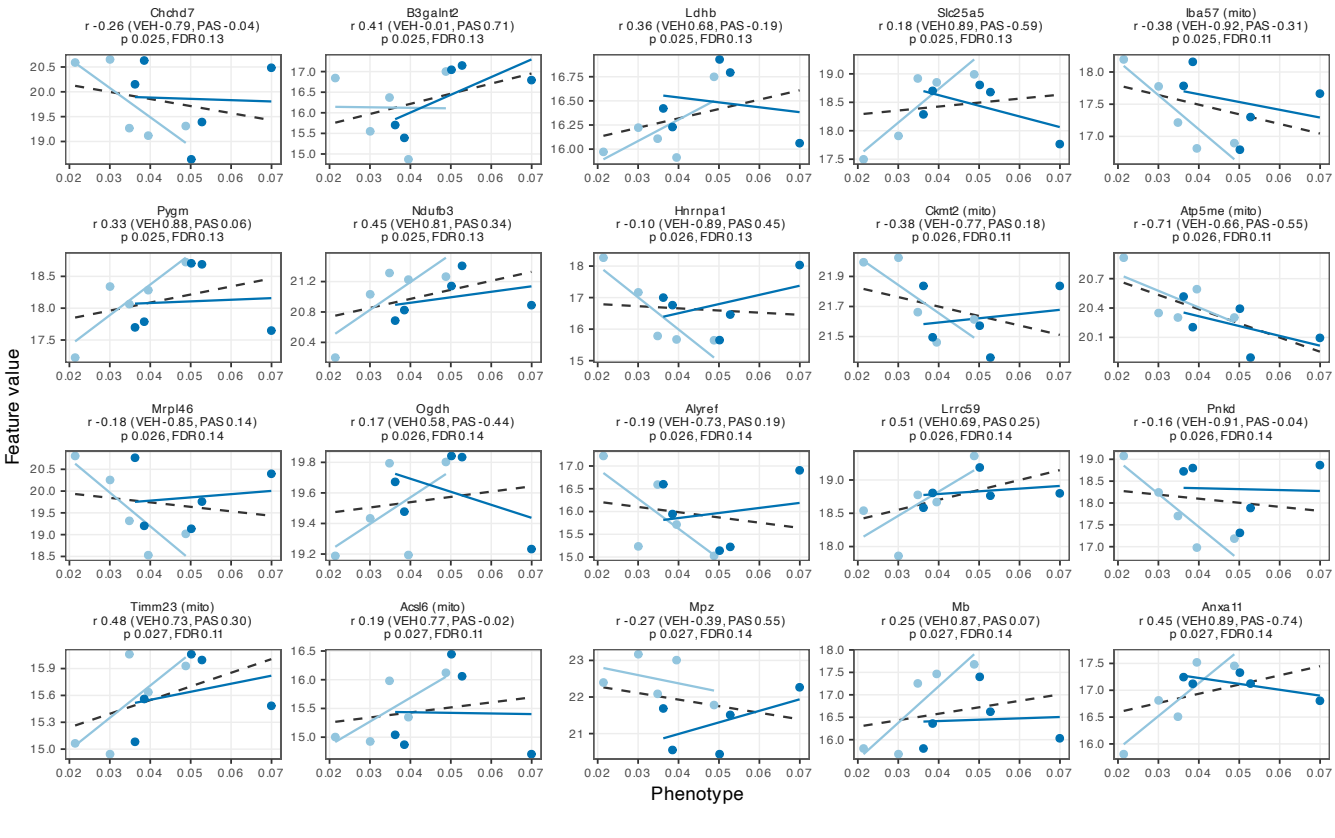
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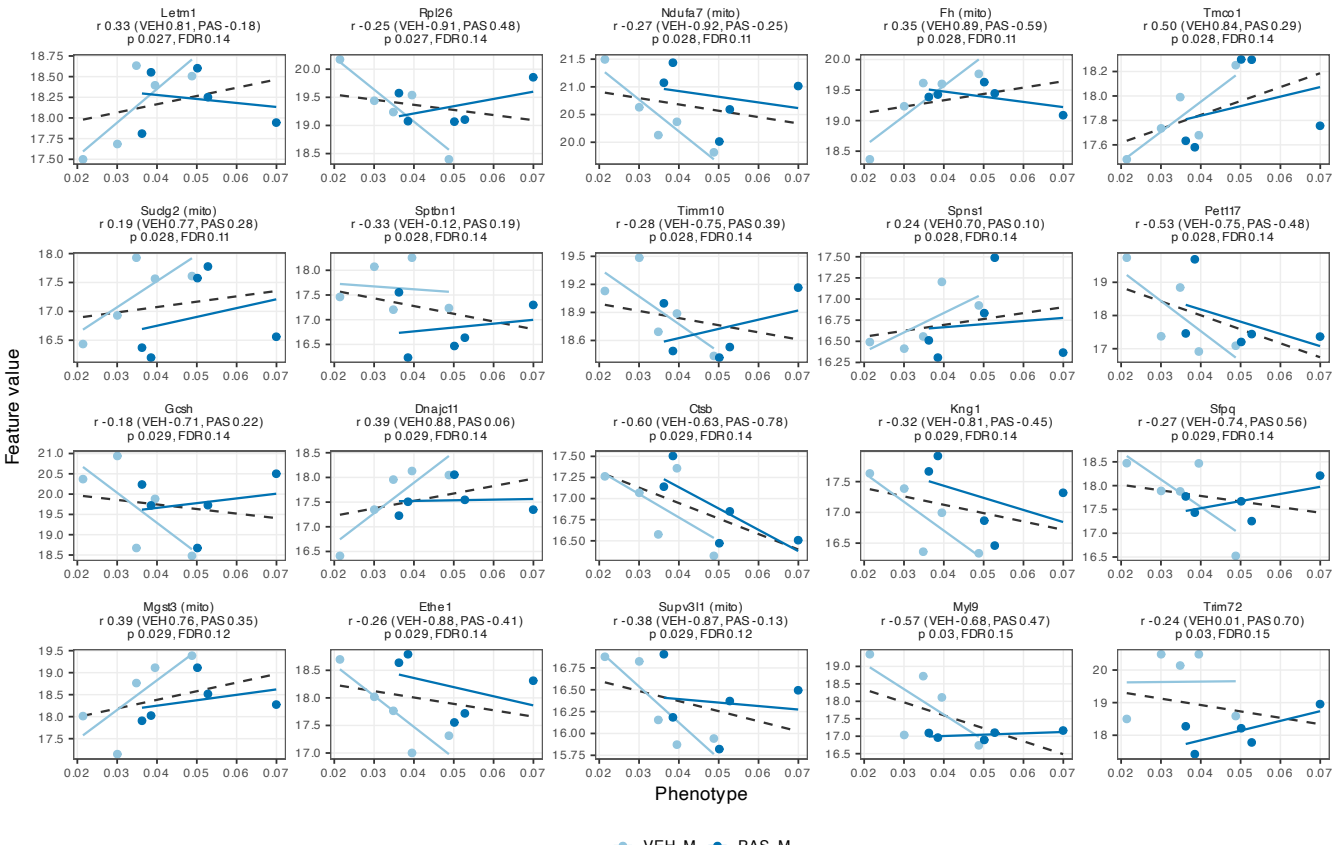
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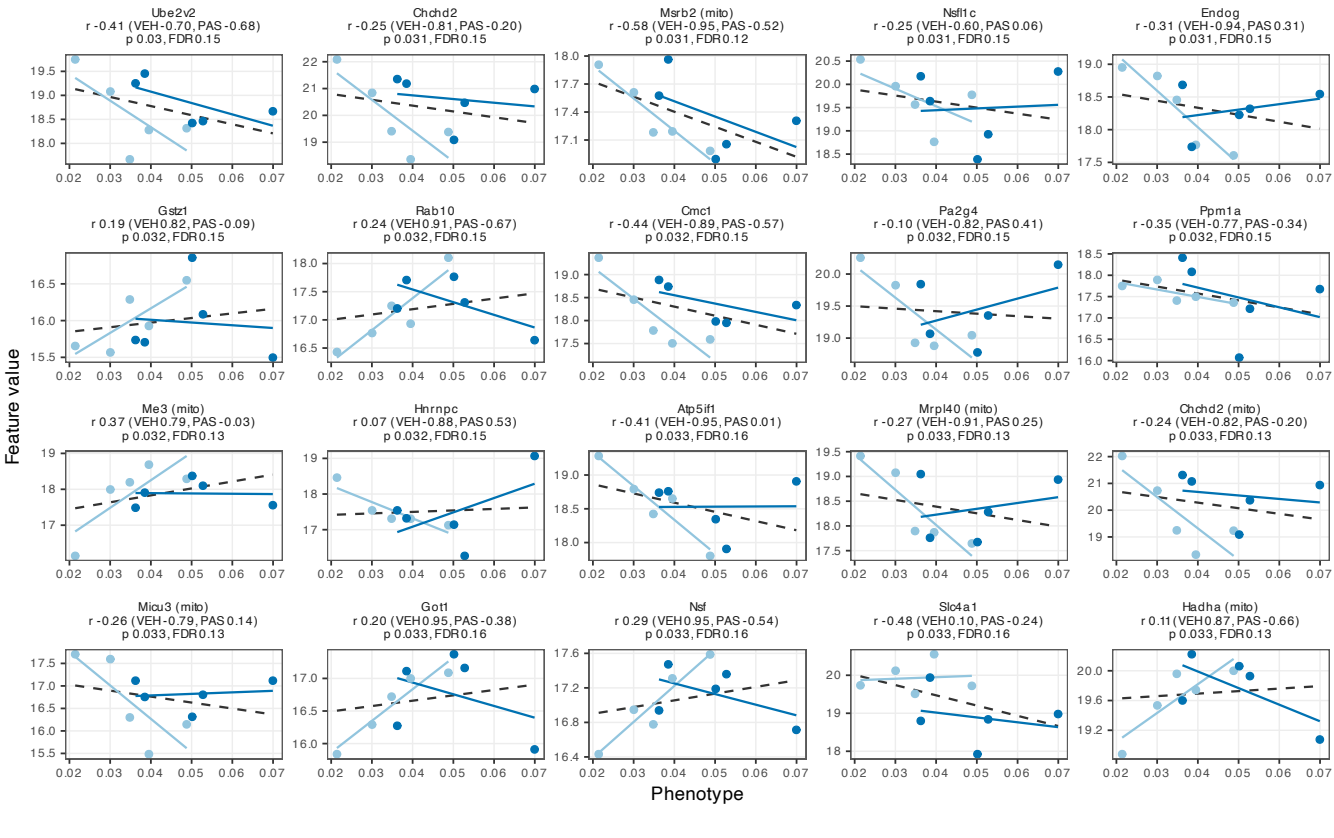
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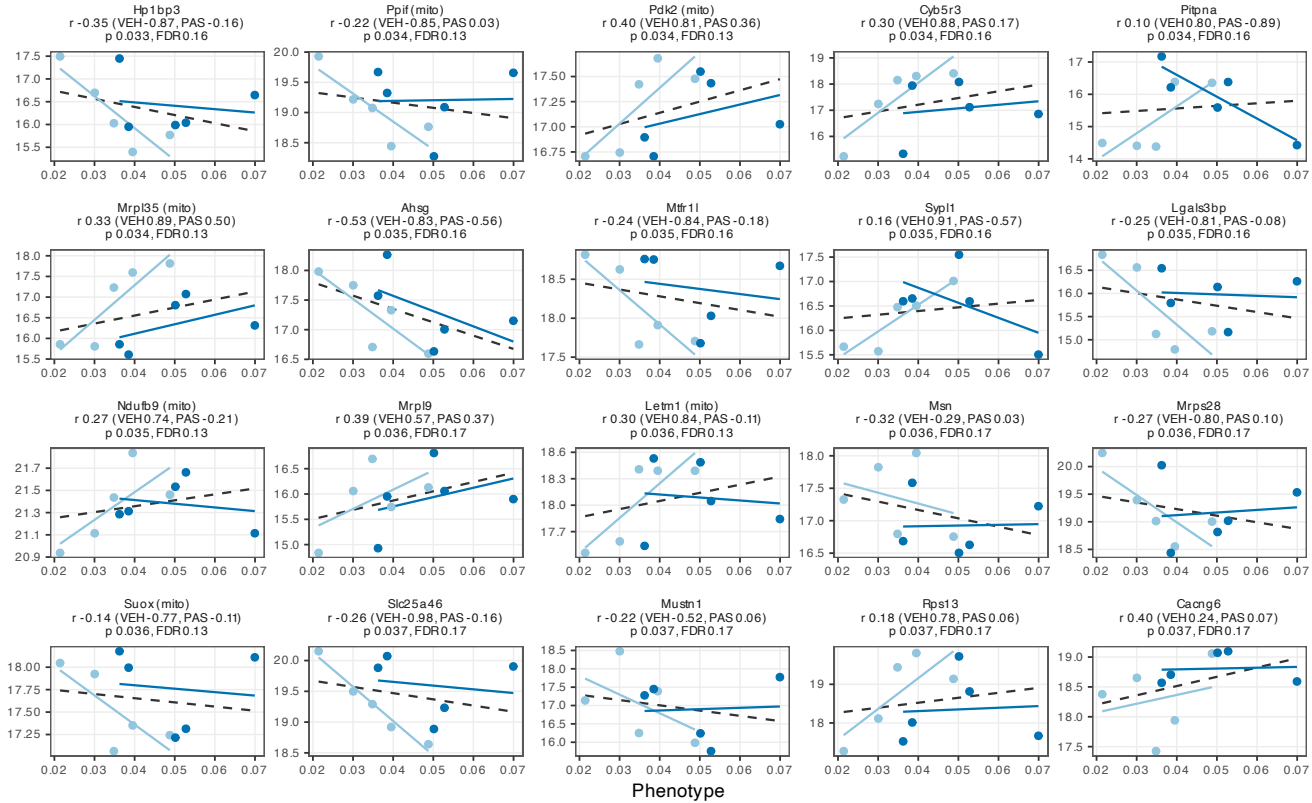
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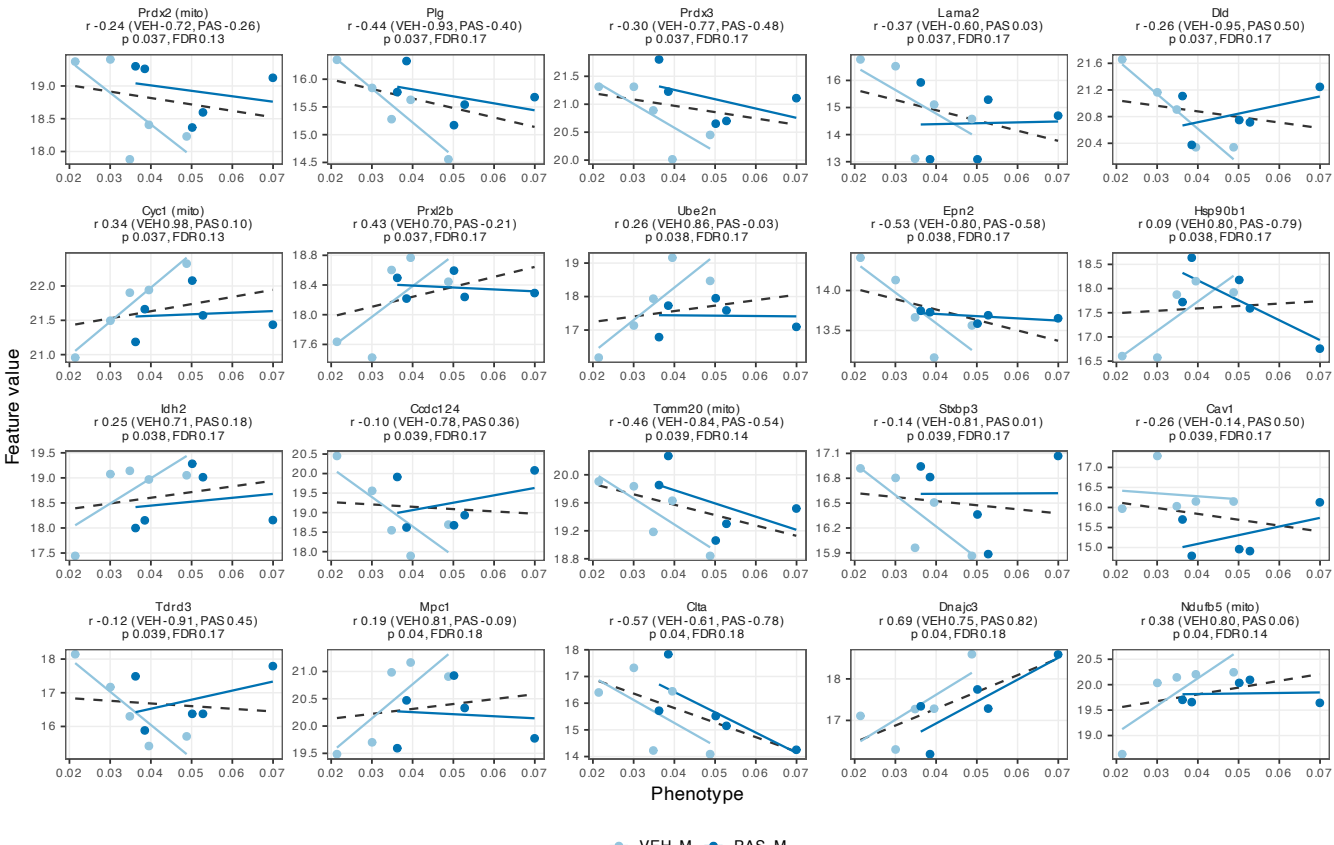
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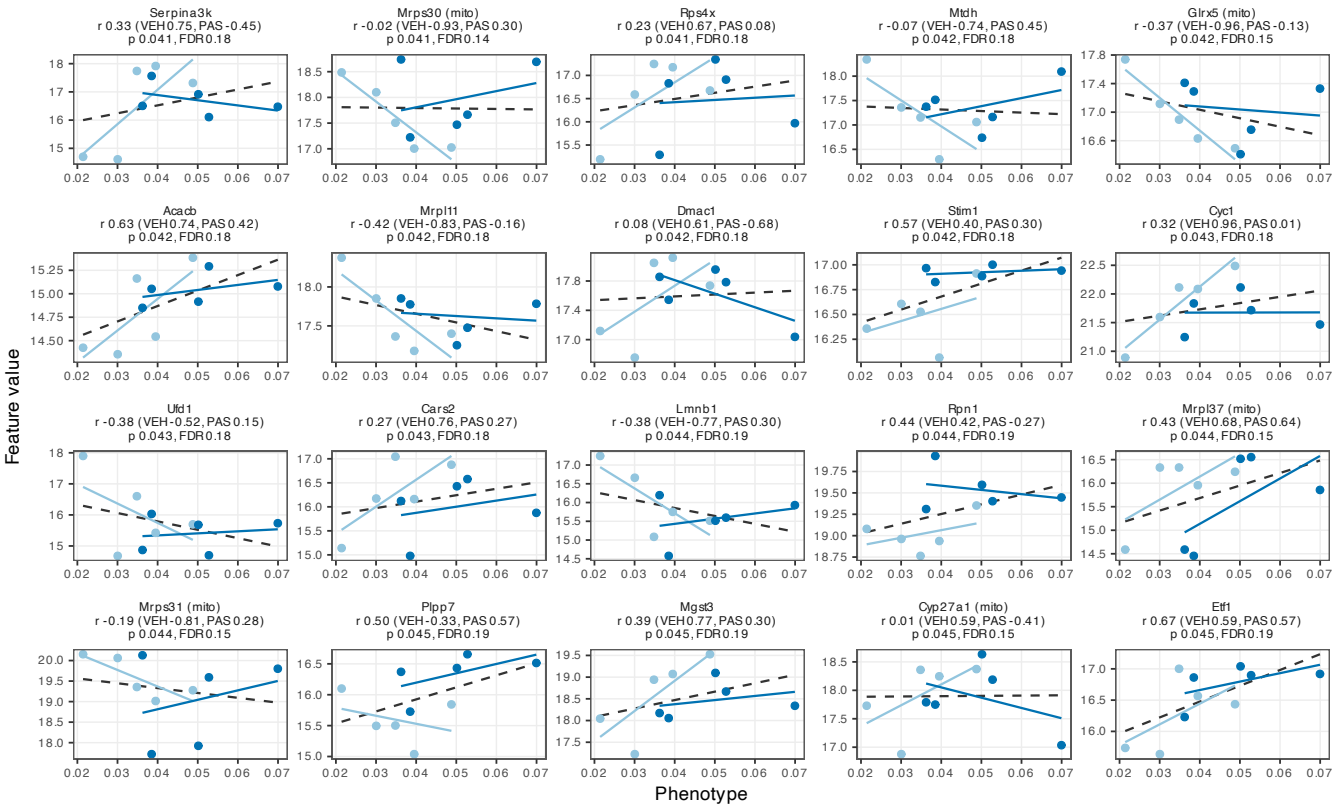
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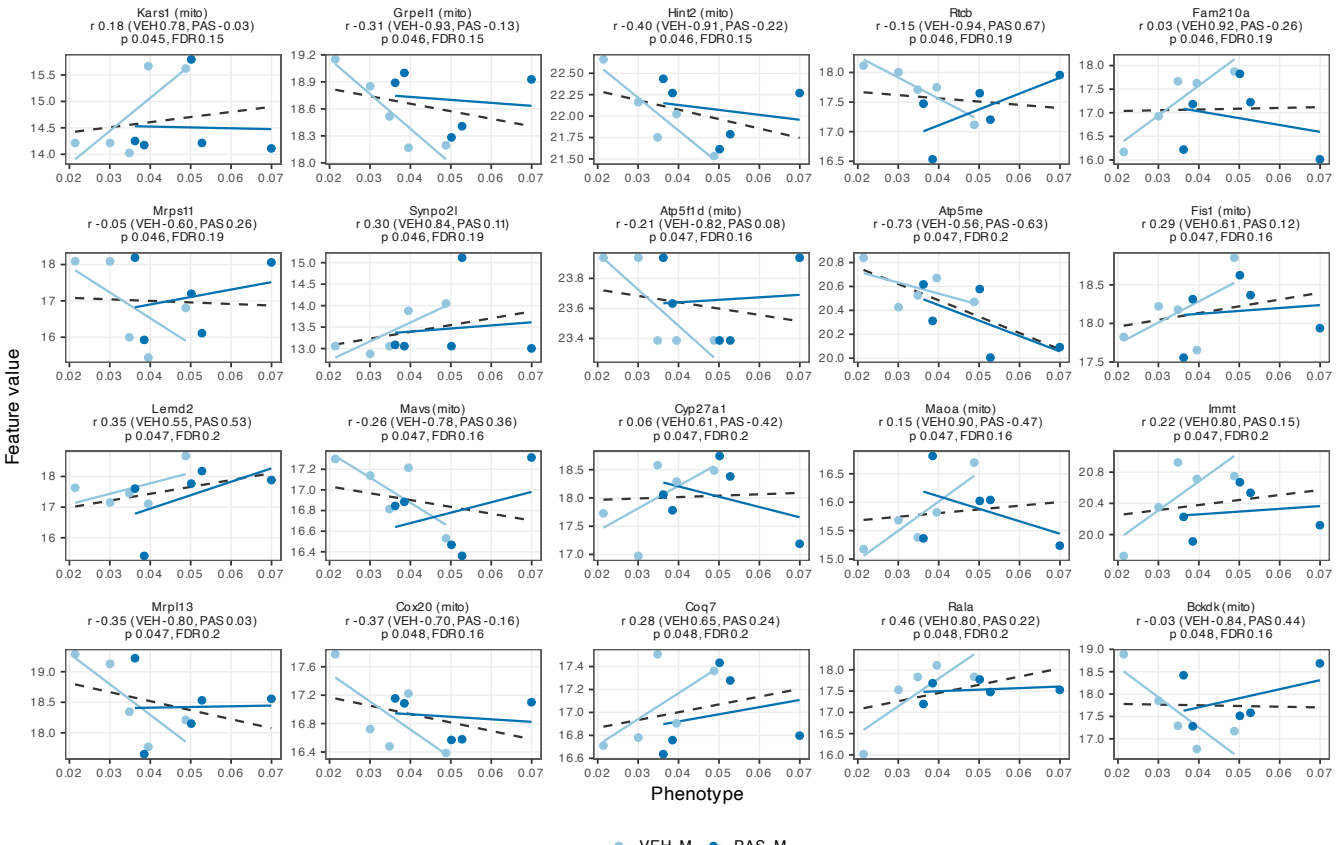
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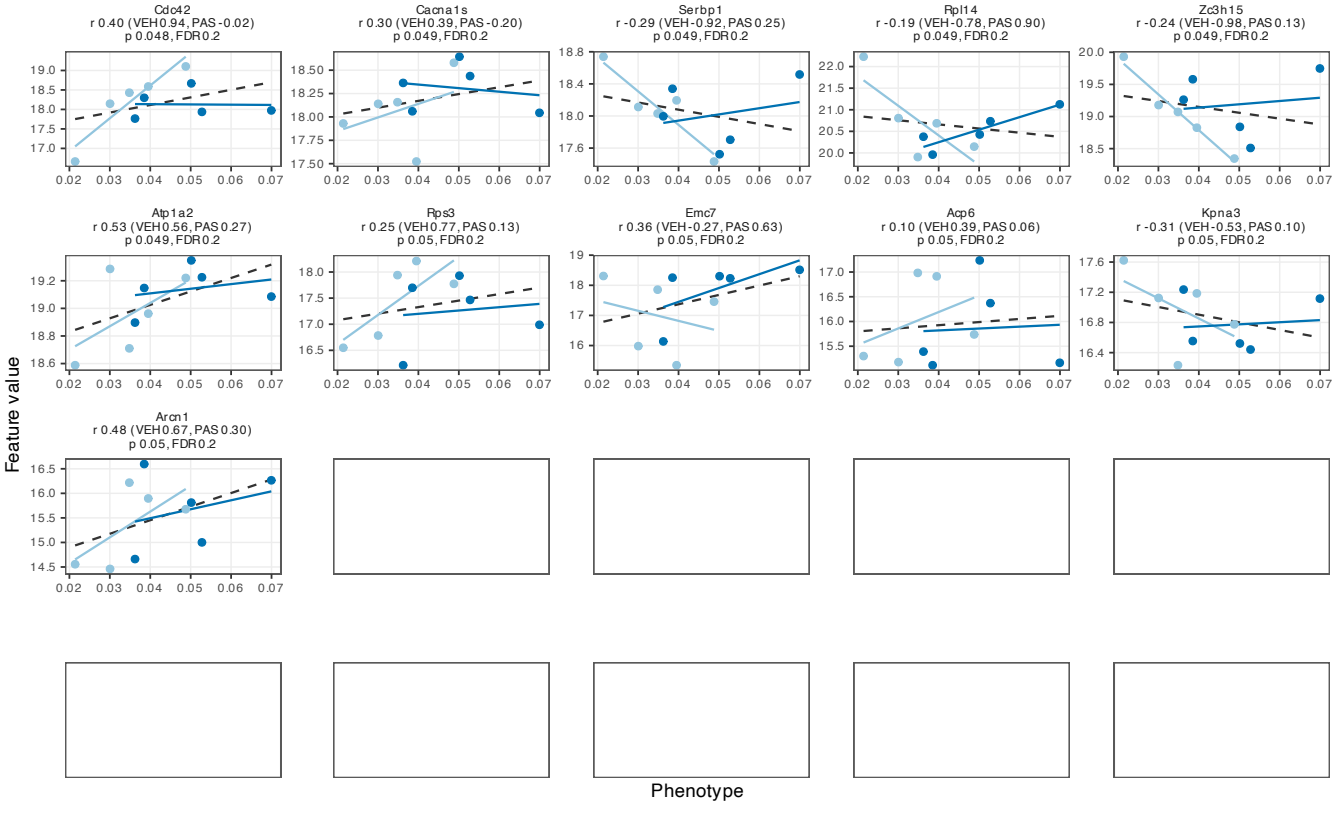
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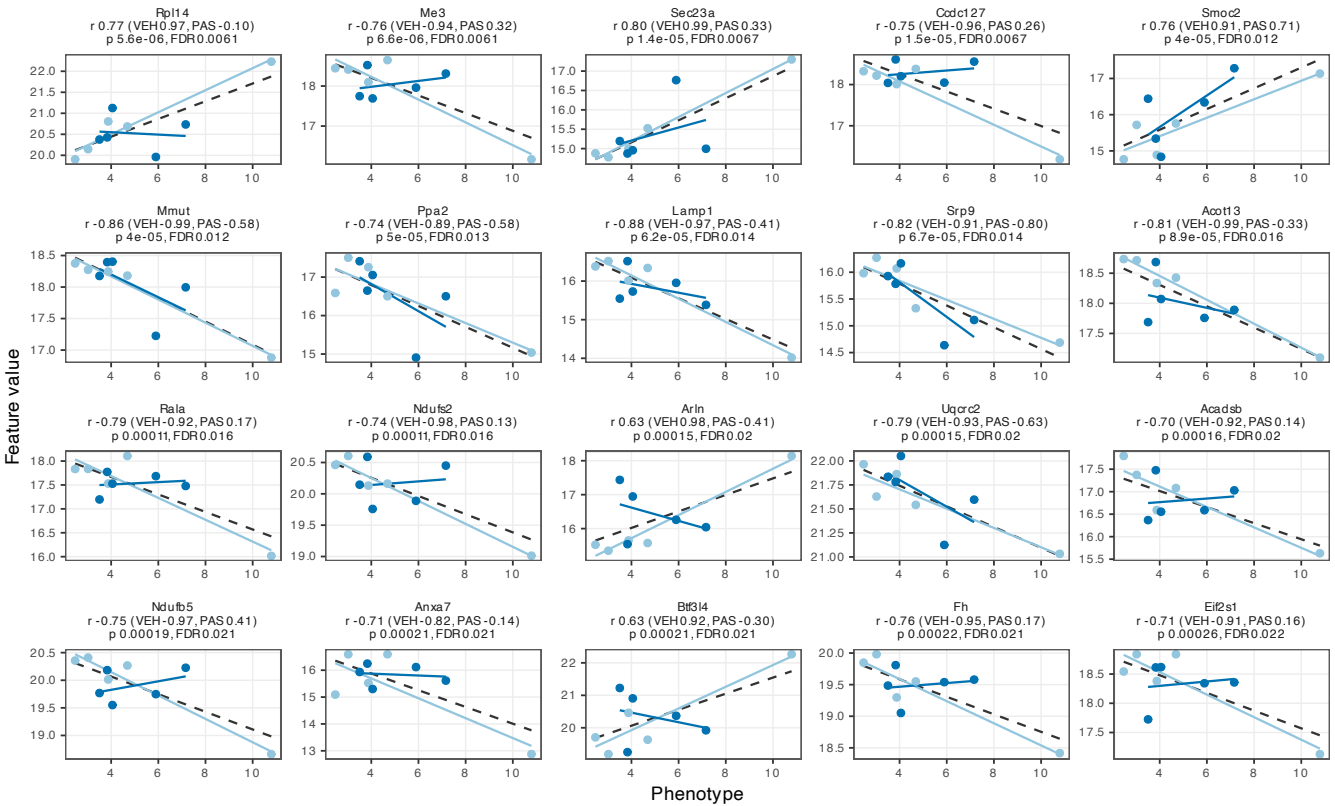
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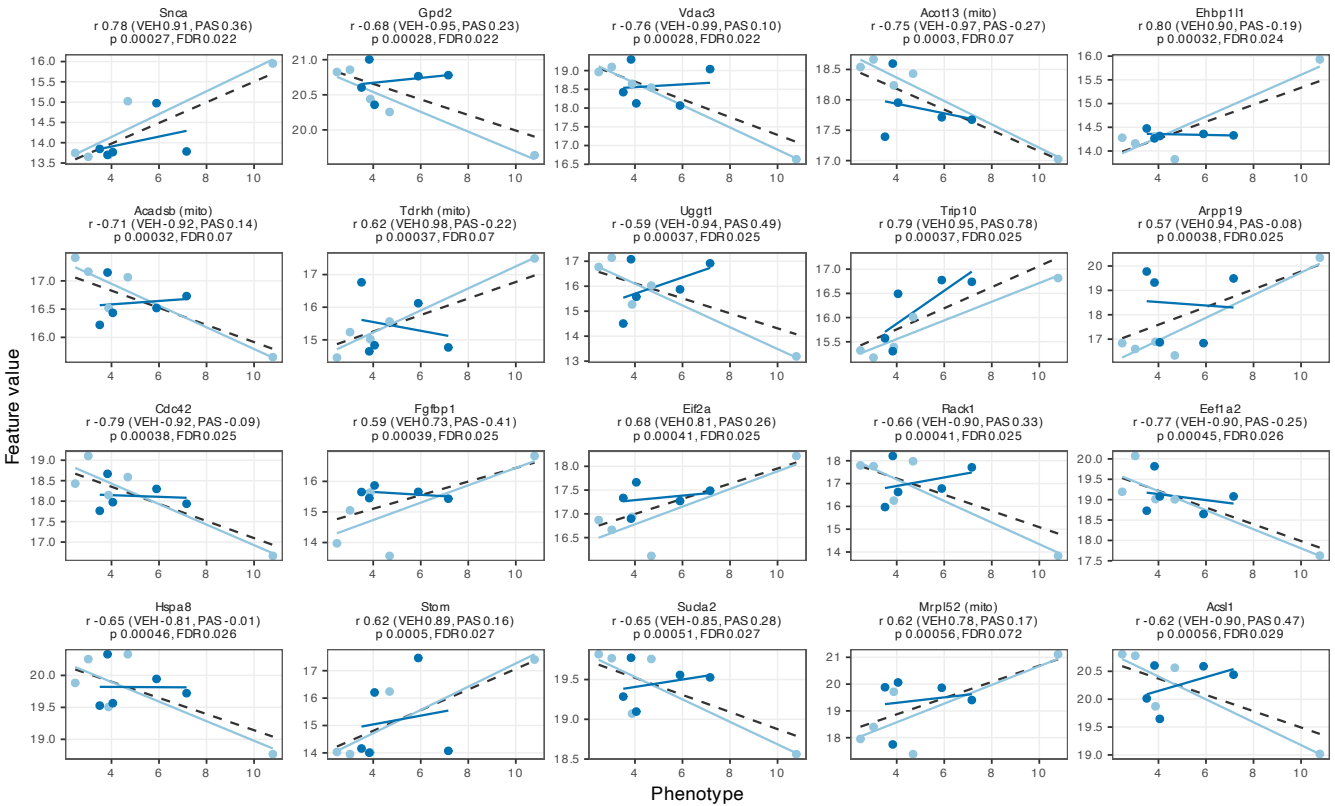
**Male: Proteins associated with Soleus pyruvate, every association at p ≤ 0.05 (n = 594)**

*Phenotype effect of PAS in this sex: +9% (p 0.77), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



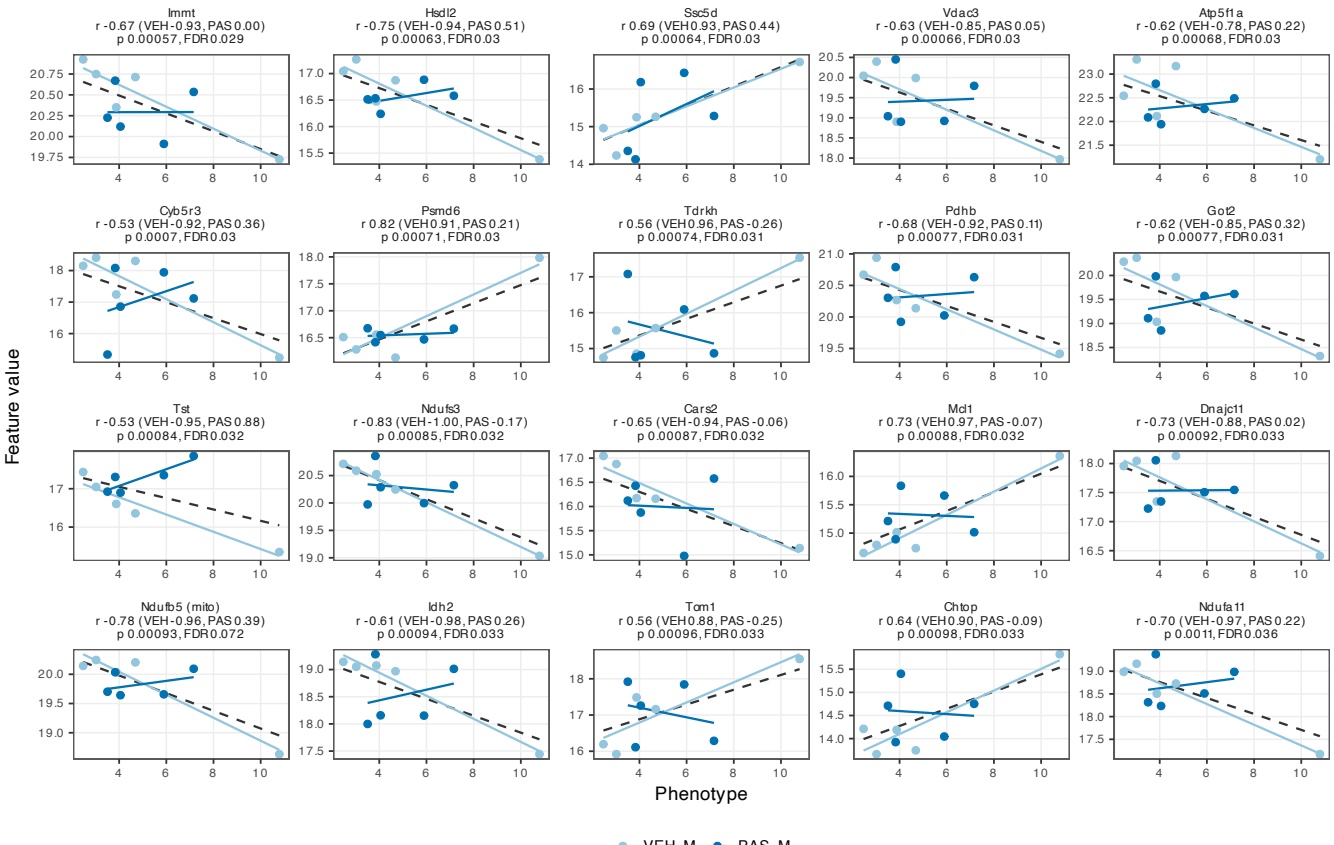
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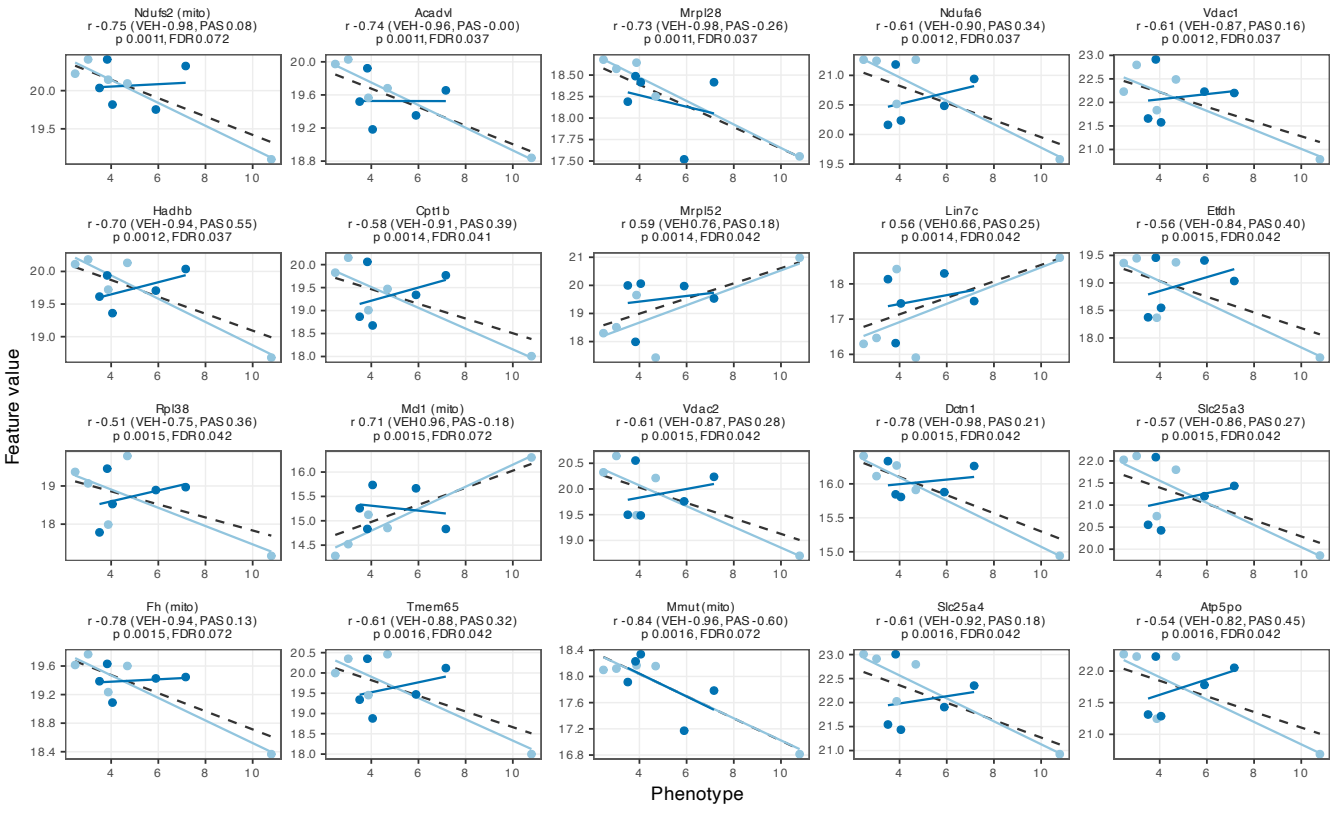
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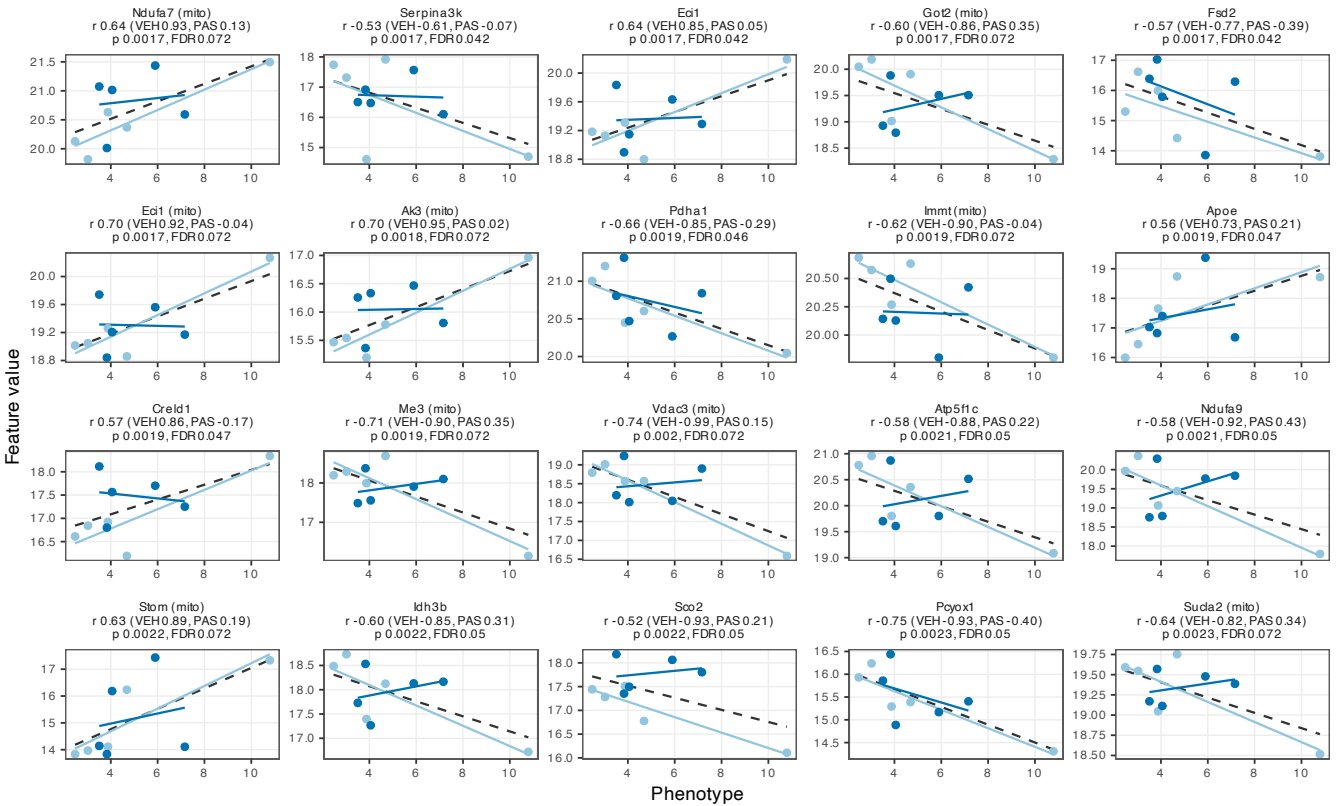
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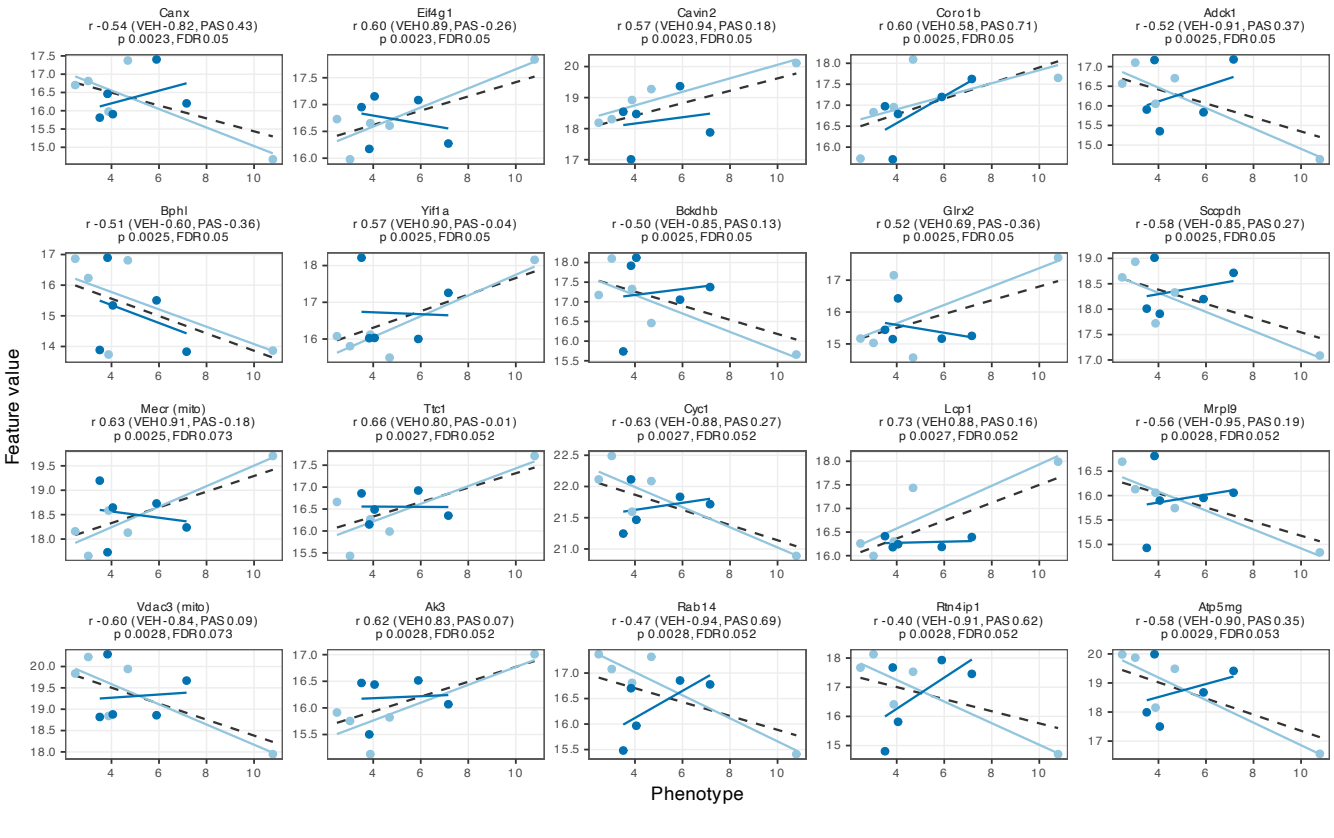
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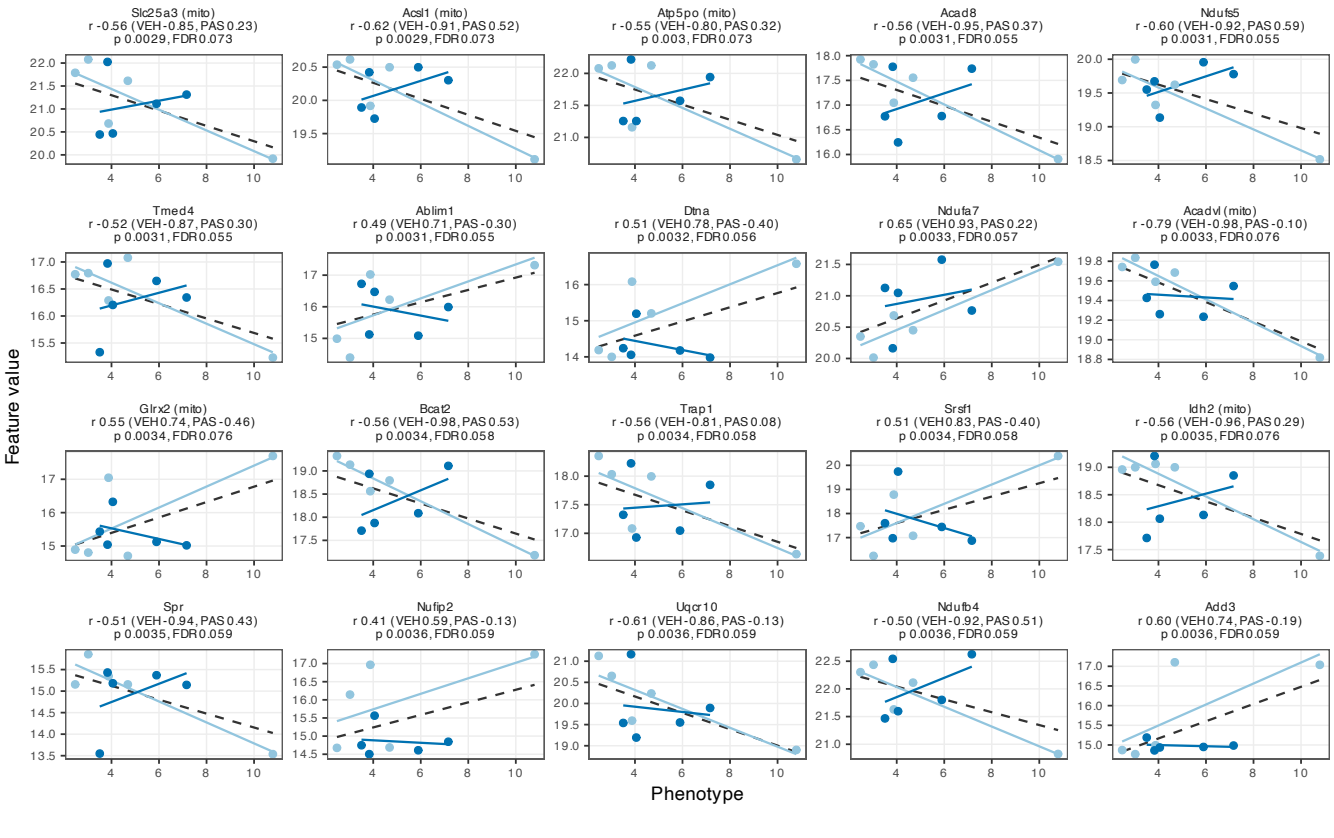
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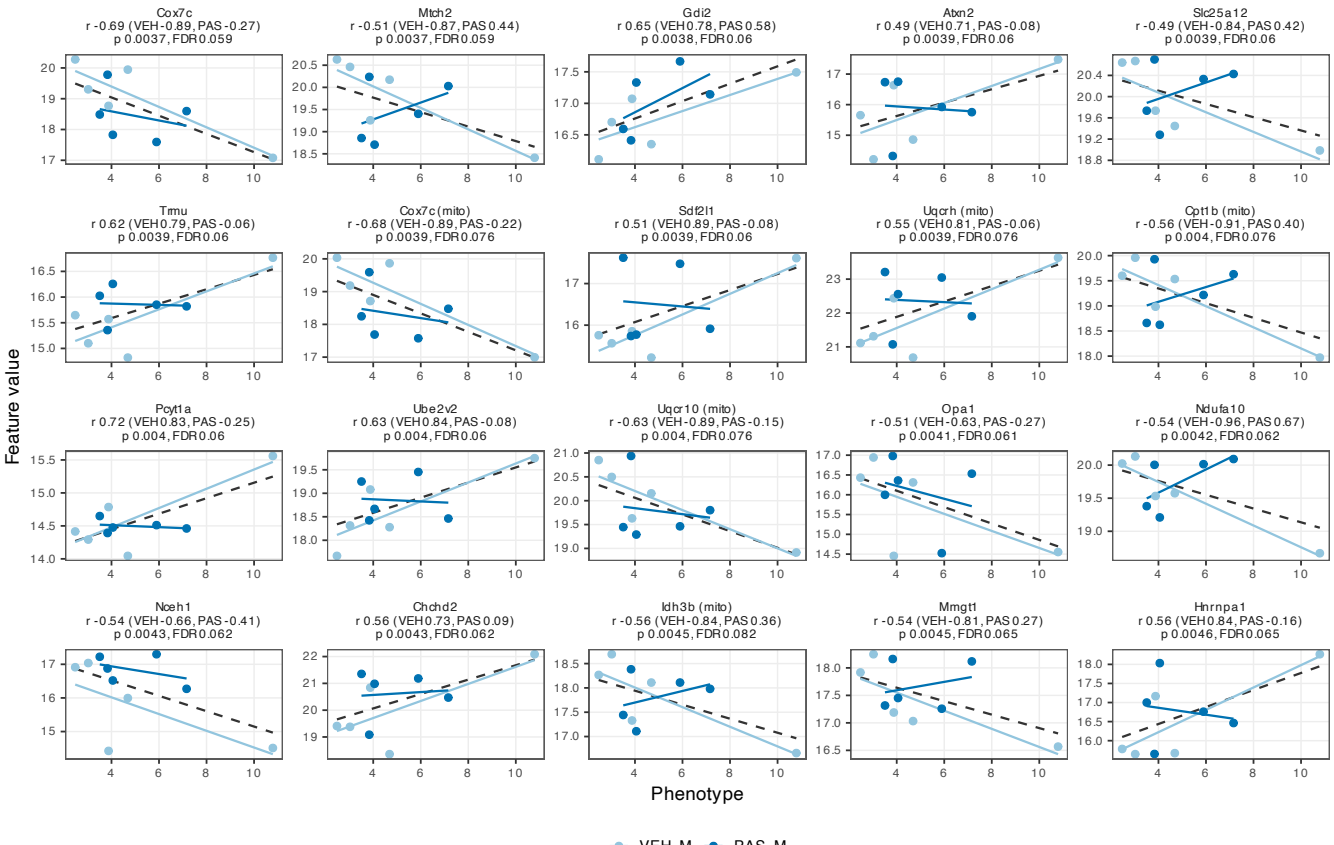
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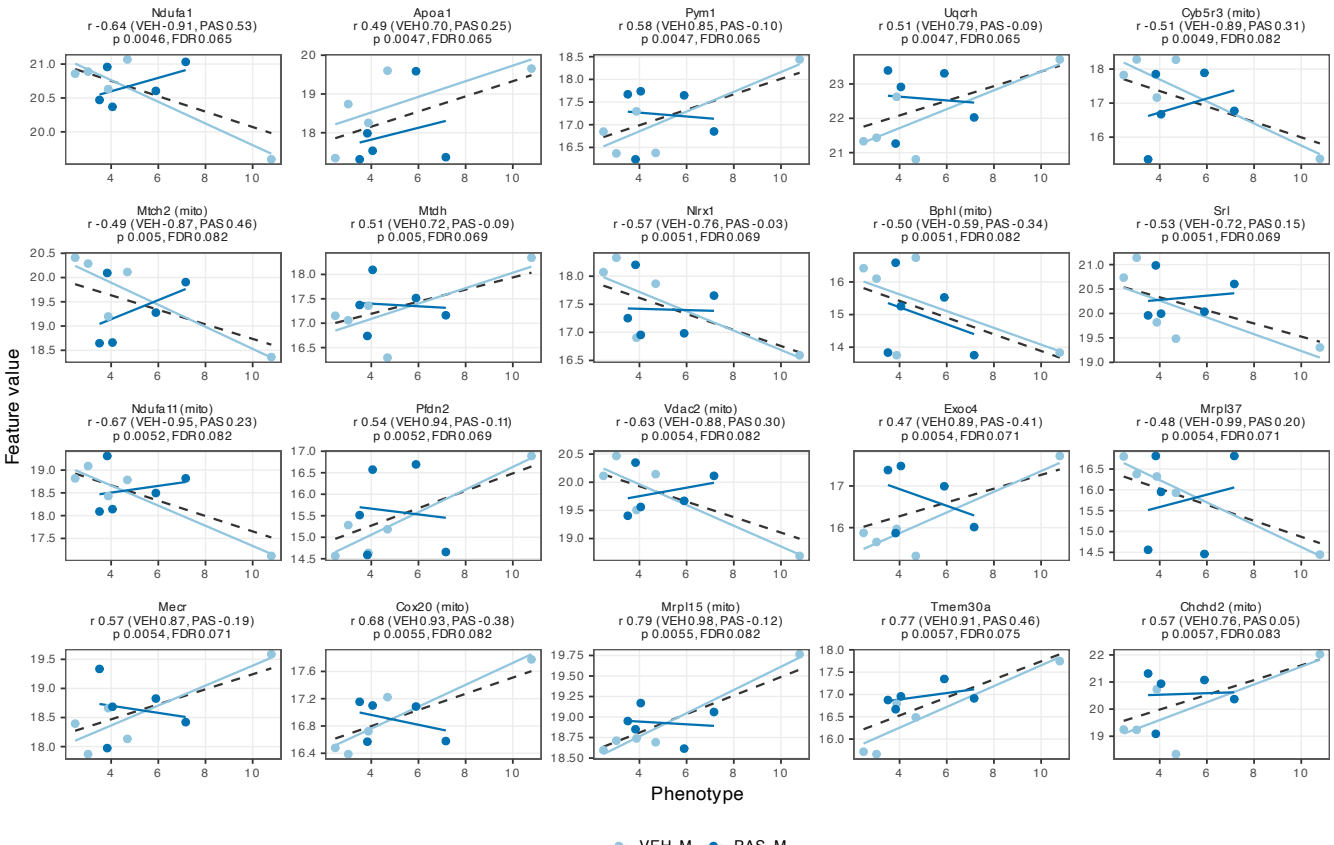
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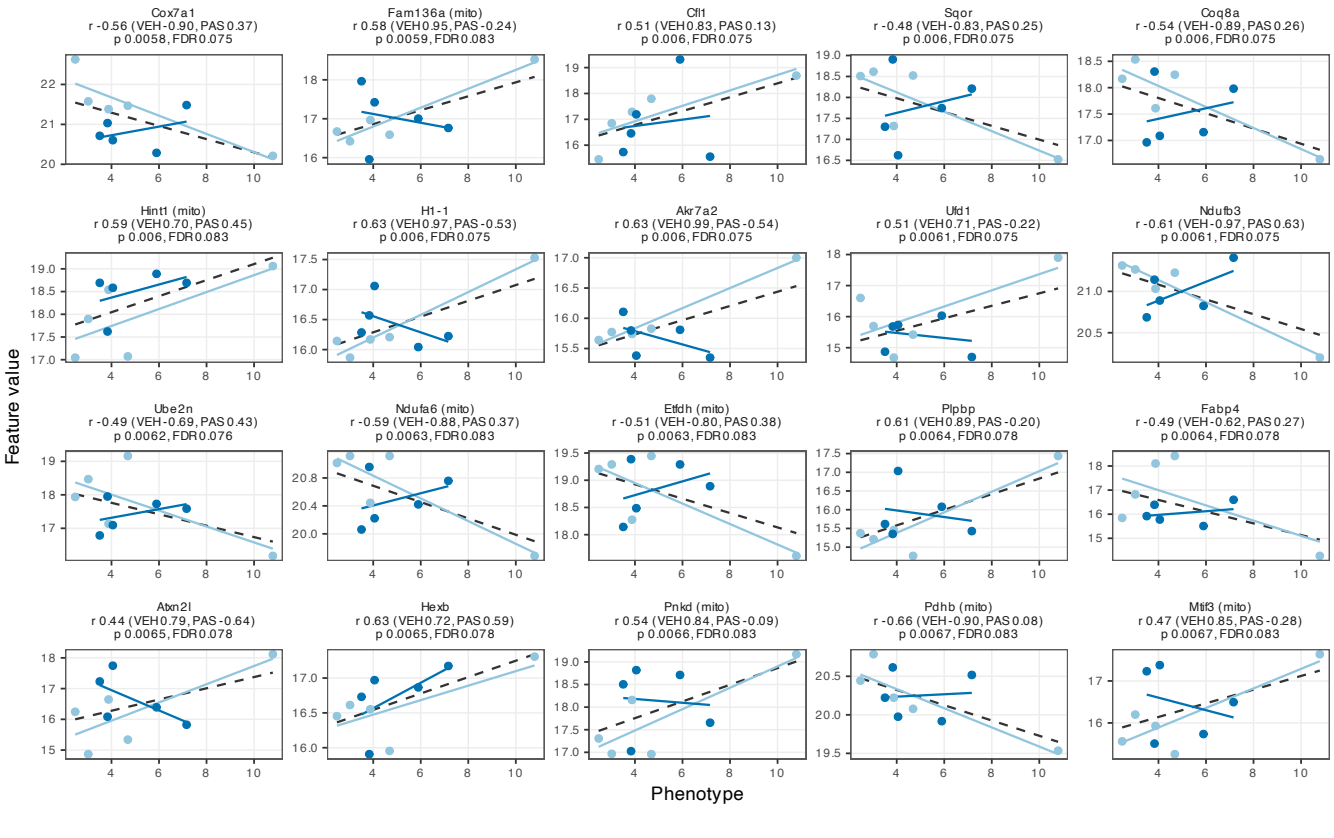
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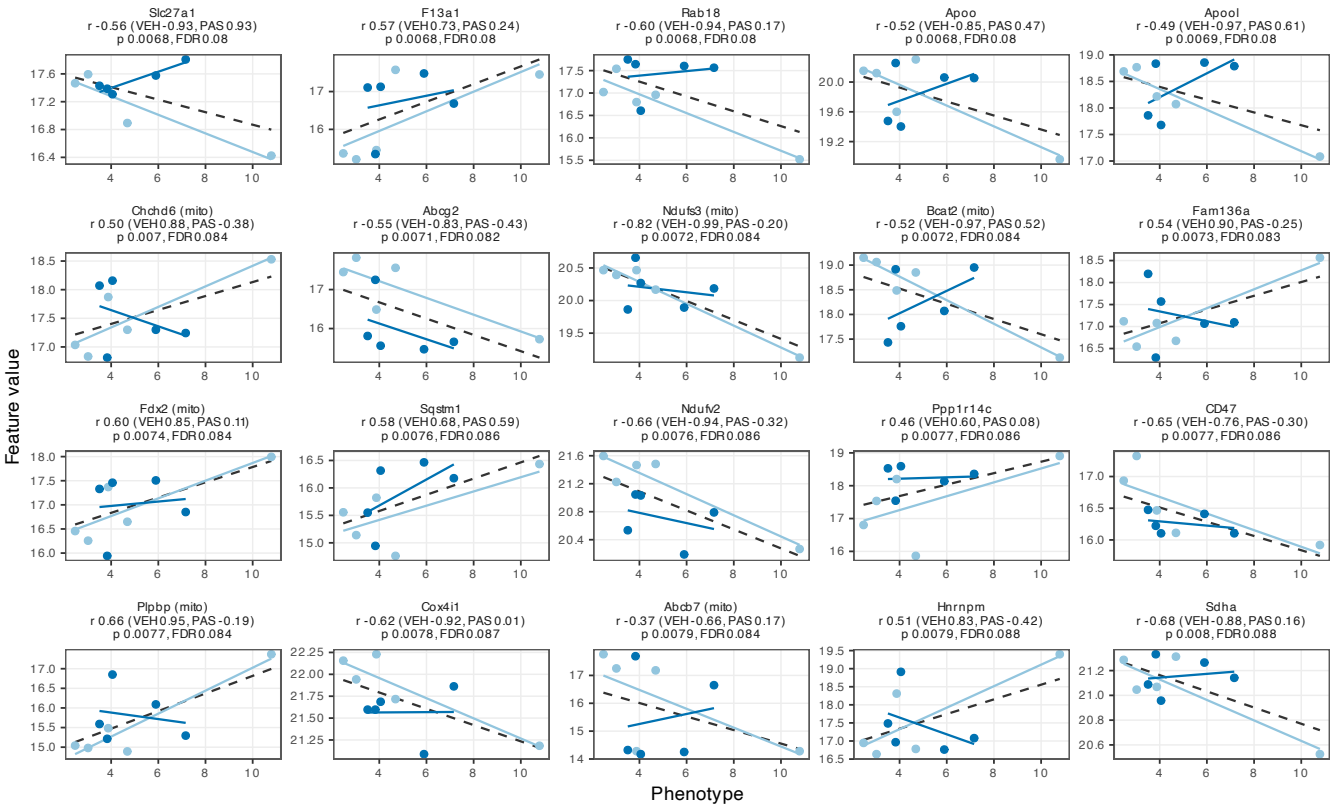
**Male: Proteins associated with Soleus pyruvate, every association at p ≤ 0.05 (n = 594)**

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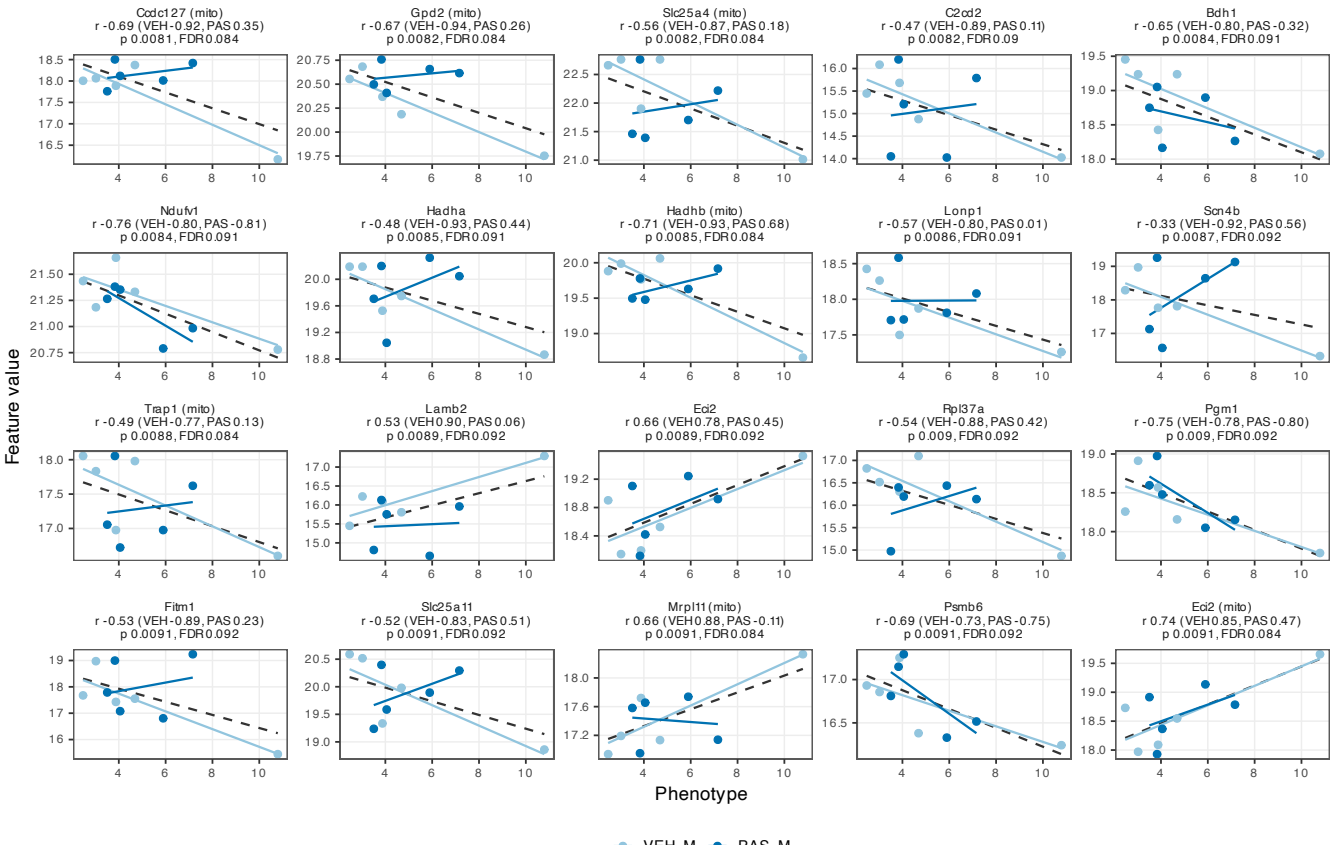
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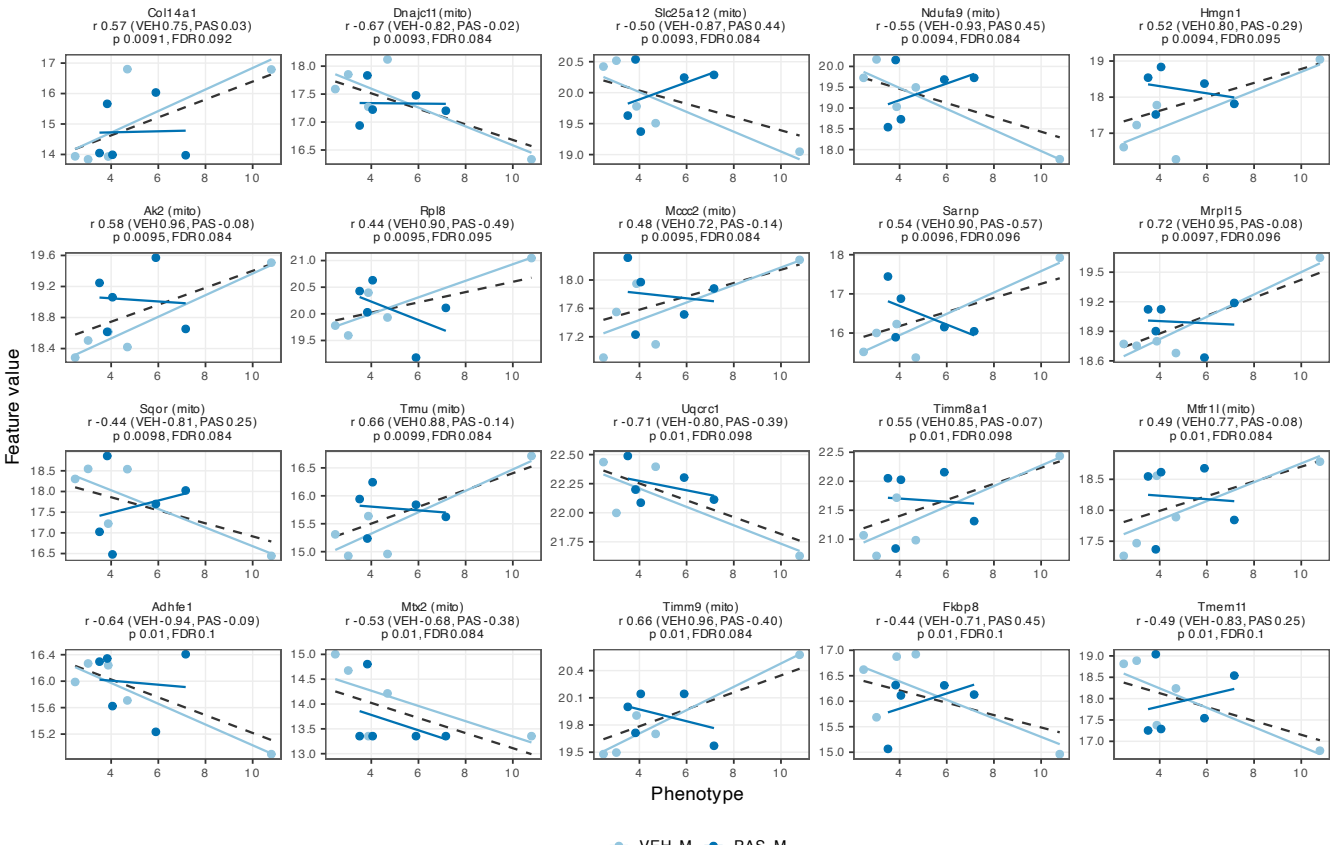
**Male: Proteins associated with Soleus pyruvate, every association at p ≤ 0.05 (n = 594)**

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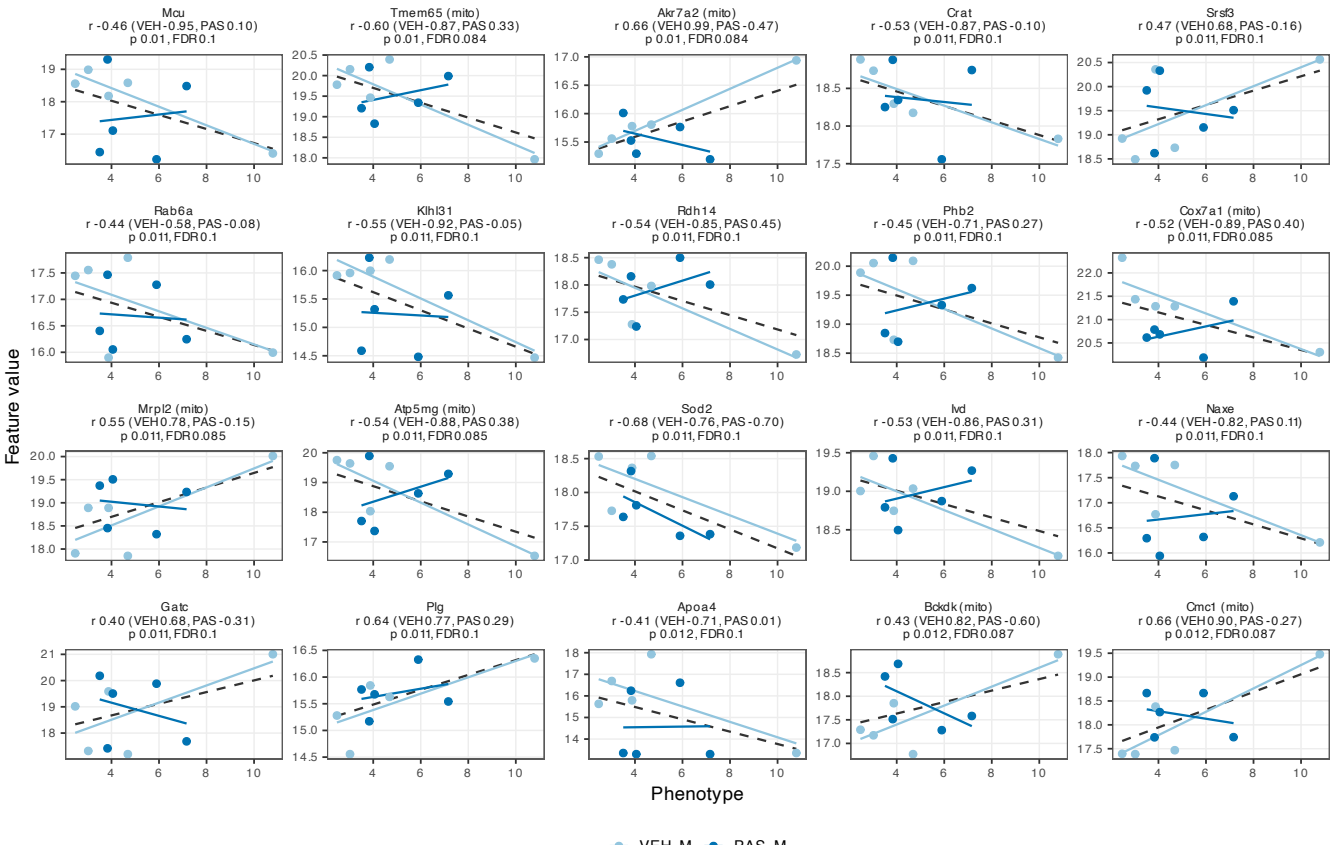
**Male: Proteins associated with Soleus pyruvate, every association at p ≤ 0.05 (n = 594)**

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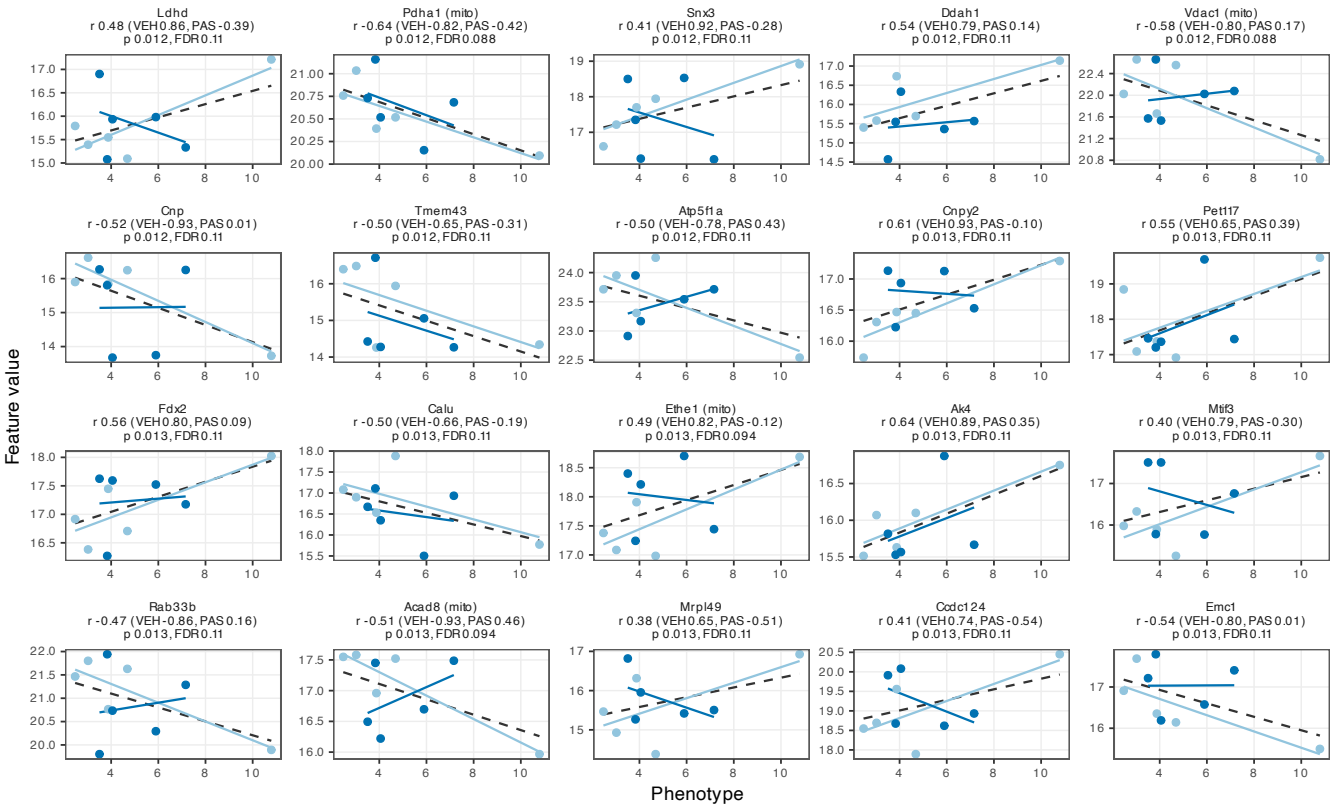
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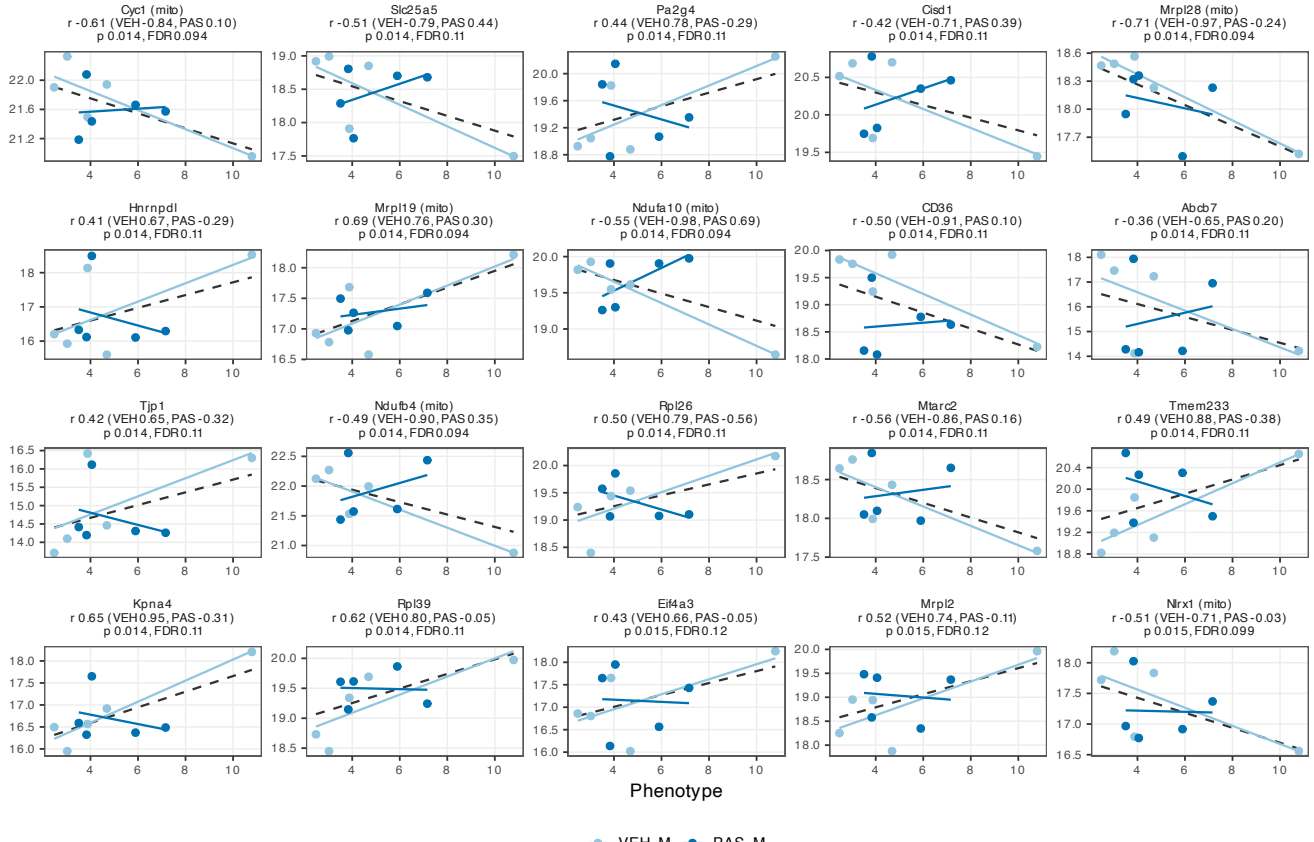
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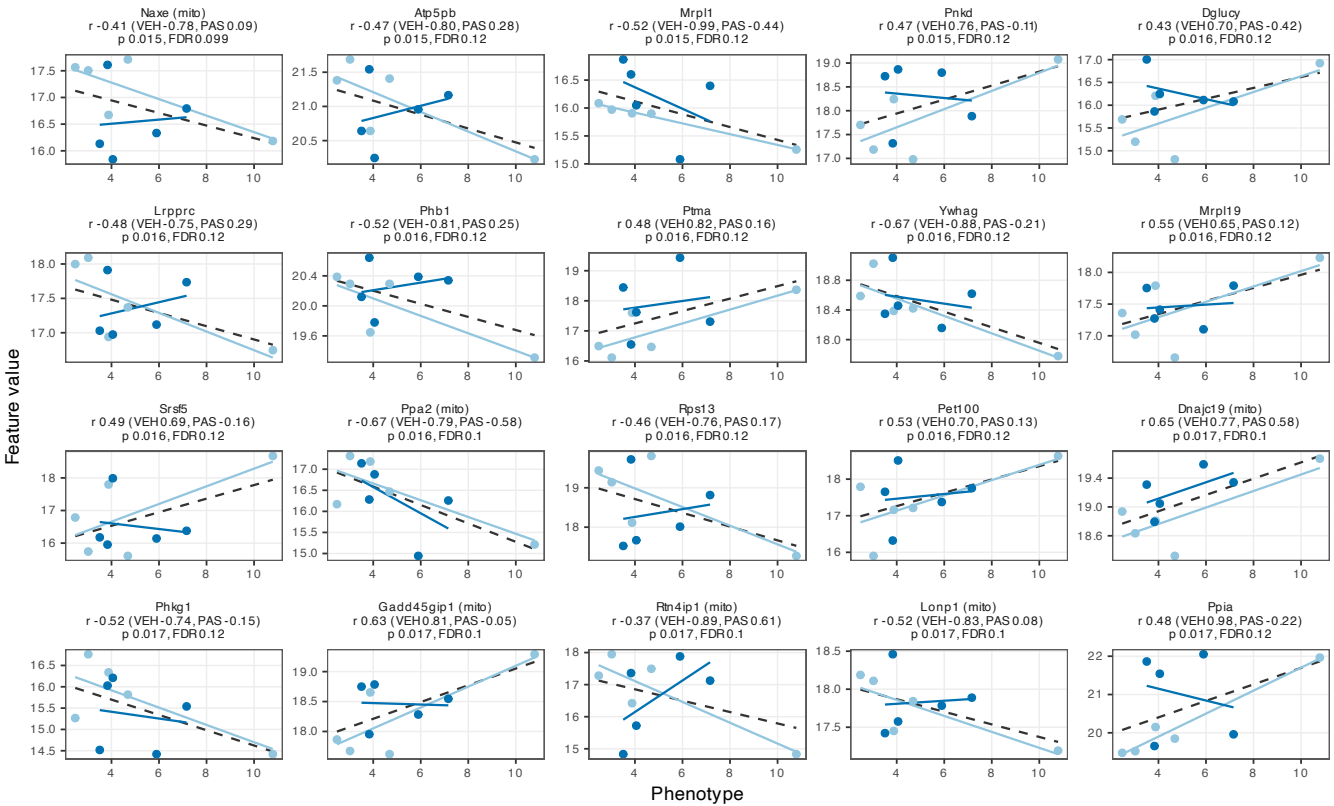
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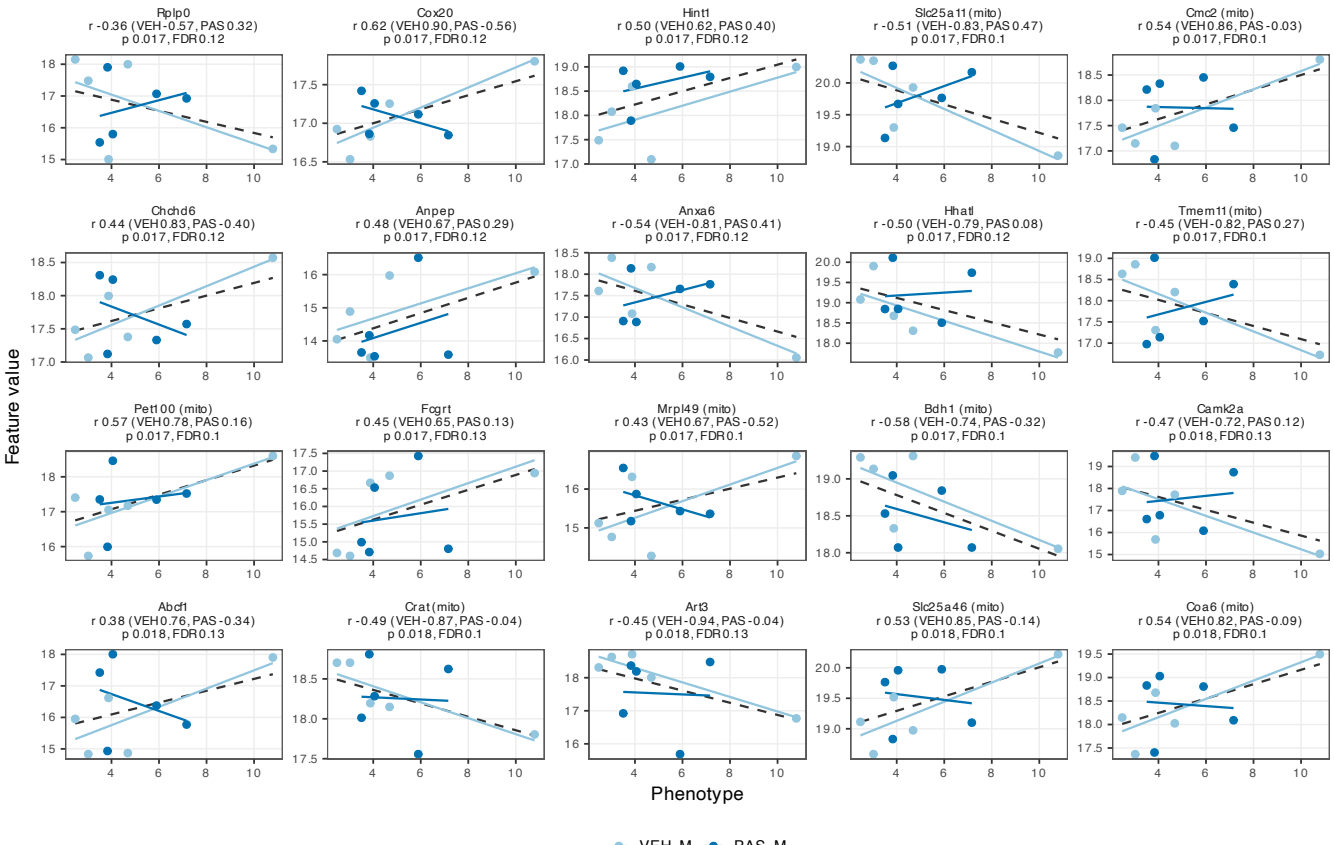
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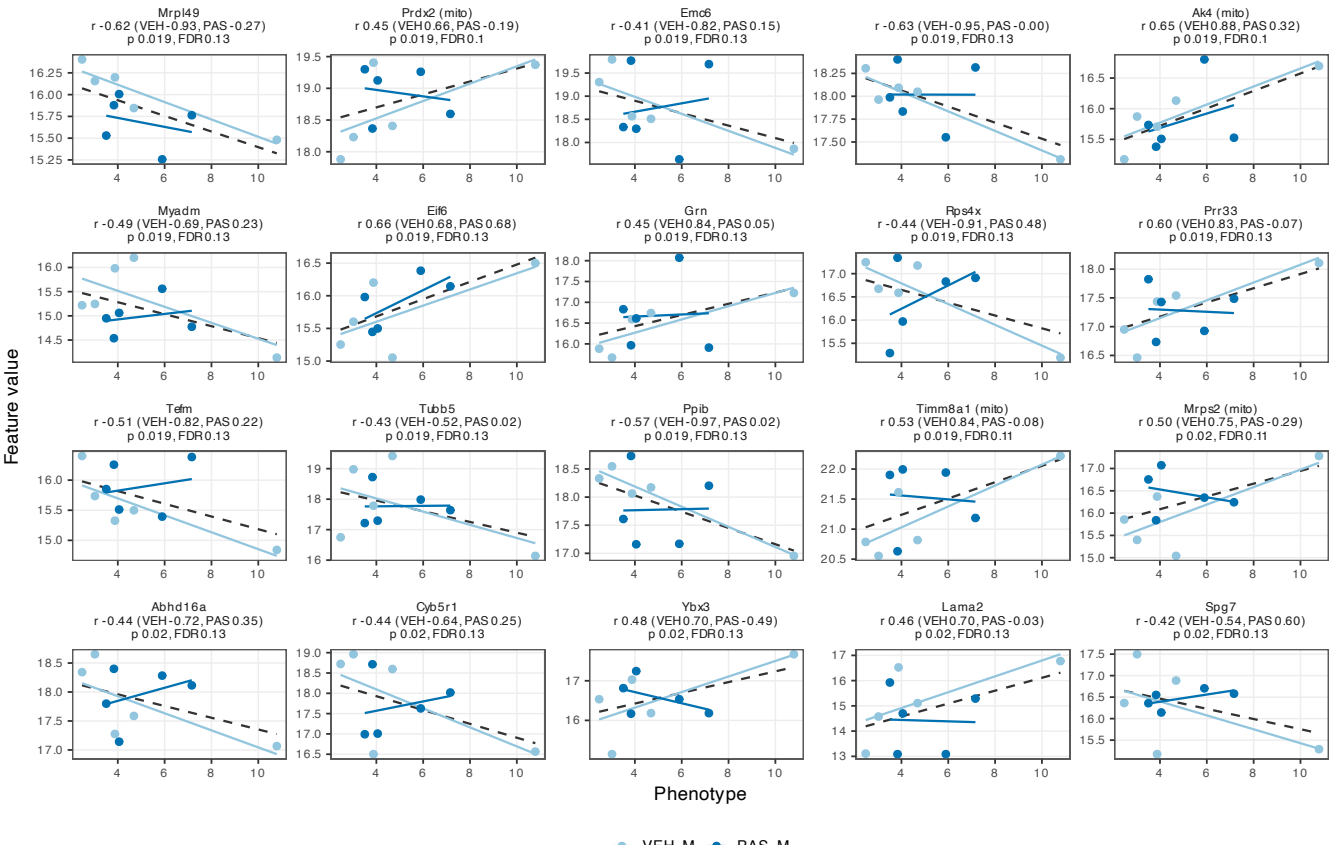
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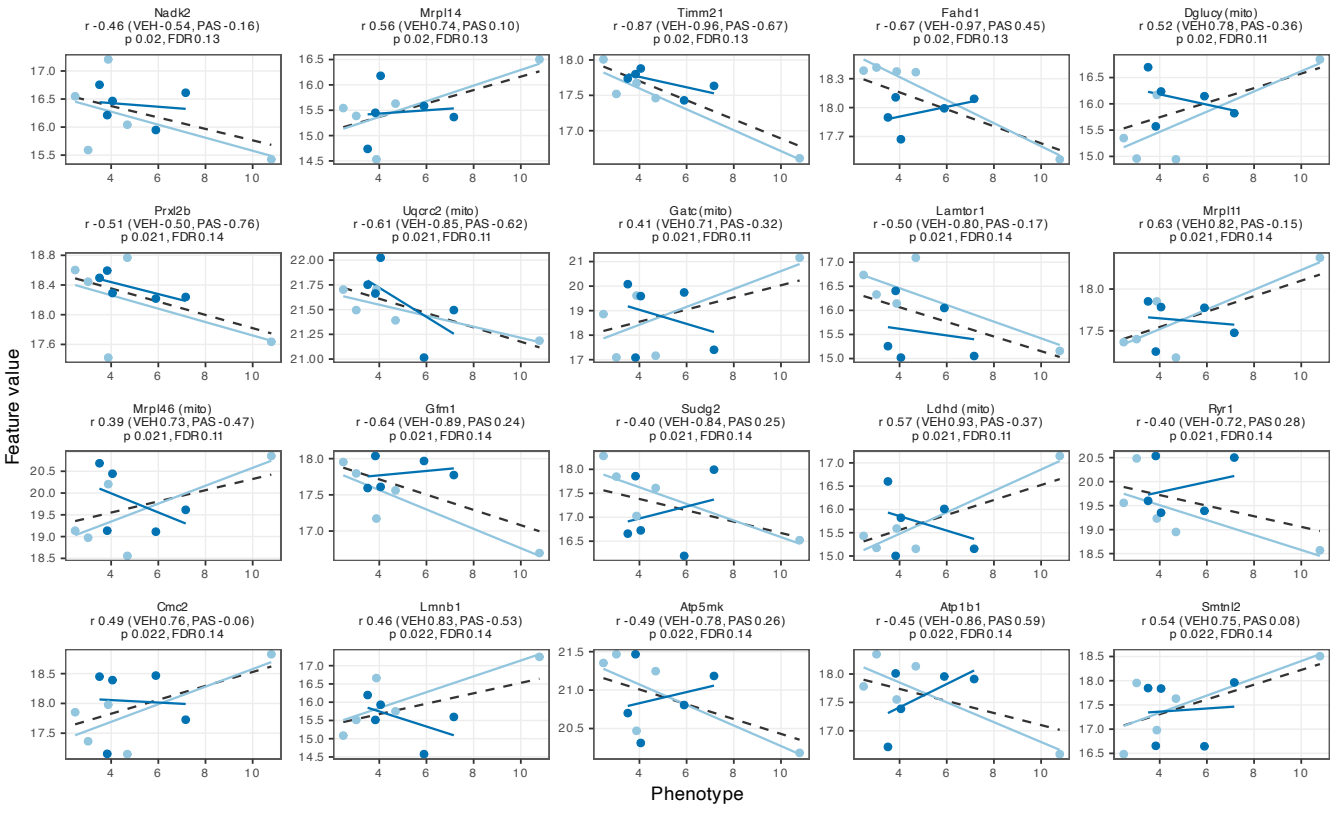
**Male: Proteins associated with Soleus pyruvate, every association at p ≤ 0.05 (n = 594)**

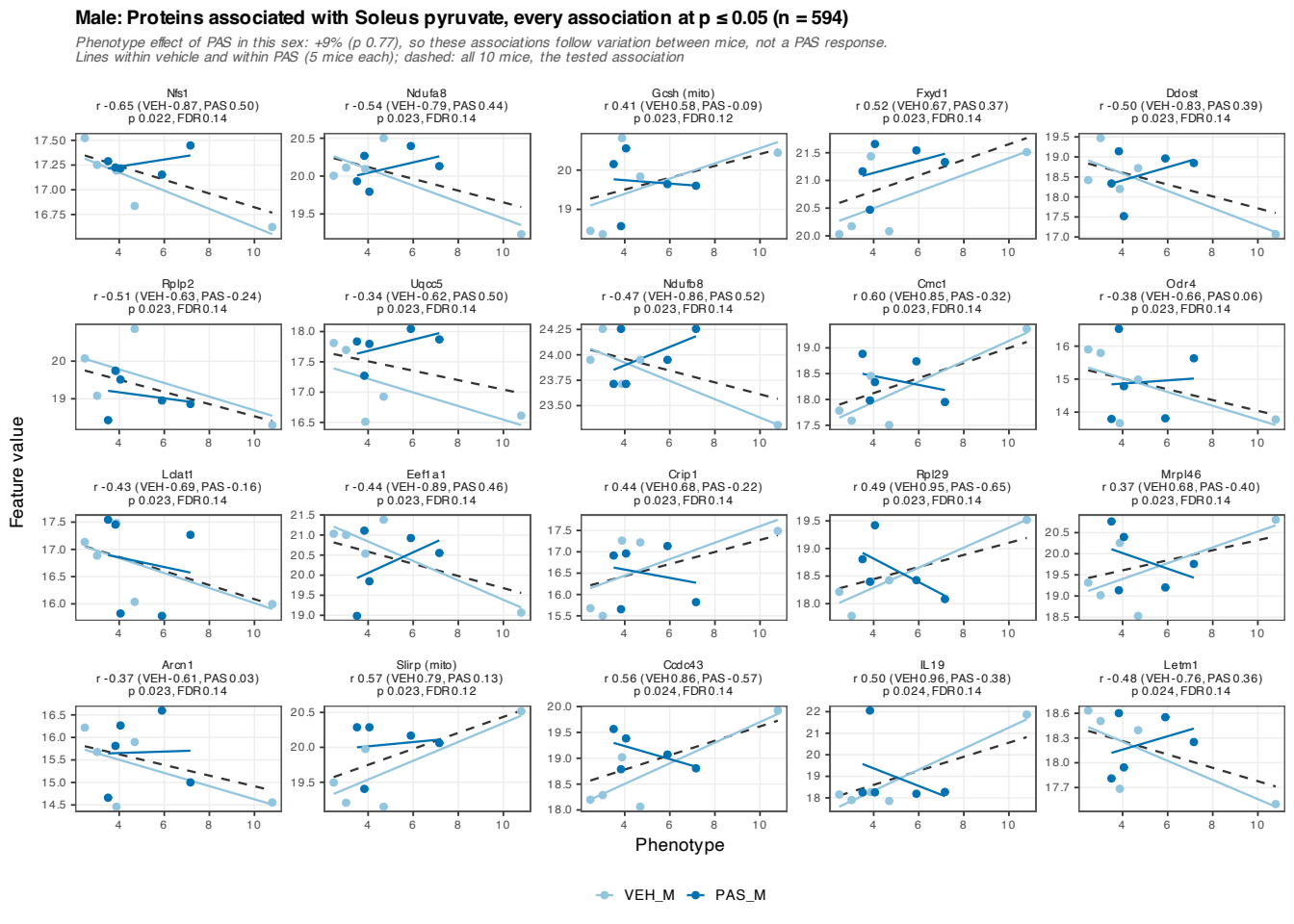
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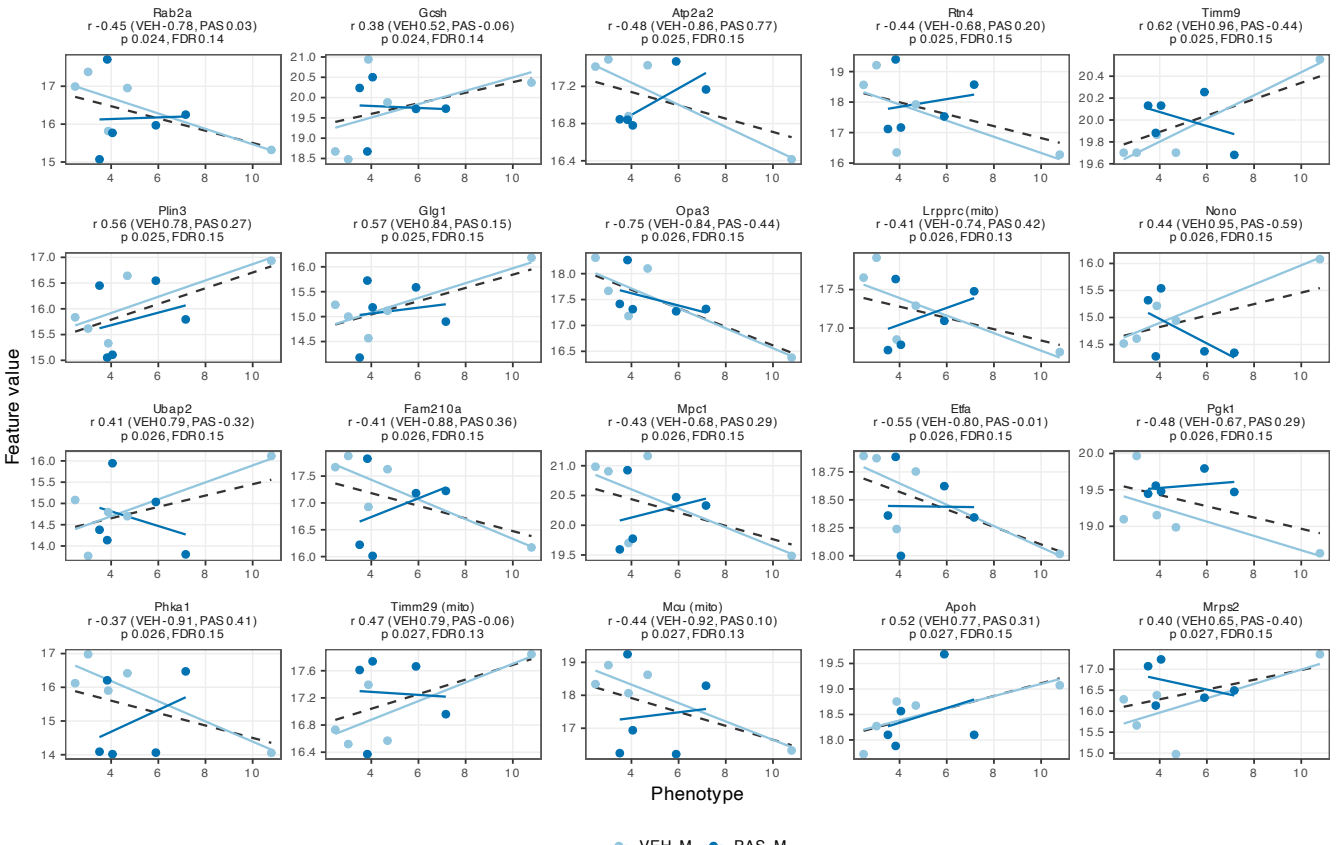
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Male: Proteins associated with Soleus pyruvate, every association at p ≤ 0.05 (n = 594)
Phenotype effect of PAS in this sex: +9% (p 0.77), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association
Nfs1
r -0.65 (VEH -0.87, PAS 0.50)
p 0.022, FDR 0.14
Ndufa8
r -0.54 (VEH -0.79, PAS 0.44)
p 0.023, FDR 0.14
Gcsh (mito)
r 0.41 (VEH 0.58, PAS -0.09)
p 0.023, FDR 0.12
Fxyd1
r 0.52 (VEH 0.67, PAS 0.37)
p 0.023, FDR 0.14
Ddost
r -0.50 (VEH -0.83, PAS 0.39)
p 0.023, FDR 0.14
Rplp2
r -0.51 (VEH -0.63, PAS -0.24)
p 0.023, FDR 0.14
Uqcc5
r -0.34 (VEH -0.62, PAS 0.50)
p 0.023, FDR 0.14
Ndufb8
r -0.47 (VEH -0.86, PAS 0.52)
p 0.023, FDR 0.14
Cmc1
r 0.60 (VEH 0.85, PAS -0.32)
p 0.023, FDR 0.14
Odr4
r -0.38 (VEH -0.66, PAS 0.06)
p 0.023, FDR 0.14
Lclat1
r -0.43 (VEH -0.69, PAS -0.16)
p 0.023, FDR 0.14
Eef1a1
r -0.44 (VEH -0.89, PAS 0.46)
p 0.023, FDR 0.14
Crip1
r 0.44 (VEH 0.68, PAS -0.22)
p 0.023, FDR 0.14
Rpl29
r 0.49 (VEH 0.95, PAS -0.65)
p 0.023, FDR 0.14
Mrpl46
r 0.37 (VEH 0.68, PAS -0.40)
p 0.023, FDR 0.14
Arcn1
r -0.37 (VEH -0.61, PAS 0.03)
p 0.023, FDR 0.14
Slirp (mito)
r 0.57 (VEH 0.79, PAS 0.13)
p 0.023, FDR 0.12
Ccdc43
r 0.56 (VEH 0.86, PAS -0.57)
p 0.024, FDR 0.14
Il19
r 0.50 (VEH 0.96, PAS -0.38)
p 0.024, FDR 0.14
Letm1
r -0.48 (VEH -0.76, PAS 0.36)
p 0.024, FDR 0.14
Feature value
Phenotype
VEH_M
PAS_M

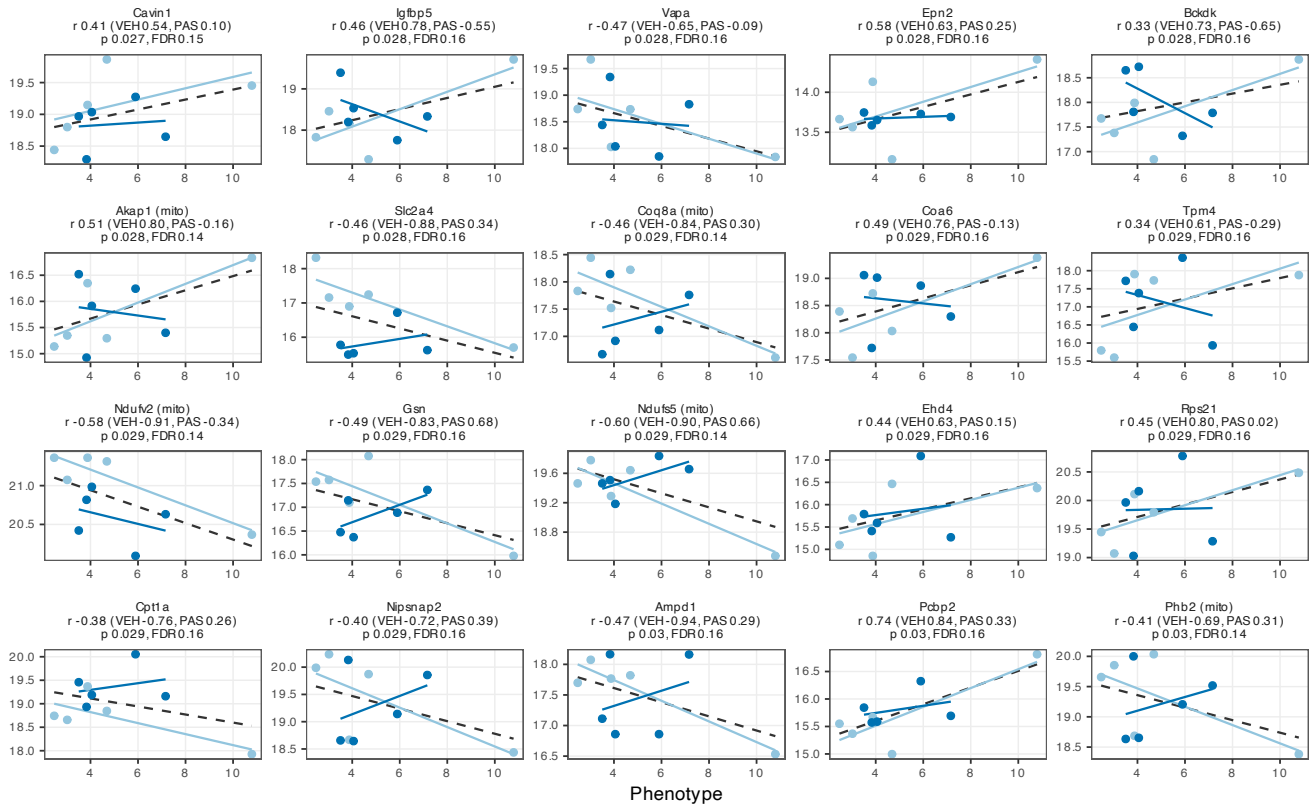
**Male: Proteins associated with Soleus pyruvate, every association at p ≤ 0.05 (n = 594)**

*Phenotype effect of PAS in this sex: +9% (p 0.77), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



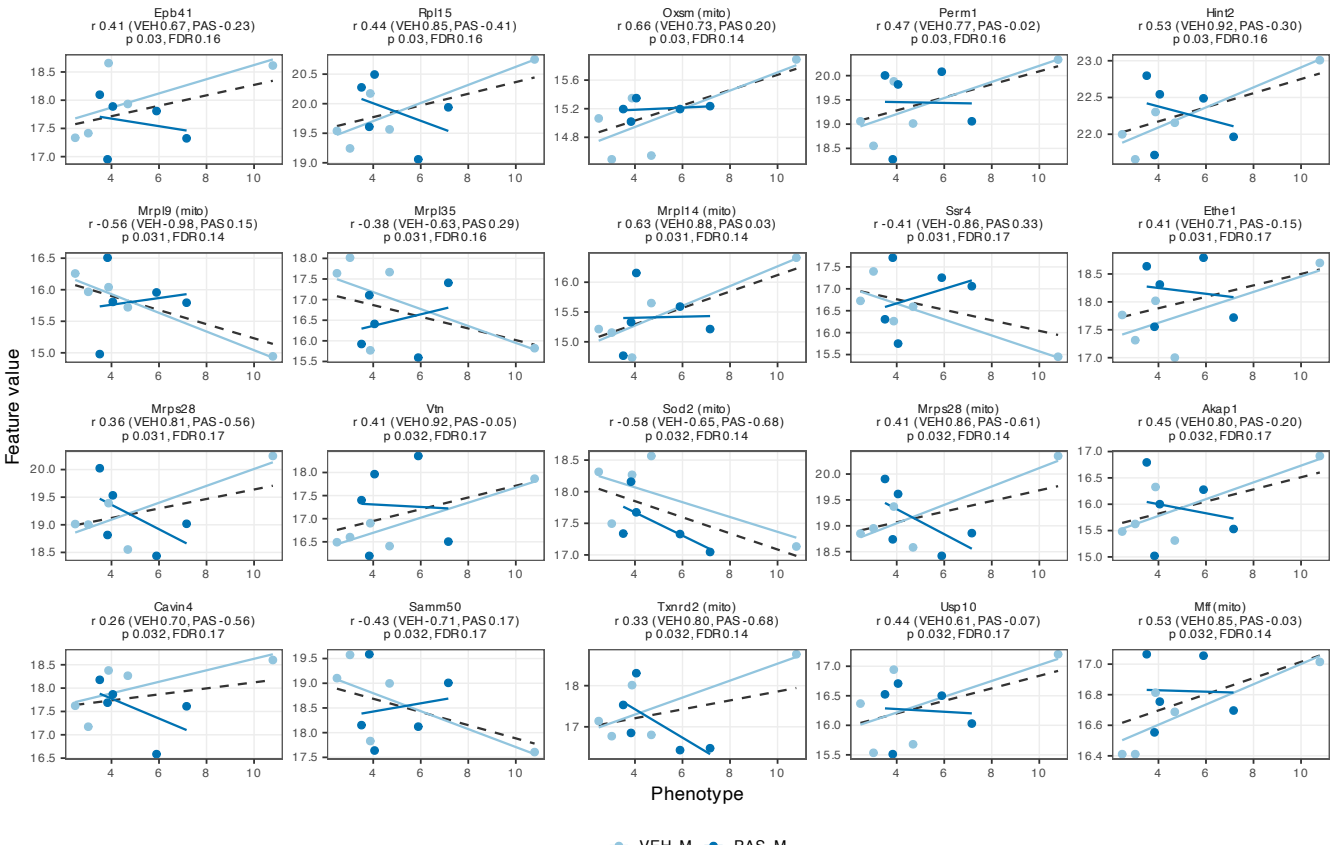
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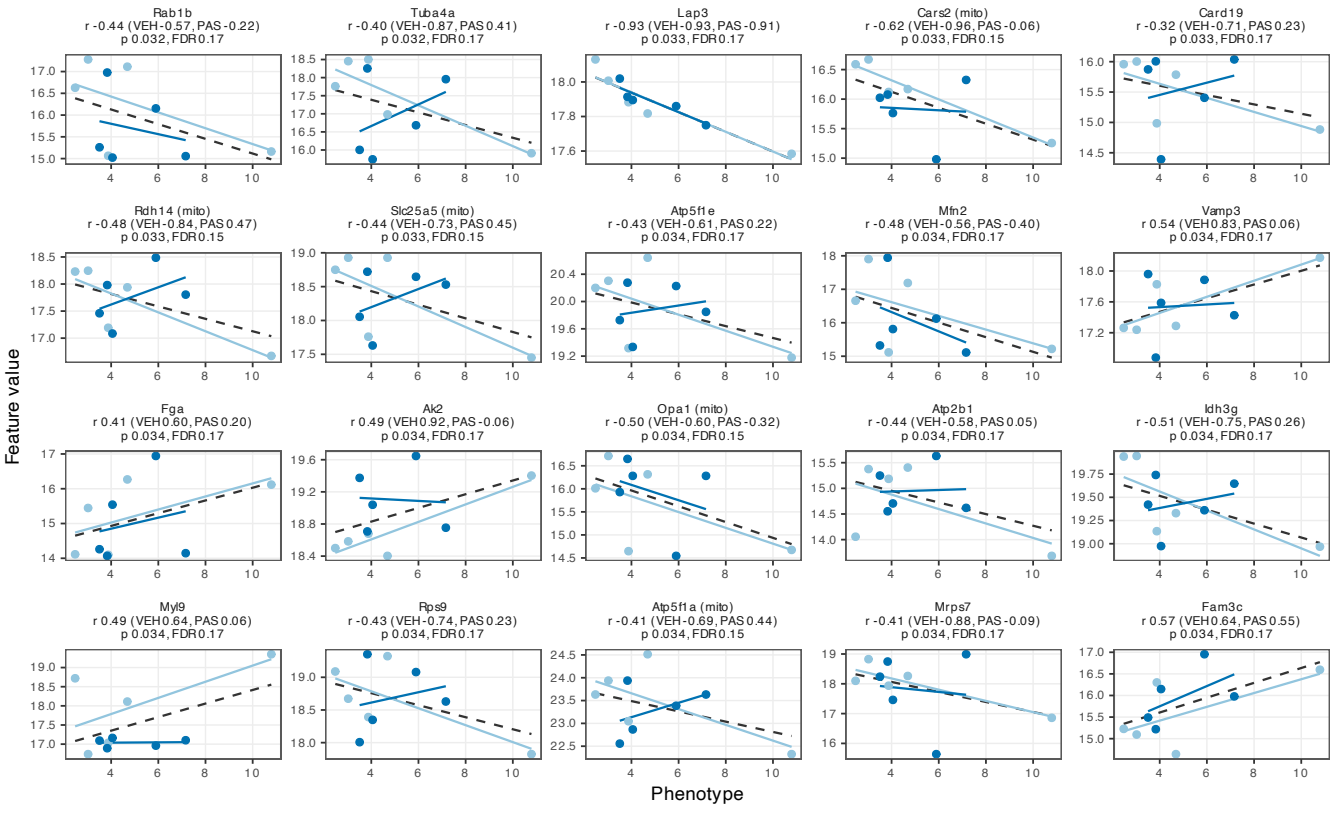
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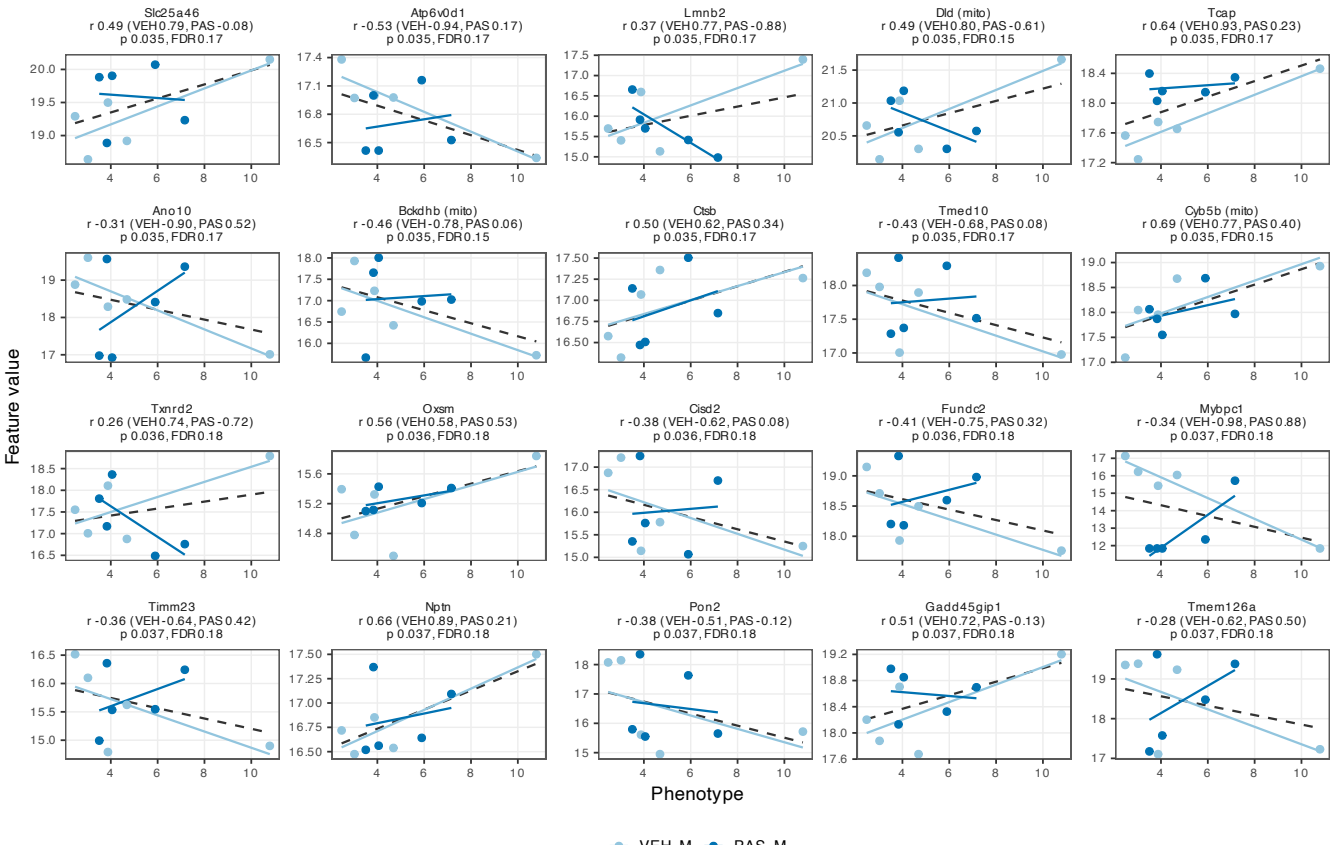
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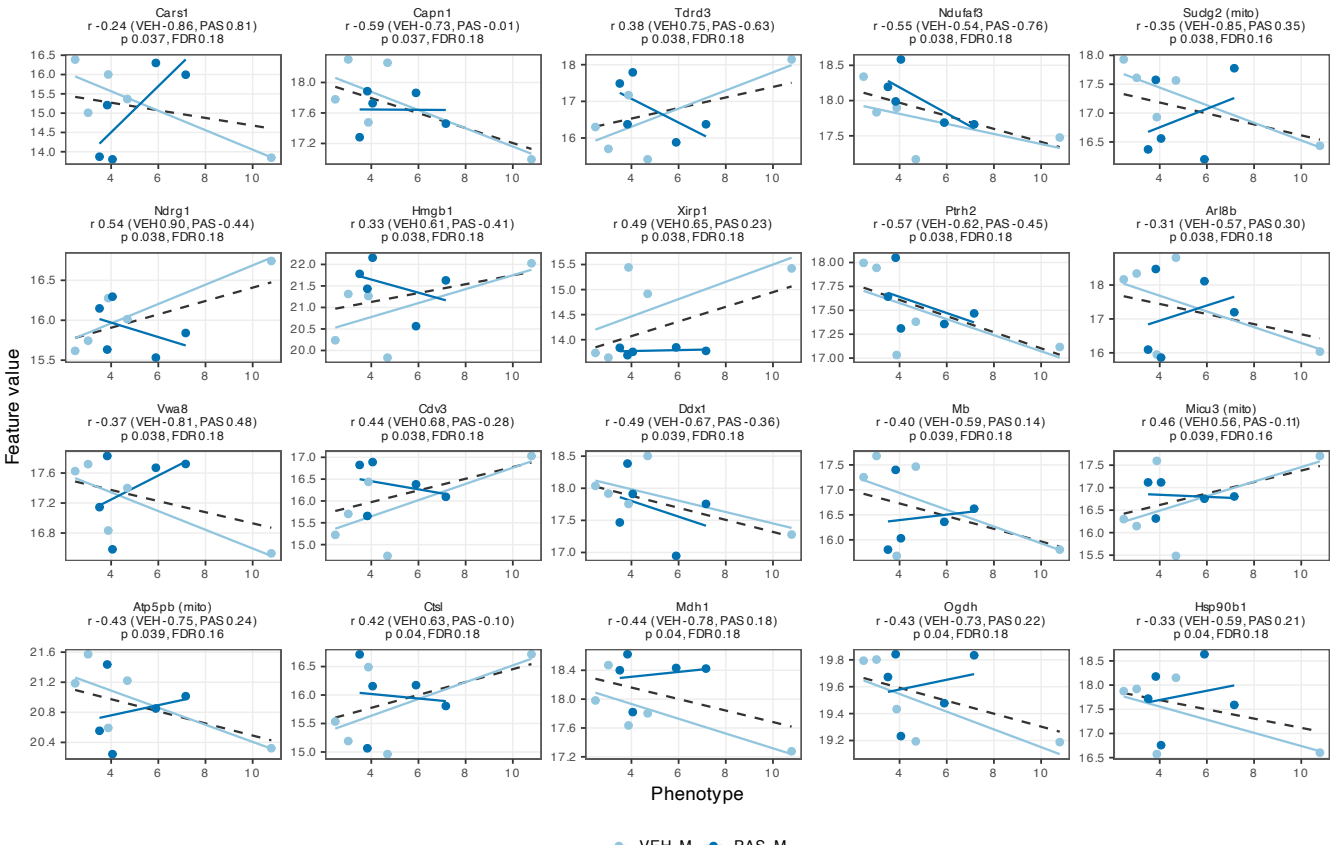
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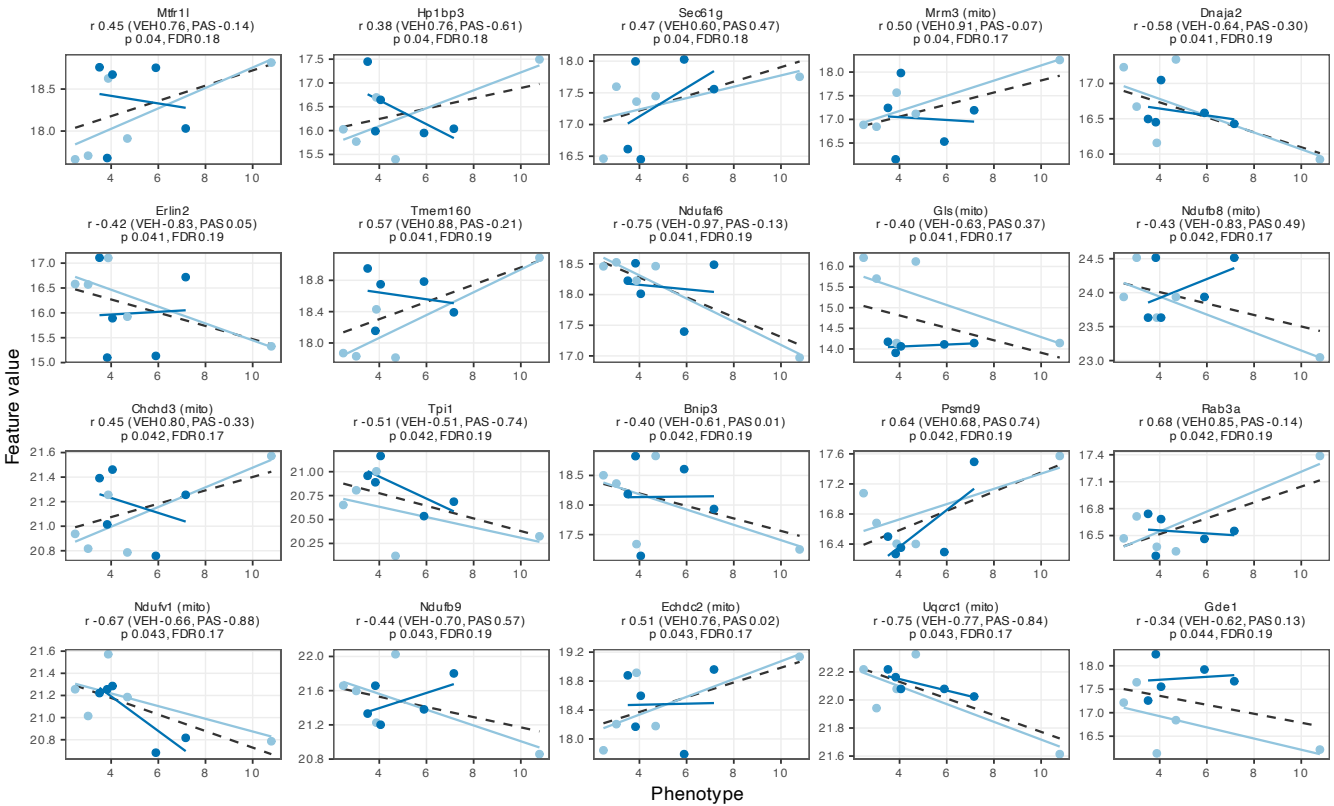
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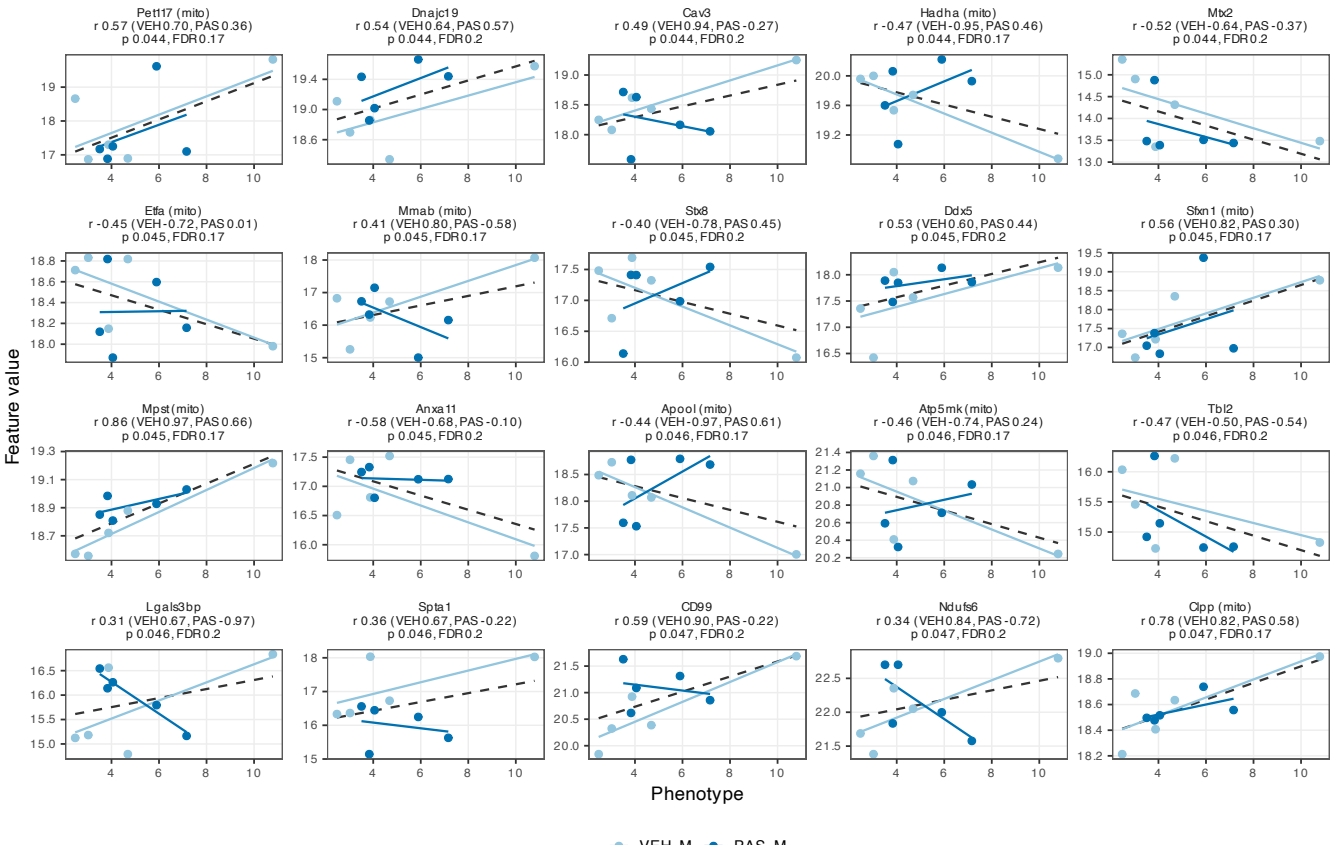
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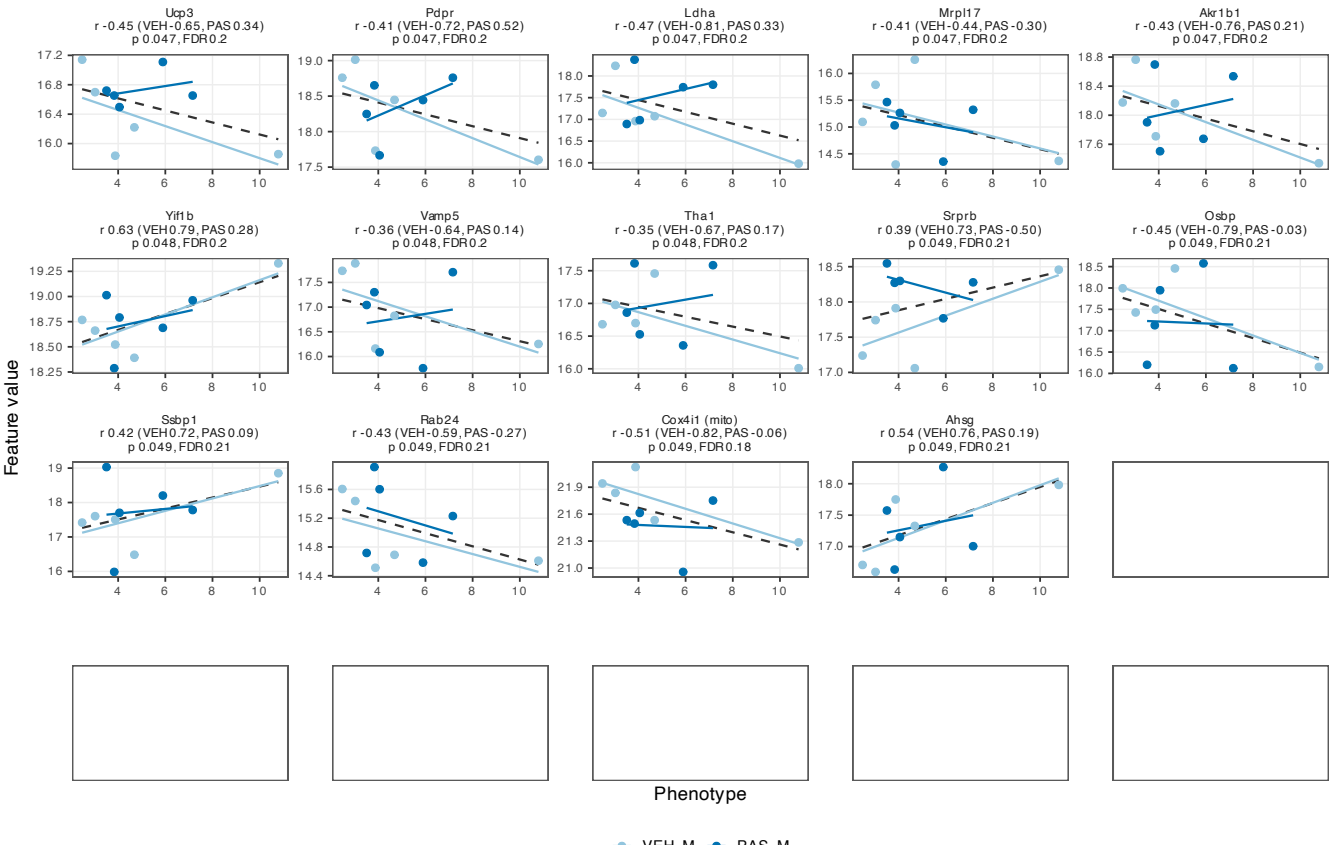
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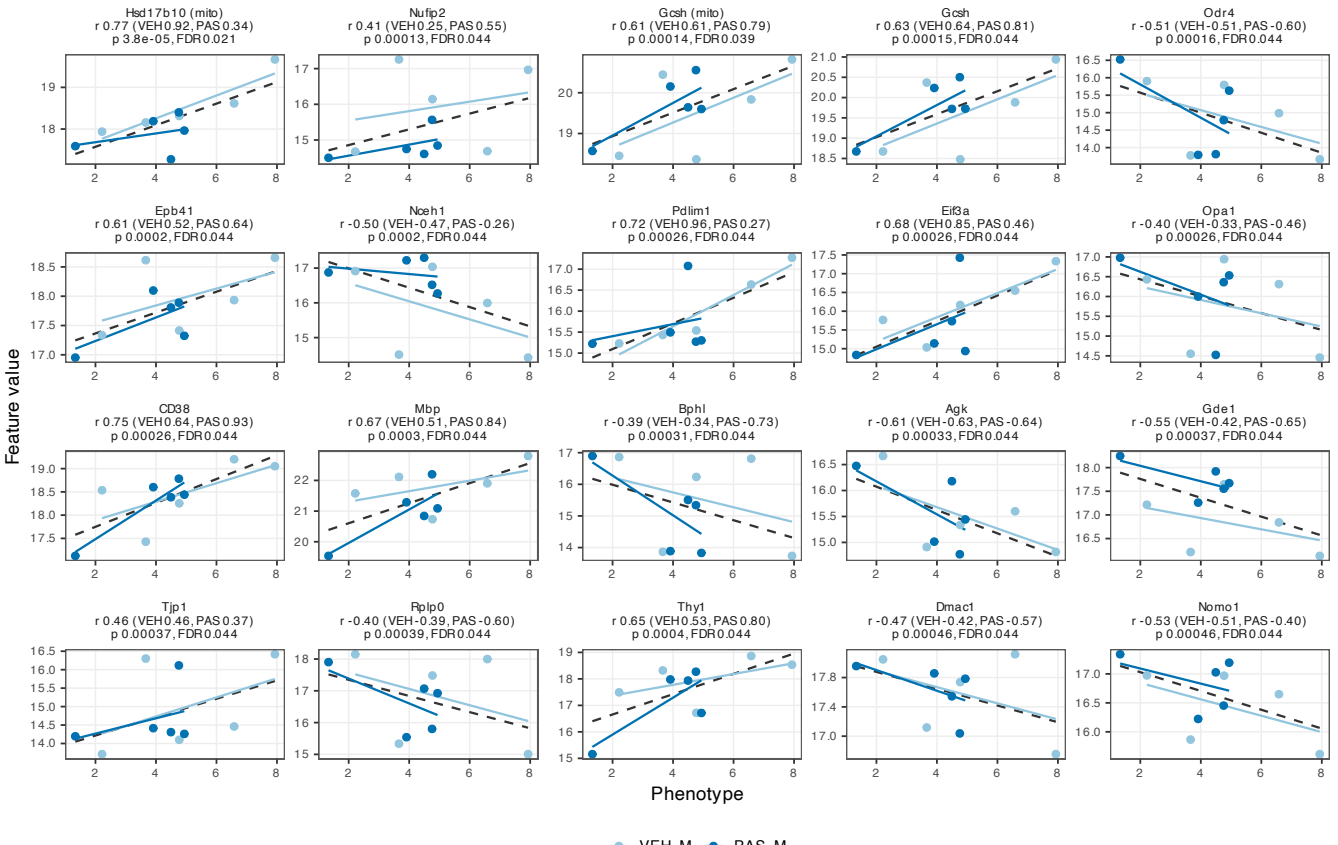
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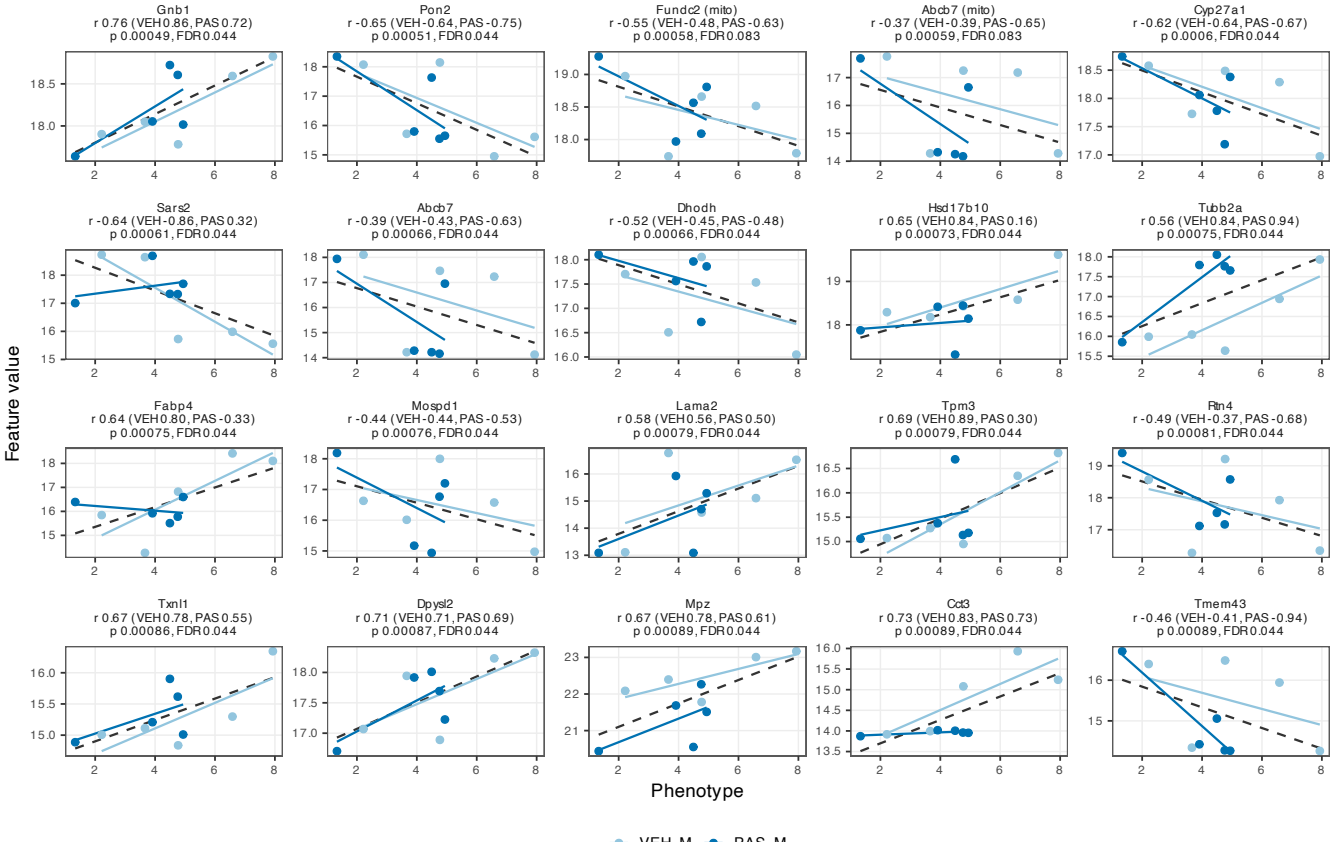
**Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



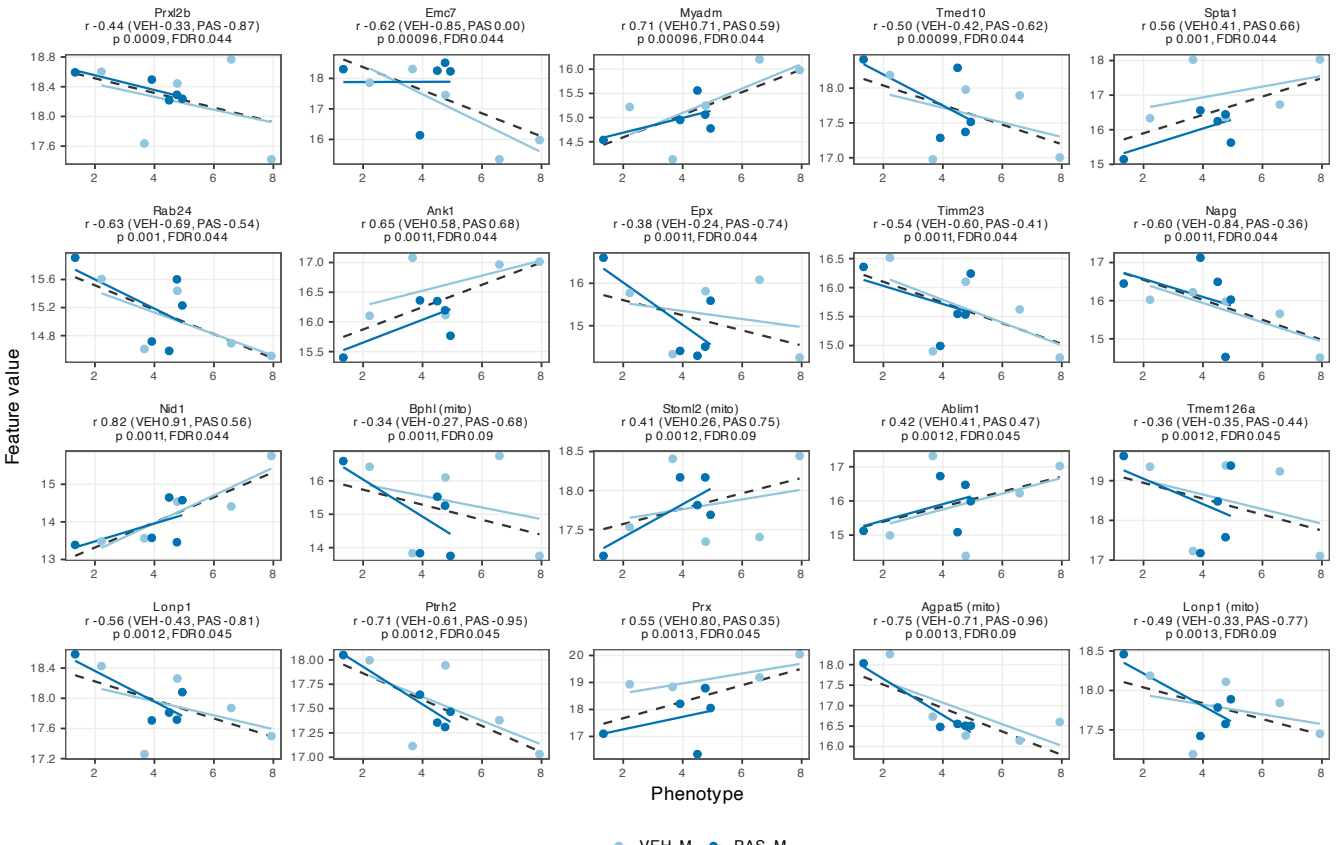
**Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)**

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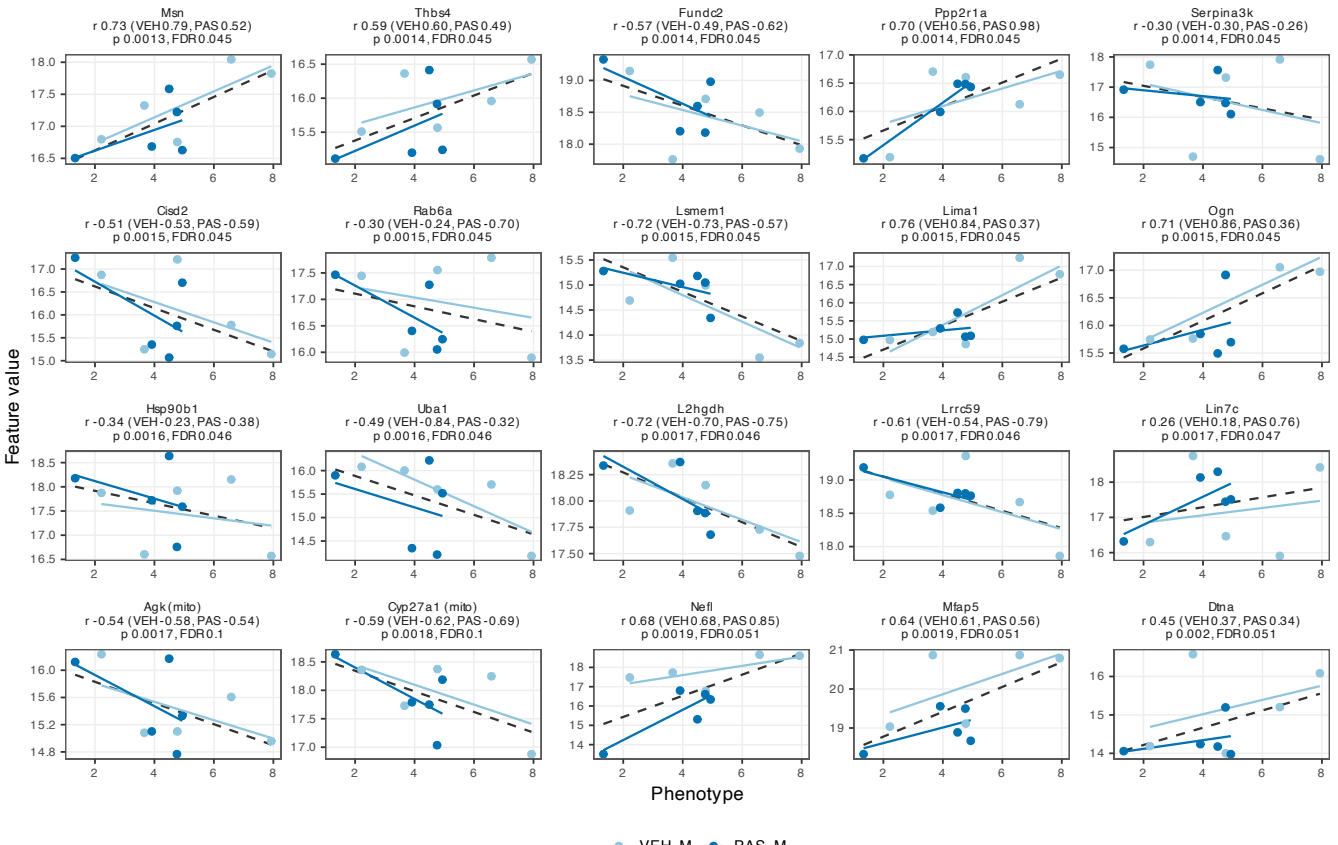
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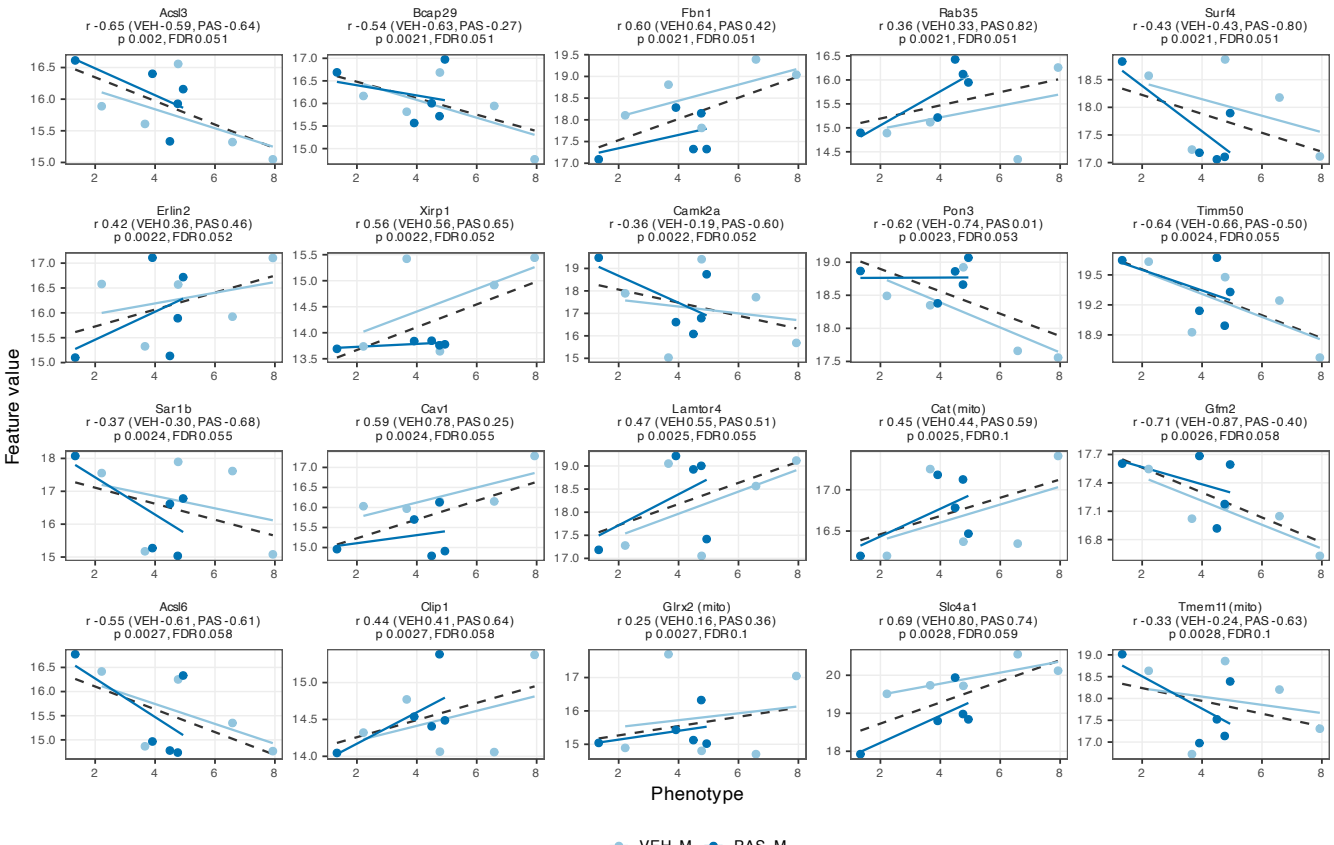
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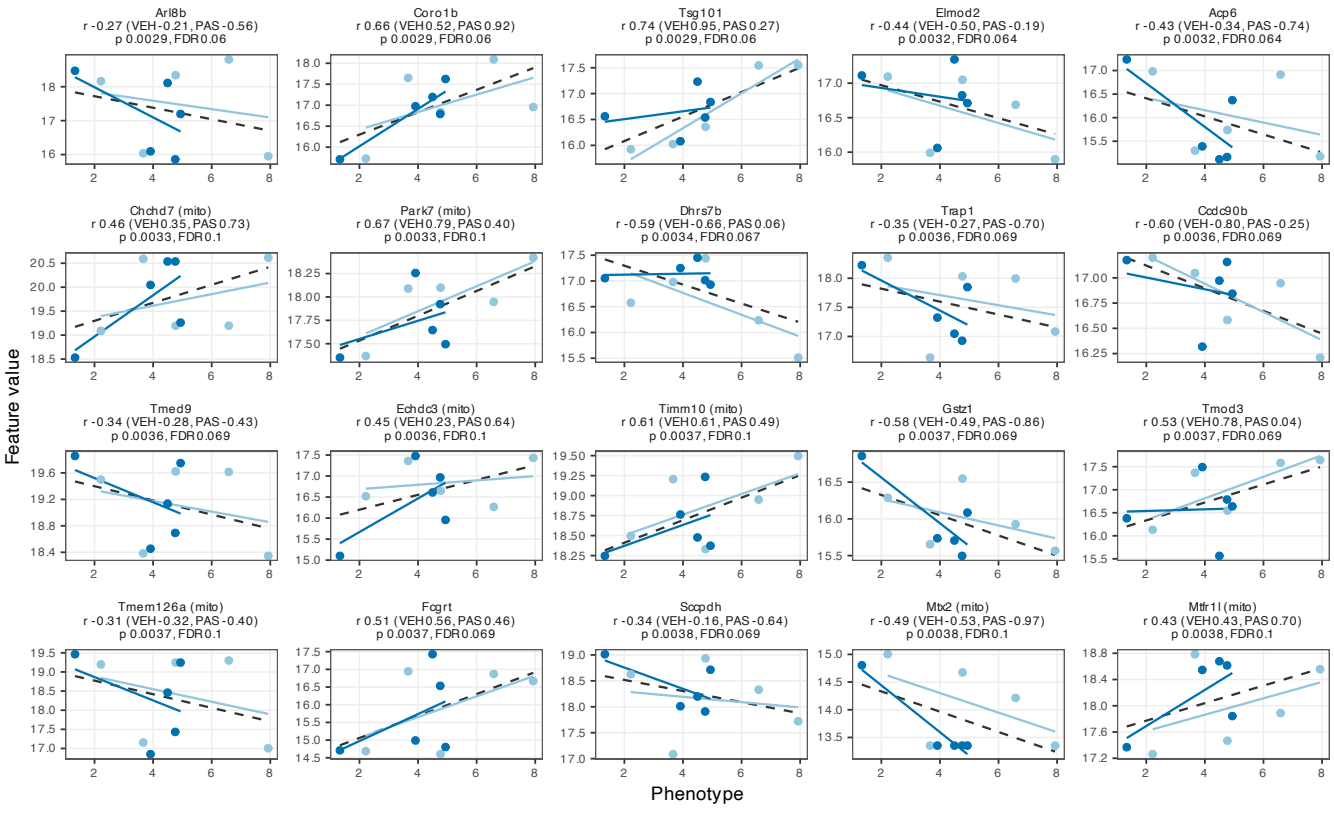
**Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)**

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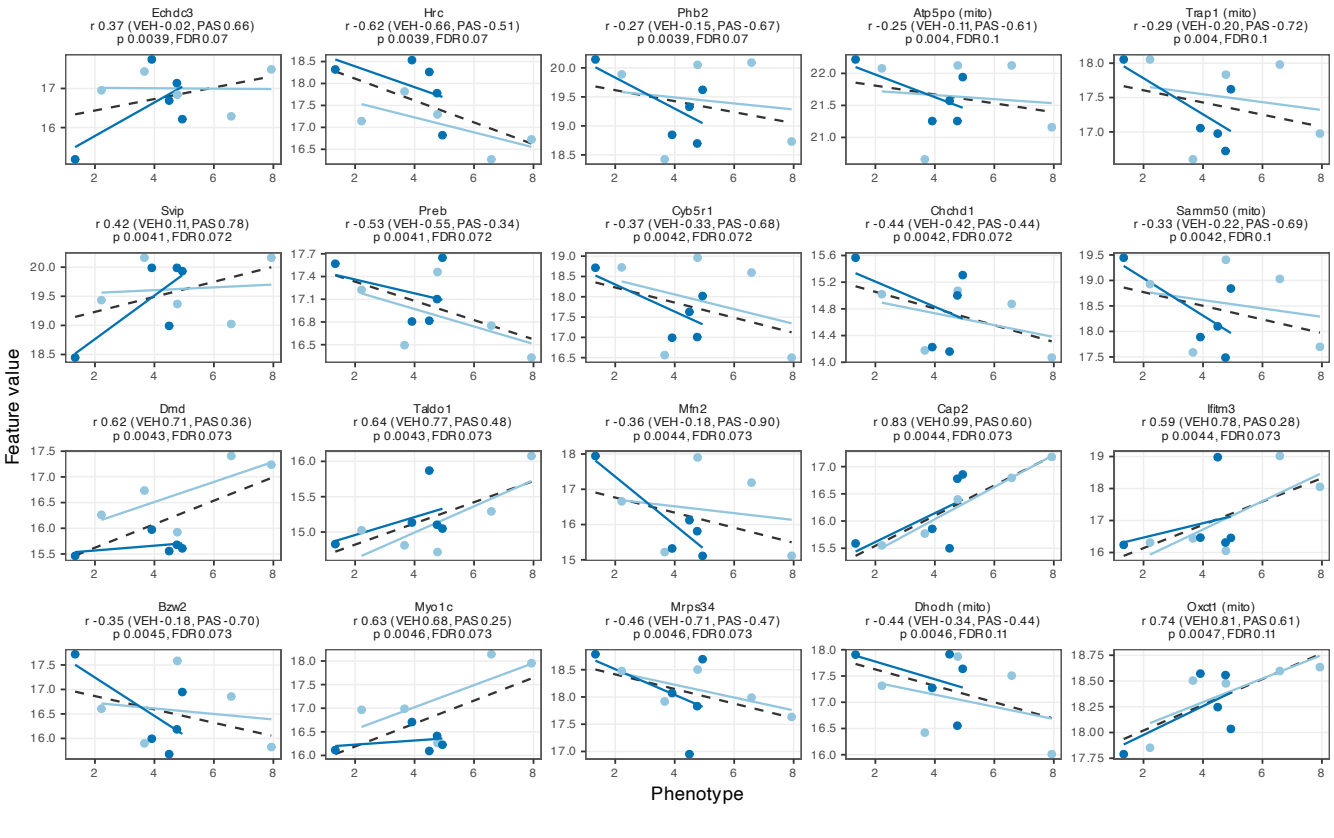
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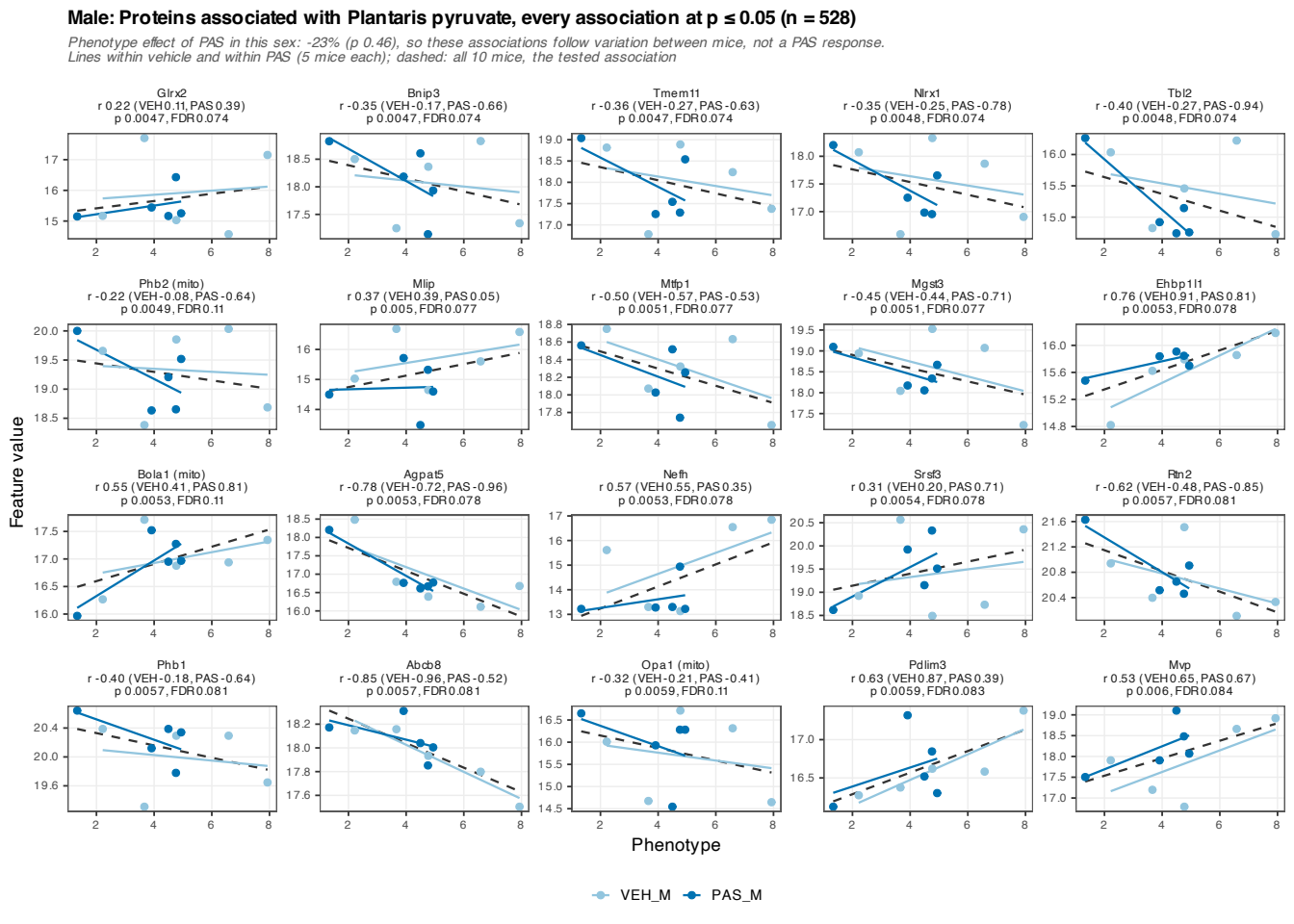
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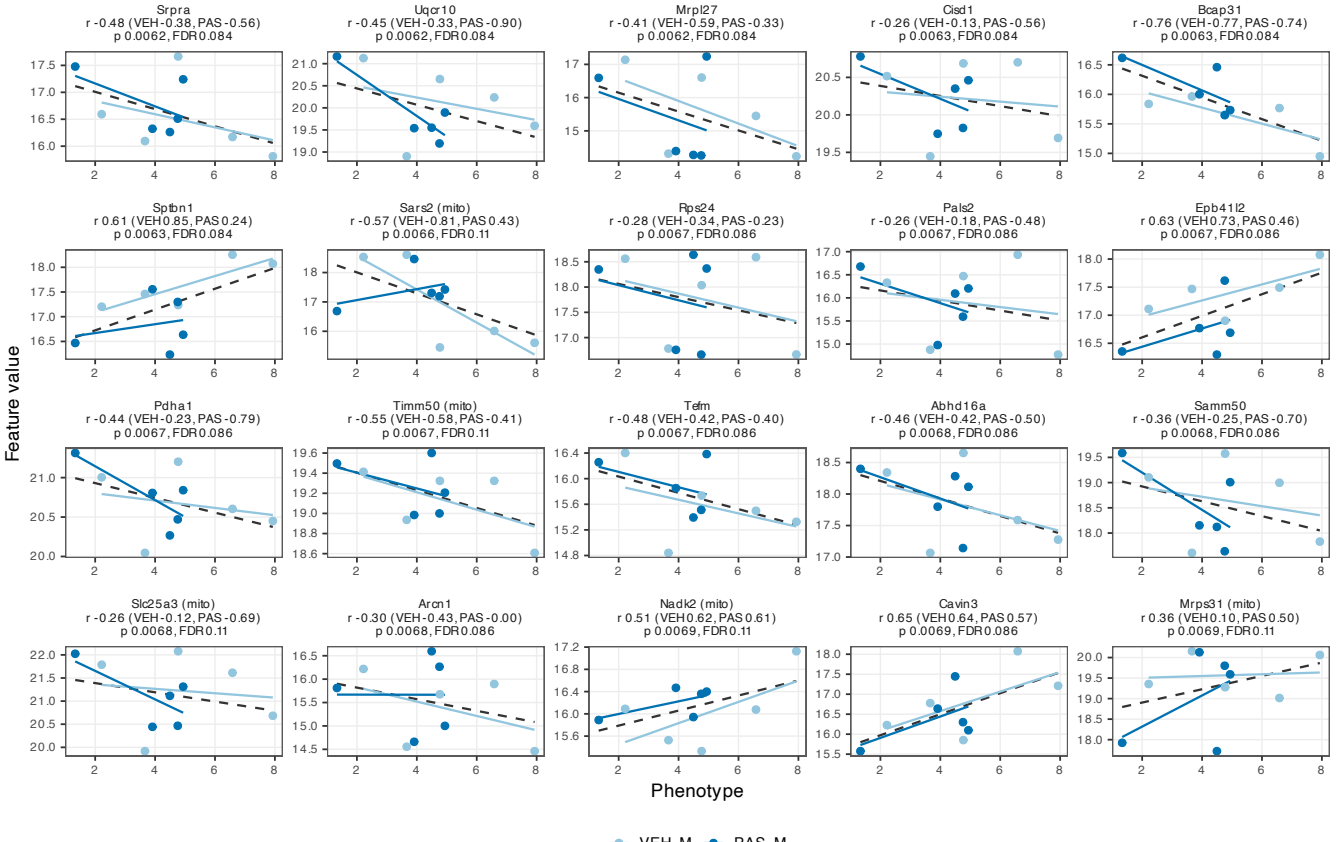
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Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)
Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association
Glrx2
r 0.22 (VEH 0.11, PAS 0.39)
p 0.0047, FDR 0.074
Bnip3
r -0.35 (VEH -0.17, PAS -0.66)
p 0.0047, FDR 0.074
Tmem11
r -0.36 (VEH -0.27, PAS -0.63)
p 0.0047, FDR 0.074
Nlrx1
r -0.35 (VEH -0.25, PAS -0.78)
p 0.0048, FDR 0.074
Tbl2
r -0.40 (VEH -0.27, PAS -0.94)
p 0.0048, FDR 0.074
Phb2 (mito)
r -0.22 (VEH -0.08, PAS -0.64)
p 0.0049, FDR 0.11
Mlip
r 0.37 (VEH 0.39, PAS 0.05)
p 0.005, FDR 0.077
Mtfp1
r -0.50 (VEH -0.57, PAS -0.53)
p 0.0051, FDR 0.077
Mgst3
r -0.45 (VEH -0.44, PAS -0.71)
p 0.0051, FDR 0.077
Ehbp1l1
r 0.76 (VEH 0.91, PAS 0.81)
p 0.0053, FDR 0.078
Bola1 (mito)
r 0.55 (VEH 0.41, PAS 0.81)
p 0.0053, FDR 0.11
Agpat5
r -0.78 (VEH -0.72, PAS -0.96)
p 0.0053, FDR 0.078
Nefh
r 0.57 (VEH 0.55, PAS 0.35)
p 0.0053, FDR 0.078
Srsf3
r 0.31 (VEH 0.20, PAS 0.71)
p 0.0054, FDR 0.078
Rtn2
r -0.62 (VEH -0.48, PAS -0.85)
p 0.0057, FDR 0.081
Phb1
r -0.40 (VEH -0.18, PAS -0.64)
p 0.0057, FDR 0.081
Abcb8
r -0.85 (VEH -0.96, PAS -0.52)
p 0.0057, FDR 0.081
Opa1 (mito)
r -0.32 (VEH -0.21, PAS -0.41)
p 0.0059, FDR 0.11
Pdlim3
r 0.63 (VEH 0.87, PAS 0.39)
p 0.0059, FDR 0.083
Mvp
r 0.53 (VEH 0.65, PAS 0.67)
p 0.006, FDR 0.084
Feature value
Phenotype
VEH_M
PAS_M

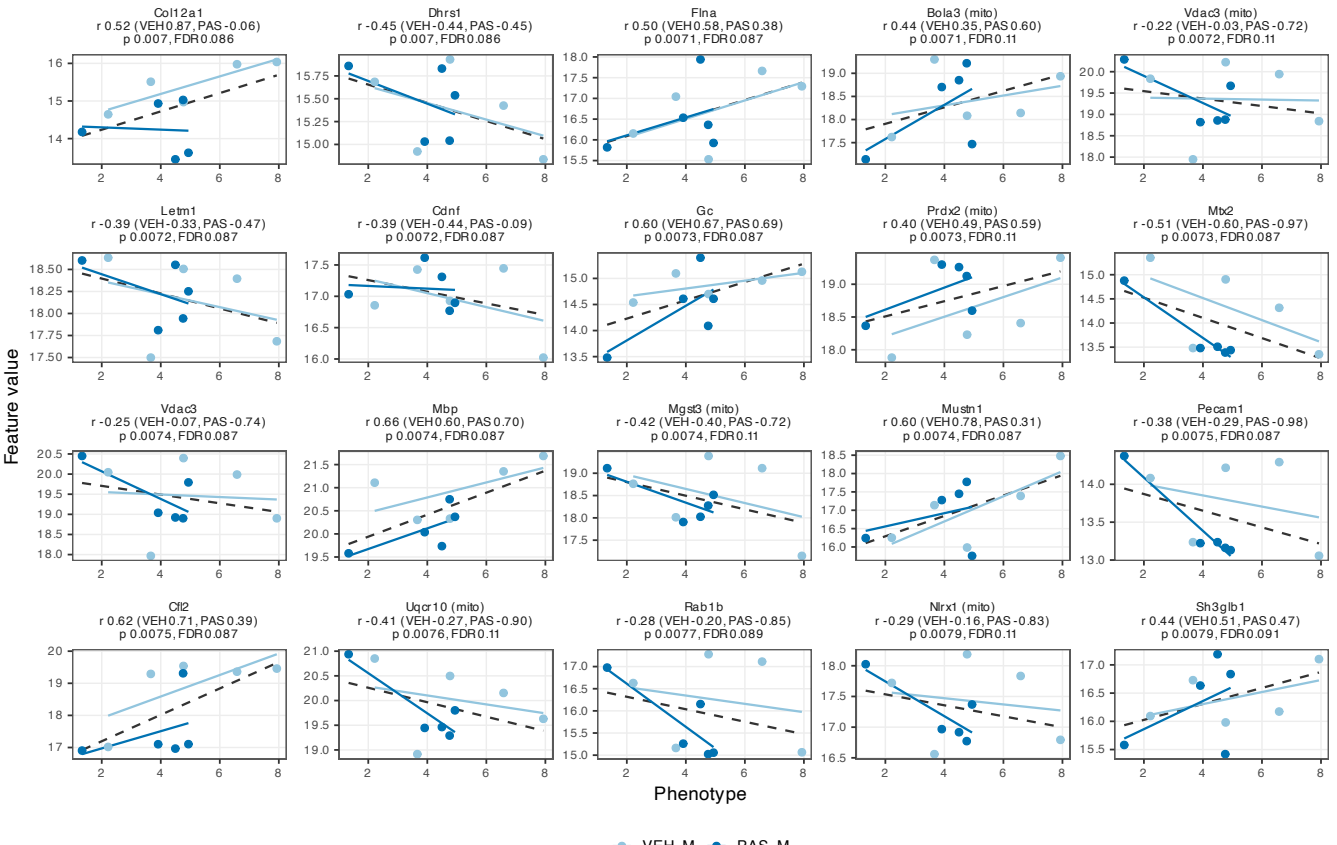
**Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



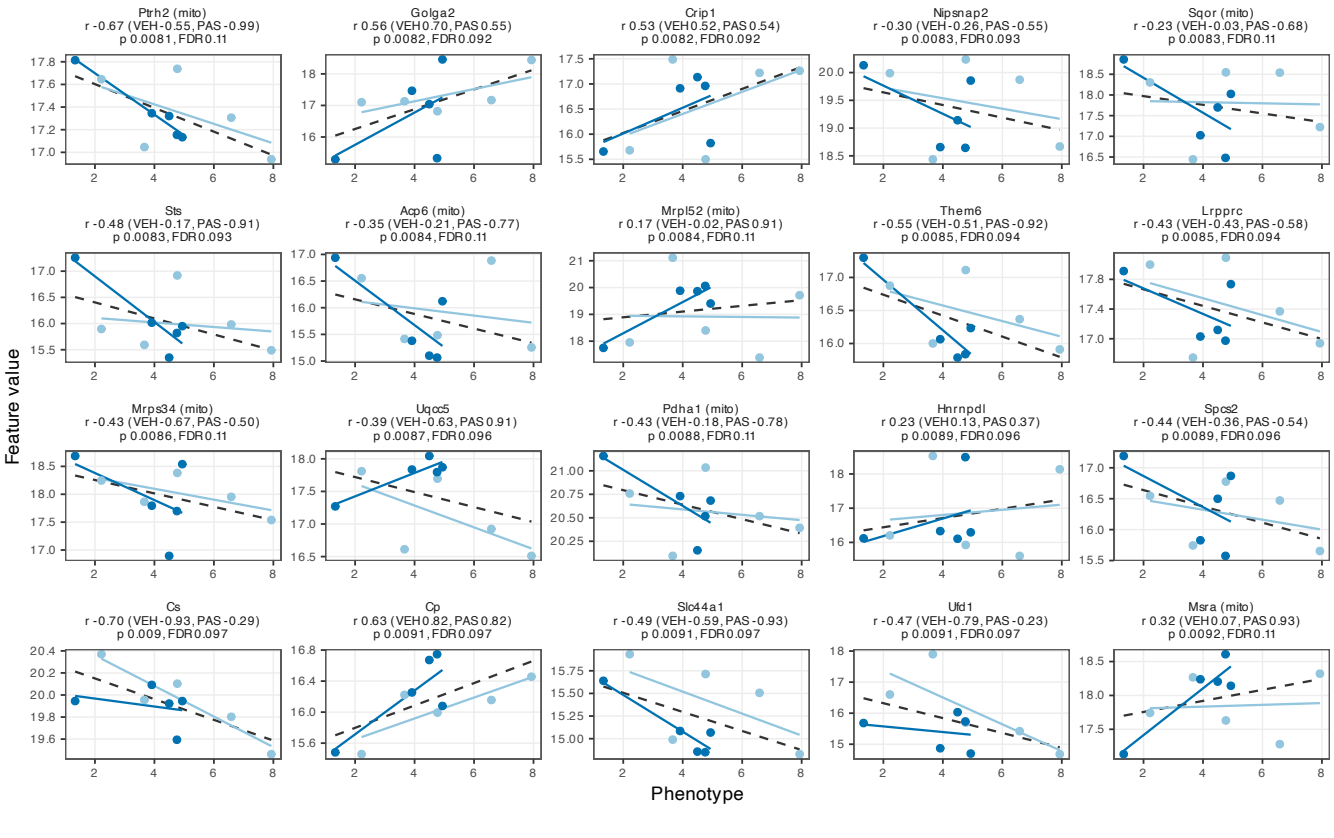
**Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



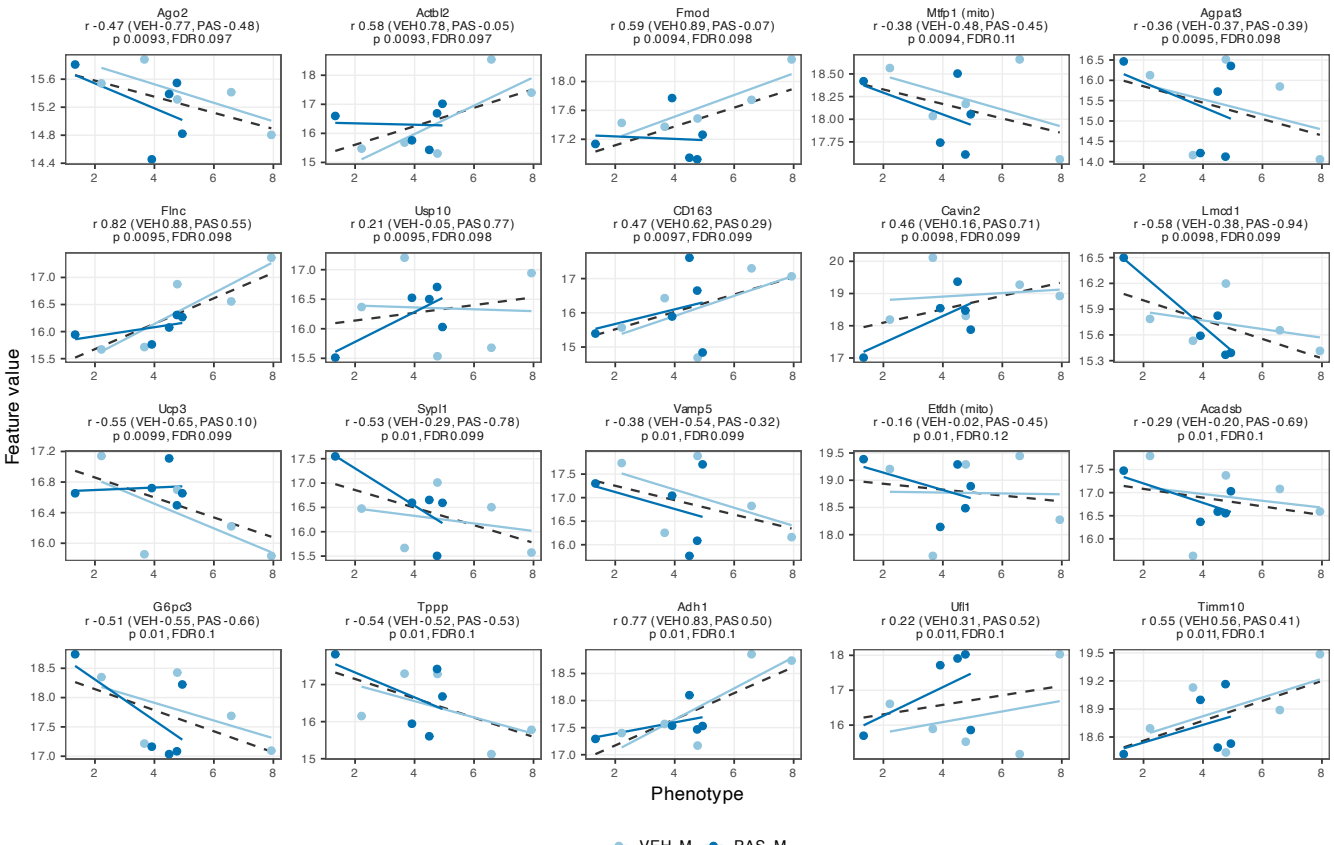
**Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



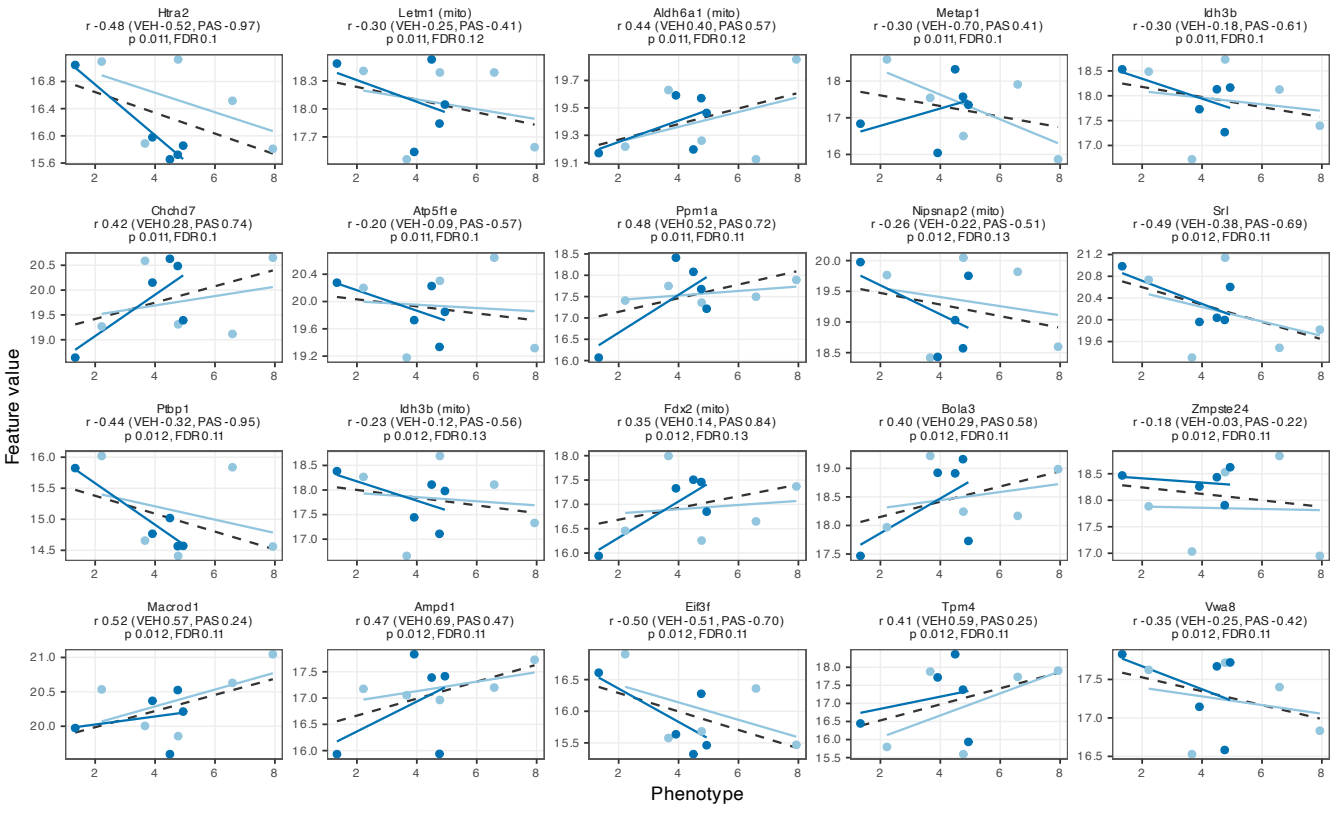
**Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



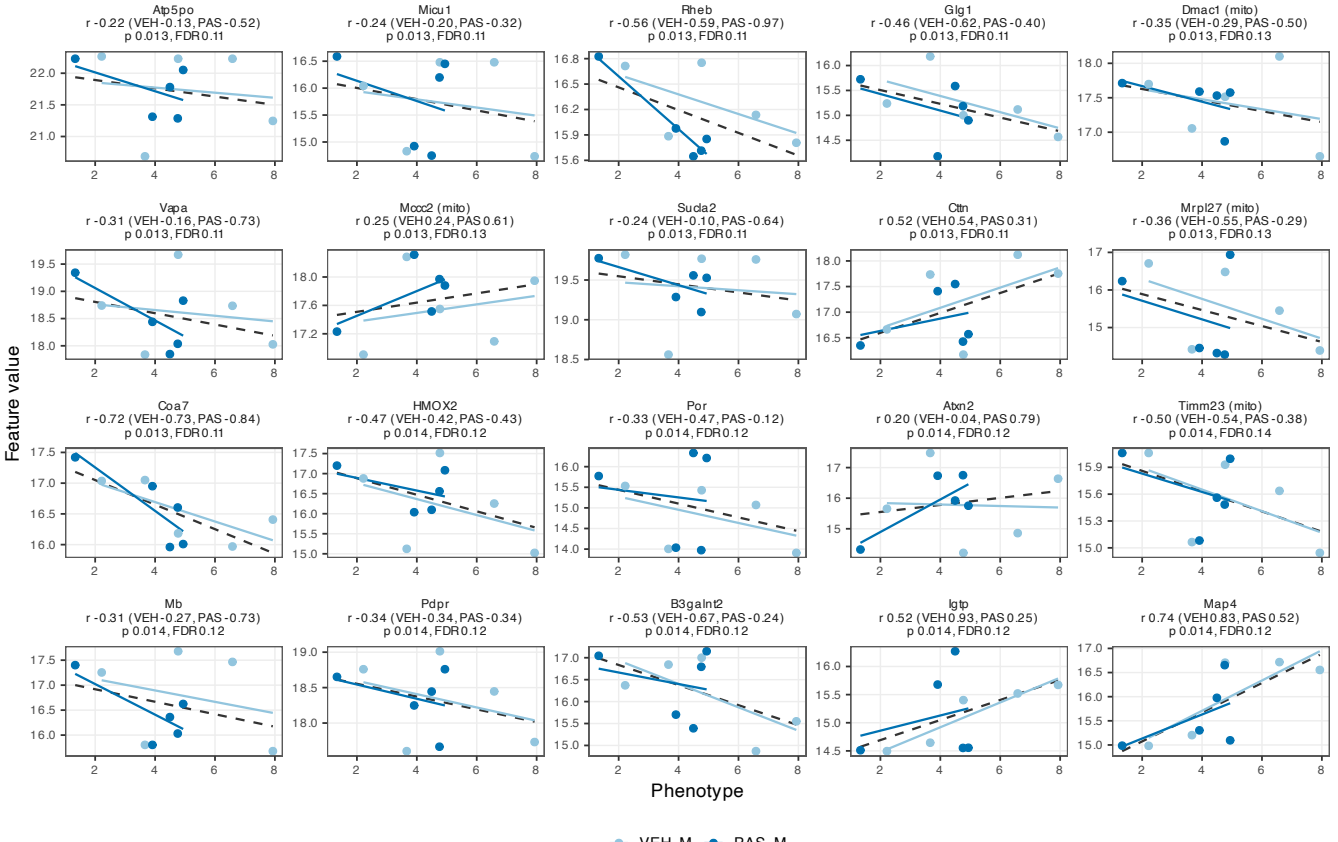
**Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



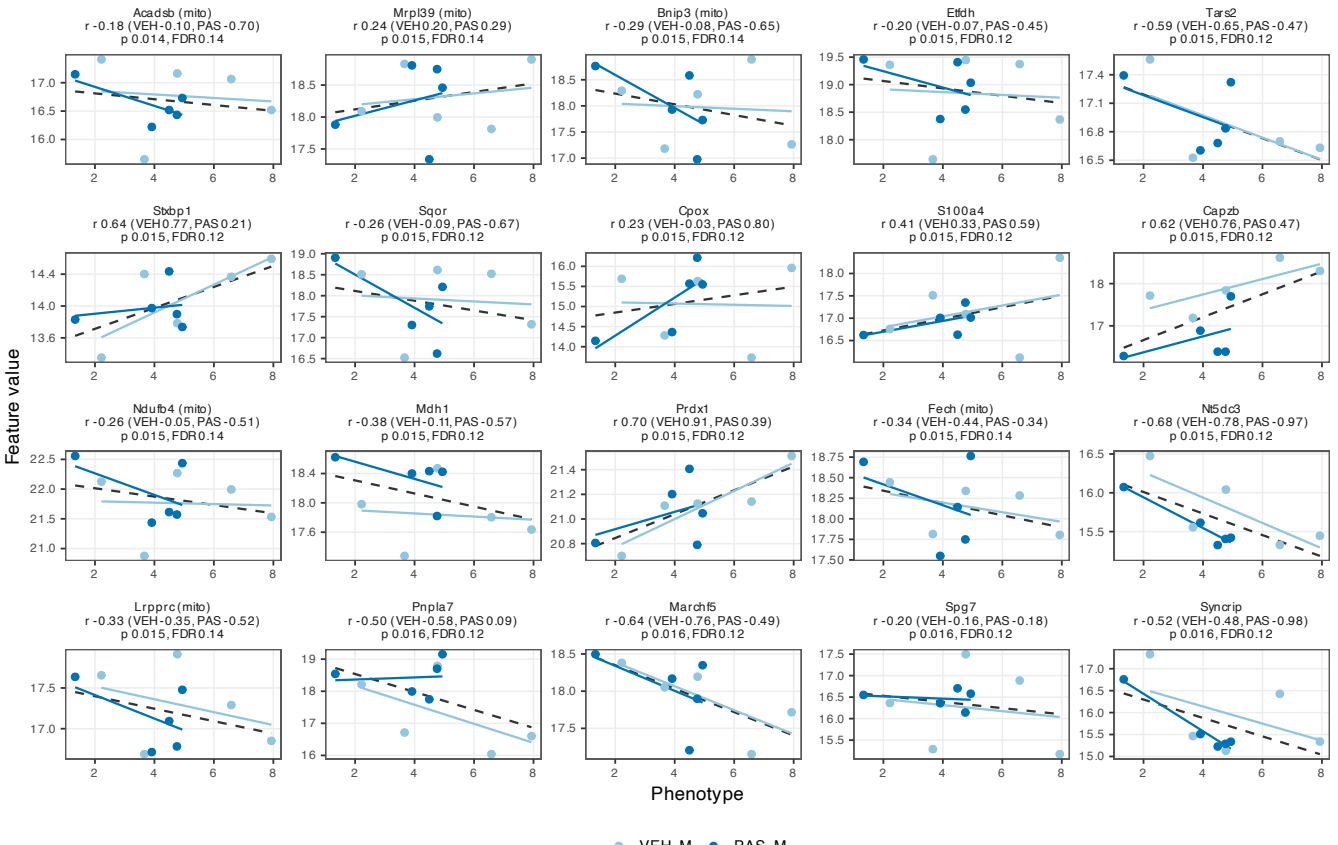
**Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



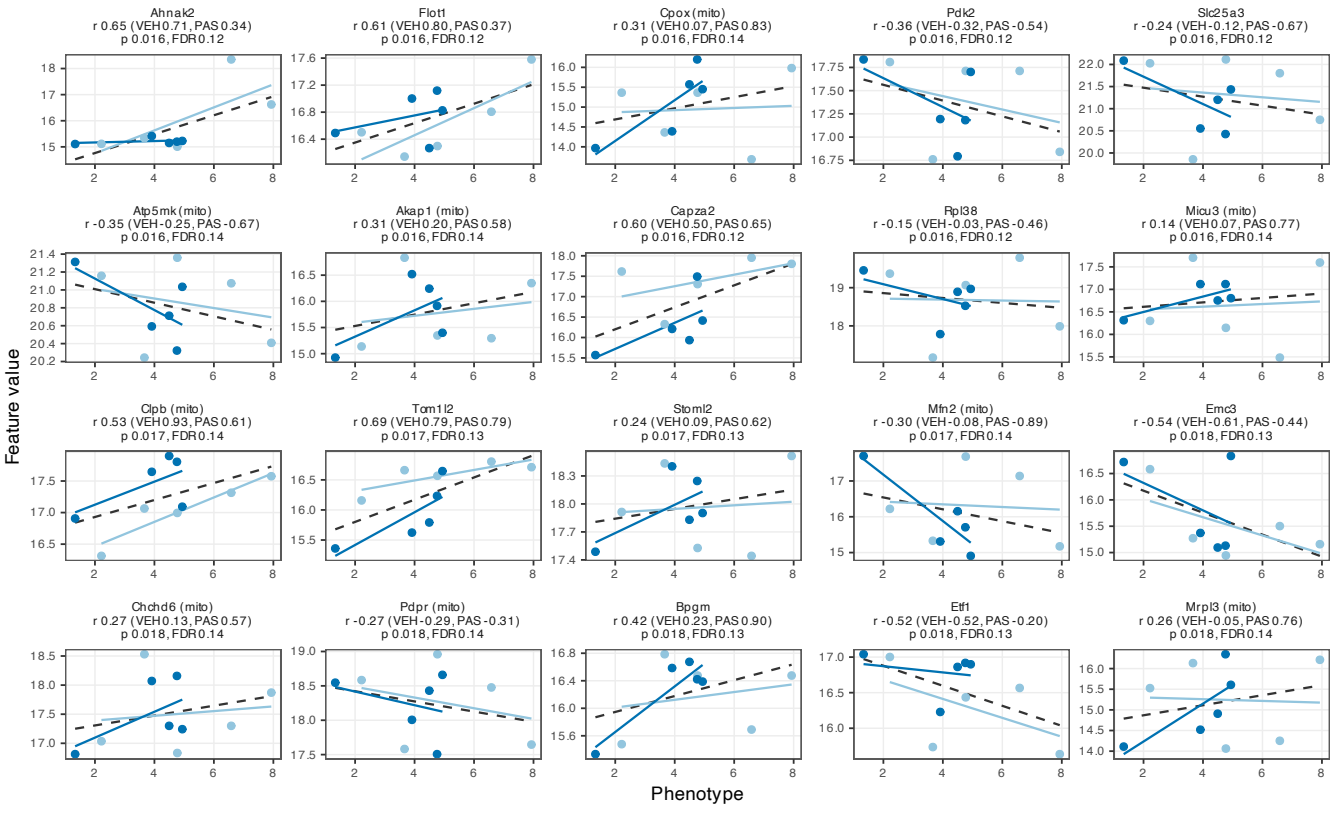
**Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



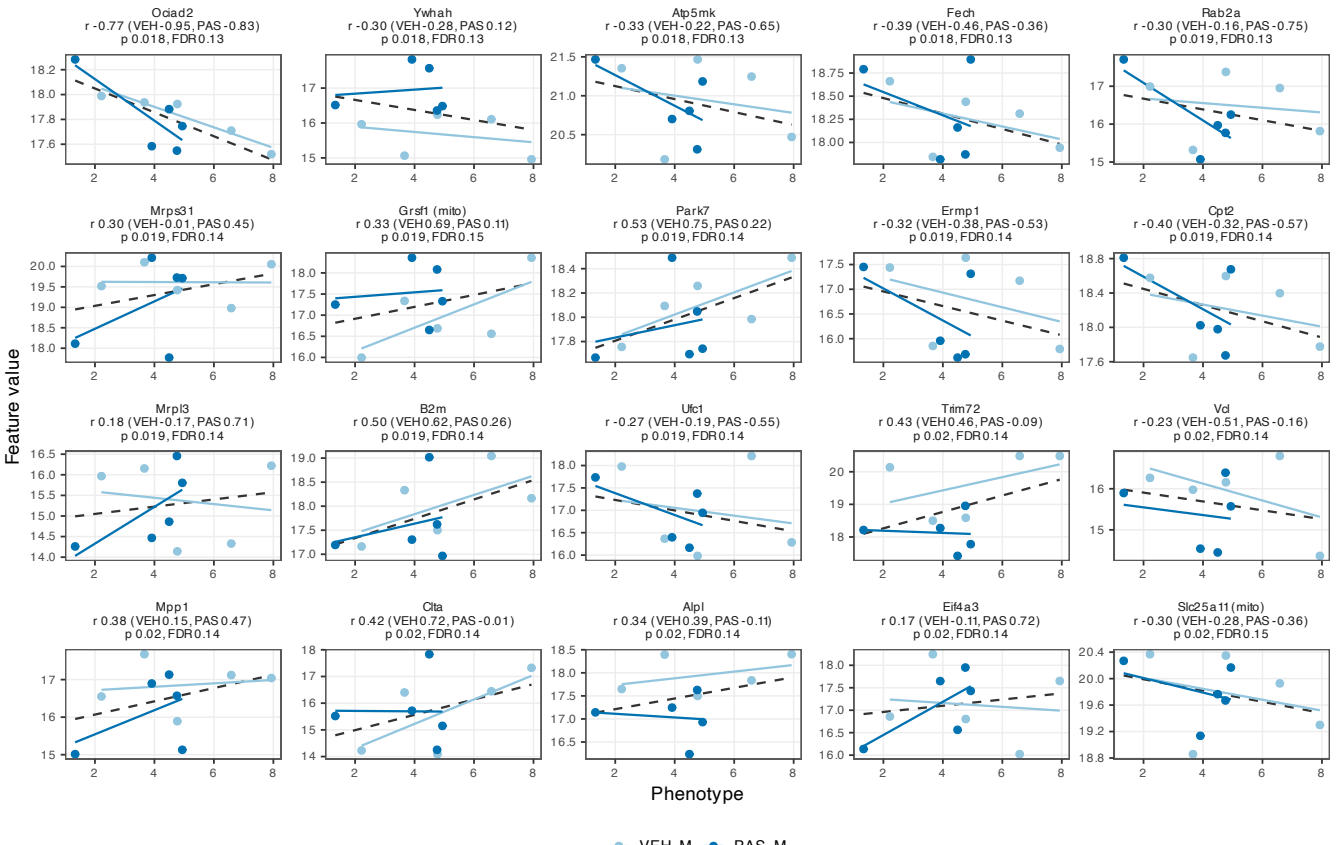
**Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



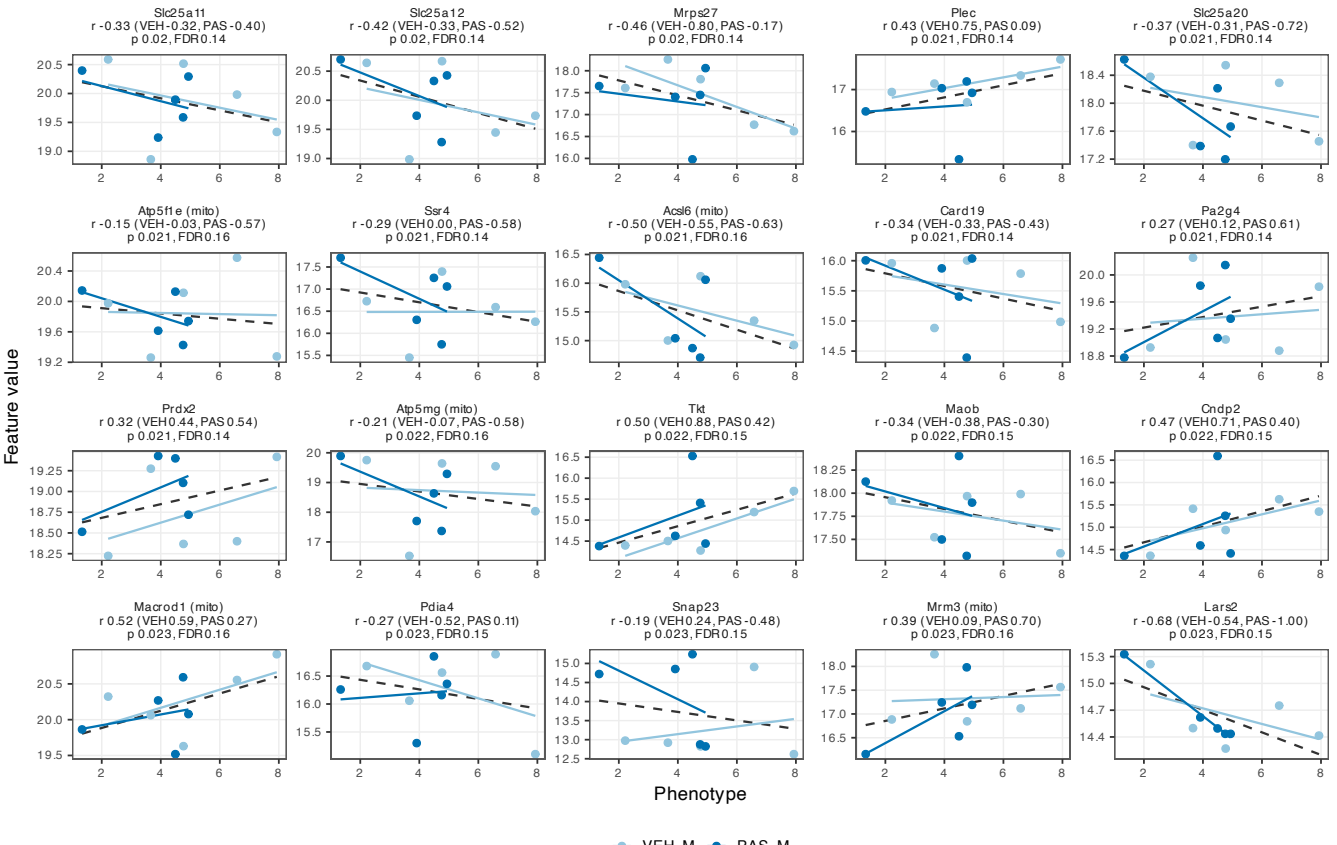
**Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



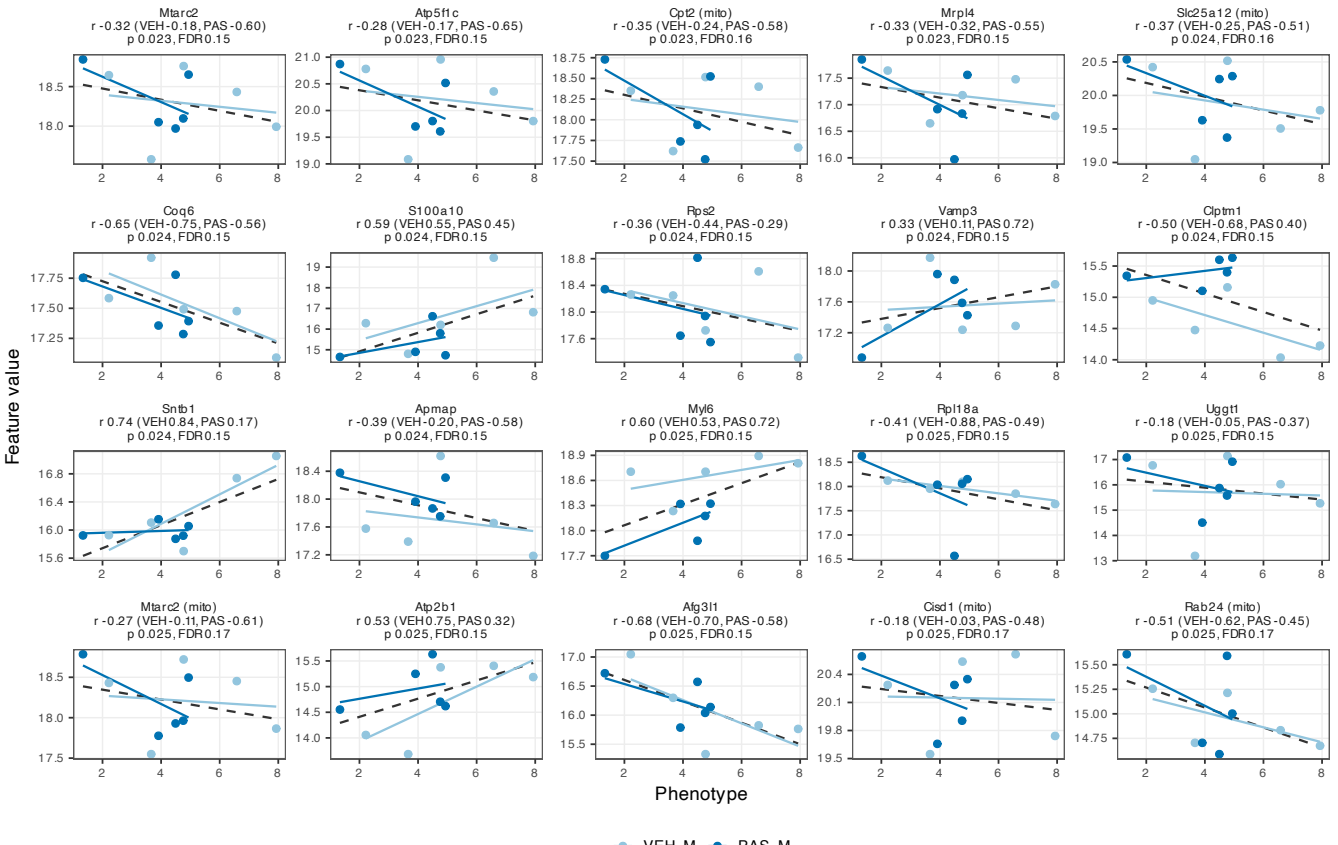
**Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



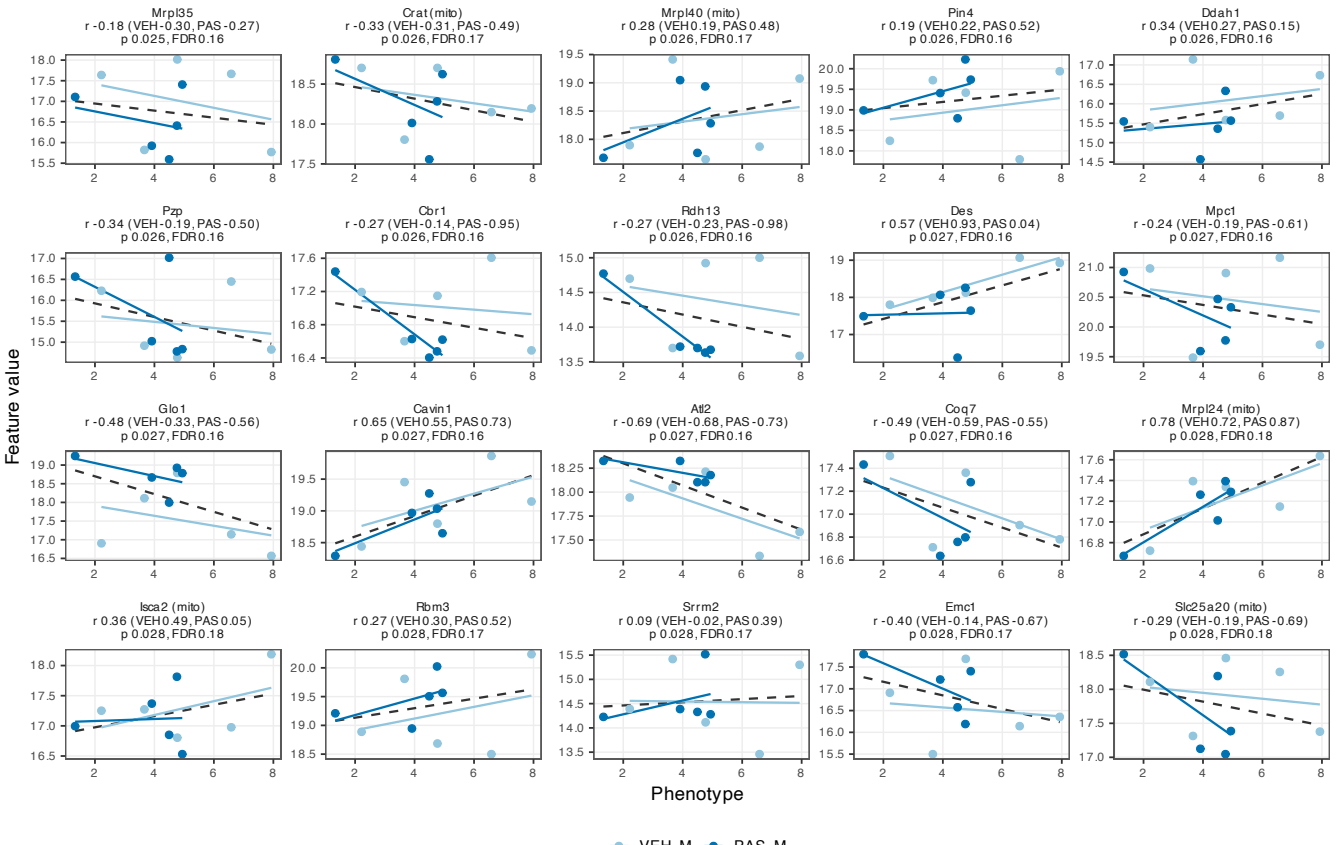
**Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



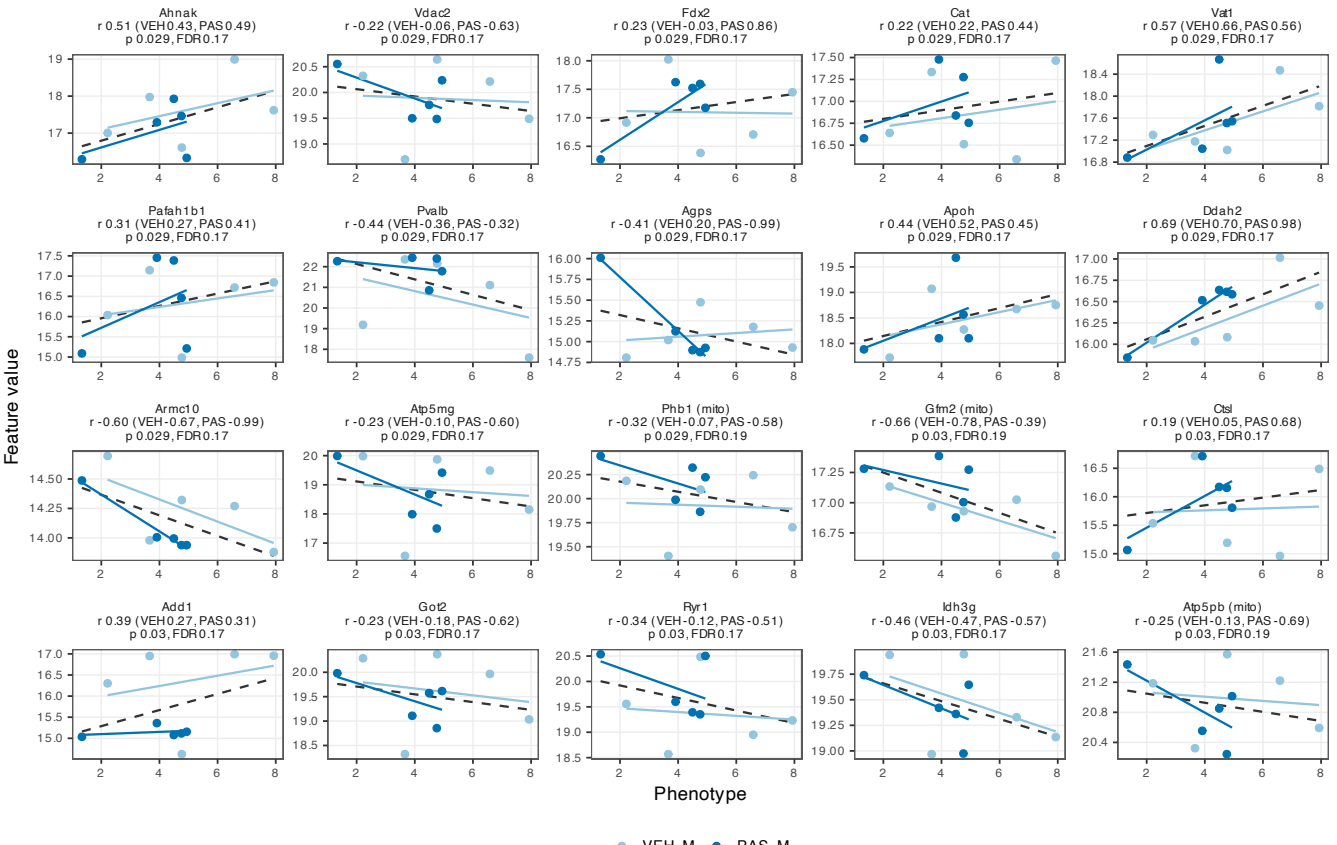
**Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



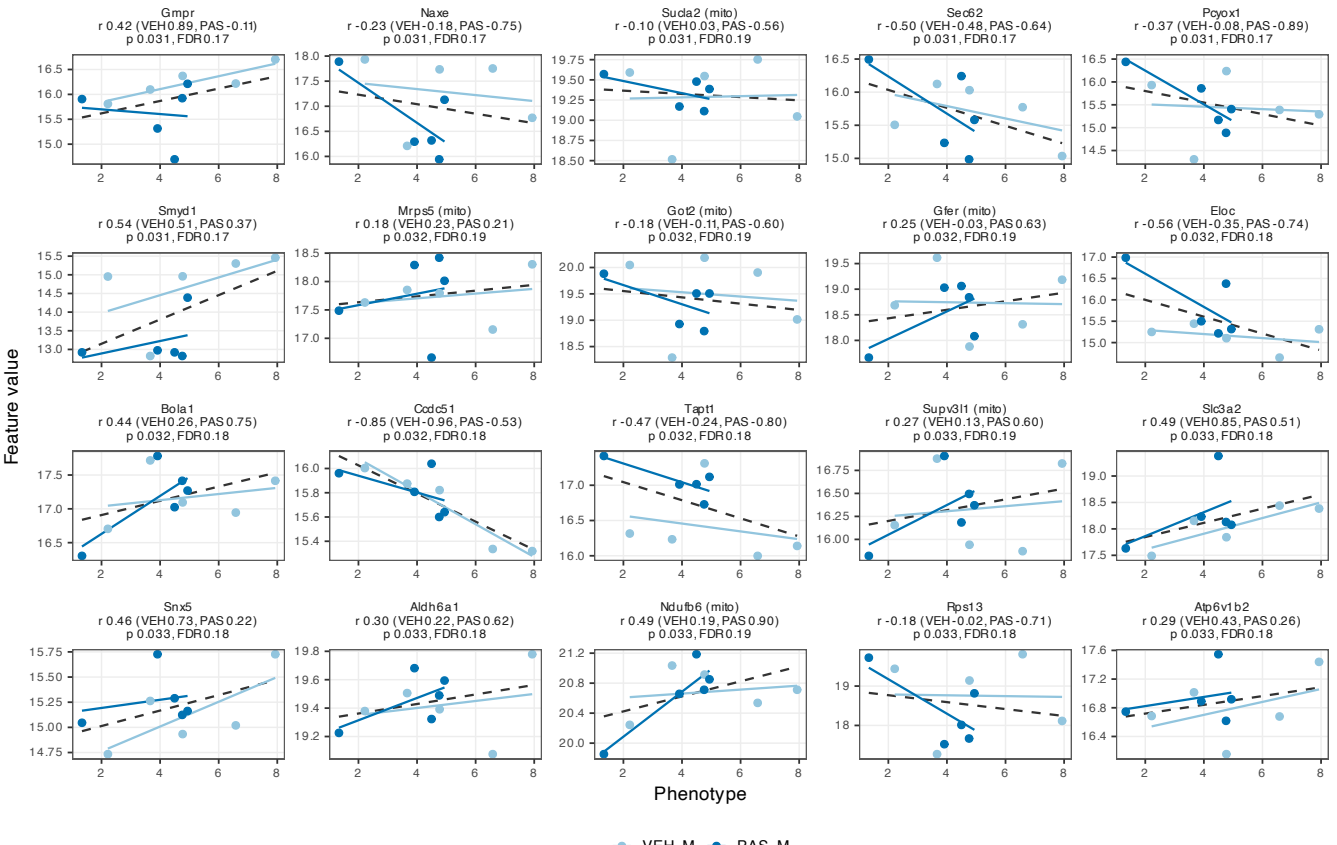
**Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



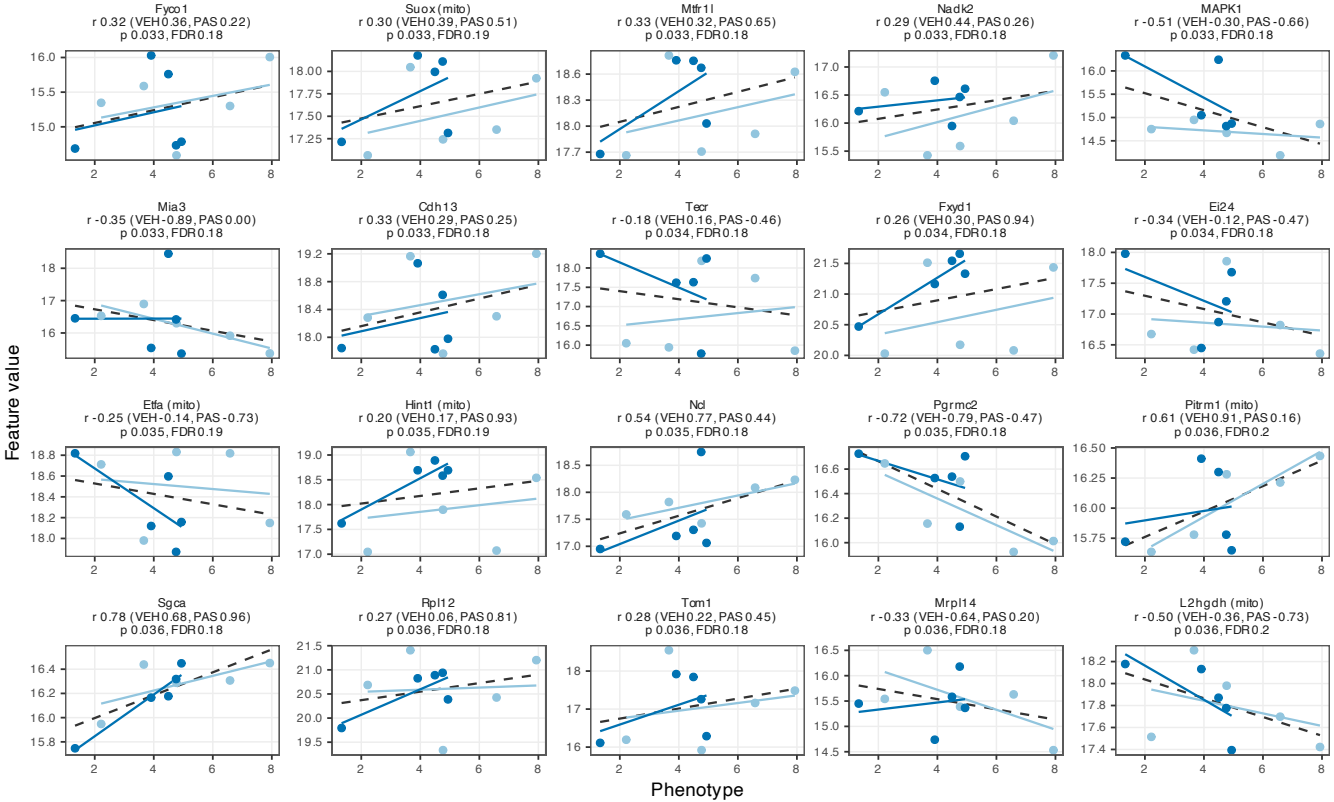
**Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



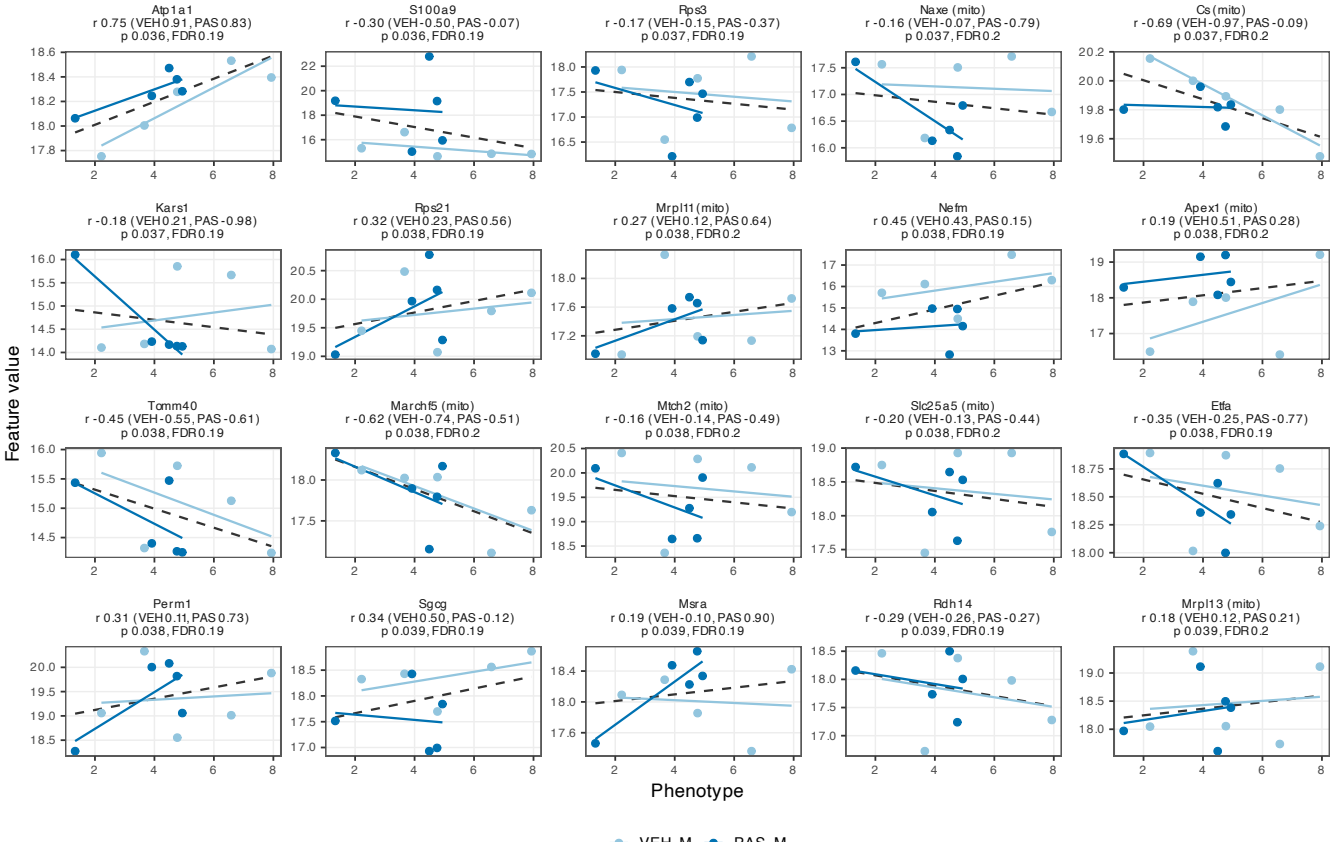
**Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



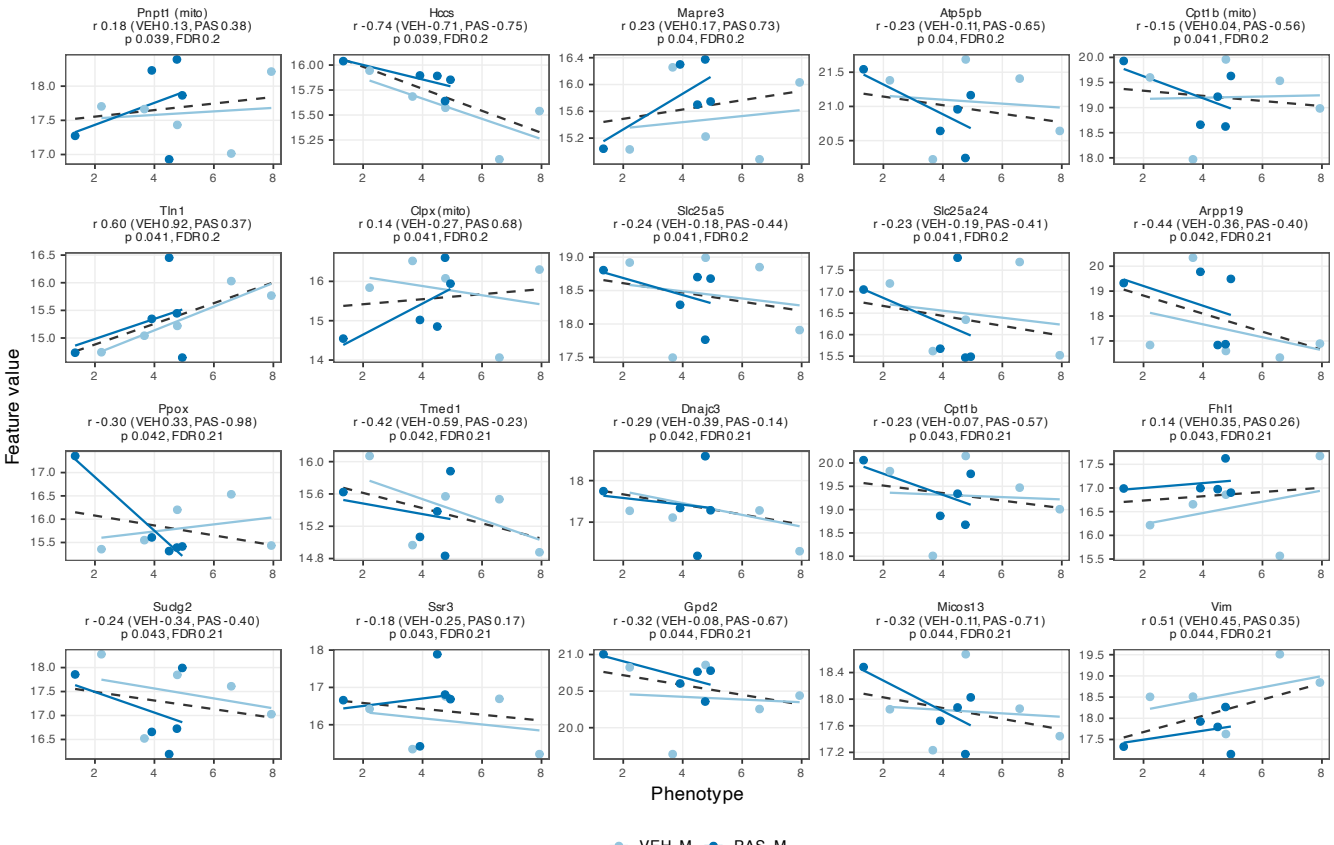
**Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



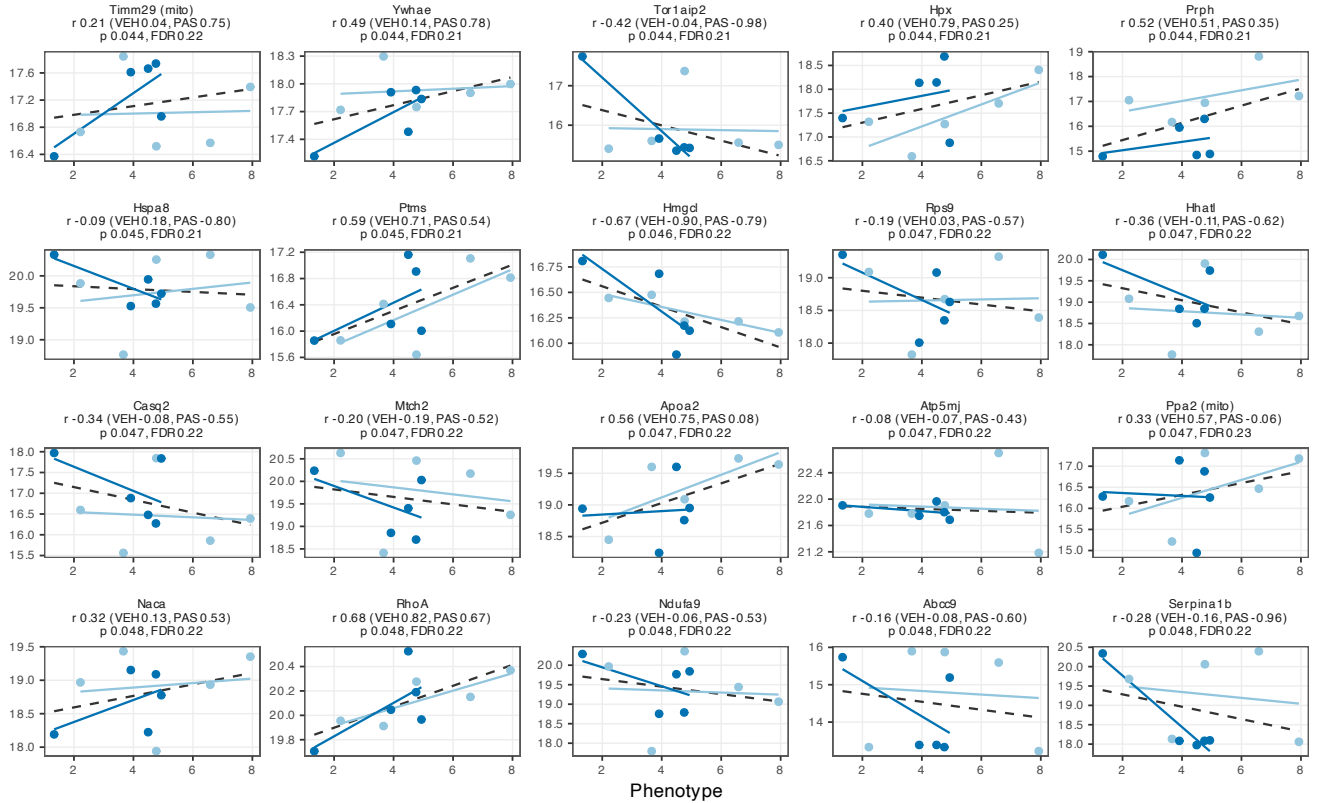
**Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



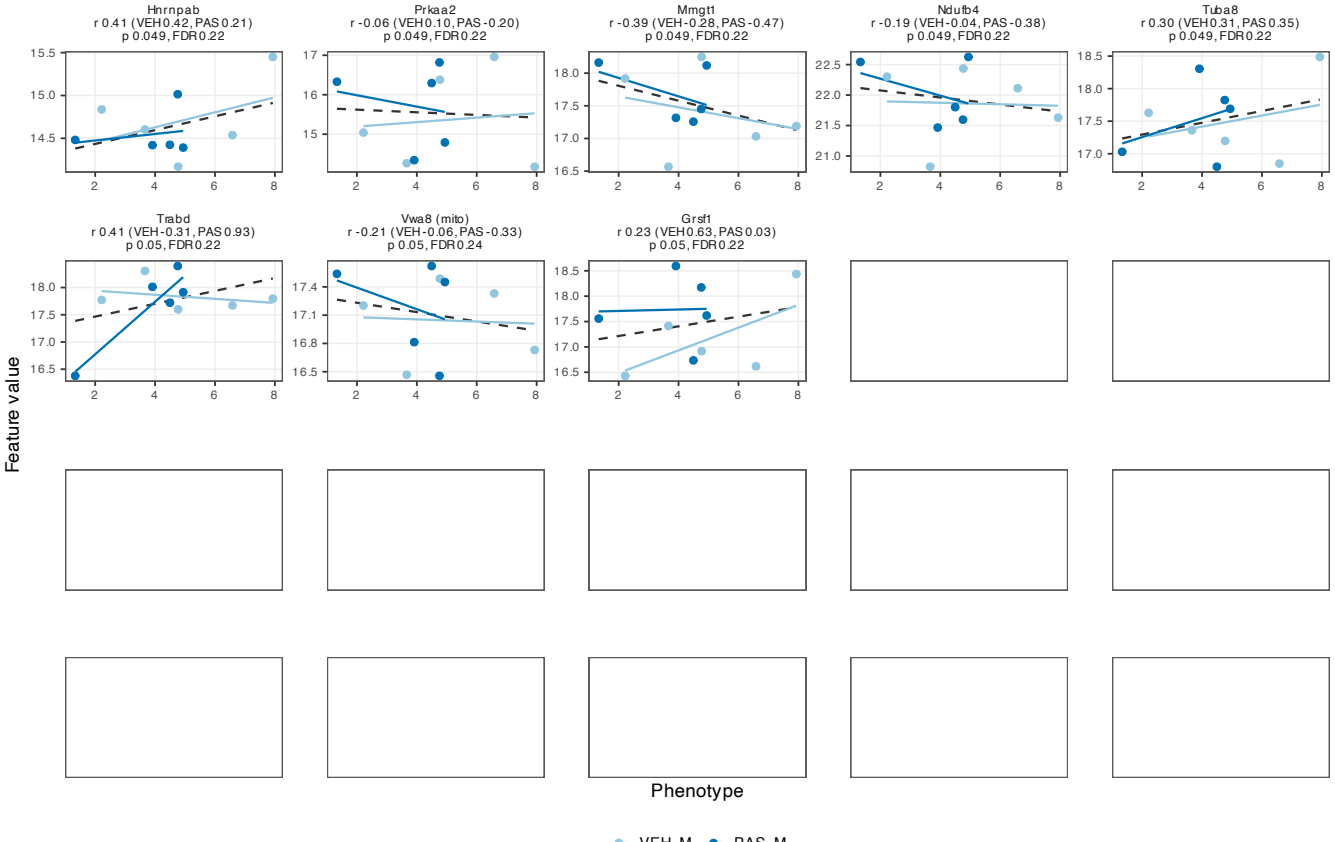
**Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



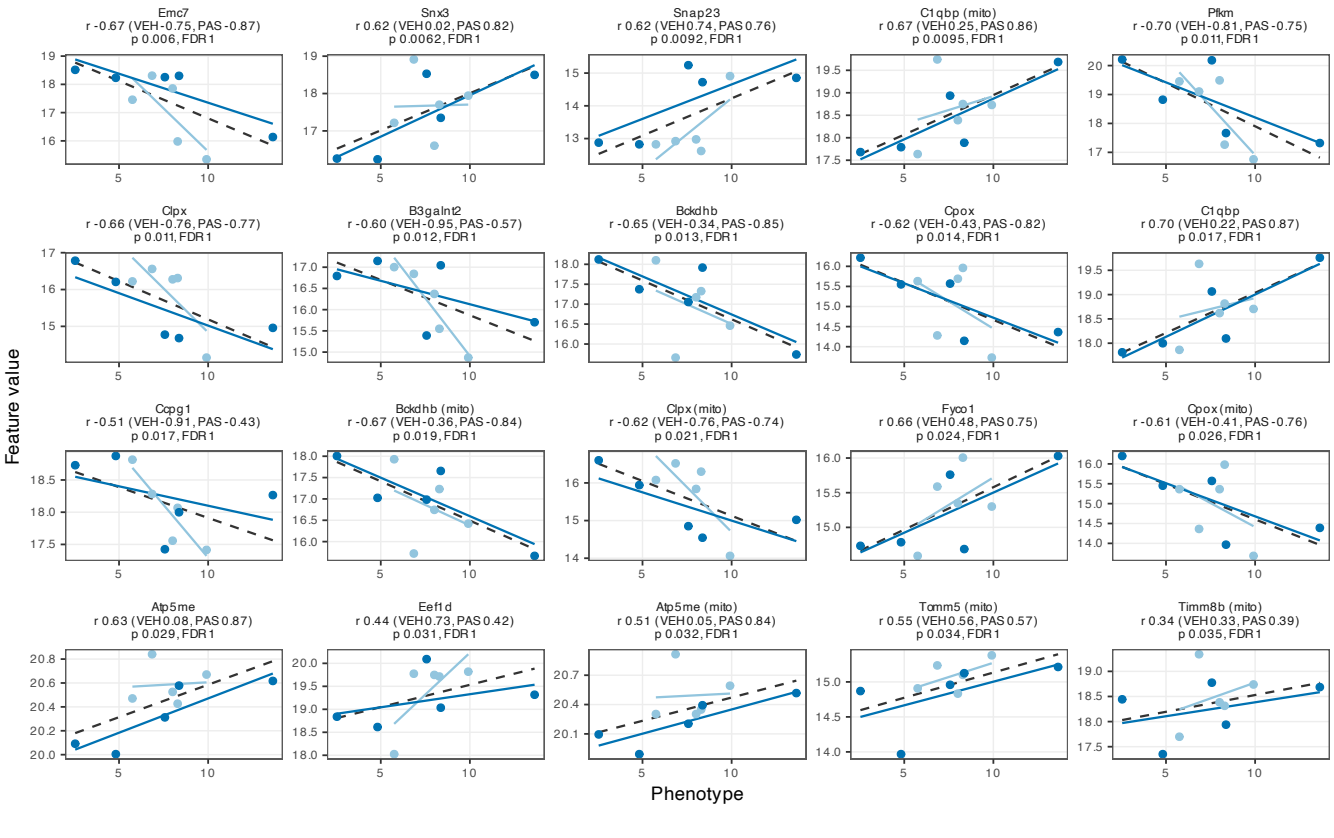
**Male: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 528)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



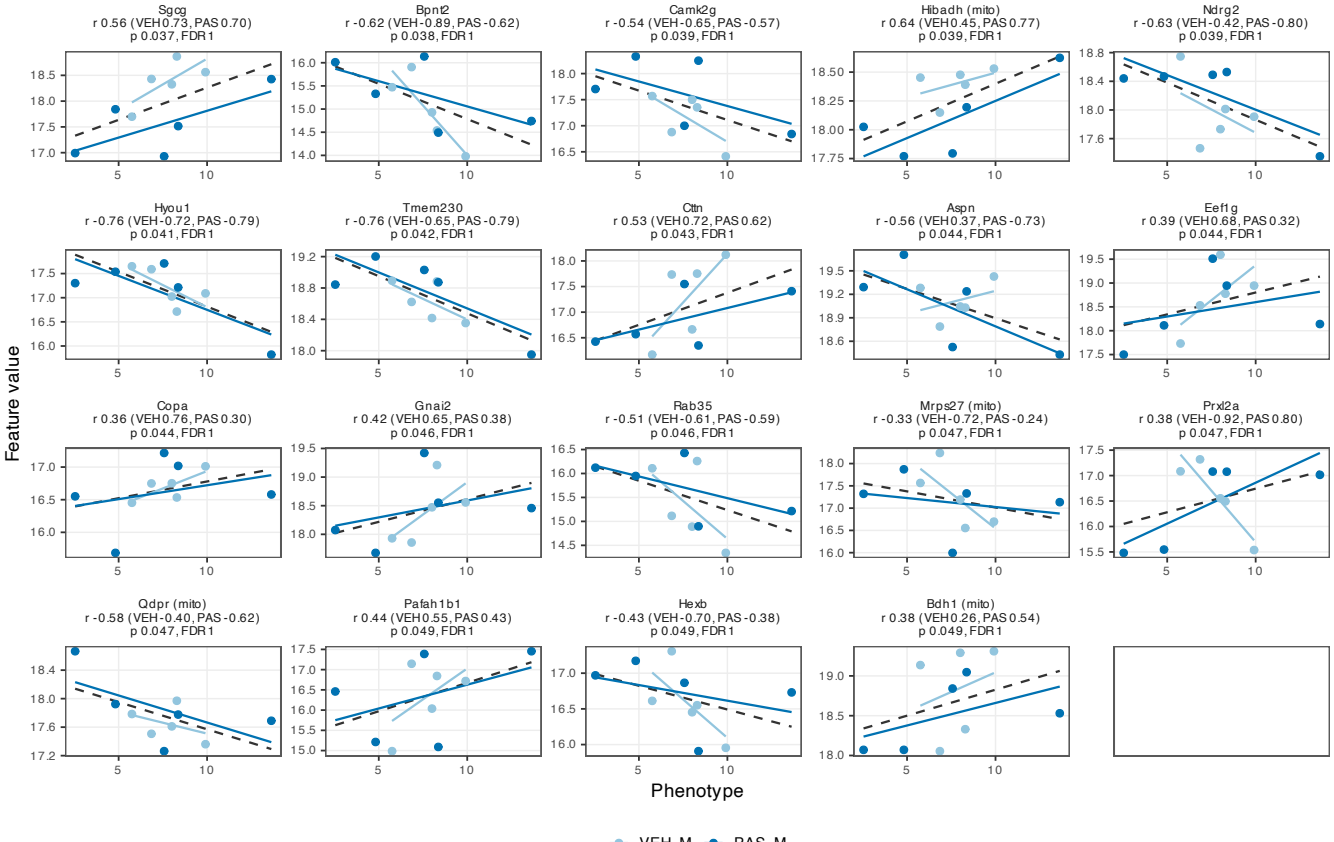
**Male: Proteins associated with Soleus octanoyl, every association at p ≤ 0.05 (n = 39)**

*Phenotype effect of PAS in this sex: -16% (p 0.57), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



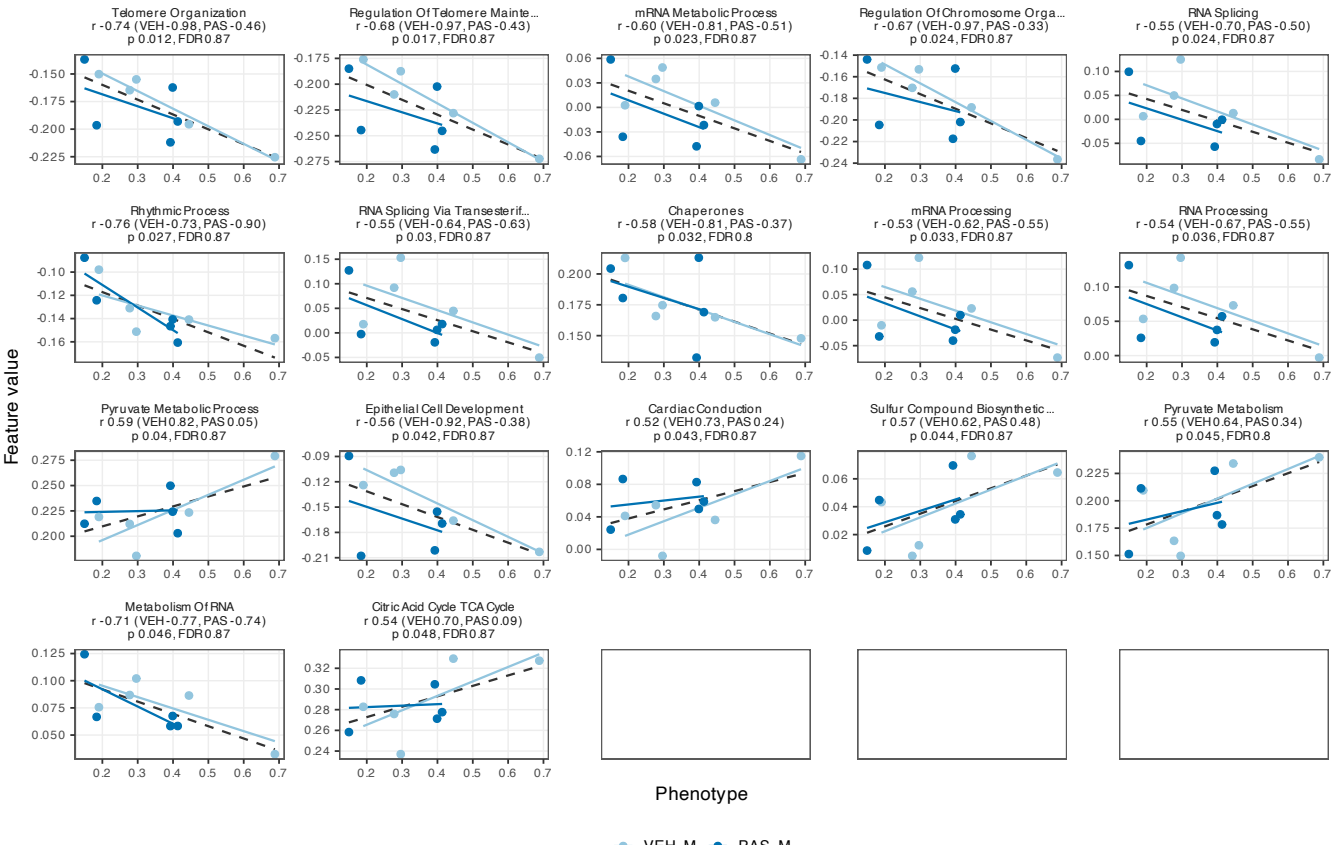
**Male: Proteins associated with Soleus octanoyl, every association at p ≤ 0.05 (n = 39)**

*Phenotype effect of PAS in this sex: -16% (p 0.57), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



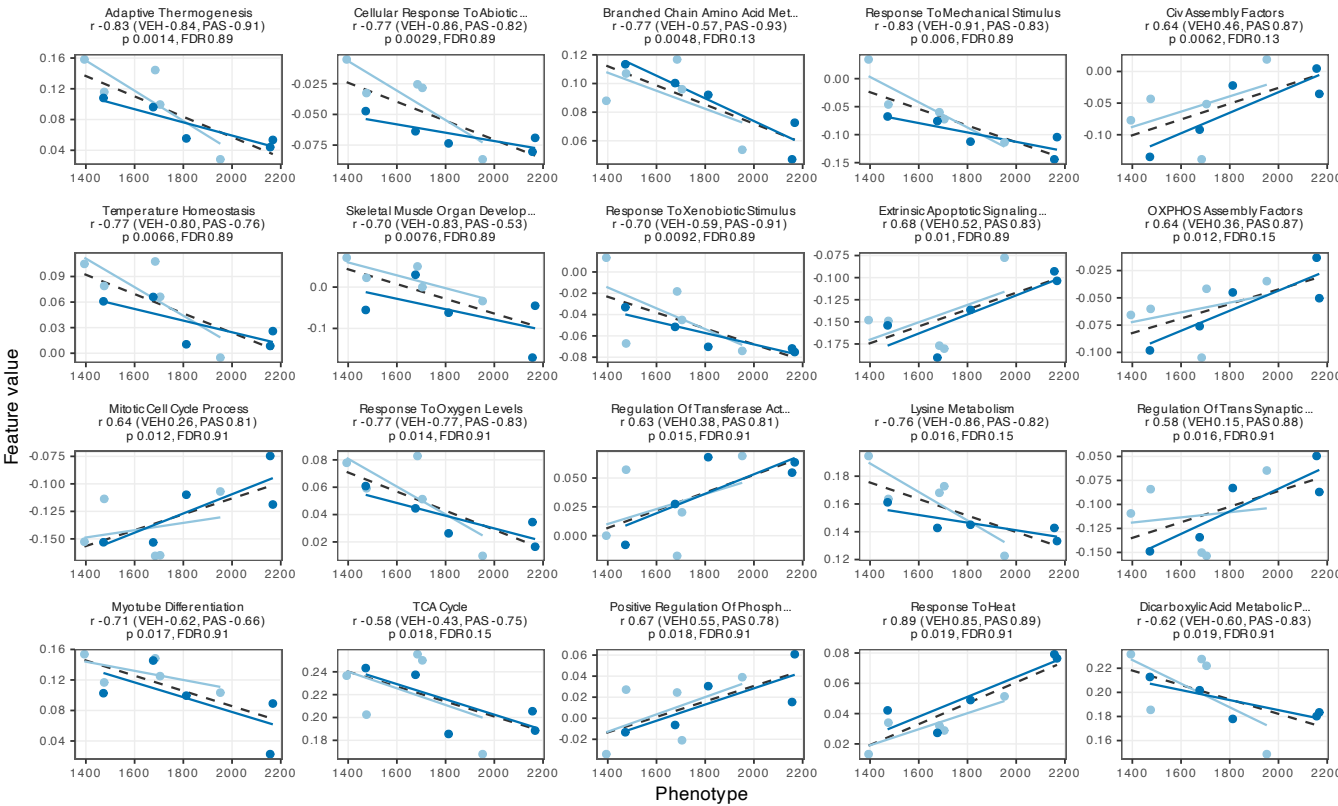
**Male: Pathways associated with Capillary density, every association at p ≤ 0.05 (n = 17)**

*Phenotype effect of PAS in this sex: -18% (p 0.54), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



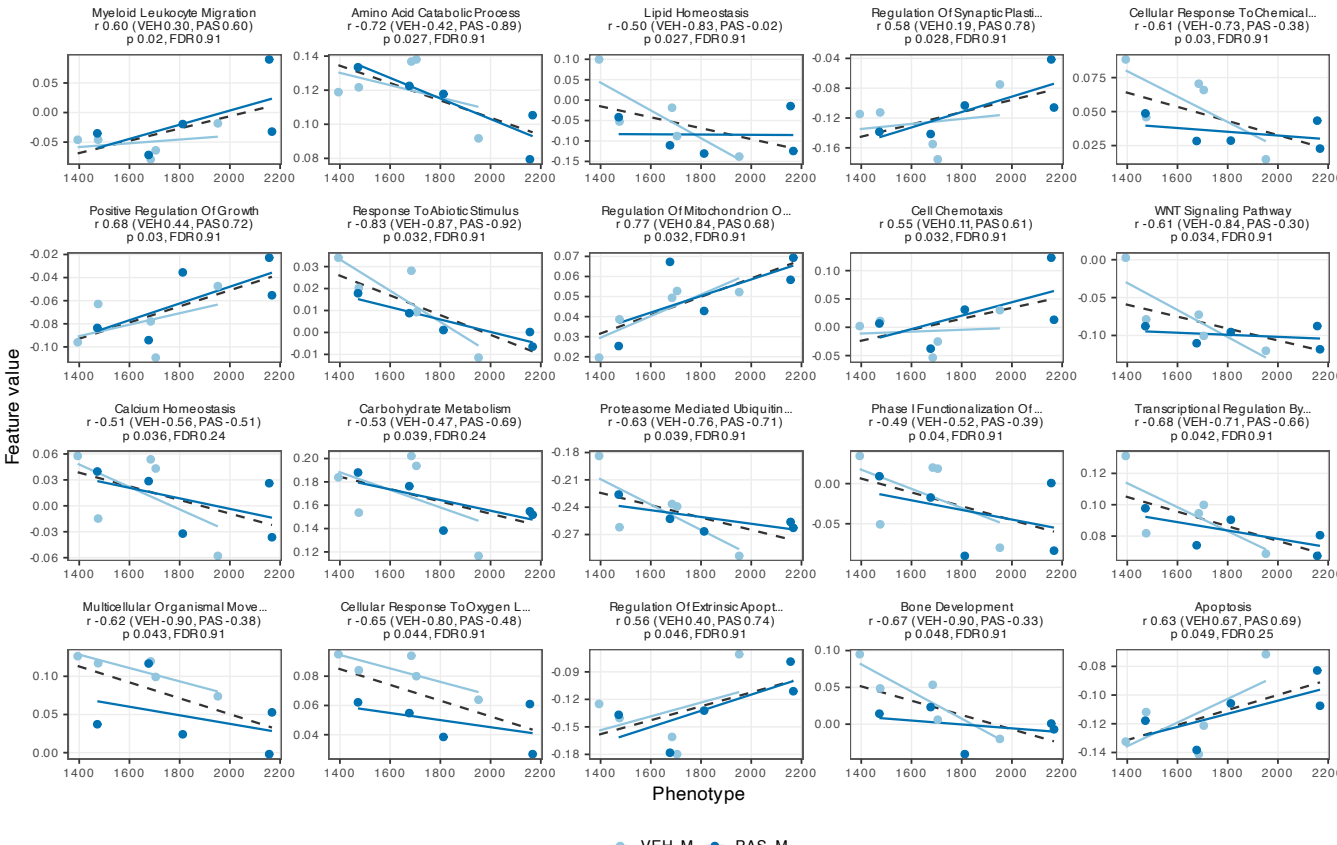
**Male: Pathways associated with CSA, every association at p ≤ 0.05 (n = 42)**

*Phenotype effect of PAS in this sex: +13% (p 0.25), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



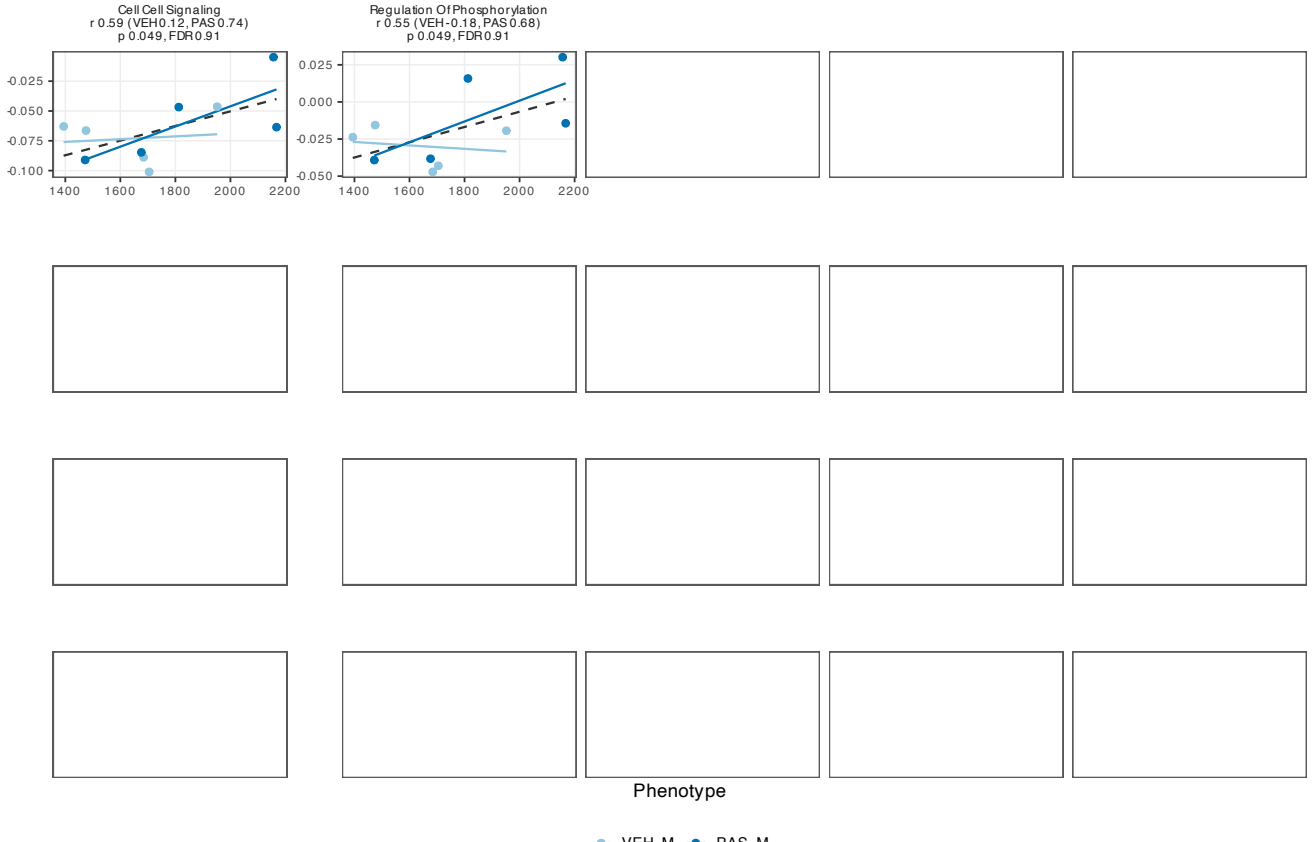
**Male: Pathways associated with CSA, every association at p ≤ 0.05 (n = 42)**

*Phenotype effect of PAS in this sex: +13% (p 0.25), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



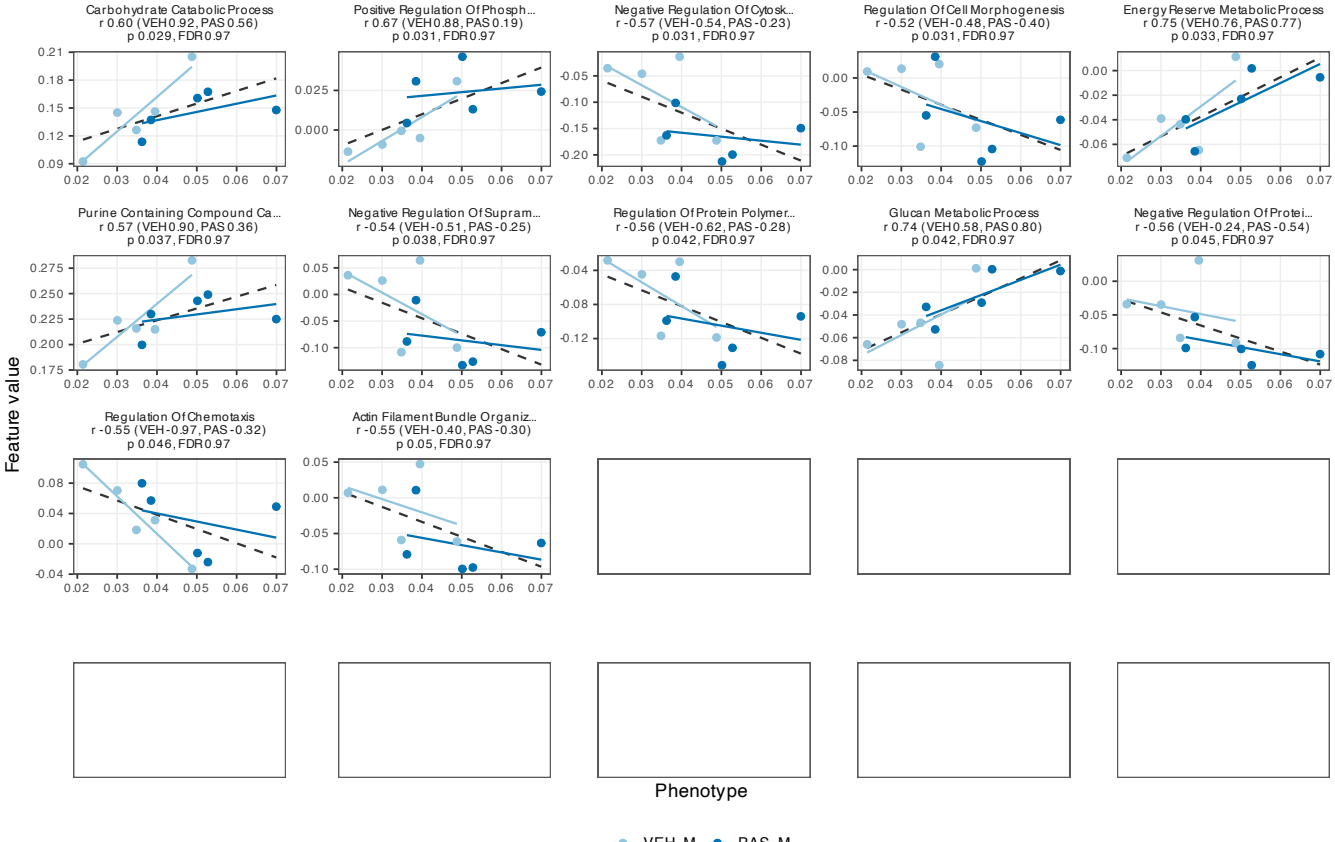
**Male: Pathways associated with CSA, every association at p ≤ 0.05 (n = 42)**

*Phenotype effect of PAS in this sex: +13% (p 0.25), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



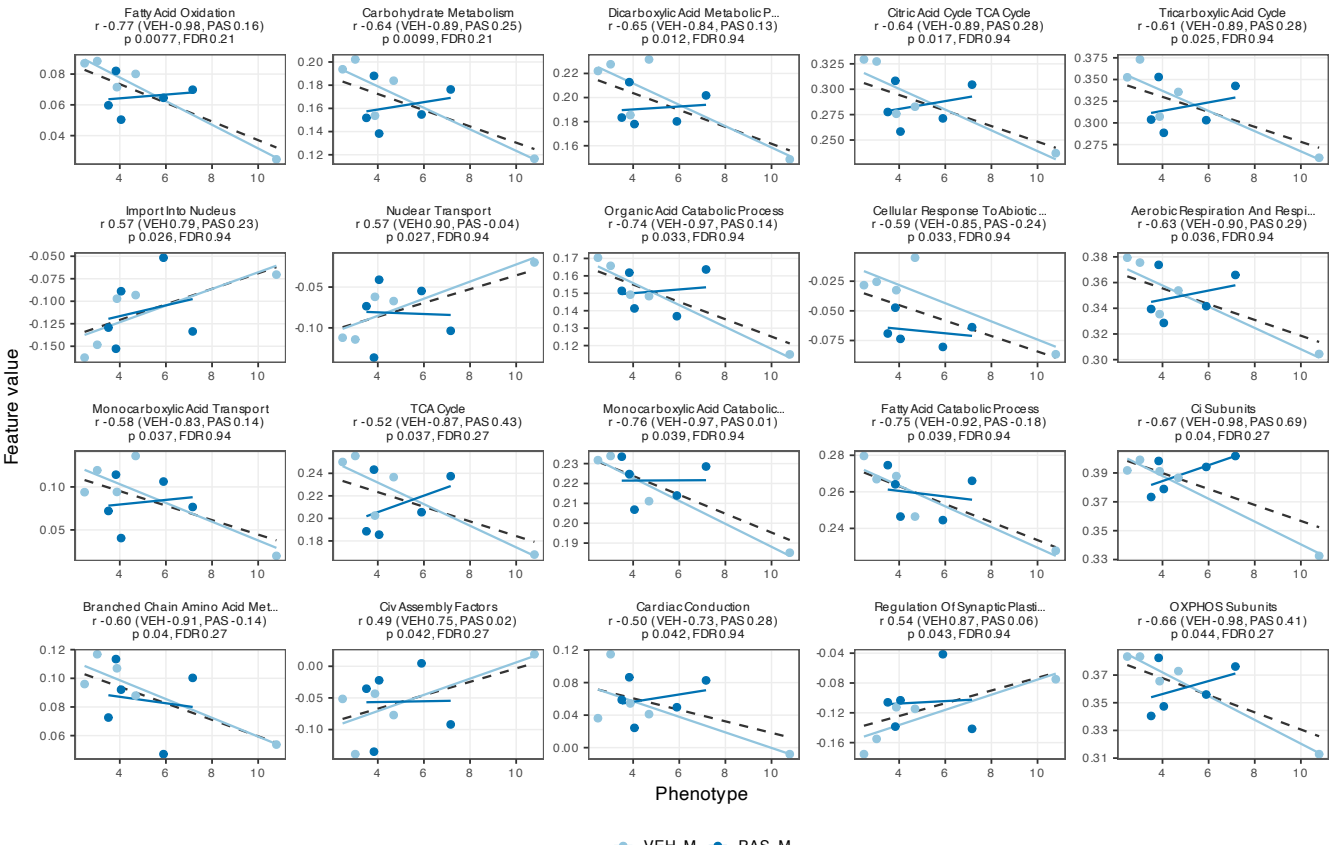
**Male: Pathways associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 12)**

*Phenotype effect of PAS in this sex: +43% (p 0.086), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



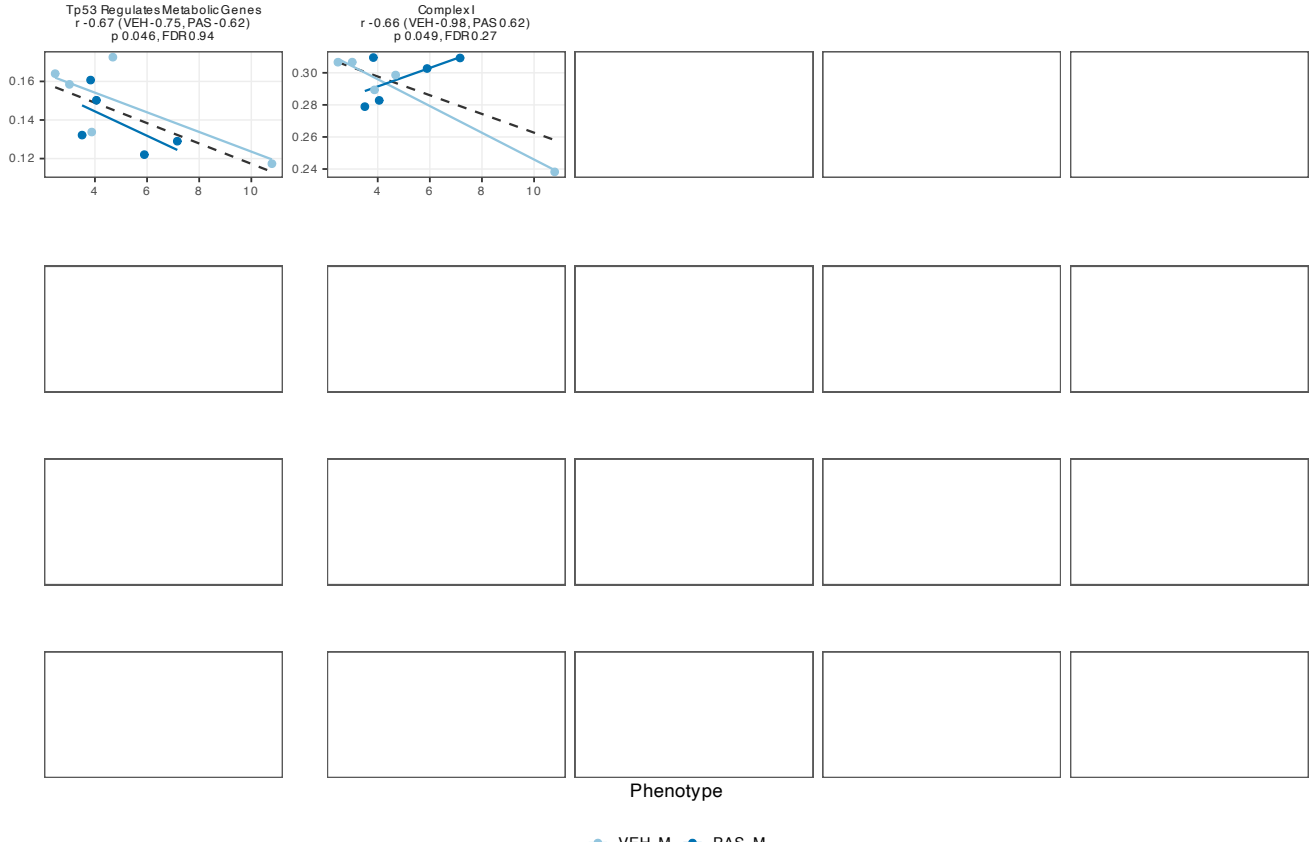
**Male: Pathways associated with Soleus pyruvate, every association at p ≤ 0.05 (n = 22)**

*Phenotype effect of PAS in this sex: +9% (p 0.77), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



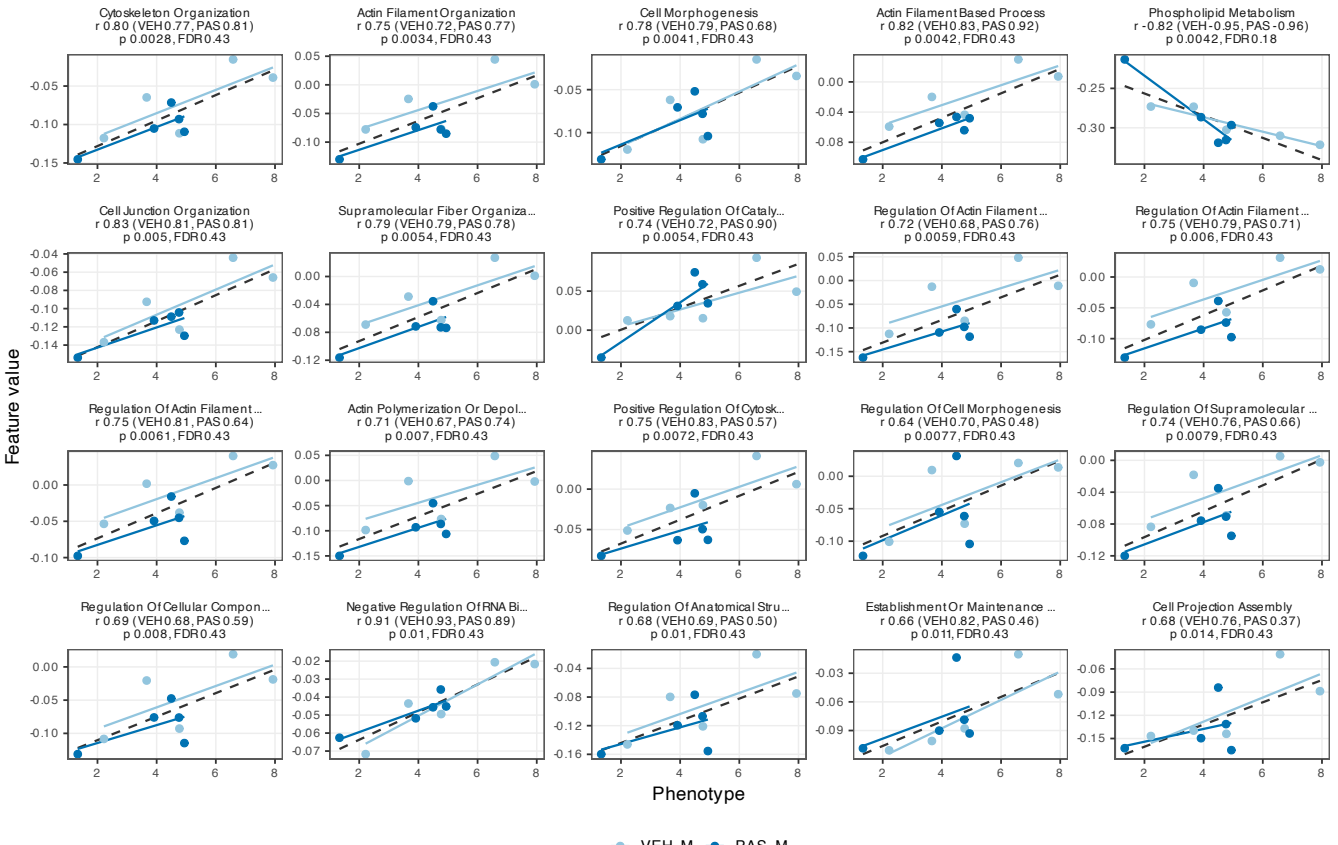
**Male: Pathways associated with Soleus pyruvate, every association at p ≤ 0.05 (n = 22)**

*Phenotype effect of PAS in this sex: +9% (p 0.77), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



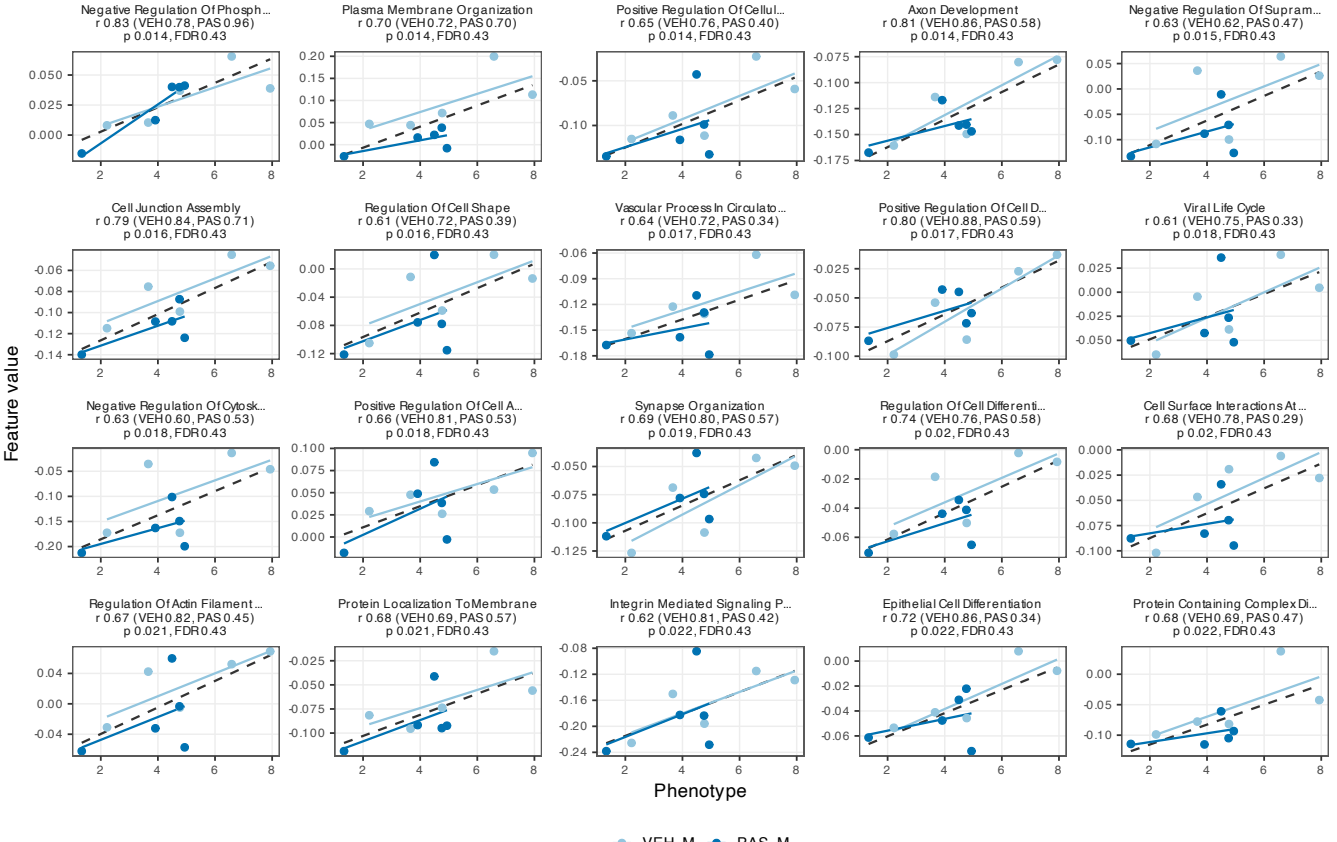
**Male: Pathways associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 93)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



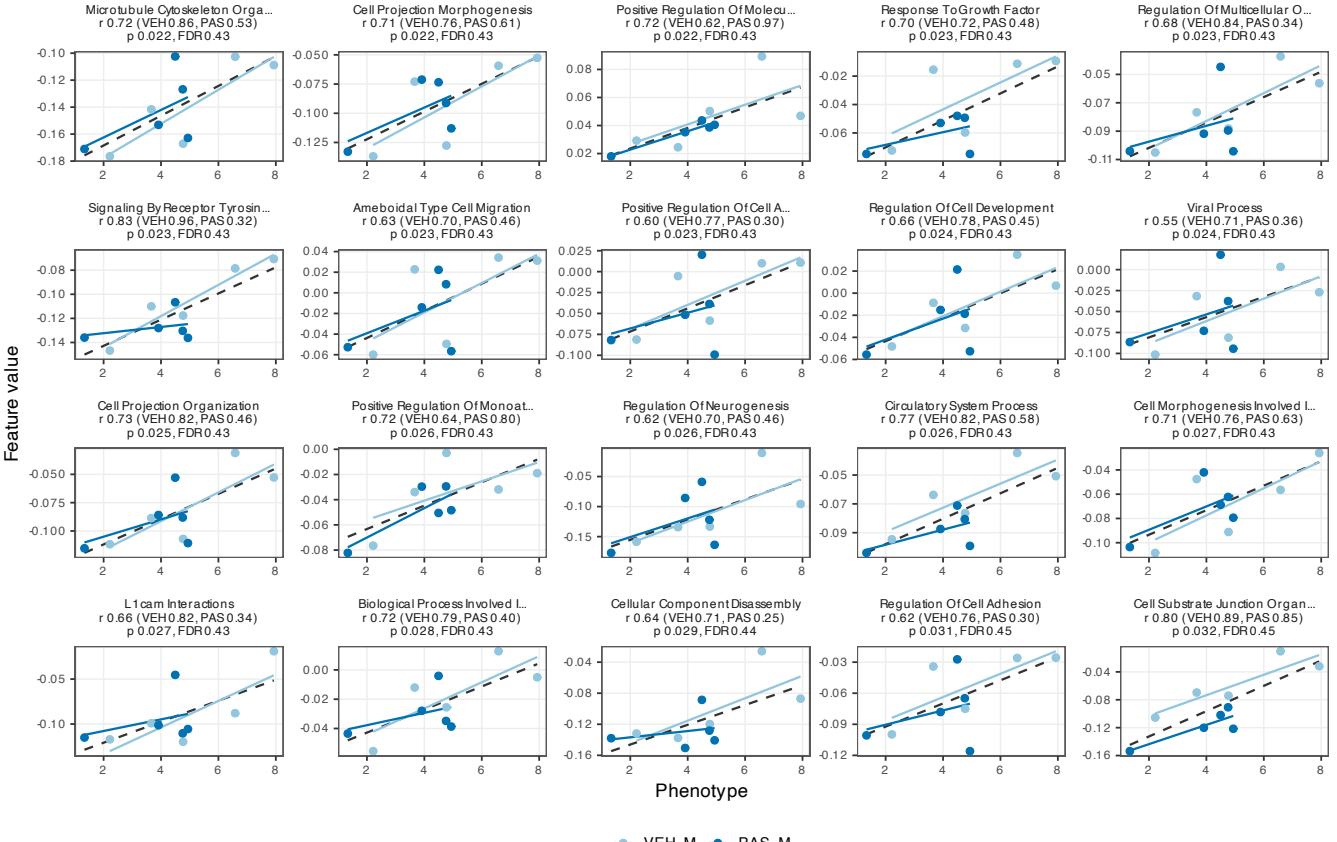
**Male: Pathways associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 93)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



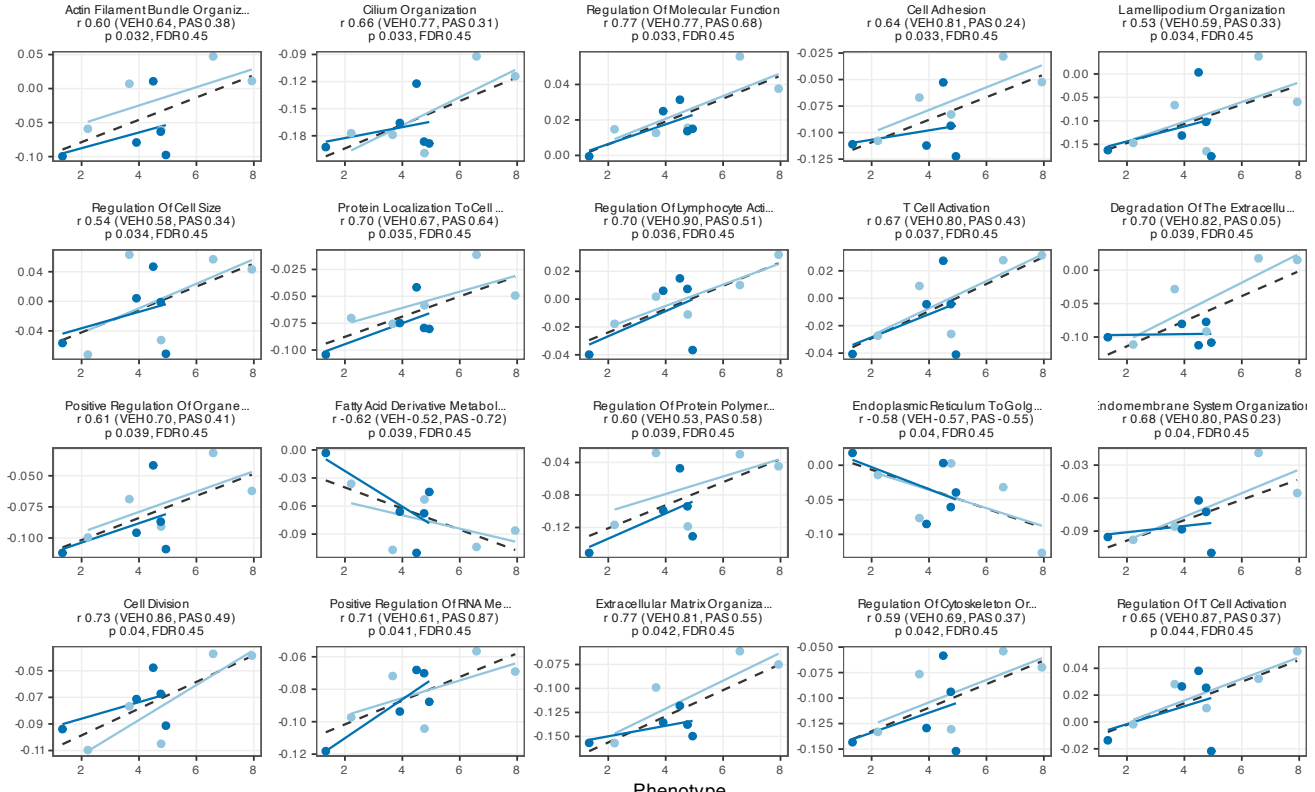
**Male: Pathways associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 93)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



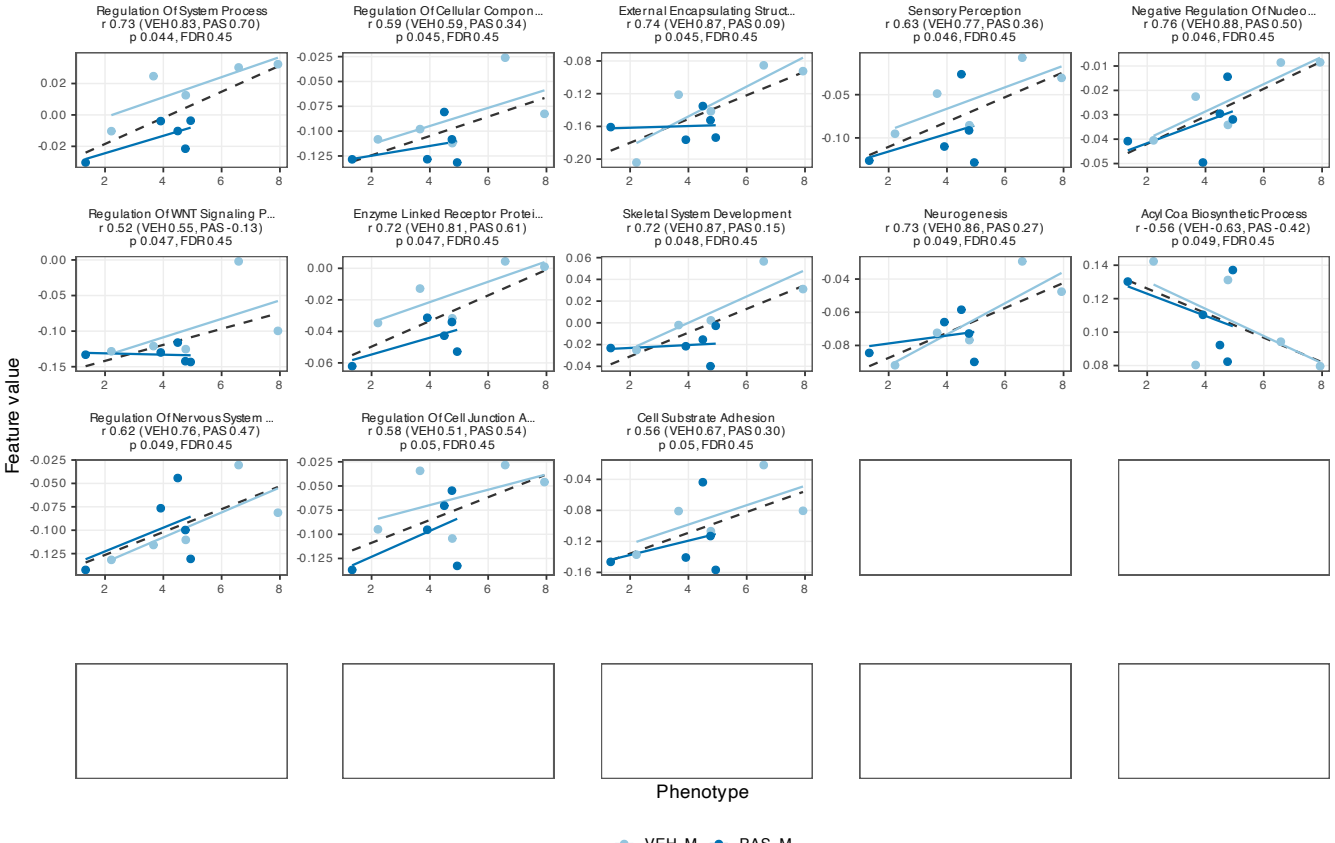
**Male: Pathways associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 93)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



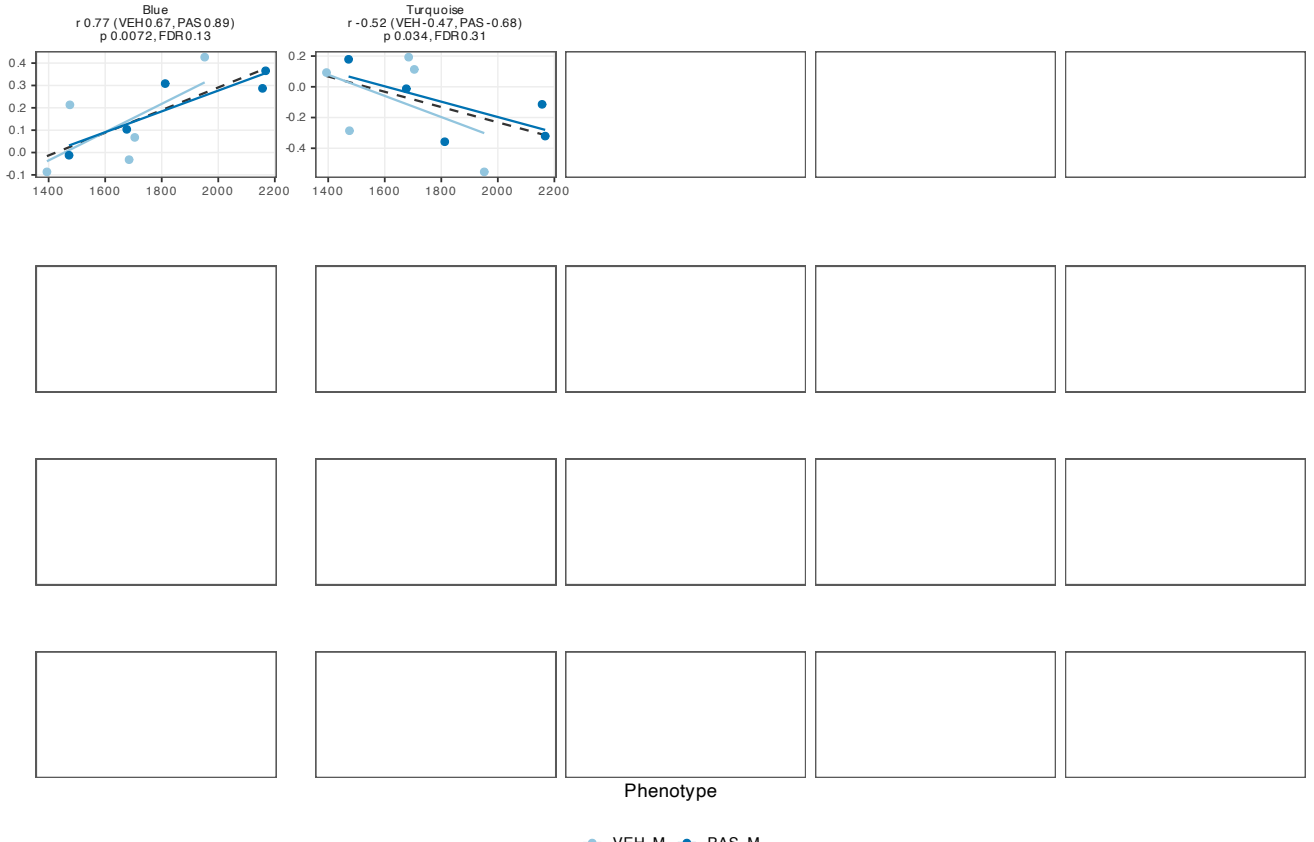
**Male: Pathways associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 93)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



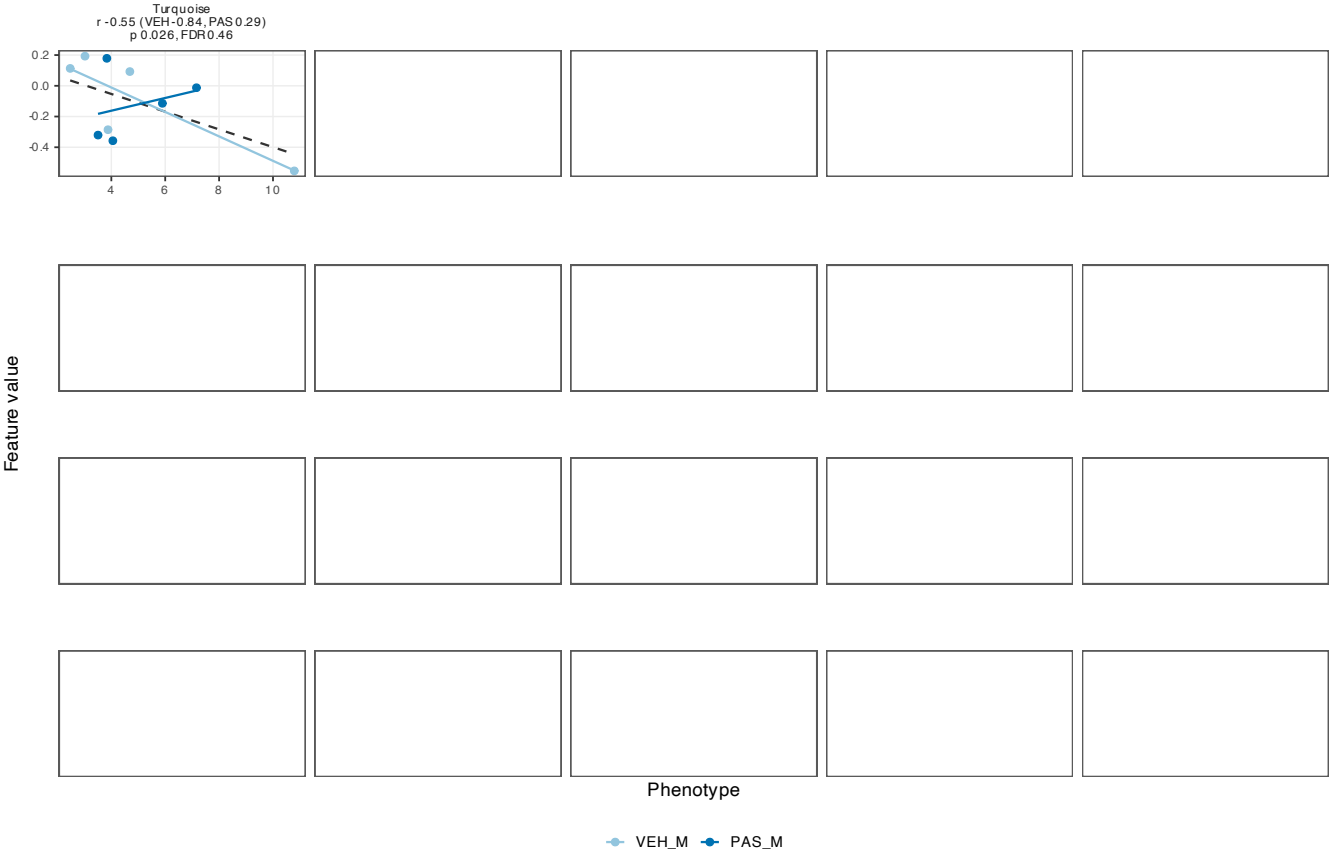
**Male: Modules associated with CSA, every association at p ≤ 0.05 (n = 2)**

*Phenotype effect of PAS in this sex: +13% (p 0.25), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



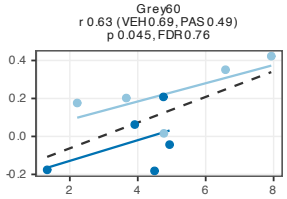
**Male: Modules associated with Soleus pyruvate, every association at p ≤ 0.05 (n = 1)**

*Phenotype effect of PAS in this sex: +9% (p 0.77), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



**Male: Modules associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 1)**

*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



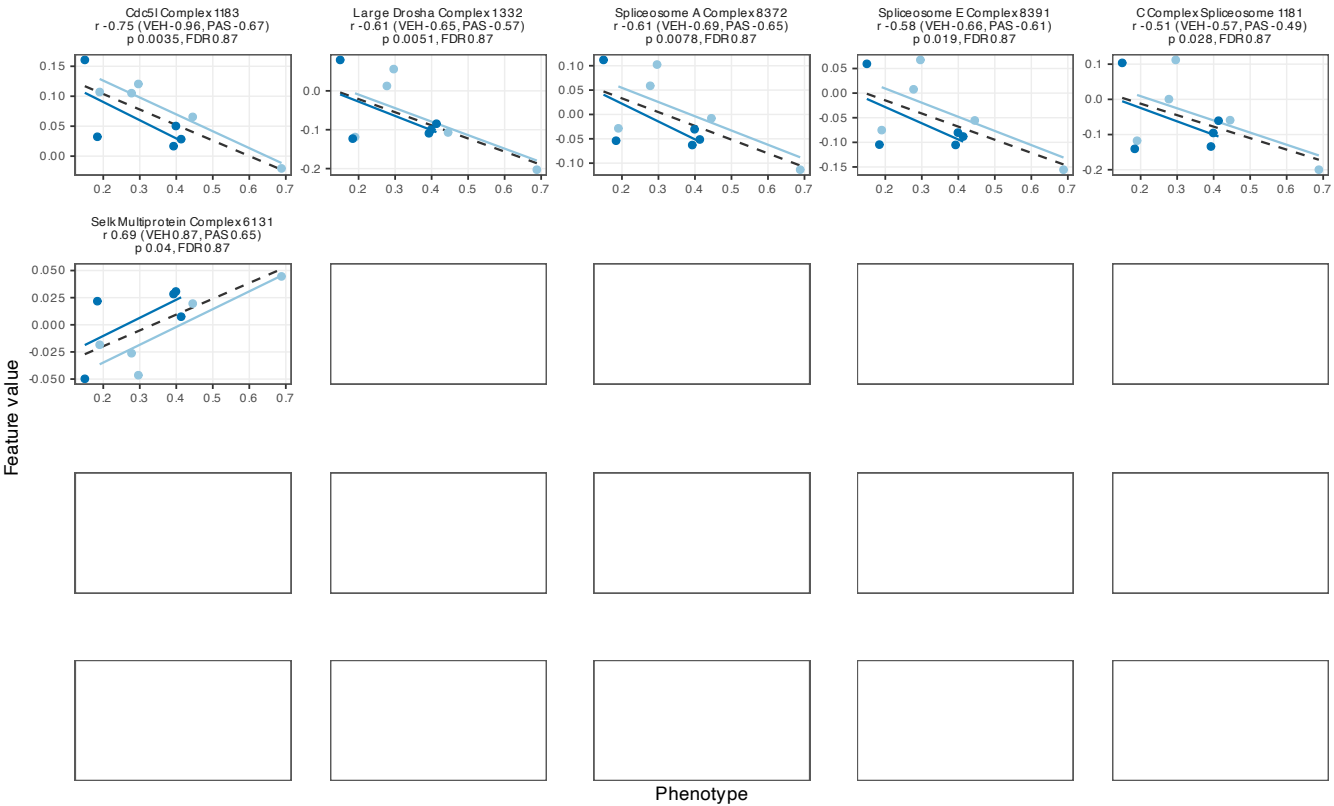

Feature value

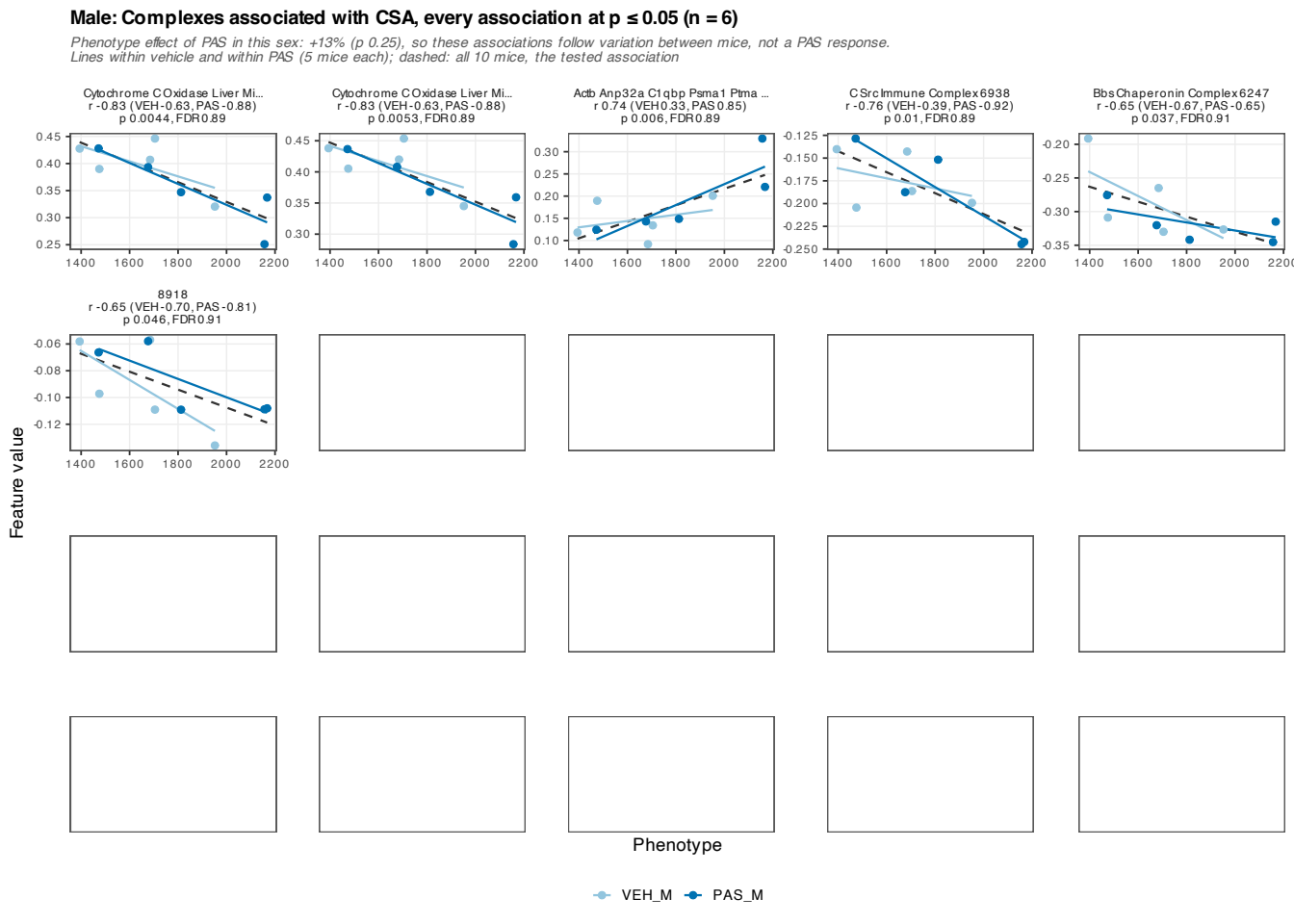
Phenotype

VEH_M PAS_M

**Male: Complexes associated with Capillary density, every association at p ≤ 0.05 (n = 6)**

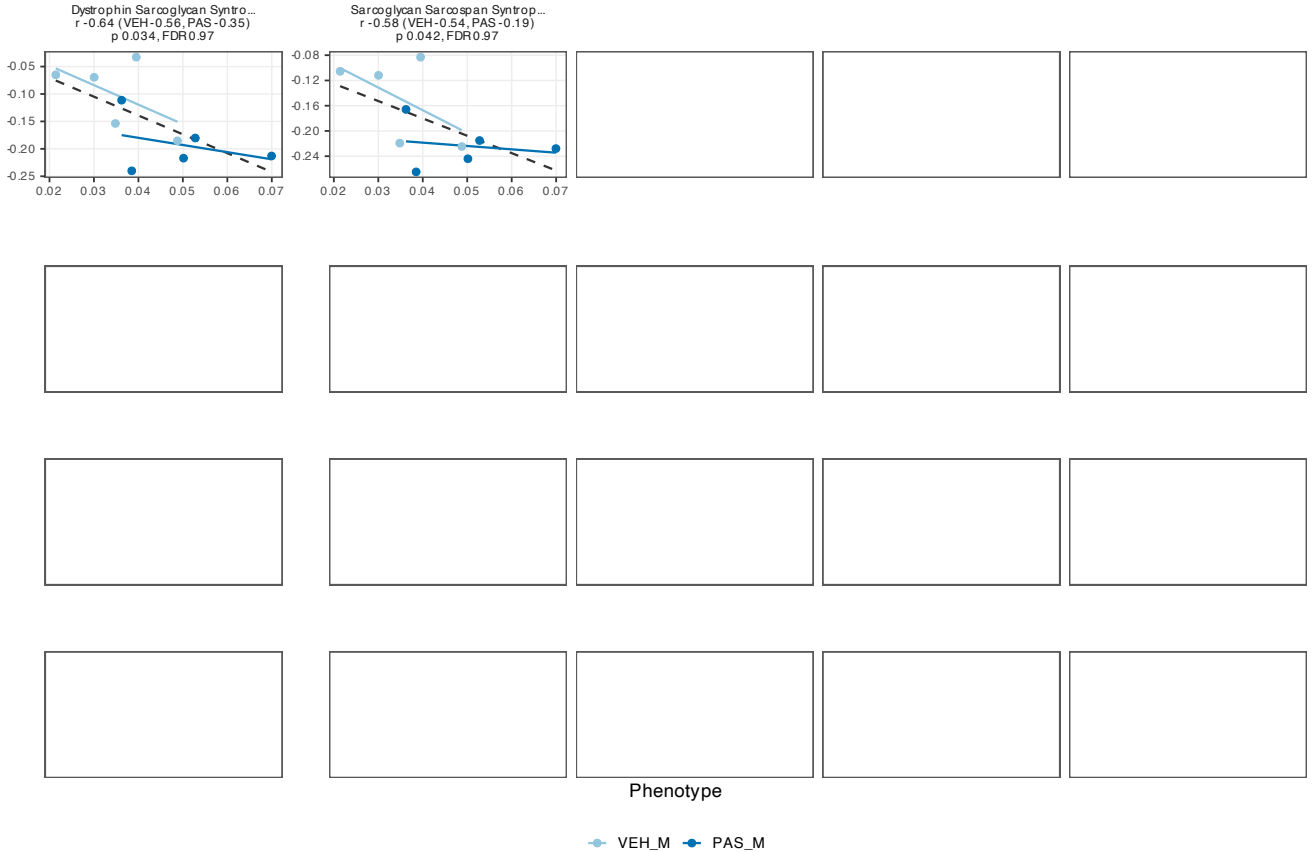
*Phenotype effect of PAS in this sex: -18% (p 0.54), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*




Male: Complexes associated with CSA, every association at p ≤ 0.05 (n = 6)
Phenotype effect of PAS in this sex: +13% (p 0.25), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association
Cytochrome C Oxidase Liver Mi...
r -0.83 (VEH -0.63, PAS -0.88)
p 0.0044, FDR 0.89
Cytochrome C Oxidase Liver Mi...
r -0.83 (VEH -0.63, PAS -0.88)
p 0.0053, FDR 0.89
Actb Anp32a C1qbp Psma1 Ptma ...
r 0.74 (VEH 0.33, PAS 0.85)
p 0.006, FDR 0.89
C Src Immune Complex 6938
r -0.76 (VEH -0.39, PAS -0.92)
p 0.01, FDR 0.89
Bbs Chaperonin Complex 6247
r -0.65 (VEH -0.67, PAS -0.65)
p 0.037, FDR 0.91
8918
r -0.65 (VEH -0.70, PAS -0.81)
p 0.046, FDR 0.91
Feature value
Phenotype
VEH_M
PAS_M

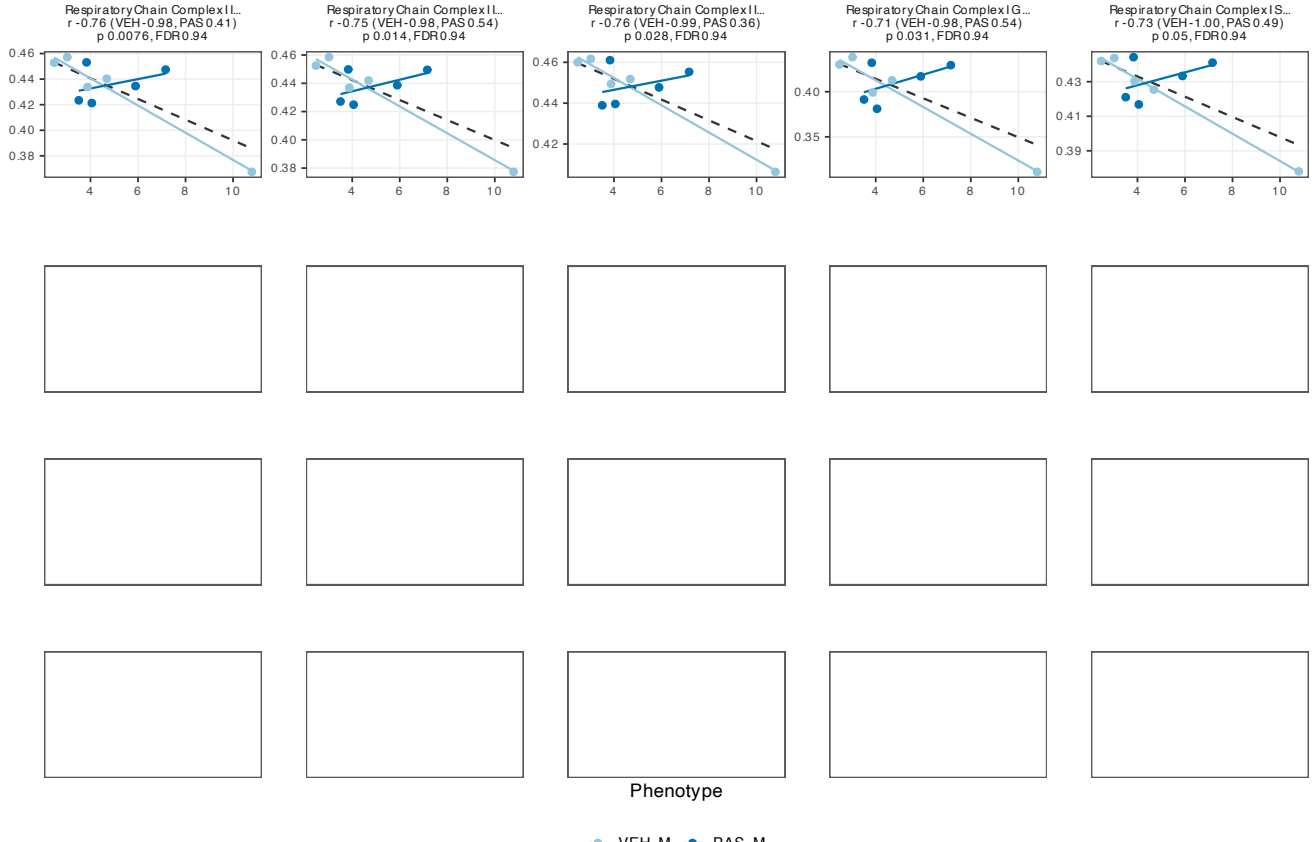
**Male: Complexes associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 2)**

*Phenotype effect of PAS in this sex: +43% (p 0.086), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



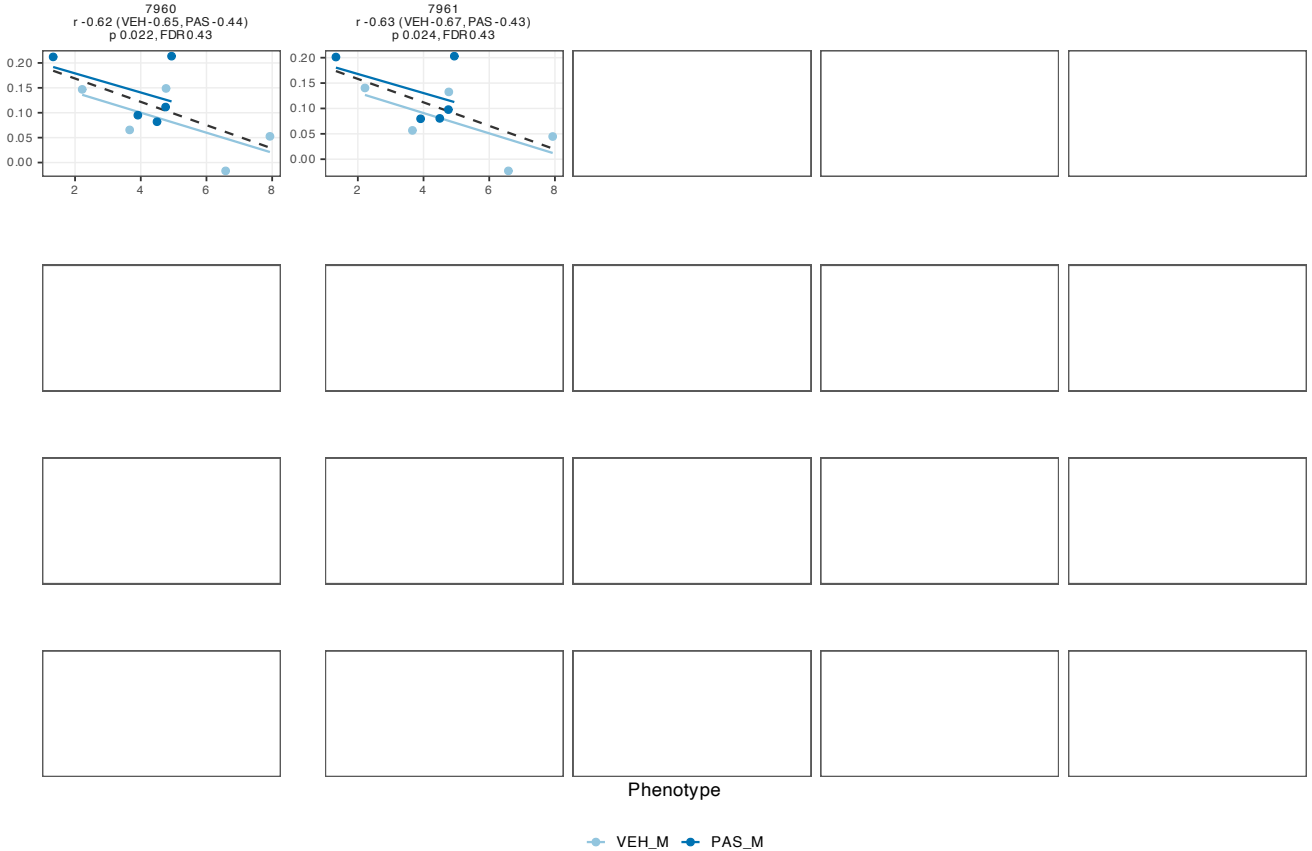
**Male: Complexes associated with Soleus pyruvate, every association at p ≤ 0.05 (n = 5)**

*Phenotype effect of PAS in this sex: +9% (p 0.77), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



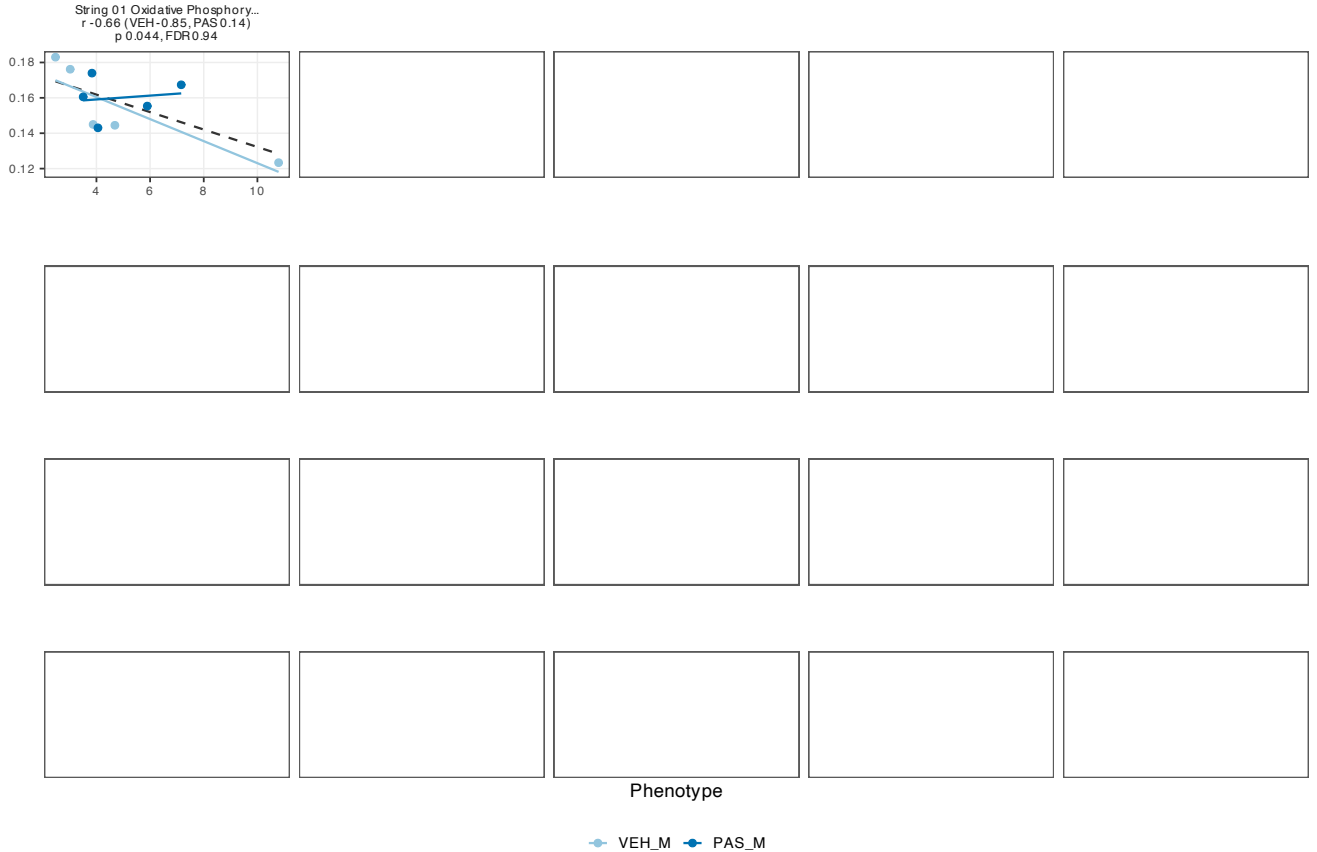
**Male: Complexes associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 2)**

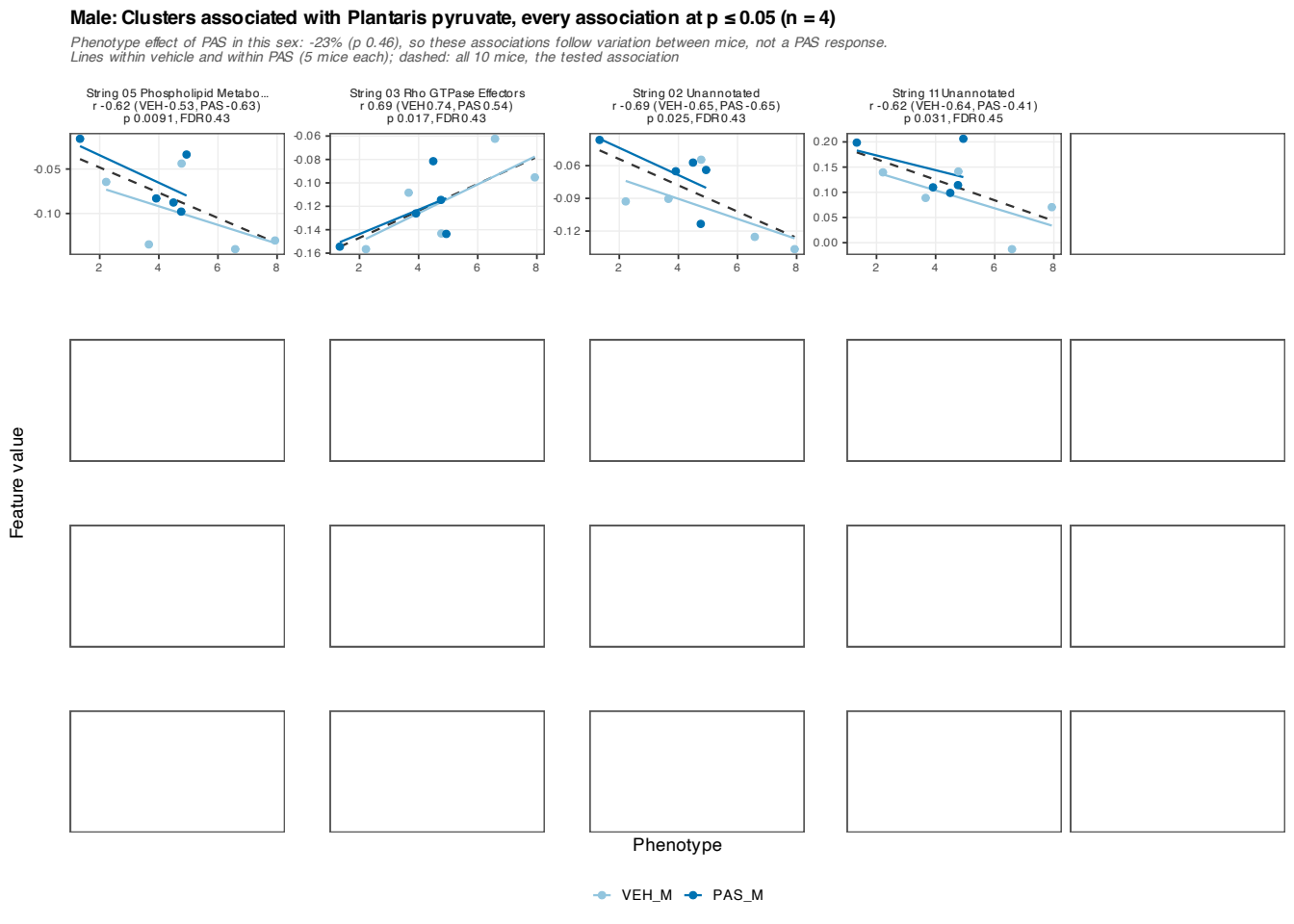
*Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.*
*Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*



**Male: Clusters associated with Soleus pyruvate, every association at p ≤ 0.05 (n = 1)**

*Phenotype effect of PAS in this sex: +9% (p 0.77), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association*




Male: Clusters associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 4)
Phenotype effect of PAS in this sex: -23% (p 0.46), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association
String 05 Phospholipid Metabo…
r -0.62 (VEH -0.53, PAS -0.63)
p 0.0091, FDR 0.43
String 03 Rho GTPase Effectors
r 0.69 (VEH 0.74, PAS 0.54)
p 0.017, FDR 0.43
String 02 Unannotated
r -0.69 (VEH -0.65, PAS -0.65)
p 0.025, FDR 0.43
String 11 Unannotated
r -0.62 (VEH -0.64, PAS -0.41)
p 0.031, FDR 0.45
Feature value
Phenotype
VEH_M
PAS_M